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(54) **MAPT ANTISENSE OLIGONUCLEOTIDE**

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(71) Applicant: **ELI LILLY AND COMPANY**,  
Indianapolis, IN (US)  
(72) Inventors: **Jeremy Simon Louis DESAPHY**,  
Indianapolis, IN (US); **Roberto EL**  
**KHOURY SEJNAUI**, Cambridge, MA  
(US); **Maire Fiona JUNG**, Lincoln,  
MA (US); **Michael Payam**  
**MOAZAMI**, Boston, MA (US); **Bum**  
**Sok SEO**, Quincy, MA (US); **Jibo**  
**WANG**, Carmel, IN (US)

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(57) **ABSTRACT**

Provided herein are MAPT antisense oligonucleotides and compositions comprising a MAPT antisense oligonucleotide. Also provided herein are methods of using the MAPT antisense oligonucleotides or compositions comprising a MAPT antisense oligonucleotide for reducing MAPT expression and/or treating tauopathy in a subject.

**Specification includes a Sequence Listing.**

**MAPT ANTISENSE OLIGONUCLEOTIDE****SEQUENCE LISTING**

**[0001]** The present application is being filed along with a Sequence Listing in ST.26 XML format. The Sequence Listing is provided as a file titled “31012\_US” created Feb. 6, 2025, and is 726 kilobytes in size. The Sequence Listing information in the ST.26 XML format is incorporated herein by reference in its entirety.

**BACKGROUND**

**[0002]** Microtubule associated protein Tau (MAPT) is encoded by the MAPT gene located on chromosome 17. Tau protein interacts with tubulin to stabilize the microtubules and promote tubulin assembly into microtubules. MAPT transcripts are differentially expressed throughout the body, predominantly in the central and peripheral nervous system.

**[0003]** The MAPT gene consists of 16 exons. Alternative mRNA splicing gives rise to multiple MAPT isoforms. At least six Tau isoforms exist in human brain, ranging from 352 to 441 amino acids long. Alternative splicing of exons 2 and/or 3 leads to inclusion of zero, one, or two copies of the N-terminal acidic domain, which are referred to as 0N, 1N, or 2N Tau, respectively. The Tau isoforms that include exon 10, which encodes an additional microtubule-binding domain, are referred to as “4R Tau”, as it has four microtubule-binding domains. The Tau isoforms without exon 10 are referred to as “3R Tau”, as it has three microtubule-binding domains.

**[0004]** Mutations in MAPT and hyperphosphorylation of Tau protein can cause aggregation and deposition of Tau in pathogenic neurofibrillary tangles, causing progressive neurodegenerative disorders such as Alzheimer’s disease, frontotemporal dementia (FTD), progressive supranuclear palsy (PSP), and other tauopathies.

**[0005]** Antisense oligonucleotides (ASOs) are powerful and versatile agents being used in a number of applications including mRNA reduction, translation arrest, miRNA inhibition, splicing modulation, and polyadenylation site selection. MAPT antisense oligonucleotides have been described before, e.g., in US 2016/0145617.

**[0006]** There is currently no FDA-approved disease-modifying therapeutic agent specifically for reducing MAPT and treating tauopathies. Accordingly, there remains a need for therapeutic agents that can reduce or adjust MAPT expression for treating tauopathies, e.g., by utilizing antisense oligonucleotides.

**SUMMARY OF INVENTION**

**[0007]** Provided herein are MAPT antisense oligonucleotides and compositions comprising a MAPT antisense oligonucleotide. Also provided herein are methods of using the MAPT antisense oligonucleotides or compositions comprising a MAPT antisense oligonucleotide for reducing MAPT expression and/or treating tauopathy in a subject.

**[0008]** In one aspect, provided herein are antisense oligonucleotides targeting MAPT. In some embodiments, provided herein are MAPT antisense oligonucleotides that comprises a nucleic acid sequence selected from any one of SEQ ID NOs: 1-11, 30-32, or 36-63, wherein optionally one or more nucleotides are independently modified nucleotides, and wherein optionally one or more internucleotide linkages

are modified internucleotide linkages. In some embodiments, the antisense oligonucleotide is single stranded.

**[0009]** The MAPT antisense oligonucleotides described herein may include modifications. The modifications can be made to one or more nucleotides (e.g., to nucleobase, sugar, or phosphate) or to one or more internucleotide linkages.

**[0010]** In some embodiments, the antisense oligonucleotide provided herein comprises one or more modified nucleotides. In some embodiments, such modified nucleotides comprise a modified nucleobase, e.g., 5-methylcytosine. In some embodiments, each C in the antisense oligonucleotide is a 5-methylcytosine. In some embodiments, such modified nucleotides comprise a modified sugar, e.g., a modified ribose or deoxyribose. In some embodiments, the modified sugar is a 2'-O-methoxyethyl (2'-O-MOE) modified sugar, 2'-O-methyl modified sugar, or 2'-fluoro modified sugar. In some embodiments, the modified sugar is a 2'-O-MOE modified sugar. In some embodiments, the modified sugar is a locked nucleic acid (LNA).

**[0011]** In some embodiments, the antisense oligonucleotides provided herein are 12 to 30 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 15 to 25 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 16 to 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 18 to 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 16 nucleotides in length.

**[0012]** In some embodiments, the MAPT antisense oligonucleotide provided herein are gapmers, wherein the first to fifth nucleotides each comprise a 2'-O-MOE modified sugar, wherein the sixth to fifteenth nucleotides each comprise a 2'-deoxynucleoside, and wherein the sixteenth to twentieth nucleotides each comprise a 2'-O-MOE modified sugar.

**[0013]** In some embodiments, the MAPT antisense oligonucleotide provided herein are gapmers, wherein the first to third nucleotides each comprise LNA, wherein the fourth to thirteenth nucleotides each comprise a 2'-deoxynucleoside, and wherein the fourteenth to sixteenth nucleotides each comprise LNA.

**[0014]** In some embodiments, the antisense oligonucleotides provided herein comprise one or more modified internucleotide linkages. In some embodiments, the modified internucleotide linkage is a phosphorothioate linkage. In some embodiments, the antisense oligonucleotides provided herein comprise phosphodiester and phosphorothioate internucleotide linkage. In some embodiments, the internucleotide linkages of the MAPT antisense oligonucleotides are sooooooooooooo, or sooooooooooooo, from 5' end to 3' end, wherein each s is a phosphorothioate linkage and each o is a phosphodiester linkage. In some embodiments, the internucleotide linkages of the MAPT antisense oligonucleotides are all phosphorothioate linkages.

**[0015]** In some embodiments, the antisense oligonucleotides provided herein comprise a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94. In some embodiments, the antisense oligonucleotides provided herein consist of a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94.

**[0016]** In another aspect, provided herein are pharmaceutical compositions comprising a MAPT antisense oligonucleotide described herein and a pharmaceutically acceptable carrier. Also provided herein are pharmaceutical

compositions comprising a means for reducing MAPT expression in a cell and a pharmaceutically acceptable carrier.

**[0017]** In another aspect, provided herein are methods of reducing MAPT expression in a patient in need thereof, and such method comprises administering to the patient an effective amount of a MAPT antisense oligonucleotide or a pharmaceutical composition described herein. The MAPT antisense oligonucleotide can be administered to the patient intrathecally, intravenously, subcutaneously, or via intracranial magna injection.

**[0018]** In another aspect, provided herein are methods of treating a tauopathy in a patient in need thereof, and such method comprises administering to the patient an effective amount of the MAPT antisense oligonucleotide or a pharmaceutical composition described herein.

**[0019]** Also provided herein are methods of reducing MAPT expression in a cell (e.g., a neuron), and such methods can include contacting the cell with a MAPT antisense oligonucleotide described herein; and incubating the cell for a time sufficient for degradation of MAPT mRNA, thereby reducing MAPT expression in the cell.

**[0020]** In another aspect, provided herein are MAPT antisense oligonucleotides or pharmaceutical compositions comprising a MAPT antisense oligonucleotide for use in reducing MAPT expression. Also provided herein are MAPT antisense oligonucleotides or the pharmaceutical composition comprising a MAPT antisense oligonucleotide for use in a therapy. Also provided herein are MAPT antisense oligonucleotides or pharmaceutical compositions comprising a MAPT antisense oligonucleotide for use in the treatment of a tauopathy. Also provided herein are uses of MAPT antisense oligonucleotides in the manufacture of a medicament for the treatment of a tauopathy.

#### DETAILED DESCRIPTION

**[0021]** Provided herein are MAPT antisense oligonucleotides and compositions comprising a MAPT antisense oligonucleotide. Also provided herein are methods of using the MAPT antisense oligonucleotides or compositions com-

prising a MAPT antisense oligonucleotide for reducing MAPT expression and/or treating tauopathy in a subject.

**[0022]** An antisense oligonucleotide usually comprises a nucleic acid sequence complementary to the sequence of a target nucleic acid, e.g., a target genomic sequence, pre-mRNA, or mRNA sequence. Hybridization occurs when hydrogen bonding (e.g., Watson-Crick, Hoogsteen or reversed Hoogsteen hydrogen bonding) forms between the complementary nucleobases of the antisense oligonucleotide and the target nucleic acid. Non-complementary nucleobases between an antisense oligonucleotide and a target nucleic acid may be tolerated provided that the antisense oligonucleotide remains able to specifically hybridize to a target nucleic acid.

**[0023]** Antisense oligonucleotides can be designed to modulate the expression of a target protein through either RNase H-dependent or RNase H-independent manners (see Watts J K, et al., J Pathol. 2012 January; 226(2): 365-379). When an antisense oligonucleotide comprising a contiguous stretch of DNA hybridizes with a target RNA, the DNA-RNA heteroduplex recruit RNase H, which cleaves the target RNA in the duplex and promotes subsequent degradation of the RNA fragments by cellular nucleases. An antisense oligonucleotide can also decrease target expression independent of RNase H by sterically blocking pre-mRNA processing or translation of mRNA into protein or modulating splicing of pre-mRNA into mature mRNA.

**[0024]** In one aspect, provided herein are antisense oligonucleotides targeting MAPT. In some embodiments, exemplary nucleic acid sequences of MAPT antisense oligonucleotides are provided in Table 1. In some embodiments, provided herein are MAPT antisense oligonucleotides that comprises a nucleic acid sequence selected from any one of SEQ ID NOs: 1-11, 30-32, or 36-63, wherein optionally one or more nucleotides are independently modified nucleotides, and wherein optionally one or more internucleotide linkages are modified internucleotide linkages. In some embodiments, the antisense oligonucleotide is single stranded. In some embodiments, the antisense oligonucleotides provided herein consist of a sequence selected from any one of SEQ ID NOs: 1-11, 30-32, or 36-63.

TABLE 1

| Nucleic Acid Sequences of MAPT Antisense Oligonucleotides (ASO) |                      |           |                                                                               |           |                                                                          |
|-----------------------------------------------------------------|----------------------|-----------|-------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------|
| MAPT ASO No.                                                    | Unmodified Sequence  | SEQ ID NO | Modified Sequence                                                             | SEQ ID NO | Target region of human MAPT sequence (range 5092-138872) (SEQ ID NO: 23) |
| 1                                                               | CTTCTTGGAAAGTATATTC  | 1         | [eCm]seToeToeTo[eCm]sdTsdTsdGsdGsdAsdAsdAsdGsdTsdAseToeAoeTseTs[eCm]          | 12        | 72350-72369                                                              |
| 2                                                               | GCTCAGAGCTACAGCAAGTC | 2         | eGs[eCm]oeTo[eCm]oeAsdGsdAsdGs[dCm]sdTsdAs[dCm]sdAsdGs[dCm]seAoeAoeGseTs[eCm] | 13        | 82367-82386                                                              |
| 3                                                               | GCTGCAGGAAAGAGTTCTT  | 3         | eGs[eCm]oeToeGo[eCm]sdAsdGsdGsdGsdAsdAsdAsdGsdAsdGseToeTo[eCm]seTseT          | 14        | 22592-22611                                                              |

TABLE 1-continued

| Nucleic Acid Sequences of MAPT Antisense Oligonucleotides (ASO) |                      |                 |                                                                                              |                 |                                                                                                        |
|-----------------------------------------------------------------|----------------------|-----------------|----------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------|
| MAPT<br>ASO<br>No.                                              | Unmodified Sequence  | SEQ<br>ID<br>NO | Modified Sequence                                                                            | SEQ<br>ID<br>NO | Target region<br>of human<br>MAPT sequence<br>NG_007398.2<br>(range<br>5092-138872)<br>(SEQ ID NO: 23) |
| 4                                                               | GATGGAGACAGTGTGTATCT | 4               | eGseAoeToeGoeGsdAsdG<br>sdAs[dCm]sdAsdGsdTsdG<br>sdTsdGseToeAoeTs[eCm]<br>seT                | 15              | 19124-19143                                                                                            |
| 5                                                               | GCAGATGGGGCCCAATCCT  | 5               | eGs[eCm]oeAoeGoeAsdT<br>dGsdGsdGsdGs[dCm]s<br>[dCm]s[dCm]sdAsdAseAoeTo<br>[eCm]s[eCm]seT     | 16              | 74462-74481                                                                                            |
| 6                                                               | GCCACAGTGACAGCAAATTC | 6               | eGs[eCm]o[eCm]oeAo<br>[eCm]sdAsdGsdTsdGsdAs<br>[dCm]sdAsdGs[dCm]sdAseAo<br>eAoeTseTs[eCm]    | 17              | 36467-36486                                                                                            |
| 7                                                               | GCCACAGACTCGAGTATCAG | 7               | eGs[eCm]o[eCm]oeAo<br>[eCm]sdAsdGsdAs[dCm]sdTs<br>[dCm]sdGsdAsdGsdTseAo<br>eTo[eCm]seAseG    | 18              | 45989-46008                                                                                            |
| 8                                                               | CCAAACGTGACTTGGCTCCT | 8               | [eCm]s[eCm]oeAoeAoeAs<br>[dCm]sdGsdTsdGsdAs<br>[dCm]sdTsdTsdGsdGs[eCm]o<br>eTo[eCm]s[eCm]seT | 19              | 105626-105645                                                                                          |
| 9                                                               | TATGGACACACATGGGTGTA | 9               | eTseAoeToeGoeGsdAs<br>[dCm]sdAs[dCm]sdAs[dCm]s<br>dAsdTsdGsdGseGoeToeG<br>seTseA             | 20              | 30079-30098                                                                                            |
| 10                                                              | ACTAAAGACGTGTCTTTTGG | 10              | eAs[eCm]oeToeAoeAsdAs<br>dGsdAs[dCm]sdGsdTsdGs<br>dTsd[dCm]sdTseToeToeTse<br>GseG            | 21              | 13464-13483                                                                                            |
| 11                                                              | GTCATCTCTAGGTGACTTCC | 11              | eGseTo[eCm]oeAoeTs[dC<br>m]sdTs[dCm]sdTsdAsdGs<br>dGsdTsdGsdAs[eCm]oeTo<br>eTs[eCm]s[eCm]    | 22              | 79347-79366                                                                                            |
| 12                                                              | GCCTTTGCGAGCGTCACAG  | 30              | eGs[eCm]o[eCm]oeToeTs<br>dTsdGs[dCm]sdGsdAsdGs<br>[dCm]sdGsdTsdGs[eCm]o<br>eAo[eCm]seAseG    | 33              | 898-917                                                                                                |
| 13                                                              | TTTCTGTGGGCAACTGGG   | 31              | eTseToeTo[eCm]oeTsdTsd<br>GsdTsdTsdGsdGsdGs[dCm]<br>sdAsdAs[eCm]oeToeGs<br>eGseG             | 34              | 80673-80692                                                                                            |
| 14                                                              | CTGGTGTGGGTGTTATTCCT | 32              | [eCm]seToeGoeGoeTsdGs<br>dTsdGsdGsdGsdTsdGsdTs<br>dTsdAseToeTo[eCm]s<br>[eCm]seT             | 35              | 88009-88028                                                                                            |
| 15                                                              | GTGGATCTAGTTCTCATCCA | 36              | eGseToeGseGseAsdTsdC<br>msdTsdAsdGsdTsdTsdCm<br>sdTsdCmseAseTo[eCm]s<br>[eCm]seA             | 64              | 75985-76004                                                                                            |
| 16                                                              | GGATAGAATCCAGTTAATTA | 37              | eGseGoeAseTseAsdGsdAs<br>dAsdTsdCmsdCmsdAsdGs<br>dTsdTseAseAoeTseTseA                        | 65              | 42171-42190                                                                                            |



TABLE 1-continued

| Nucleic Acid Sequences of MAPT Antisense Oligonucleotides (ASO) |                      |                 |                                                                                          |                 |                                                                                                        |
|-----------------------------------------------------------------|----------------------|-----------------|------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------|
| MAPT<br>ASO<br>No.                                              | Unmodified Sequence  | SEQ<br>ID<br>NO | Modified Sequence                                                                        | SEQ<br>ID<br>NO | Target region<br>of human<br>MAPT sequence<br>NG_007398.2<br>(range<br>5092-138872)<br>(SEQ ID NO: 23) |
| 17                                                              | TGTCACATCTAGAAACCACC | 38              | eTseGoeTs[eCm]seAsdCm<br>sdAsdTsdCmsdTsdAsdGsd<br>AsdAsdAs[eCm]s[eCm]oe<br>As[eCm]s[eCm] | 66              | 79820-79839                                                                                            |
| 18                                                              | GCCTGGACAGAGGGATGAGC | 39              | eGs[eCm]o[eCm]seTseGsd<br>GsdAsdCmsdAsdGsdAsd<br>GsdGsdGsdAseTseGoeAse<br>Gs[eCm]        | 67              | 13357-13376                                                                                            |
| 19                                                              | AAGCTCCAGGTGATTGAGAC | 40              | eAseAoeGs[eCm]seTsdC<br>msdCmsdAsdGsdGsdTsdG<br>sdAsdTsdTseGseAoeGseA<br>s[eCm]          | 68              | 124341-124360                                                                                          |
| 20                                                              | TTCTGATACATTGTCAAAGT | 41              | eTseTo[eCm]seTseGsdAsd<br>TsdAsdCmsdAsdTsdTsdG<br>sdTsdCmseAseAoeAseGse<br>T             | 69              | 10260-10279                                                                                            |
| 21                                                              | AGCAGCTTGAAGGAGCCGGC | 42              | eAseGo[eCm]seAseGsdC<br>msdTsdTsdGsdAsdAsdGs<br>dGsdAsdGs[eCm]s[eCm]o<br>eGseGs[eCm]     | 70              | 132887-132906                                                                                          |
| 22                                                              | TGGCTGATGATACCATTGG  | 43              | eTseGoeGs[eCm]seTsdGs<br>dAsdTsdGsdAsdTsdAsdC<br>msdCmsdAseTseToeTseG<br>seG             | 71              | 39953-39972                                                                                            |
| 23                                                              | CTTTCTTGAAAGTAT      | 44              | [CmL]s[TL]s[TL]sdTsdC<br>msdTsdTsdGsdGsdAsdAs<br>dAsdGs[TL]s[AL]s[TL]                    | 72              | 72354-72369                                                                                            |
| 24                                                              | TTTCTTGAAAGTATA      | 45              | [TL]s[TL]s[TL]sdCmsdT<br>dTsdGsdGsdAsdAsdAsdG<br>sdTs[AL]s[TL]s[AL]                      | 73              | 72353-72368                                                                                            |
| 25                                                              | TTCTTGAAAGTATAT      | 46              | [TL]s[TL]s[CmL]sdTsdTs<br>dGsdGsdAsdAsdAsdGsdT<br>sdAs[TL]s[AL]s[TL]                     | 74              | 72352-72367                                                                                            |
| 26                                                              | TCTTGAAAGTATATT      | 47              | [TL]s[CmL]s[TL]sdTsdGs<br>dGsdAsdAsdAsdGsdTsdA<br>sdTs[AL]s[TL]s[TL]                     | 75              | 72351-72366                                                                                            |
| 27                                                              | CTTGAAAGTATATTC      | 48              | [CmL]s[TL]s[TL]sdGsdGs<br>dAsdAsdAsdGsdTsdAsdT<br>s[TL]s[TL]s[CmL]                       | 76              | 72350-72365                                                                                            |
| 28                                                              | GTGGATCTAGTTCTCA     | 49              | [GL]s[TL]s[GL]sdGsdAsd<br>TsdCmsdTsdAsdGsdTsdTs<br>dCms[TL]s[CmL]s[AL]                   | 77              | 75989-76004                                                                                            |
| 29                                                              | TGGATCTAGTTCTCAT     | 50              | [TL]s[GL]s[GL]sdAsdTsd<br>CmsdTsdAsdGsdTsdTsdC<br>msdT[CmL]s[AL]s[TL]                    | 78              | 75988-76003                                                                                            |
| 30                                                              | GGATCTAGTTCTCATC     | 51              | [GL]s[GL]s[AL]sdTsdCms<br>dTsdAsdGsdTsdTsdCmsd<br>TsdCms[AL]s[TL]s[CmL]                  | 79              | 75987-76002                                                                                            |
| 31                                                              | GATCTAGTTCTCATCC     | 52              | [GL]s[AL]s[TL]sdCmsdT<br>sdAsdTsdTsdCmsdTsd<br>CmsdAs[TL]s[CmL]s<br>[CmL]                | 80              | 75986-76001                                                                                            |

TABLE 1-continued

| Nucleic Acid Sequences of MAPT Antisense Oligonucleotides (ASO) |                     |                 |                                                                                         |                 |                                                                                                        |
|-----------------------------------------------------------------|---------------------|-----------------|-----------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------|
| MAPT<br>ASO<br>No.                                              | Unmodified Sequence | SEQ<br>ID<br>NO | Modified Sequence                                                                       | SEQ<br>ID<br>NO | Target region<br>of human<br>MAPT sequence<br>NG_007398.2<br>(range<br>5092-138872)<br>(SEQ ID NO: 23) |
| 32                                                              | ATCTAGTTCTCATCCA    | 53              | [AL]s[TL]s[CmL]sdTsdAs<br>dGsdTsdTsdCmsdTsdCms<br>dAsdTsd[CmL]s[CmL]s[AL]               | 81              | 75985-76000                                                                                            |
| 33                                                              | TTCTGATACATTGTCA    | 54              | [TL]s[TL]s[CmL]sdTsdGs<br>dAsdTsdAsdTsdCmsdTsd<br>TsdGs[TL]s[CmL]s[AL]                  | 82              | 10264-10279                                                                                            |
| 34                                                              | TCTGATACATTGTCAA    | 55              | [TL]s[CmL]s[TL]sdGsdAs<br>dTsdAsdTsdCmsdTsdTsd<br>GsdTs[CmL]s[AL]s[AL]                  | 83              | 10263-10278                                                                                            |
| 35                                                              | CTGATACATTGTCAAA    | 56              | [CmL]s[TL]s[GL]sdAsdTsd<br>dAsdTsdCmsdTsdTsdGsd<br>TsdCms[AL]s[AL]s[AL]                 | 84              | 10262-10277                                                                                            |
| 36                                                              | TGATACATTGTCAAAG    | 57              | [TL]s[GL]s[AL]sdTsdAsdTsd<br>CmsdTsdTsdTsdGsdTsdC<br>msdTsd[AL]s[AL]s[GL]               | 85              | 10261-10276                                                                                            |
| 37                                                              | GATACATTGTCAAAGT    | 58              | [GL]s[AL]s[TL]sdAsdTsdCms<br>dTsdTsdTsdGsdTsdCmsdTsd<br>AsdTsd[AL]s[GL]s[TL]            | 86              | 10260-10275                                                                                            |
| 38                                                              | AGCAGCTTGAAGGAGC    | 59              | [AL]s[GL]s[CmL]sdAsdTsdG<br>sdCmsdTsdTsdGsdAsdTsd<br>GsdGs[AL]s[GL]s[CmL]               | 87              | 132891-132906                                                                                          |
| 39                                                              | GCAGCTTGAAGGAGCC    | 60              | [GL]s[CmL]s[AL]sdGsdC<br>msdTsdTsdGsdAsdTsdGs<br>dTsdAs[GL]s[CmL]s[CmL]                 | 88              | 132890-132905                                                                                          |
| 40                                                              | CAGCTTGAAGGAGCCG    | 61              | [CmL]s[AL]s[GL]sdCmsdTsd<br>TsdTsdGsdAsdTsdGsdGsd<br>AsdTsd[CmL]s[CmL]s[GL]             | 89              | 132889-132904                                                                                          |
| 41                                                              | AGCTTGAAGGAGCCGG    | 62              | [AL]s[GL]s[CmL]sdTsdTs<br>dGsdAsdTsdGsdGsdAsdTsd<br>sdCms[CmL]s[GL]s[GL]                | 90              | 132888-132903                                                                                          |
| 42                                                              | GCTTGAAGGAGCCGGC    | 63              | [GL]s[CmL]s[TL]sdTsdGs<br>dTsdAsdTsdGsdAsdTsdC<br>msdTsdCms[GL]s[GL]s[CmL]              | 91              | 132887-132902                                                                                          |
| 43                                                              | CTTTCTTGAAAGTATATTC | 1               | [eCm]seToeTseTs[eCm]sd<br>TsdTsdGsdGsdAsdTsdAsdTsd<br>GsdTsdAseTseAoeTseTs<br>[eCm]     | 92              | 72350-72369                                                                                            |
| 44                                                              | GCTGCAGGAAAGAGTTCTT | 3               | eGs[eCm]oeTseGs[eCm]sd<br>AsdTsdGsdGsdAsdTsdAsdTsd<br>dTsdAsdTsdGseTseTo[eCm]s<br>eTseT | 93              | 22592-22611                                                                                            |
| 45                                                              | GATGGAGACAGTGTATCT  | 4               | eGseAoeTseGseGsdAsdTsd<br>dTsd[dCm]sdAsdTsdTsdGs<br>dTsdGseTseAoeTs[eCm]se<br>T         | 94              | 19124-19143                                                                                            |

Abbreviations- "Cm" stands for 5-methylcytidine; "d" indicates 2'-deoxynucleotides; "e" indicates a 2'-O-MOE modified nucleotide; "L" indicates a LNA modified nucleotide; "s" indicates phosphorothioate linkage; "o" indicates phosphodiester linkage.

**[0025]** In some embodiments, the MAPT antisense oligonucleotide described herein can comprise a sequence that has 1, 2, or 3 differences from the sequences listed in Table 1. Such MAPT antisense oligonucleotides may still be capable of binding to the target nucleic acid, with 1, 2, or 3 mismatches, when a sufficient number of nucleobases of the antisense oligonucleotide form hydrogen bond with the corresponding nucleobases of the target nucleic acid.

**[0026]** In some embodiments, the antisense oligonucleotides provided herein are 12 to 30 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 15 to 25 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 16 to 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 18 to 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 20 nucleotides in length. In some embodiments, the MAPT antisense oligonucleotides are 16 nucleotides in length.

**[0027]** The MAPT antisense oligonucleotides described herein may include modifications. The modifications can be made to one or more nucleotides (e.g., to nucleobase, sugar, or phosphate) or to one or more internucleotide linkages, which are the bonds between two nucleotides. Modifications can improve stability, efficacy, and/or reduce immunogenicity of the antisense oligonucleotides. For example, antisense oligonucleotides can be modified to have increased resistance to nucleases, increased stability, increased half-life, enhanced binding affinity for nucleic acid target, enhanced cellular uptake, and/or increased inhibitory activity when compared to the unmodified oligonucleotides.

**[0028]** In some embodiments, the antisense oligonucleotide provided herein comprises one or more modified nucleotides. In some embodiments, such modified nucleotides comprise a modified nucleobase, e.g., 5-methylcytosine. In some embodiments, each C in the antisense oligonucleotide is a 5-methylcytosine. In some embodiments, such modified nucleotides comprise a modified sugar, e.g., a modified ribose or deoxyribose. In some embodiments, the modified sugar is a 2'-O-methoxyethyl (2'-O-MOE) modified sugar, 2'-O-methyl modified sugar, or 2'-fluoro modified sugar. In some embodiments, the modified sugar is a 2'-O-MOE modified sugar. Oligonucleotides containing 5-methylcytosine or 2'-O-MOE modifications have been shown to exhibit decreased immune stimulation in mice (Henry S. et al., *J Pharmacol. Exp Ther.* 2000 February; 292(2):468-79). In some embodiments, the modified sugar is a locked nucleic acid (LNA).

**[0029]** In some embodiments, the antisense oligonucleotide provided herein are gapmers, which comprise a central gap segment consisting of a stretch of contiguous 2'-deoxyribonucleotides, positioned between two 5' and 3' wing segments consisting of modified nucleotides. The central gap segment of DNA can serve as the substrate for cellular endonuclease RNase H, recruit it to the target RNA:DNA heteroduplex and cleave the target RNA in the RNA:DNA duplex. The 5' and 3' wing segments with modified nucleotides confer increased resistance to other nuclease degradation. The gapmers can be in "X-Y-Z" format, where "X" represents the 5' wing region, "Y" represents the central gap region, and "Z" represents the 3' wing region. In some embodiments, the 5' and 3' wing regions comprise 2'-O-MOE modified nucleotides, and the central gap region comprises 2'-deoxynucleotides. In some embodiments, the 5'

and 3' wing regions comprise LNA modified nucleotides, and the central gap region comprises 2'-deoxynucleotides.

**[0030]** In some embodiments, the MAPT antisense oligonucleotide provided herein are gapmers, wherein the first to fifth nucleotides each comprise a 2'-O-MOE modified sugar, wherein the sixth to fifteenth nucleotides each comprise a 2'-deoxynucleotide, and wherein the sixteenth to twentieth nucleotides each comprise a 2'-O-MOE modified sugar.

**[0031]** In some embodiments, the MAPT antisense oligonucleotide provided herein are gapmers, wherein the first to third nucleotides each comprise LNA, wherein the fourth to thirteenth nucleotides each comprise a 2'-deoxynucleoside, and wherein the fourteenth to sixteenth nucleotides each comprise LNA.

**[0032]** In some embodiments, the antisense oligonucleotides provided herein comprise one or more modified internucleotide linkages. In some embodiments, the modified internucleotide linkage is a phosphorothioate linkage. In some embodiments, the antisense oligonucleotides provided herein comprise phosphodiester and phosphorothioate internucleotide linkage. In some embodiments, the internucleotide linkages of the MAPT antisense oligonucleotides are `sooooooooooooo`, or `sooooooooooooo`, from 5' end to 3' end, wherein each s is a phosphorothioate linkage and each o is a phosphodiester linkage. In some embodiments, the internucleotide linkages of the MAPT antisense oligonucleotides are all phosphorothioate linkages.

**[0033]** In some embodiments, the antisense oligonucleotides provided herein comprise a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94. In some embodiments, the antisense oligonucleotides provided herein consist of a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94.

**[0034]** Antisense oligonucleotide can be synthesized using any nucleic acid polymerization methods known in the art, for example, solid-phase synthesis by employing phosphoramidite chemistry methodology (e.g., *Current Protocols in Nucleic Acid Chemistry*, Beaucage, S. L. et al. (Eds.), John Wiley & Sons, Inc., New York, NY, USA), H-phosphonate, phosphotriester chemistry, or enzymatic synthesis. Automated commercial synthesizers can be used, for example, MerMade™ 12 from LGC Biosearch Technologies, or other synthesizers from BioAutomation or Applied Biosystems. Phosphorothioate linkages can be introduced using a sulfurizing reagent such as phenylacetyl disulfide or DDTT (((dimethylaminomethylidene) amino)-3H-1,2,4-dithiazol-3-thione). It is well known to use similar techniques and commercially available modified amidites and controlled-pore glass (CPG) products to synthesize modified oligonucleotides or conjugated oligonucleotides.

**[0035]** Purification methods can be used to exclude the unwanted impurities from the final oligonucleotide product. Commonly used purification techniques for single stranded oligonucleotides include reverse-phase ion pair high performance liquid chromatography (RP-IP-HPLC), capillary gel electrophoresis (CGE), anion exchange HPLC (AX-HPLC), and size exclusion chromatography (SEC). After purification, oligonucleotides can be analyzed by mass spectrometry and quantified by spectrophotometry at a wavelength of 260 nm.

**[0036]** The activity of antisense oligonucleotides can be tested in vitro and/or in vivo. For in vitro testing, the antisense oligonucleotides can be introduced into cultured cells by natural uptake, transfection or electroporation. Fol-

lowing a period of treatment, MAPT (tau) expression level in the ASO-treated cells can be determined and compared to MAPT (tau) expression level in the untreated control cells. MAPT expression level can be determined by any appropriate method, for example, by quantitation of MAPT mRNA level, by measuring the quantity of cDNA produced from reverse transcription of MAPT mRNA, or by determining the quantity of tau protein. These methods can be performed on a sample-by-sample basis or modified for high throughput analysis. For example, MAPT mRNA level can also be detected and quantitated by polymerase chain reaction (PCR), using a pair of primers that recognize MAPT transcript. General procedures for PCR are taught in MacPherson et al., *PCR: A Practical Approach*, (IRL Press at Oxford University Press (1991)). Tau protein level can be quantitated by Western blot analysis (immunoblotting), enzyme-linked immunosorbent assay (ELISA), immunohistochemistry, immunoassays, immunoprecipitation, immunofluorescent assays, immunocytochemistry, fluorescence-activated cell sorting (FACS), radioimmunoassays, immunoradiometric assays, high-performance liquid chromatography (HPLC), mass spectrometry, confocal microscopy, enzymatic assays, or surface plasmon resonance (SPR).

**[0037]** In another aspect, provided herein are pharmaceutical compositions comprising a MAPT antisense oligonucleotide described herein and a pharmaceutically acceptable carrier. Also provided herein are pharmaceutical compositions comprising a means for reducing MAPT expression in a cell and a pharmaceutically acceptable carrier. Such pharmaceutical compositions can also comprise one or more pharmaceutically acceptable excipient, diluent, or carrier. Pharmaceutical compositions can be prepared by methods well known in the art (e.g., Remington: *The Science and Practice of Pharmacy*, 23rd edition (2020), A. Loyd et al., Academic Press).

**[0038]** In a further aspect, provided herein are methods of reducing MAPT expression in a cell (e.g., a neuron), such methods can include contacting the cell with a MAPT antisense oligonucleotide described herein; and incubating the cell for a time sufficient for degradation of MAPT mRNA, thereby reducing MAPT expression in the cell.

**[0039]** In another aspect, provided herein are methods of reducing MAPT expression in a patient in need thereof, and such method comprises administering to the patient an effective amount of a MAPT antisense oligonucleotide or a pharmaceutical composition described herein. Aggregation of MAPT can be caused by overexpression of the MAPT protein or a mutation that affects the structure of the protein, resulting in an increased tendency of the MAPT protein to self-associate. Therefore, reducing MAPT expression level can be beneficial to the patient with tauopathy.

**[0040]** In another aspect, provided herein are methods of treating a tauopathy in a patient in need thereof, and such method comprises administering to the patient an effective amount of the MAPT antisense oligonucleotide or a pharmaceutical composition described herein. Exemplary tauopathy includes, but are not limited to, Alzheimer's disease (AD), frontotemporal dementia (FTD), frontotemporal dementia with parkinsonism linked to chromosome 17 (FTDP-17), frontotemporal lobar degeneration (FTLD), behavioral variant frontotemporal dementia (bvFTD), non-fluent variant primary progressive aphasia (nfvPPA), Parkinson's disease, Pick's disease (PiD), primary progressive

aphasia-semantic (PPA-S), primary progressive aphasia-logopenic (PPA-L), multiple system tauopathy with presenile dementia (MSTD), neurofibrillary tangle (NFT) dementia, FTD with motor neuron disease, progressive supranuclear palsy (PSP), amyotrophic lateral sclerosis/parkinsonism-dementia complex (ALS-PDC), argyrophilic grain dementia (AGD), British type amyloid angiopathy, cerebral amyloid angiopathy, chronic traumatic encephalopathy (CTE), corticobasal degeneration (CBD), Creutzfeldt-Jakob disease (CJD), dementia pugilistica, diffuse neurofibrillary tangles with calcification, Down's syndrome, epilepsy, Gerstmann-Straussler-Scheinker disease, Hallervorden-Spatz disease, Huntington's disease, inclusion body myositis, lead encephalopathy, Lytico-Bodig disease (Parkinson-dementia complex of Guam), meningioangiomas, multiple system atrophy, myotonic dystrophy, Niemann-Pick disease type C (NP-C), non-Guamanian motor neuron disease with neurofibrillary tangles, postencephalitic parkinsonism, prion protein cerebral amyloid angiopathy, progressive subcortical gliosis, tangle only dementia, tangle-predominant dementia, ganglioglioma, gangliocytoma, subacute sclerosingpan encephalitis, tuberous sclerosis, lipofuscinosis, primary age-related tauopathy (PART), globular glial tauopathies (GGT). In some embodiments, the tauopathy is Alzheimer's disease (AD), frontotemporal dementia (FTD), or progressive supranuclear palsy (PSP).

**[0041]** The MAPT antisense oligonucleotide can be administered to the patient intrathecally, intravenously, subcutaneously, or via intracisternal magna injection. In some embodiments, the MAPT antisense oligonucleotide is administered to the patient intrathecally.

**[0042]** Dosage regimens of antisense oligonucleotide may be adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus may be administered, several divided doses may be administered over time, or the dose may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation.

**[0043]** Dosage values may vary with the type and severity of the condition to be alleviated. It is further understood that for any particular subject, specific dosage regimens should be adjusted over time according to the individual need and the professional judgment of the person administering or supervising the administration of the compositions.

**[0044]** In another aspect, provided herein are MAPT antisense oligonucleotides or pharmaceutical compositions comprising a MAPT antisense oligonucleotide for use in reducing MAPT expression. Also provided herein are MAPT antisense oligonucleotides or the pharmaceutical composition comprising a MAPT antisense oligonucleotide for use in a therapy. Also provided herein are MAPT antisense oligonucleotides or pharmaceutical compositions comprising a MAPT antisense oligonucleotide for use in the treatment of a tauopathy. Also provided herein are uses of MAPT antisense oligonucleotides in the manufacture of a medicament for the treatment of a tauopathy.

**[0045]** As used herein, the terms "a," "an," "the," and similar terms used in the context of the present disclosure (especially in the context of the claims) are to be construed to cover both the singular and plural unless otherwise indicated herein or clearly contradicted by the context.

**[0046]** The term "antisense oligonucleotide" as used herein refers to an oligonucleotide comprising a nucleobase

sequence that is complementary to a corresponding segment of a target nucleic acid, e.g., a target genomic sequence, pre-mRNA, or mRNA molecule.

**[0047]** As used herein, “complementary” or “complementarity” means a structural relationship between two nucleotides, nucleosides, or nucleobases (e.g., on two opposing nucleic acids or on opposing regions of a single nucleic acid strand e.g., a hairpin) that permits the two nucleotides to form base pairs with one another. For example, a purine nucleotide of one nucleic acid that is complementary to a pyrimidine nucleotide of an opposing nucleic acid may base pair together by forming hydrogen bonds with one another. Complementary nucleotides can base pair in the canonical Watson-Crick manner, which means adenine pairing with thymine or uracil, and guanine pairing with cytosine, or in any other manner that allows for the formation of stable duplexes. Likewise, two nucleic acids may have regions of multiple nucleotides that are complementary with each other to form regions of complementarity.

**[0048]** An “effective amount” refers to an amount necessary (for periods of time and for the means of administration) to achieve the desired therapeutic result. An effective amount of an antisense oligonucleotide may vary according to factors such as the disease state, age, sex, and weight of the individual, and the ability of the antisense oligonucleotide to elicit a desired response in the individual. An effective amount is also one in which any toxic or detrimental effects of the antisense oligonucleotide are outweighed by the therapeutically beneficial effects.

**[0049]** The term “knockdown” or “expression knockdown” refers to reduced mRNA or protein expression of a gene after treatment of a reagent, e.g., an antisense oligonucleotide.

**[0050]** As used herein, “modified internucleotide linkage” means an internucleotide linkage having one or more chemical modifications when compared with a reference internucleotide linkage having a phosphodiester bond. A modified internucleotide linkage can be a non-naturally occurring linkage.

**[0051]** As used herein, “modified nucleotide” refers to a nucleotide having one or more chemical modifications when compared with a corresponding reference nucleotide selected from: adenine ribonucleotide, guanine ribonucleotide, cytosine ribonucleotide, uracil ribonucleotide, adenine deoxyribonucleotide, guanine deoxyribonucleotide, cytosine deoxyribonucleotide, and thymidine deoxyribonucleotide. A modified nucleotide can be a non-naturally occurring nucleotide. A modified nucleotide can have, for example, one or more chemical modification in its sugar, nucleobase, and/or phosphate group. Additionally, or alternatively, a modified nucleotide can have one or more chemical moieties conjugated to a corresponding reference nucleotide.

**[0052]** As used herein, “nucleotide” means an organic compound having a nucleoside (a nucleobase, e.g., adenine, cytosine, guanine, thymine, or uracil, and a pentose sugar, e.g., ribose or 2'-deoxyribose) linked to a phosphate group, which can serve as a monomeric unit of nucleic acid polymers such as deoxyribonucleic acid (DNA) and ribonucleic acid (RNA).

**[0053]** As used herein, “oligonucleotide” means a polymer of linked nucleotides, each of which can be modified or unmodified. An oligonucleotide is typically less than about 100 nucleotides in length.

**[0054]** The term “patient”, as used herein, refers to a human patient.

**[0055]** The term “% sequence identity” or “percentage sequence identity” with respect to a reference nucleic acid sequence is defined as the percentage of nucleotides, nucleosides, or nucleobases in a candidate sequence that are identical with the nucleotides, nucleosides, or nucleobases in the reference nucleic acid sequence, after optimally aligning the sequences and introducing gaps or overhangs, if necessary, to achieve the maximum percent sequence identity. Alignment for purposes of determining percent nucleic acid sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software programs, for example, those described in Current Protocols in Molecular Biology (Ausubel et al., eds., 1987, Supp. 30, section 7.7.18, Table 7.7.1), and including BLAST, BLAST-2, ALIGN, Clustal W2.0, Clustal X2.0, or Megalign (DNASTAR) software. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared. Percentage of “sequence identity” can be determined by comparing two optimally aligned sequences over a comparison window, where the fragment of the nucleic acid sequence in the comparison window may comprise additions or deletions (e.g., gaps or overhangs) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage can be calculated by determining the number of positions at which the identical nucleotide, nucleoside, or nucleobase occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison, and multiplying the result by 100 to yield the percentage of sequence identity. The output is the percent identity of the subject sequence with respect to the query sequence.

**[0056]** As used herein, the term “tauopathy” refers to a disease associated with abnormal tau protein expression, secretion, phosphorylation, cleavage, and/or aggregation.

**[0057]** As used herein, “treatment” or “treating” refers to all processes wherein there may be a slowing, controlling, delaying, or stopping of the progression of the disorders or disease disclosed herein, or ameliorating disorder or disease symptoms, but does not necessarily indicate a total elimination of all disorder or disease symptoms. Treatment includes administration of a protein or nucleic acid or vector or composition for treatment of a disease or condition in a patient, particularly in a human.

**[0058]** As used herein, “MAPT” refers to a human MAPT pre-mRNA or mRNA transcript, encoding a microtubule associated protein Tau. The human MAPT gene is located at chromosomal location 17q21.1, and the genomic sequence of human MAPT gene can be found in GenBank at NG\_007398.2 (SEQ ID NO: 23). The nucleotide sequences of human MAPT transcript variants and amino acid sequences of human Tau protein isoforms can be found at:

**[0059]** i. MAPT transcript variant 1→Tau protein isoform 1: NM\_016835.5 (nucleotide sequence)→NP\_058519.3 (amino acid sequence);

**[0060]** ii. MAPT transcript variant 2→Tau protein isoform 2: NM\_005910.6 (nucleotide sequence)→NP\_005901.2 (amino acid sequence);

- [0061] iii. MAPT transcript variant 3→Tau protein isoform 3: NM\_016834.5 (nucleotide sequence)→NP\_058518.1 (amino acid sequence);
- [0062] iv. MAPT transcript variant 4→Tau protein isoform 4: NM\_016841.5 (nucleotide sequence)→NP\_058525.1 (amino acid sequence);
- [0063] v. MAPT transcript variant 5→Tau protein isoform 5: NM\_001123067.4 (nucleotide sequence)→NP\_001116539.1 (amino acid sequence);
- [0064] vi. MAPT transcript variant 6→Tau protein isoform 6: NM\_001123066.4 (nucleotide sequence)→NP\_001116538.2 (amino acid sequence);
- [0065] vii. MAPT transcript variant 7→Tau protein isoform 7: NM\_001203251.2 (nucleotide sequence)→NP\_001190180.1 (amino acid sequence);
- [0066] viii. MAPT transcript variant 8→Tau protein isoform 8: NM\_001203252.2 (nucleotide sequence)→NP\_001190181.1 (amino acid sequence);
- [0067] ix. MAPT transcript variant 9→Tau protein isoform 9: NM\_001377265.1 (nucleotide sequence)→NP\_001364194.1 (amino acid sequence);
- [0068] x. MAPT transcript variant 10→Tau protein isoform 10: NM\_001377266.1 (nucleotide sequence)→NP\_001364195.1 (amino acid sequence);
- [0069] xi. MAPT transcript variant 11→Tau protein isoform 11: NM\_001377267.1 (nucleotide sequence)→NP\_001364196.1 (amino acid sequence);
- [0070] xii. MAPT transcript variant 12→Tau protein isoform 4: NM\_001377268.1 (nucleotide sequence)→NP\_001364197.1 (amino acid sequence).
- [0071] The nucleotide sequence of the human MAPT transcript variant 6 (encoding 2N4R Tau) can be found at NM\_001123066.4 (SEQ ID NO: 24). The corresponding amino acid sequence of human Tau protein isoform 6 can be found at NP\_001116538.2 (SEQ ID NO: 25).
- [0072] The nucleotide sequence of a human MAPT transcript variant 5 (encoding 1N4R Tau) can be found at NM\_001123067.4 (SEQ ID NO: 26). The corresponding amino acid sequence of human Tau protein isoform 5 can be found at NP\_001116539.1 (SEQ ID NO: 27).
- [0073] The nucleotide sequence of the human MAPT transcript variant 4 (encoding 0N3R Tau) can be found at NM\_016841.5 (SEQ ID NO: 28). The corresponding amino acid sequence of human Tau protein isoform 4 can be found at NP\_058525.1 (SEQ ID NO: 29).

## EXAMPLES

### Example 1. Synthesis of MAPT Antisense Oligonucleotides

- [0074] The oligonucleotide sequences shown in Table 1 were synthesized on solid support via phosphoramidite chemistry using the Shasta Plate Synthesizer (Sierra Biosystems) for 100 nmol scale syntheses and the H8-SE Synthesizer (K&A) for 10  $\mu$ mol scale syntheses.
- [0075] Standard reagents were used in the oligonucleotide synthesis (Table 2), where 0.1M DDTT in pyridine was used as the sulfurization reagent. All phosphoramidites were commercially available (ChemGenes) and were prepared at 0.1M in Acetonitrile and contained a molecular sieves trap bag.
- [0076] Oligonucleotides synthesized at the 0.1 nmol scale on 96-well plates were cleaved and deprotected (C/D) using ammonium hydroxide in the gas phase at 80° C. and at 40

psi. Wells were washed twice with 100  $\mu$ L of anhydrous acetonitrile and then dried. Oligonucleotides were eluted in 300  $\mu$ L of water, analyzed for concentration using absorbance at 260 nm recorded by the Lunatic (Unchained Labs), and characterized by LC-QTOF for mass purity of >85%.

[0077] Oligonucleotides synthesized at the 10  $\mu$ mol scale were cleaved and deprotected (C/D) using a 1:1 solution of 25% aqueous ammonium hydroxide and 40% aqueous methylamine at room temperature for 2 hours. The solution containing cleaved oligonucleotide was decanted and concentrated in the Genevac for the ammonia and methylamine removal. The CPG was filtered via 0.45  $\mu$ m PVDF filter.

[0078] The crude oligonucleotides were purified via AKTA™ Pure purification system using anion-exchange (AEX) with a HiScale 16 Capto Q ImpRes column (Cytiva Life Sciences) with MPA: 20 mM NaOH, 20% ACN, pH 12.4 and MPB: 20 mM NaOH, 1.5M NaBr, 20% ACN, pH 12.4.

[0079] The purified oligonucleotides were desalted in 15 mL 3K MWCO centrifugal spin tubes by rinsing with RNase free water at 3500 $\times$ g for ~30 min. The final desalted oligonucleotides were analyzed for concentration (nano drop at A260), characterized by LC-QTOF for mass purity and UPLC for UV-purity.

TABLE 2

| Oligonucleotide Synthesis Reagents<br>Reagents                    |
|-------------------------------------------------------------------|
| Activator Solution (0.45M ETT in ACN)                             |
| Cap A (Acetic Anhydride, Pyridine in THF, 1:1:8)                  |
| Cap B (1-Methylimidazole in THF, 16:84)                           |
| Oxidation Solution (0.02M Iodine in THF/Pyridine/Water, 70:20:10) |
| Deblock Solution, 3% TCA in DCM (w/v)                             |
| Acetonitrile (Anhydrosolv, Water max. 10 ppm)                     |
| DDTT (0.1M in Pyridine)                                           |

### Example 2. In Vitro Characterization of MAPT Antisense Oligonucleotides

[0080] Selected MAPT antisense oligonucleotides were tested in vitro for MAPT inhibition.

### Materials and Methods

[0081] SH-SY5Y Cell Culture and MAPT Antisense Oligonucleotide Treatment: SH-SY5Y cells (ATCC CRL-2266) were derived from the SK-N-SH neuroblastoma cell line (Ross, R. A., et al., 1983. J Natl Cancer Inst 71, 741-747). The base medium was composed of a 1:1 mixture of ATCC-formulated Eagle's Minimum Essential Medium, (Cat No. 30-2003), and F12 Medium. The complete growth medium was supplemented with 10% fetal bovine serum, and cells were incubated at 37° C. in a humidified atmosphere of 5% CO<sub>2</sub>. On Day One, antisense oligonucleotide and Lipofectamine RNAiMax transfection reagent (ThermoFisher 13778030) complexes were prepared inside the wells and SH-SY5Y cells were plated in 96 well tissue culture plates and incubated for 24 hrs. On Day Two, analysis of changes in gene expression in the treated SH-SY5Y cells was measured using TaqMan™ Gene Expression Cells-to-CT™ Kits following the manufacturer's protocol (ThermoFisher AM1729). Predesigned gene expression assays (supplied as 60 $\times$  or 20 $\times$  mixtures) were

selected from TaqMan Real-Time PCR Assays. The efficiencies of these assays for MAPT (ThermoFisher Hs00902193\_m1) and TBP (ThermoFisher #4326322E) were characterized with a dilution series of cDNA. RT-QPCR was performed in MicroAmp Optical 384-well reaction plates using QuantStudio 7 Pro system. The delta-delta CT method of normalizing to the transfection only control sample was used to determine relative amounts of MAPT gene expression. GraphPad Prism v10.2 was used to determine absolute or relative IC<sub>50</sub> with a four-parameter logistic fit.

**[0082]** Human induced pluripotent stem cells (hiPSCs) derived glutamatergic neurons (iCell GlutNeurons) (CDI, 01279) was obtained from Fujifilm Cellular Dynamics International (CDI). iCell GlutNeurons were handled and maintained following CDI's guidelines. Tissue culture treated 96 well plates (Thermo Scientific, 167574) were coated with poly-L-ornithine (PLO) and Matrigel (Corning 354230). 80,000 cells were plated in each well and were grown in Complete BrainPhys Medium consisting of BrainPhys Neuronal Medium (STEMCELL Technologies, 05790), iCell Neural Supplement B (CDI, M1029), iCell Nervous System Supplement (CDI, M1031), N-2 Supplement (Thermo Scientific, 17502-048) and Laminin (Sigma-Aldrich, L2020) in an incubator (37°/5% CO<sub>2</sub>). Starting the day after plating, half of media was replenished every other day for seven days. On the eighth day, antisense oligonucleotide was serially diluted in Complete BrainPhys Medium, and cells were treated with MAPT antisense oligonucleotide by aspirating half of media from each well (100  $\mu$ L) and adding 100  $\mu$ L of 2 $\times$ MAPT antisense oligonucleotide concentration for a final of 1 $\times$ MAPT antisense oligonucleotide according to dilutions. Cells were half-fed either every three or four days after treatment by removing half of media and adding back fresh Complete BrainPhys Medium with 2 $\times$ MAPT antisense oligonucleotide concentration. Cell lysates were harvested after seven days of treatment (14 days post seeding) and RT-qPCR was performed using TaqMan Fast Advanced Cells-to CT Kit (Thermo Scientific, A35377) to determine mRNA knock down level. MAPT probe (ThermoFisher, Hs00902193\_m1) was used as the gene of interest while TBP probe was used as the reference gene (ThermoFisher, 4326322E). RT-QPCR was performed in MicroAmp Optical 384-well reaction plates using the QuantStudio 7 Pro system. The delta-delta CT method was employed to normalize to the non-treated control sample and determine relative MAPT gene expression. GraphPad Prism v10.2 was used to calculate or relative IC<sub>50</sub> value with a four-parameter logistic fit.

**[0083]** Table 3 summarizes the in vitro activities of selected MAPT antisense oligonucleotides in SH-SY5Y cells. As shown in Table 3, the tested antisense oligonucleotides knock down MAPT expression in vitro in SH-SY5Y cells.

TABLE 3

| In vitro activities of selected MAPT antisense oligonucleotides in SH-SY5Y cells |                                                                   |                                                                           |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|
| MAPT ASO No                                                                      | SH-SY5Y 1 d<br>Absolute IC <sub>50</sub><br>(nM),<br>Transfection | SH-SY5Y 1 d<br>% KD (knockdown) of MAPT<br>mRNA at 10 nM,<br>Transfection |
| 1                                                                                | 0.41                                                              | 93.9 $\pm$ 0.4                                                            |
| 2                                                                                | 2.06                                                              | 90.9 $\pm$ 0.6                                                            |
| 3                                                                                | 0.73                                                              | 90.3 $\pm$ 2.4                                                            |
| 4                                                                                | 1.69                                                              | 89.5 $\pm$ 2.5                                                            |

TABLE 3-continued

| In vitro activities of selected MAPT antisense oligonucleotides in SH-SY5Y cells |                                                                   |                                                                           |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|
| MAPT ASO No                                                                      | SH-SY5Y 1 d<br>Absolute IC <sub>50</sub><br>(nM),<br>Transfection | SH-SY5Y 1 d<br>% KD (knockdown) of MAPT<br>mRNA at 10 nM,<br>Transfection |
| 5                                                                                | 0.37                                                              | 89.1 $\pm$ 1.8                                                            |
| 6                                                                                | 1.23                                                              | 89.1 $\pm$ 1.7                                                            |
| 7                                                                                | 0.74                                                              | 87.5 $\pm$ 1.9                                                            |
| 8                                                                                | 21.95                                                             | 86.1 $\pm$ 3.9                                                            |
| 9                                                                                | 1.94                                                              | 84.8 $\pm$ 1.9                                                            |
| 10                                                                               | 4.90                                                              | 84.1 $\pm$ 2.1                                                            |
| 11                                                                               | 0.70                                                              | 84.1 $\pm$ 1.9                                                            |
| 12                                                                               | 0.25                                                              | 77.41 $\pm$ 2.1                                                           |
| 13                                                                               | 1.55                                                              | 82.4 $\pm$ 3.3                                                            |
| 14                                                                               | 10.38                                                             | 83.2 $\pm$ 0.5                                                            |

**[0084]** Table 4 summarizes the in vitro activities of selected MAPT antisense oligonucleotides in human iPSC cells. As shown in Table 4, the tested antisense oligonucleotides knock down MAPT expression in human iPSC cells.

TABLE 4

| In vitro activities of selected MAPT antisense oligonucleotides in human iPSC cells |                                                     |                                                       |
|-------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|
| MAPT ASO No                                                                         | iPSC 7 d<br>Relative IC <sub>50</sub><br>( $\mu$ M) | iPSC 7 d<br>% KD (knockdown) of MAPT<br>mRNA at 10 nM |
| 15                                                                                  | 2.916                                               | 70.11                                                 |
| 16                                                                                  | 10.4                                                | 44.69                                                 |
| 17                                                                                  | 3.562                                               | 54.88                                                 |
| 18                                                                                  | 10.66                                               | 26.92                                                 |
| 19                                                                                  | 9.743                                               | 33.38                                                 |
| 20                                                                                  | 1.782                                               | 78.32                                                 |
| 21                                                                                  | 2.782                                               | 64.02                                                 |
| 22                                                                                  | 7.396                                               | 43.72                                                 |
| 23                                                                                  | 1.08                                                | 80.31                                                 |
| 24                                                                                  | 0.6954                                              | 81.3                                                  |
| 25                                                                                  | 2.214                                               | 64.58                                                 |
| 26                                                                                  | 0.956                                               | 79.54                                                 |
| 27                                                                                  | 0.05902                                             | 98                                                    |
| 28                                                                                  | ND*                                                 | 94.83                                                 |
| 29                                                                                  | ND*                                                 | 85.99                                                 |
| 30                                                                                  | ND*                                                 | 96.75                                                 |
| 31                                                                                  | ND*                                                 | 61.96                                                 |
| 32                                                                                  | 0.057                                               | 97.35                                                 |
| 33                                                                                  | 0.018                                               | 99.37                                                 |
| 34                                                                                  | 0.016                                               | 98.41                                                 |
| 35                                                                                  | ND*                                                 | 97.04                                                 |
| 36                                                                                  | ND*                                                 | 90.86                                                 |
| 37                                                                                  | 0.012                                               | 98.39                                                 |
| 38                                                                                  | ND*                                                 | 68.99                                                 |
| 39                                                                                  | ND*                                                 | 56.67                                                 |
| 40                                                                                  | ND*                                                 | 58.94                                                 |
| 41                                                                                  | ND*                                                 | 34.01                                                 |
| 42                                                                                  | ND*                                                 | 67.05                                                 |
| 43                                                                                  | 6.04                                                | 53.89                                                 |
| 44                                                                                  | 6.64                                                | 57.38                                                 |
| 45                                                                                  | 2.19                                                | 70.57                                                 |

ND\* indicates not determined.

Example 3. In Vivo Characterization of MAPT Antisense Oligonucleotides

[0085] Selected MAPT antisense oligonucleotides were tested in vivo for MAPT inhibition in human MAPT transgenic mice.

Materials and Methods

[0086] Mouse Stereotaxic Intracerebroventricular (ICV) Surgery and MAPT Antisense Oligonucleotide Treatment: C57/BL6 transgenic mice expressing human MAPT were placed in an induction chamber (1-5% Isoflurane using a calibrated vaporizer) after administering 5 mg/kg of keto-profen (subcutaneous). Hair is removed from the surgical site using electric clippers. The surgical area is disinfected with iodine and alcohol three times to create an aseptic surgical field. Intracerebroventricular (ICV) injection was performed, and the animal is returned to its cage. 6 mice per group received ICV injection of 150 µg of the MAPT ASO or PBS (phosphate buffered saline).

[0087] After a 14-day incubation, animals were euthanized. The tissues of interest were harvested and weighed. The tissues are then stored in RNAlater™ stabilization solution (Invitrogen™ catalog number AM7024). Once all the tissues were collected, RNA was extracted from the tissues using the protocol for the RNeasy Kit (Qiagen™ catalog number. 74182. Reverse transcription was then performed using the High-Capacity cDNA RT Master mix (Applied Biosystems™ catalog number. 4374967). Analysis of changes in hMAPT expression from the tissue homogenates was measured using Fast-Advanced TaqMan™ Master Mix (Applied Biosystems™ catalog number. 4444557) along with validated pre-designed TaqMan primers for human MAPT (Hs00902193\_m1) and mouse GAPDH (Mm99999915\_g1) as a housekeeping gene. The qPCR was performed in MicroAmp Optical 384-well reaction plates using QuantStudio 7 Pro system. The delta-delta CT method of normalizing against samples that were treated with PBS control was used to determine relative amounts of MAPT gene expression.

TABLE 5

| summarizes the in-vivo activities of selected MAPT antisense oligonucleotides in different tissues of transgenic mice expressing the human MAPT gene. Antisense |                |               |               |               |               |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------|---------------|---------------|---------------|--|
| MAPT ASO No                                                                                                                                                     | Frontal Cortex | Striatum      | Cerebellum    | Brain Stem    | Spinal Cord   |  |
| PBS                                                                                                                                                             | 100.0 ± 7.825  | 100.0 ± 8.807 | 100.0 ± 5.194 | 100.0 ± 4.056 | 100.0 ± 3.726 |  |
| 1                                                                                                                                                               | 40.53 ± 8.643  | 55.08 ± 6.910 | 102.3 ± 6.262 | 57.91 ± 9.432 | 79.44 ± 7.965 |  |
| 2                                                                                                                                                               | 49.14 ± 4.246  | 65.28 ± 5.187 | 100.3 ± 3.412 | 66.55 ± 4.031 | 91.97 ± 1.882 |  |
| 3                                                                                                                                                               | 58.81 ± 4.871  | 66.55 ± 6.022 | 91.99 ± 4.492 | 67.61 ± 5.137 | 89.58 ± 5.040 |  |
| 4                                                                                                                                                               | 56.30 ± 8.123  | 63.68 ± 7.089 | 88.19 ± 4.695 | 65.54 ± 4.362 | 95.85 ± 9.168 |  |
| 5                                                                                                                                                               | 50.69 ± 8.700  | 46.86 ± 4.532 | 99.39 ± 4.039 | 48.22 ± 4.727 | 98.84 ± 6.570 |  |
| 6                                                                                                                                                               | 56.41 ± 5.546  | 52.97 ± 3.938 | 101.2 ± 4.742 | 44.13 ± 3.688 | 94.31 ± 2.996 |  |
| 7                                                                                                                                                               | 95.84 ± 6.839  | 95.15 ± 10.50 | 110.8 ± 6.958 | 137.1 ± 10.63 | 96.31 ± 3.672 |  |
| 8                                                                                                                                                               | 69.89 ± 6.987  | 74.51 ± 10.57 | 113.2 ± 4.399 | 134.6 ± 11.97 | 101.7 ± 4.444 |  |
| 9                                                                                                                                                               | 69.67 ± 6.117  | 78.68 ± 6.420 | 124.8 ± 7.841 | 97.14 ± 13.59 | 96.24 ± 6.407 |  |
| 10                                                                                                                                                              | 76.27 ± 9.847  | 78.33 ± 4.310 | 115.9 ± 2.247 | 91.91 ± 8.329 | 103.6 ± 6.060 |  |
| 11                                                                                                                                                              | 62.31 ± 7.683  | 61.20 ± 4.366 | 97.20 ± 6.065 | 23.91 ± 2.600 | 71.36 ± 9.038 |  |

Sequence Listing

| SEQ ID NO | Sequence                                                                      |
|-----------|-------------------------------------------------------------------------------|
| 1         | CTTTCTTGAAAGTATATTC                                                           |
| 2         | GCTCAGAGCTACAGCAAGTC                                                          |
| 3         | GCTGCAGGGAAAGAGTTCTT                                                          |
| 4         | GATGGAGACAGTGTGTATCT                                                          |
| 5         | GCAGATGGGGCCAAATCCT                                                           |
| 6         | GCCACAGTGACAGCAAATTC                                                          |
| 7         | GCCACAGACTCGAGTATCAG                                                          |
| 8         | CCAAACGTGACTTGGCTCCT                                                          |
| 9         | TATGGACACACATGGGTGTA                                                          |
| 10        | ACTAAAGACGTGTCTTTTGG                                                          |
| 11        | GTCACTCTAGGTGACTTCC                                                           |
| 12        | [eCm]seToeToeTo[eCm]sdTsdTsdGsdGsdAsdAsdAsdGsdTsdAseToeAoeTseTs[eCm]          |
| 13        | eGs[eCm]oeTo[eCm]oeAsdGsdAsdGs[dCm]sdTsdAs[dCm]sdAsdGs[dCm]seAoeAoeGseTs[eCm] |



-continued

| SEQ ID NO | Sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14        | eGs[eCm]oeToeGo[eCm]sdAsdGsdGsdGsdAsdAsdAsdGsdAsdGseToeTo[eCm]seTseT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 15        | eGseAoeToeGoeGsdAsdGsdAs[dCm]sdAsdGsdTsdGsdTsdGseToeAoeTs[eCm]seT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 16        | eGs[eCm]oeAoeGoeAsdTsdGsdGsdGsdGs[dCm]s[dCm]s[dCm]sdAsdAseAoeTo[eCm]s[eCm]seT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 17        | eGs[eCm]o[eCm]oeAo[eCm]sdAsdGsdTsdGsdAs[dCm]sdAsdGs[dCm]sdAseAoeAoeTseTs[eCm]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 18        | eGs[eCm]o[eCm]oeAo[eCm]sdAsdGsdAs[dCm]sdTs[dCm]sdGsdAsdGsdTseAoeTo[eCm]seAseG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 19        | [eCm]s[eCm]oeAoeAoeAs[dCm]sdGsdTsdGsdAs[dCm]sdTsdTsdGsdGs[eCm]oeTo[eCm]s[eCm]seT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 20        | eTseAoeToeGoeGsdAs[dCm]sdAs[dCm]sdAs[dCm]sdAsdTsdGsdGseGoeToeGseTseA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 21        | eAs[eCm]oeToeAoeAsdAsdGsdAs[dCm]sdGsdTsdGsdTs[dCm]sdTseToeToeTseGseG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 22        | eGseTo[eCm]oeAoeTs[dCm]sdTs[dCm]sdTsdAsdGsdGsdTsdGsdAs[eCm]oeToeTs[eCm]s[eCm]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 23        | 1 GCAGTACACG CCACCCACCA GCTCCGGCAC CAACAGCAGC GCCGCTGCCA CCGCCACCT<br>61 TCTGCCGCCG CCACACAGC CACCTTCTCC TCCTCCGCTG TCCTCTCCCG TCCTCGCCTC<br>121 TGTCTGACTAT CAGGTAAGCG CCGCGGCTCC GAAATCTGCC TCGCCGTCCG CCTCTGTGCA<br>181 CCCCCTGCCG GCGGCCCTC GCCCTCCCTC TCCGCAGACT GGGGCTTCGT GCGCCGGGCA<br>241 TCGGTCCGGG CCACCGCAGG GCCCTCCCTC GCCTCCCTG CTCGGGGGCT GGGGCCAGGG<br>301 CGGCCTGGAA AGGACCTGA GCAAGGGATG CACGCACGCG TGAGTGC GCGTGTGTGTG<br>361 TGCTGGAGGG TCTTACCAC CAGATTGCGG CAGACCCAG GTGGAGGCTG TGC CGCAGG<br>421 GTGGGGCGCG GCGGCGGTGA CTTGGGGGAG GGGGCTGCC TCACTCTCG ACTGCAGCCT<br>481 TTTGCCGCAA TGGGCGTGTG TGTGTGTGTG TGTGTGTGTG TGTGTGTGTG TGTGTGTGGA<br>541 GGGGTCCGAT AACGACCCC GAAACCGAAT CTGAAATCCG CTGTCCCTGC CGTGTTCGC<br>601 CATCAGCTCT AAGAAAGACG TGGATCGGGT TCTAGAAAAG ATGACTCCCT GCACGCCCCT<br>661 CCCTGCACCT CCCGAGCAGT GATTCCGACA GGGCCTTCAC TGCCCTGAT TTAGGCGGG<br>721 GGGCGGCCCT CTCCCCTTTT CCTCCTTCAG AAACCCGTAG GGGACATTG GGGCTGGGA<br>781 GAAATCGAGG AGATGGGGAG GGTCCACGC GCTGTCACTT TAGTTGCCCT TCCCCTGCG<br>841 CACGCTGGC ACAGAGACGC GAGCAGCGCC GTGCCTGAGA ACAGTGC GCGATCCCACTG<br>901 TGCACGCTCG CAAAGGCAGG GTTACCTGG CCTGGCGATG TGGACGGACT CCGCGCCGC<br>961 TGGTCCCCGT TCGCGGCAC GCACAGCCGC AGCCACGCAC GGATGGGCGC GGGGCTGCAG<br>1021 GTGCATCTCG GGGCGGATTT CTTTCTCAGC GCTCGGAGCG CAGGGCGCCC GCGTGTGCG<br>1081 CTCCCTGCCG GAGGCGCGGG GCTGGCGCGC AGGGCTCGCC CCTCACTGCG GCAGTGGGTG<br>1141 TGGACCTTGG TGGCGAGGA AGGGGAGGA TAGGCTGTGC TCCTCCAC TCCCGCCCC<br>1201 AGCCCCCTT TTTTCCCTC TCGAACGCG AGGIGCCATC TTTTTCGCG GTGTACGCT<br>1261 TTTACGGTGC CATGCCAAAC CGGGTGGCCG GGTTCATAG GACAGGGCGG GGCCTGGCAT<br>1321 TAAAGGGAGG GGGACAATCA GCGCTGAAAT CTTGGCGTTT TGCTGTGCG GCGTGAGCA<br>1381 CTGGGGGCGT TCGCCAGCA CCTTCTTCG GGGCTCTTTG CTTTGTCTGT AGAGGTTACG<br>1441 TGATCTGCGC TCCAGCCCT GGTTCCTGCG TTTTATTCTG AGGGTGTCA GTCAACCTC<br>1501 CCCCTACGCC CATGCGCCTC TCTTCTCTTT TTCGCTCCTC ATTTCCGAGC CCATTGTTGG<br>1561 ATCTCGAGGC TTGCTGGGTT CGATGAACTC GAGTCAACCC CCCGACCCC GGCACGCATG<br>1621 GAACGGGCGT GACCGCGCGC AGCCTCGTCT CGGAGTCTGC CCGCGCCGGG AAGCTTCTGA<br>1681 AGGGATGGGA TTCGAGTCTC CGTGC GCGCGCGC GCAGAGGGAT CTCGCCCTC<br>1741 CCTACACCCC AAGTGTCTG AGGGCCACGC CACACCAAGT TGCCACGCA GGGACGCTGG<br>1801 CTACCCATTC GGGATGGGT GGGAGCCCT GCGGGGCGCT CTCGCGCTT ACGCCCTGTT<br>1861 GCTTCGCTG GCCGGAAT GTGAGGAAG GGCATAAGGT TACTGGTGT TCGGCCACAC<br>1921 CCATCTTTCT GAGCCACTG GACTGGGCGC AGAGGGGGGA TTGCCATGGA AACCAAGGT<br>1981 GTCCGGAGAG GGGATCTTGG GGCTGGCCTC ACCCTTCCC TGCGGAGATT GGGGACCTG<br>2041 GGGTAGGGGG AGCCGCGCCC AGTCGCGCTC CTGGAGGACA CGGGAGGAAG CCCCGAACCC<br>2101 CCGCGCCTGA GGCTGTTTCT GATTGGCCCC TGGAGGCCGC AGACACGCAG ATAGGCGGCC<br>2161 CTGGGTGTAT TTTTATTAAT ATTATGTCCG TACTGATTAA TATTATTAT CTTAAATAAA<br>2221 TTTACCCGT GTCCAAGTTC ACCGCGCCCC CAAAACCGAG TCTGGGCGCG CAGGGGGAAC<br>2281 TCCTGGCCAA CGAATCCATG CCTCGCCCTC CTGTGATGAA CCTGGTACGC ACGGTTTTCT<br>2341 GGTTAATTCT ATCGCTGAAA ACTGGTGCGG GGGGCGCACT TCTGAGACGG AAGAGCATCT<br>2401 AGGAGCTGAA TCCTCCACGC GGTTCGCCCC GGTGATCTG AATTTCCTGG GAATGGCTTG<br>2461 GCTGCCGCC CCGGACCAGG CCGACCTCC TTGACGGTGG CGTAGAGGGC TGGAGCCTGG<br>2521 GTACTGCGAG GCTCTCGCA TGGCTGGGCC CGCCGCGAGG GGTTCAGAG CCGCTCAGGG<br>2581 ATCGATTCAA GCATCGTCTC TCCTCCCTCG CCCCAGACA GAGCTGGGCG CGGGGTTCCC<br>2641 CTTCAGATG GAGCGAGGGT CTCGGGGTGG CCCCGGAAA GGGGAGCCCG CCGCCACGGC<br>2701 TACGTATTGC CATCTCGCGA GCAGAGATGT CACCTCTGCG CTTTGGAGGA AAGGAGCCC<br>2761 GTTGGGGATG AGCGCATTTA GCCAATGCT GGGAAACAAAG CGCACTCCGC GCTTCTGCGA<br>2821 TTTGCTCCA TTTTGAAATG TGTGCGCGCT TTGGTGGGGC CGCTGCGGTG GGCAAGGCCG<br>2881 GGGGCGCTGT TAATGGAGGA ACCTCAGGGG GACGGTCTCT CGTAGGAAAC TCTATCCTGG<br>2941 CTCTGCGCGC GCTTAAAGGA AATGGCTTCC CTCAGGACC TCGAGGGATG CAGCTTTTGC<br>3001 GCGGATGACG GTGGGGTGCT GAACCAGCCG GTGCGCTCT GGAATGTCT GGGCACGGAT<br>3061 CCTGGGGCCA TCGACGACTC CTCCCATTG CCAGCAGGCG GGAGCTCTTA CATTCGAGC<br>3121 GAGTGACCCC TCTCACCTC TGGCGCTCAC ACACCTGTAA CTCCAACCT CCGTCTCAGA<br>3181 ATGGTCCAGG CTGGAAGGGA TGATGGGGGC TCCGACAGCG ACTGCCTAGC TCACCCCTCT<br>3241 GCGTGCTCAG GCTCCAGGCT CAGCAGGACC AATTGTAGTT CTATCTGATC CCCCTCGGCC |

-continued

SEQ ID NO Sequence

|      |         |      |      |      |       |      |     |     |     |     |     |      |      |     |     |     |     |     |
|------|---------|------|------|------|-------|------|-----|-----|-----|-----|-----|------|------|-----|-----|-----|-----|-----|
| 3301 | CCTTAAC | TGA  | CCC  | ATC  | TCTAC | AGG  | GAG | CAC | AGG | GAA | ATG | TCTT | TCCT | ACC | GCG | GTG | ATT | CTG |
| 3361 | GGGTG   | TCT  | CATT | TTGT | GTTT  | TG   | AT  | GG  | TG  | CT  | GC  | TTAT | ATTT | TAC | TGT | ATA | AAG | CA  |
| 3421 | TGTATA  | AAG  | CA   | TTG  | TATT  | TATA | ATT | ACT | GT  | AT  | AAG | CTG  | CTT  | A   | ATA | AAG | CAT | CT  |
| 3481 | CCAAAT  | CTCT | CCT  | CT   | TAC   | GT   | AAC | AA  | ATT | AA  | TGG | ATA  | AA   | CA  | GAT | AAG | TG  | TA  |
| 3541 | CCACCC  | CTGC | TAC  | G    | CAG   | GT   | C   | G   | G   | A   | G   | CT   | C    | T   | T   | G   | A   | A   |
| 3601 | TGCTT   | CTCT | ATA  | CAG  | AT    | GTTT | AT  | ATA | G   | CA  | AT  | AAC  | CT   | G   | G   | CT  | T   | T   |
| 3661 | GACGAT  | TTTG | GG   | A    | AGG   | GG   | CT  | T   | G   | T   | G   | G    | CT   | T   | T   | G   | G   | T   |
| 3721 | CTGTT   | CC   | TG   | G    | AGG   | GG   | CT  | T   | G   | A   | CT  | T    | T    | T   | A   | G   | T   | T   |
| 3781 | GACAGG  | AACA | AAA  | G    | AA    | TGG  | AT  | ACT | G   | CC  | CA  | TAA  | AT   | T   | G   | T   | A   | A   |
| 3841 | ATTGAG  | GAAA | GGT  | T    | AG    | ATTT | AT  | T   | C   | G   | G   | TT   | G    | AAA | A   | AG  | T   | G   |
| 3901 | TTTGAC  | CC   | AT   | G    | CC    | T    | T   | G   | G   | A   | G   | G    | T    | G   | A   | G   | A   | T   |
| 3961 | TCCTT   | CT   | T    | G    | C     | C    | T   | T   | T   | A   | C   | A    | T    | T   | T   | A   | A   | A   |
| 4021 | TTTTAT  | T    | C    | T    | T     | C    | C   | T   | C   | A   | G   | A    | T    | T   | T   | T   | T   | T   |
| 4081 | TACGA   | A    | C    | A    | C     | G    | A   | G   | A   | G   | A   | G    | A    | G   | A   | G   | A   | G   |
| 4141 | TCTATT  | G    | A    | T    | T     | T    | A   | T   | T   | T   | A   | G    | A    | T   | T   | T   | A   | T   |
| 4201 | AACAA   | A    | T    | A    | A     | A    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 4261 | CATAC   | C    | A    | A    | A     | A    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 4321 | GGAA    | A    | A    | A    | A     | A    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 4381 | AGAC    | C    | C    | C    | C     | C    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 4441 | TGGC    | C    | A    | G    | A     | G    | A   | G   | A   | G   | A   | G    | A    | G   | A   | G   | A   | G   |
| 4501 | CAGG    | T    | G    | A    | C     | G    | A   | G   | A   | G   | A   | G    | A    | G   | A   | G   | A   | G   |
| 4561 | CTCAT   | G    | T    | T    | C     | C    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 4621 | GCCAC   | T    | G    | A    | G     | A    | G   | A   | G   | A   | G   | A    | G    | A   | G   | A   | G   | A   |
| 4681 | ATCTT   | G    | T    | G    | G     | A    | G   | A   | G   | A   | G   | A    | G    | A   | G   | A   | G   | A   |
| 4741 | CTCCT   | A    | C    | A    | T     | C    | A   | T   | C   | A   | T   | C    | A    | T   | C   | A   | T   | C   |
| 4801 | TGCTA   | T    | C    | T    | T     | A    | A   | T   | A   | T   | A   | T    | A    | T   | A   | T   | A   | T   |
| 4861 | AGATT   | A    | T    | A    | T     | A    | T   | A   | T   | A   | T   | A    | T    | A   | T   | A   | T   | A   |
| 4921 | CGTG    | A    | T    | G    | A     | T    | G   | A   | T   | G   | A   | T    | G    | A   | T   | G   | A   | T   |
| 4981 | TGGT    | G    | T    | G    | T     | G    | T   | G   | T   | G   | T   | G    | T    | G   | T   | G   | T   | G   |
| 5041 | ACTT    | G    | G    | C    | C     | A    | T   | G   | T   | G   | G   | A    | C    | C   | A   | T   | G   | T   |
| 5101 | AATAT   | T    | G    | A    | T     | T    | G   | A   | T   | T   | G   | A    | T    | T   | G   | A   | T   | T   |
| 5161 | CATG    | A    | C    | C    | C     | C    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 5221 | CCGAC   | A    | C    | A    | C     | A    | G   | A   | G   | A   | G   | A    | G    | A   | G   | A   | G   | A   |
| 5281 | AAAT    | A    | T    | A    | T     | C    | T   | C   | T   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 5341 | TGCC    | A    | T    | A    | T     | G    | A   | G   | A   | G   | A   | G    | A    | G   | A   | G   | A   | G   |
| 5401 | GGAT    | C    | A    | A    | G     | A    | C   | A   | G   | A   | C   | A    | G    | A   | C   | A   | G   | A   |
| 5461 | TCCC    | C    | A    | G    | A     | T    | T   | G   | A   | T   | T   | G    | A    | T   | T   | G   | A   | T   |
| 5521 | TGCT    | C    | T    | G    | A     | G    | A   | G   | A   | G   | A   | G    | A    | G   | A   | G   | A   | G   |
| 5581 | ATCT    | T    | T    | T    | T     | G    | G   | C   | T   | G   | G   | A    | G    | A   | G   | A   | G   | A   |
| 5641 | ATAA    | G    | A    | T    | A     | T    | T   | C   | A   | T   | T   | C    | A    | T   | T   | C   | A   | T   |
| 5701 | TGCT    | T    | G    | C    | C     | T    | T   | G   | A   | G   | A   | G    | A    | G   | A   | G   | A   | G   |
| 5761 | TCCT    | G    | C    | C    | A     | C    | C   | A   | G   | A   | G   | A    | G    | A   | G   | A   | G   | A   |
| 5821 | GAAT    | T    | T    | G    | A     | G    | A   | G   | A   | G   | A   | G    | A    | G   | A   | G   | A   | G   |
| 5881 | GCAG    | A    | T    | A    | C     | A    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 5941 | ATTT    | T    | C    | T    | T     | T    | T   | G   | A   | G   | A   | G    | A    | G   | A   | G   | A   | G   |
| 6001 | TAGAA   | A    | T    | C    | C     | A    | T   | G   | T   | C   | C   | A    | T    | G   | T   | C   | C   | A   |
| 6061 | CAAG    | G    | A    | A    | G     | A    | G   | A   | G   | A   | G   | A    | G    | A   | G   | A   | G   | A   |
| 6121 | GCAGG   | C    | A    | G    | T     | G    | C   | A   | T   | G   | A   | G    | T    | G   | C   | A   | T   | G   |
| 6181 | ATCGG   | C    | C    | T    | C     | A    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 6241 | TTT     | A    | G    | C    | T     | G    | C   | T   | G   | C   | T   | G    | C    | T   | G   | C   | T   | G   |
| 6301 | CGGG    | T    | T    | G    | C     | C    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 6361 | ACTG    | C    | T    | C    | A     | G    | T   | C   | A   | T   | C   | A    | G    | T   | C   | A   | T   | C   |
| 6421 | ATG     | A    | C    | C    | A     | G    | A   | G   | A   | G   | A   | G    | A    | G   | A   | G   | A   | G   |
| 6481 | CTG     | A    | G    | T    | A     | G    | A   | G   | A   | G   | A   | G    | A    | G   | A   | G   | A   | G   |
| 6541 | GCAG    | A    | A    | G    | T     | G    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 6601 | TGCC    | A    | C    | A    | G     | A    | G   | A   | G   | A   | G   | A    | G    | A   | G   | A   | G   | A   |
| 6661 | CACT    | C    | C    | A    | T     | G    | C   | T   | C   | A   | T   | G    | C    | T   | C   | A   | T   | G   |
| 6721 | CCAG    | C    | C    | A    | T     | G    | C   | T   | C   | A   | T   | G    | C    | T   | C   | A   | T   | G   |
| 6781 | TACT    | C    | G    | T    | T     | G    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 6841 | GGAG    | G    | T    | A    | A     | A    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 6901 | TGG     | C    | A    | A    | A     | A    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 6961 | AAT     | T    | T    | G    | T     | G    | T   | G   | A   | T   | T   | G    | A    | T   | T   | G   | A   | T   |
| 7021 | ATTA    | A    | T    | A    | A     | A    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 7081 | ATC     | A    | G    | T    | T     | T    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 7141 | GAT     | T    | T    | T    | T     | A    | C   | C   | G   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 7201 | GCT     | C    | A    | G    | G     | T    | C   | T   | T   | T   | G   | T    | G    | T   | G   | T   | G   | T   |
| 7261 | TGCC    | A    | C    | A    | T     | T    | A   | A   | T   | G   | C   | A    | T    | T   | A   | A   | A   | A   |
| 7321 | TTGT    | G    | G    | A    | T     | G    | C   | A   | T   | G   | C   | A    | T    | G   | C   | A   | T   | G   |
| 7381 | GGT     | C    | T    | T    | T     | A    | C   | C   | A   | T   | T   | T    | G    | A   | A   | A   | A   | A   |
| 7441 | ATT     | C    | A    | A    | C     | T    | A   | A   | A   | A   | A   | A    | A    | A   | A   | A   | A   | A   |
| 7501 | TACT    | C    | C    | A    | T     | C    | A   | C   | T   | A   | T   | T    | T    | T   | T   | A   | A   | A   |
| 7561 | GCCC    | A    | G    | G    | C     | T    | T   | A   | A   | G   | T   | G    | C    | A   | A   | T   | G   | A   |
| 7621 | AAG     | C    | A    | A    | T     | T    | C   | C   | A   | T   | T   | C    | C    | A   | T   | T   | C   | C   |
| 7681 | GCT     | G    | A    | T    | T     | T    | T   | T   | G   | T   | G   | T    | G    | T   | G   | T   | G   | T   |

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SEQ ID NO Sequence

|       |             |            |             |             |             |             |
|-------|-------------|------------|-------------|-------------|-------------|-------------|
| 7741  | CTTCTGACCC  | AAGTGATCCA | CCCACCTCGG  | CCTCCCAAAG  | TGCTGGGATT  | ACAGGTGTGA  |
| 7801  | GCCACCCGCG  | CCGGCCAGCT | TGCACCTTAT  | TTAGGATATG  | TGATTATTAT  | AGCAAGTCTG  |
| 7861  | GTGTACATAC  | AAGATTTTGA | ATGGGCACAG  | ATGACCTTTA  | GTAAGTGTCT  | GGCTGTGATA  |
| 7921  | AGAGGCGAGT  | CTGACTGCAG | ATCAGGCTGT  | GTGGACCCCA  | GCCTTGCAATG | TTTACAGACC  |
| 7981  | TTCATGTCTT  | ATTCTTACAG | GGTATCAGAA  | GAACACCTAC  | TGGGGAAACT  | TATAAATTAG  |
| 8041  | TAAAAGGTGG  | GCATTCTCCC | CGCCCATCTT  | CTGTCTGTCT  | GCCAGGACTA  | GCACAGCACT  |
| 8101  | TTGAAGTCAT  | TCACATAGAA | TCCCAACTTA  | AGAGGGTAAA  | ATCCTCCTCA  | ACAGACTGAA  |
| 8161  | AATAAGTTTA  | AATTCCTTTT | GCTATATTAA  | CTCCCTGAG   | GAAAGAGTCT  | TAGATCAATG  |
| 8221  | TCCAACACTA  | AAAACAGTTT | TAAATCAGCA  | AGTGAGAAAT  | AAATCTGAAG  | CAATTGATAA  |
| 8281  | TAATGTTTCA  | TTCAATTCCT | TCCTTTGGCC  | CCGTCCACCC  | TACTGCTAAA  | TCCAGGCATC  |
| 8341  | AAAGAGAAGA  | GGGACATAAT | TATCTCTAGT  | CCCAGCTGCT  | GGTTTTCTCT  | CCAGCCTATG  |
| 8401  | GCCCAGTTTT  | CTGTTTTACT | GAGAAGGCTG  | GTGATGTTAT  | CTTGGGATCT  | AAGTCTGCAG  |
| 8461  | TTTCACCACA  | AAAAGTCCAG | GGATGCACCT  | TCATGCTTGT  | GTCCTCCTCC  | CTGGGATAGC  |
| 8521  | AAGGATGTTA  | GAAGACCCCT | GGCTCTGTAA  | TTGCTTGTTA  | TGTGCTCTAC  | AGACGCCACA  |
| 8581  | GAATGCCAAG  | AACGAAGTGC | TGGGAAGGAC  | AAATTCATGG  | AACCGTGGGA  | CGGTGCTCCT  |
| 8641  | CCCCCAGCGT  | AAAGGACAGC | TCCTCCTCCT  | GAATGGGAGC  | CAGCGTTCTA  | AATCATGTGT  |
| 8701  | CAACAGAGTT  | GTCTTGATC  | GGATCCAGTT  | CTGCCATTGA  | TTTGCAGGTC  | ATTTCAGTGG  |
| 8761  | TACCTGTTTC  | CAGTTGTTCT | TAATTGAACA  | GTGGCACCAC  | ACTATTGTCT  | TGCCTCATCC  |
| 8821  | CCCTCCCCTG  | GCCTGTCCCC | CAAAAAGAGA  | CTTCTTGGGT  | AATTAATCAG  | GGCAACATCA  |
| 8881  | GGCAGTCTGG  | GCGCGGTGGC | TCACGCTGTG  | AATCCCAGCA  | CTTTGGGAGG  | CCGAGGCGGG  |
| 8941  | CAGATCATGA  | GGTTAGGAGA | TTGAGACCAT  | CCTGGCTTTG  | TGAAACCCCG  | TCTCTACTAA  |
| 9001  | AAATACAAAA  | AATTAGCCGG | GCGTGGTGGC  | GGGCGCCTGT  | AGTCCCAGCT  | ACTCGAGAGG  |
| 9061  | CTGAGGCAGG  | GGATGGCGT  | GAACCCGGGA  | GGTGGAGGTT  | GCAGTGAGCC  | GAGATCGCAC  |
| 9121  | CACTGCACCT  | TAGCCTGGGC | GACAGAGCTA  | GACTTCTTCT  | CAAAAAAAGG  | AAAAAAGG    |
| 9181  | AATCTCTTTG  | GTTTTATATA | TATTTTTTTT  | ATATATATAA  | TATATATTAA  | AATATAATAT  |
| 9241  | ATATATTTAT  | ATAATATAAT | ATATAAATAT  | ATTATATATT  | ATATATTTTA  | TATATTATAT  |
| 9301  | ATTATATATA  | TTATATATTA | TATATTATTA  | TATTTATATA  | TTATATATAT  | TTATATATTA  |
| 9361  | TATATTTTAT  | TATATTATAT | ATTTATATAT  | AATATATATT  | ATATATTTAT  | TATTTATATAT |
| 9421  | TATATATTAT  | ATATTATAT  | ATATTATATA  | TTATATATAT  | TATATATTAT  | ATATTATATAT |
| 9481  | ATTATATATT  | TATATATATT | ATATATTATA  | TATTATATAT  | TTATATATTA  | TATATTTATA  |
| 9541  | TATTATATAT  | ATTTATATAT | ATTATATATT  | ATATATTTAT  | TATGTATATA  | TTATATATAT  |
| 9601  | TATATATTAT  | ATATATTAT  | ATATATAATA  | TATTGTATAT  | ATTATATATC  | TAATATATTA  |
| 9661  | TATATATTAT  | ATATATTATA | TATTATAATA  | TATATTATAT  | ATTATATATT  | ATATATATTT  |
| 9721  | TTATATATAT  | AATATGTATA | ATATATAATA  | TATATAAAAA  | CATATATAAT  | ATATATTATA  |
| 9781  | TATTATATAT  | ATATTATATA | TATTATATAT  | ATTAAATATA  | TTTTATATAT  | ATTATATATA  |
| 9841  | TATACACATA  | TATATATATA | AATGAGGCCA  | GGCTCGGTGG  | CTCACACTTG  | TAATCCCAGC  |
| 9901  | ACTGTGGGAG  | GATCACTTGA | AGCCAGGAGT  | CTGAGACTAG  | CCTGGGCAAC  | AAAACAAGAT  |
| 9961  | CCTGTCTCTA  | CAAAAGGAAA | CTGTAAAAAT  | TAGCTGGGCA  | TGATGGCATG  | TGCTGTAGC   |
| 10021 | CCTAGCTACT  | TGGGAGGCCG | AAGCAGGAGG  | ATCGCTTGAG  | CCCAGGAGTT  | CAAGGCTACA  |
| 10081 | GTGAGCTATG  | ATTGTCCCAT | AGCACTCCAG  | CCTGGGTAA   | ACAGCAAGGC  | CCTGTCTCTA  |
| 10141 | AACCTTTTTT  | TTTTAATCT  | ATTTATATTT  | ACATGTATTT  | AAATGTGAAT  | ATTCACTACC  |
| 10201 | TATTTGTTGC  | ATGCCTGCAT | TTTTTATACT  | GGGCTTGCCA  | AAAACCCGAA  | CAGCTTTCTA  |
| 10261 | CTTTGACAAT  | GTATCAGAAT | TTAAATCAGC  | AATATGTTAA  | TAGCCCAAGC  | AAAGGTTATA  |
| 10321 | TATGCAAAAT  | AAACTGTTGT | CTATAACCTC  | CTGTTACACT  | GGGGCACAGC  | AAAAGTCATG  |
| 10381 | GTGTAGTCGC  | ATGTGAACCT | GTCCCTTTCA  | TAGCTGCTCA  | TTGCCAGGAA  | ACATCAGGAA  |
| 10441 | TAGCCATTGG  | GAAGAGTCAT | CAGCCCTCCC  | ACCATCCGTT  | TCTGTCTTGT  | TCTTTTCCCT  |
| 10501 | ATGAGCAGGG  | GAAATTCAC  | GCTGGCCCA   | ATCCCAGTGT  | CAGCGGCTCA  | GCCTCTGCCT  |
| 10561 | CTGCTGCTGG  | TCCCCATGAG | GCCAGCTTAG  | AAACGGAGGA  | TTTTGCAGAA  | CATCCCTAAA  |
| 10621 | TCCGCTTGAA  | TAATGAAGTG | ATCATTCTTA  | AACTCACCTG  | AACTTATATTA | AAACCTATTT  |
| 10681 | AATATTTTTT  | CTGGATAATC | CTATAGGGAT  | AACTTGCCCT  | CTGGGCTTCT  | CTCCACCGGG  |
| 10741 | TTCAAGTCTT  | CCTTTAGTGG | TGAAGTTCCT  | CCCTTCTTAG  | CATCTCAACT  | GTGCTGAGA   |
| 10801 | AAAGGCCAGT  | GGCGGCTGCA | CTCTGTTCCT  | TGTGGAGTGT  | TAATAAAGAC  | TGAATAAAT   |
| 10861 | GAAATAAATC  | CCTTTCAATG | TCATTAAGTG  | CTATAAATAA  | TCATGAACCA  | ATGTTTCGATG |
| 10921 | GCTGATGAGA  | AATGCAAGAA | AAAATTTTTA  | ATCAGTAGGA  | TTTATAAGTT  | GACAATCTGG  |
| 10981 | GCCAAAGTTAA | AAAAAATAAA | AATAAAAAAG  | CTTTTAAAAA  | GATCTTATCG  | TTTGTACCA   |
| 11041 | GTAAGACTGA  | ATTCAGAAAG | CAAGCTACTC  | CCTCATTTGT  | GGGCCCTGT   | TATCACTGGC  |
| 11101 | TGCTTAGGGT  | TGCCAAGCCC | TGAATTCATT  | TGTCAACTAA  | GAGATTTTTG  | GCCAAGATTA  |
| 11161 | AGATTTCCCA  | TGCCTCCATA | TTTCCATCTG  | AGAAATGGAG  | ATTATACTGT  | CTTCCCCCTC  |
| 11221 | AGAAATGGATG | ATAATGTGGT | CTCTCTTCTG  | TTCGCATAGT  | CATAGAACTG  | AAATAAAACA  |
| 11281 | ACTTAAGAGA  | ATTCTTTTGA | GCTTCTCAGA  | AGTGCTGCAG  | GGCTGGGGGA  | TGCCTCCAG   |
| 11341 | GAGCCGCGAGT | CAGGTGCTGA | TCTGAAAGTCT | TTGGTGGGCT  | GACTTTAGCC  | TGACCTGAAA  |
| 11401 | TAGTATAGCT  | GCTGCCACCT | GGCTCCCTTA  | GCGTCAGTCA  | GACGGTGCAG  | CTGGTTCCCTA |
| 11461 | GGGGTGAGGG  | CTGAGCCAGC | AGGGTCCGTG  | CCCAGGAGGG  | ATGCATGGGT  | GGCCACAGCC  |
| 11521 | CAGCCTGCAC  | TGATCTTGTC | TGTCCCTTTC  | TTTGGAAAGGA | AGGAGCCCCA  | AACCAGGGTG  |
| 11581 | CAAGACAGTG  | GGTGGGGGTG | CCTTGAGCAT  | GACCTCAAGT  | GATTTCCAGC  | CCCTGCCAGT  |
| 11641 | GCTGACTTCT  | CTGGGGGAAG | GCTGGGACTT  | CCTTCTGGGC  | TCAAGTCACG  | ACCCTTGGAT  |
| 11701 | GGAATTTCTT  | GGGAGCTTTT | CTGTTTTTTC  | TGGAGTTTTT  | AGTTTTTTTC  | TAACCAGACA  |
| 11761 | GGGACTTGGT  | ACAGAATCTC | ATATTTCTAAT | TATGCTTAGG  | AGCAGCTCT   | CCCCACCACT  |
| 11821 | CACAGTGTGT  | AGCATGTGAC | AGGAATCGAT  | TAAGGCATGA  | GTGATTAAAT  | TAAAGCCAGG  |
| 11881 | CATTGACTTG  | GATGGTGTAA | TATTTCTGACA | TCTGTTTGGT  | GTCAAAGGCA  | CGGGGCAGGC  |
| 11941 | GCGTTAATTT  | AACTGCTTGC | ACCTGGCATT  | TGAATTGAGC  | CAGAGCGGGG  | CTAAAGTCAG  |
| 12001 | TTTGCCCTTCA | CCCTGTAAAT | GGAGGGTTTC  | TCCGGAGCGT  | GGATGGTGGG  | AGGTATTTC   |
| 12061 | GGGTGTATGC  | ATAACCCCA  | CCCTGACAA   | GGCCCATCTC  | TTCTCCAGCG  | TGGCCAGGTT  |
| 12121 | TGAGTGCCAG  | TCCTGGGTGT | CCAGTGGCCC  | CATAGCCTTG  | CGTTTTAGTA  | AAATGCTGCC  |

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SEQ ID NO Sequence

|       |             |             |            |             |             |            |
|-------|-------------|-------------|------------|-------------|-------------|------------|
| 12181 | CCCATTACCA  | CCTGGTCTGT  | GCACTTCGGT | CACTGGAATT  | TGCCATCTTC  | CAGTCCCCGA |
| 12241 | TGTGGCAAGC  | CATGGAGCCT  | TAAGCTCTTC | TCCCTCCACA  | TCCTGGAACA  | GACCCGCCAG |
| 12301 | TTTCTTCCAG  | GCATTGCCCTC | AGTTTGCCCC | TCTGTTTCCA  | GTCACACTCT  | CACCAGCGAT |
| 12361 | AAAAATGATT  | TAGACCTTAT  | CATCTCACCC | TCGGATCCTT  | ATGGAAACAA  | TAATGAGTTG |
| 12421 | TTCCCTGTTT  | CAATTCCAAA  | ATTATATATC | AATCCGTTTT  | GCATGCCATT  | GCCAAATTCC |
| 12481 | TCCCAGAGCA  | ACCCCGTCAC  | CTGCCCTGGC | CCTCTCCAAG  | TGTGGTCCTG  | CCATGGGCAT |
| 12541 | CGCCTGCATA  | GCCAAGCTGG  | CCTCGAGCTG | CCTGCCCGGG  | TCCCCACACC  | TTGGCTCACC |
| 12601 | TCCCTGCCCA  | GTCCCGCTCT  | CTGCCAGCCT | GCCCTGTGGC  | TCCTTCATAG  | ATGCCGTGCT |
| 12661 | CTTTCTGCCC  | CTTGCTCACC  | CATGGCAGCC | TTGCCCTCTC  | CTCCCTGCCC  | CACCCCTTAT |
| 12721 | TTAAATTGAC  | CTGACCTTCC  | TCAGTGTCCA | TCTTCCCGGA  | AGCTTTCCCC  | AGCCTTGGGA |
| 12781 | CTCAAGGTCC  | AGAGGCTACG  | CGTTTCTCTC | CACCTGTGGC  | AGCGCCGTGC  | TCCCCAGTGC |
| 12841 | CTCACAGTTT  | CCTTCTTGCC  | CCCGCTTCTC | GTGTAGGACT  | CATCTGCCCC  | CAGGTTGCAC |
| 12901 | GTCCTGTGAG  | GGCAAGGACT  | GTGTCTTATG | TGACTTTCCT  | TCTCCAGTCA  | CAGAGCTGGG |
| 12961 | CACATAGATA  | GCTCAAAACC  | CTCTTTATTA | ACACAGTTGG  | ATGTTGAGAA  | ATCAAAACAG |
| 13021 | CCAATGTCAA  | ATGAGCTCTC  | CTTATTTAAA | TCAAGTCAGT  | TCTCCACCTC  | CTAGCACTCA |
| 13081 | GTTCCAGTAC  | TCTATATACA  | TGGAAATAAT | AAAAAACACA  | TTTCCTTTGA  | AACATTCTAT |
| 13141 | AATCCTGTCT  | TTGCCCTACT  | TCAGACCAAC | TTAACGCAC   | CCCCATTGGT  | CCAATAGAGT |
| 13201 | TTTGCTATAC  | GAAGATGCTG  | ATAATAATAG | CAGCAGTGGG  | TTATTCTGCT  | AAAACCATTG |
| 13261 | CCTCGTTAAT  | CCTCAGTCCC  | GAGGTGGGGA | TTATTATCCT  | CATTTTGCAG  | AGAAGCAAA  |
| 13321 | TGAGACTCAG  | AGATTTTACA  | GCTGGGGAGG | GAGCCAGCTC  | ATCCCTCTGT  | CCAGGCCCAA |
| 13381 | GCTCTCTCCC  | GCTTGCCTTC  | CTGCCCTCTG | AACCTCAGAG  | CATCCCCCAT  | CTGGTTCTAC |
| 13441 | TGCCTGTGCT  | AGTCGTGCAG  | GAGCCAAAAG | ACACGCTCTT  | AGTGCTAAGG  | ACTGGAGAAG |
| 13501 | CATAGCCCTC  | CAGCCTCTGT  | GAATGGGTCA | TATGTAACAT  | GAGCCTGGAG  | AAATTATTTG |
| 13561 | AAACCAAAGG  | CAAGCCTCTA  | AACCAGGCTG | CTGCTTCATG  | GCGCCGGTGA  | CGGCAGAACC |
| 13621 | AAATTTAGTG  | CTGTGGGCAG  | GTCCACACTT | ATCAAAATAGA | GAAAGCTCATT | TTTCTTCCGG |
| 13681 | CTCACATCAA  | GCATGAAAAA  | TGTTTACACA | TACCCCCCAC  | ACACACATGC  | TTTCCGGAGG |
| 13741 | GGTCCATGTG  | GCTAGAGGCT  | GGAAGATGTG | GATGAGAGGA  | GCCTGGCAGG  | TAAGCCCAGG |
| 13801 | GAAGATGACA  | TTCAGCTTCC  | CAGACAGCAT | CTACAGGGAG  | AAATTTTAAT  | AAAAGTGGGG |
| 13861 | CGGTTTCCCT  | GAGCAAGGCA  | GACAAAGTCA | GCCCTCTACT  | GTTAAGAAAA  | AGGGTCACAG |
| 13921 | TGAGAGGGGA  | GGTGAGGAGA  | CTGAGTCTGT | ATTTTCTAGT  | CTGTTGGGCT  | ACACTACCTG |
| 13981 | ATCCCCCTTC  | CTCAAAAATC  | CACCTTACTT | TCCCATATGC  | TACACCAATG  | TGGTTCACAC |
| 14041 | CTCTGGGACA  | GGAAAAGGGG  | GAGTGATGGG | GAACAGAGAA  | GGGAGGAGCT  | CACACAGCTG |
| 14101 | AGGCTGGGGT  | TATGCATATC  | GAATTACTTA | GAATTTGCAA  | CCTCACAGGG  | TACTTTCATG |
| 14161 | GCCTTGAAT   | ACACTTCCCA  | CAGCCACCC  | CCCTCTAAT   | AAAAGCAAGA  | GTCAATTTCT |
| 14221 | AGTTCTGTGC  | TTGCCTCCCA  | CGTTCTCCTC | CACATTTAAG  | AAAATCCACC  | AGTACAAAAG |
| 14281 | TGAAGATACC  | ATATGTGATA  | TCCACCCCTA | GTTTCTGTTT  | TATCAGGGTT  | TGGAGCAGGT |
| 14341 | GGAGCAGGCA  | GAGGGATCAT  | TTTCCAGCTT | AAATTTGATT  | AAGGGTGAGT  | ACTGAGTCAT |
| 14401 | CTCTTCAAGAA | AAGTTTGTGA  | AGCATCCAAA | ACTGAAGGGT  | GGAGCCACCT  | GGAGACAGTA |
| 14461 | TCATCAGTCC  | TGGCCCCGAG  | CATGGCCTGC | ATAGGCCCCC  | ATGGATCCCA  | GCGGGAGCTG |
| 14521 | CAGAGTGGCG  | GCACCTTGGC  | ACACAGCCCT | GAGTGCAAAA  | TTAGGAGCTG  | GGCAGAGGGC |
| 14581 | ATCTCTCTGT  | CGCCATTGGG  | CAGCCAGGGG | CACACTGGTC  | ATAGCCTTAG  | ACCACGAACA |
| 14641 | CCCTGTGCCC  | GGGGGACAGA  | TGCAACCAGT | GTGCCCTGGG  | CTGCCCAATG  | GCAACAGAGA |
| 14701 | GATCGACACC  | TGGACCCCAT  | GTCACGGGGA | CTCCACTACT  | AAGGCTCCTA  | AGACTGCCAC |
| 14761 | CTTCCAGTGG  | GATAAGCCCT  | GCCTCCTACT | GGGCCCAACA  | TGTGCAGAGA  | ACACTTGGGA |
| 14821 | CTACCTGGCT  | TTCTGGATAC  | ACAAATATTG | ATCCAATCTG  | GACTAATTAG  | AAGGTCAGTC |
| 14881 | CCAATAACAA  | ATCGAAGTCA  | GCTGGGCGTG | ATGGCTCACT  | CCTATAATCC  | CAGCACTTTG |
| 14941 | GGAGGCTGAG  | GTGGGCAGAT  | CATTTGAAGC | CAGAAGTTCA  | AGACCAGCCT  | GGGCAACATA |
| 15001 | GCAAAACCC   | GTCTCTACTA  | AAAATACAAA | TAATTAGGCT  | GGGTGTGGTG  | GCTCATGCCT |
| 15061 | GTAATCCCAA  | CAGTTTGGGA  | GGCTGAGGCA | GGTGGTCACC  | TGAGGTCAAG  | AGTTTGAGAC |
| 15121 | CAGCCTGGCC  | AACAGGGTGA  | AACCCCGTGT | CTACTAAAAA  | CATAAAAAAT  | AGCCAAGCAT |
| 15181 | GATGGCATGT  | GCCTATAATC  | CTGGCTACTA | GGGAGGCTGA  | GACAGGAGAG  | AATCGCTTGA |
| 15241 | ATCCAGGAGG  | TGGTTGCAGT  | GAGCTGAGAT | GGTGCCACTG  | CACCTCAGCC  | TGGTTGACAG |
| 15301 | AGCAAGACTC  | TGTCTCAAAA  | AAAAAATAAA | AAAAAATAAA  | GCCATGCCTG  | GTGGAGCACT |
| 15361 | ACGTGTAATC  | TCAGCTATTT  | GGGAGGCTGA | GGCACGAGAA  | TCACTTGAAC  | CTGGGAGGCA |
| 15421 | GTGGTTGCAG  | TGAGCTGAGA  | TCGCGCCACT | GCACTCCAGC  | CTGGGCGACA  | GAGTGAGTGA |
| 15481 | GACTCCATTT  | CAAAAAAATA  | ATAAATCTGA | GTCACCTTAA  | TATTGTTTAT  | TGGATGTCAA |
| 15541 | CCTCTAGGTG  | TTTGAGACAG  | GAGAGTGATA | TGGGGGCACT  | GGAACACAC   | AGGCACGGGG |
| 15601 | TGTCCTCACA  | CTTGGGTAGC  | CCACACGATG | TGATTTTCAGG | GTGCTGGGAG  | GTCCCCCAC  |
| 15661 | TCCCAAAAT   | ACTAACAGT   | GGATAGTACT | TTACAGTTTA  | TATGATCTCA  | TTTGATTCTT |
| 15721 | AACATGAGCC  | TGTGAGTGAA  | AAATTCCTTC | CCCTCTTCTA  | CAGATTAGGA  | CGTTGAGATT |
| 15781 | CAGGAGGTT   | CAGAGGGATG  | CAGGGAAGTC | AAGTGGCACC  | TGGAGTCCCG  | TGGCTAATTT |
| 15841 | GAGGCCGCTG  | GGGGATTTCG  | ACCCAGGATT | TGTGCTTCTT  | ATGCCCTGGG  | TTCTGCTCCC |
| 15901 | TGGGCGATGG  | TCTTCCCCCT  | AGCTTTCCCA | TTCACTGCTT  | TAGCCTAGGG  | GTCTTACCCT |
| 15961 | TTATTAAACT  | GCCAGTGCTT  | CACCTGCTTT | CTCCCCCAAA  | GACAAAAAAA  | AAGTGTTTTT |
| 16021 | GCTTTTGT    | TGTTTTTCAT  | GGGCAGAGAC | CTGGAATTTT  | AGCTTGAGAA  | TTTGTGCCAT |
| 16081 | ATGATAAATA  | AATCAACAGA  | TGGCTTTTTT | CTTAAAAAAA  | AAAAAATAAA  | AAACTAAGAT |
| 16141 | GTATTGCGAG  | TGAGGCATAA  | TTTGTACCAA | AAAGTGCTCA  | CCACACTGTA  | GTGATGGGGG |
| 16201 | CAGGAGGCG   | CCGCGGGTGA  | AGGGAGAAAT | CTTGGAGTCC  | AGGCAGCCCC  | CTTCTGGGCT |
| 16261 | GAACCTGGGA  | GCTGGGGGTG  | CTGCCAGCCC | TGCCAGGTTC  | TCCTAGGAGT  | CGGCAGCTCA |
| 16321 | TATGGCTGTG  | GGAGGAGGCA  | GAGGGAGCCT | CATATGCACC  | CACATTTCCA  | GGGATCTAGA |
| 16381 | AGACAGAAGG  | AGGAAAACCA  | CCATCATGTT | AAAGCAGACA  | GTTAGGTAA   | ACATCCTGTA |
| 16441 | ATACAAGTTA  | TTTTTTCCAC  | ATCTAAAGGC | TAAAAATAGT  | TGTTAGAATT  | TAAAGATAAT |
| 16501 | TGGTAAATGA  | GTTTCTATCC  | TTCTAGTTTC | ACATCAAAATG | GAATCATGCT  | GCCTTCACAT |
| 16561 | CACTAGTGCC  | CGTTATTTGT  | GTTTAAATTT | CACAATGTTG  | TCTAATTTCA  | CTCTTTGGGC |

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SEQ ID NO Sequence

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| 16621 | TTCCCCAGGG  | ATCCAGCCTC  | CCTCACTCGC  | CCATCGCAGG  | GAGATGCTTT | ATTCACTCTT |
| 16681 | GTGTCTTCTG  | TGCCGGGCAT  | AGCGCATGGC  | ACAGAATAAG  | CACCTAGTAA | TTGATTCACG |
| 16741 | AGTGAATAAA  | TGGATGAGTG  | GGTGAGTTCA  | ATATTGACTA  | CAAAAAACCT | AAGGCCACAC |
| 16801 | TGGTGAGTGG  | CTGCGCCTGT  | AGTCCCAGCT  | GCTGGGGAAT  | CTGAGGCAGG | AGGATCTCTT |
| 16861 | GAGCCCAAGG  | GTTTGAACCT  | AGCCTGGGCG  | ATATAGCGAG  | AACCTGTCTC | AAATGACAAA |
| 16921 | AACAGGGCCA  | GGTGCAGTGG  | CTCACGCCCTG | GAATCCCAGC  | ACTTTAGGAG | GCCAAGATGG |
| 16981 | GAGGATCACT  | TGAGGCCAGG  | AGTCCGAGAC  | CAGCCTGGGC  | AACATAGGGA | GACCCCTGCT |
| 17041 | GTACAAAAAA  | TTTTTTAAAA  | ATTAGCTGGG  | CATGGCGGTG  | TGCGCTTGTA | GTCCCAGCTA |
| 17101 | CTCAGGAGGC  | TGAGGCAGGA  | GGATCACTTG  | AGCCCAGGAA  | ATTGAGGCTG | CAGCGAGCCA |
| 17161 | TGATGGCACC  | ACTGCACTGC  | AGCCTGGGCG  | TCAGAACGAG  | ACCTGTCTCT | AAAAAAACAA |
| 17221 | ACAAACAACA  | AAAAAAAGG   | CTTTCTTAAA  | GAGACTTGAG  | AACAGAAAGG | GGAACAGATA |
| 17281 | CATAACTTAT  | ATATTTATTT  | GTTTCTCTTT  | CCACCTTCCT  | GGAGGGTGGA | GGGGAACAGG |
| 17341 | TCTGTATTTG  | GAGTTTTGAA  | TGCTAAAAAGT | GGGAATACAT  | GTACTGTTTG | CCATGATCTG |
| 17401 | TTCAAAAGTT  | AAGCCAAATG  | CCTTAGATTG  | TCCTGAAAAC  | TGGAATGCCA | CTGTAAACTA |
| 17461 | TAAGCCCCAC  | TTCAAAGATA  | AAAGATCTTG  | ATGAACAGGG  | CTGGGTCTGT | GGACTGGGCC |
| 17521 | TCTCCCCACC  | ACACAAGGAA  | GGGTGGTGCC  | AGTTGAAGGA  | AAATCACTTA | AATCCTTGCT |
| 17581 | CTCTCTTAAT  | AAGGTGTGGT  | CCCAGGTAGG  | GCTGTCAGAA  | TTAGCAAAAT | AAAACACAGG |
| 17641 | GCATCTGTGA  | AAATTAGAAT  | TTCAGATAAC  | AACAAATAAT  | TGGCATAGGC | TGCATAATGT |
| 17701 | CCCTCAAAGA  | TATCAGGTCC  | TAATCTCCAG  | AACCTGTAAA  | TGTGATCTTA | TTTGGAAAAG |
| 17761 | TGGTCTTTGT  | AGATGTGGTT  | AAATTAAGGA  | TTTTGAGATG  | GGGGGATTAT | CCTGTATTAT |
| 17821 | CTAGGTAGGT  | CCTAAATGCA  | GTCACACTCA  | TCCTTGTAAG  | AGGAAGGAAG | AGAGAGATGG |
| 17881 | AAAAACACAGA | AGAGAAGACA  | ATGTGGTGAT  | GGAGGCAGAG  | ATTGGAGTGA | GGTGGCCACA |
| 17941 | AGCCCAAGGAC | TGCTGGCAGC  | TACCAGCAGC  | CAGAAAAGTC  | CAGGAACCAA | TTCTCTCTTG |
| 18001 | GAGCTCCAGA  | GGGAGTGTGG  | CCCTGCTGAC  | ACCTTAGCTT  | CAACCTAGTG | ATCCTGATTT |
| 18061 | TGGACTTTGG  | CCTTCAGAAG  | TGTGAGGGAA  | TGAATATCTG  | TTGTTTTAAG | CCACCAAGTT |
| 18121 | TATGTCTATT  | TCCTACAGCA  | GCCACAGGAA  | TCAAAAACAG  | TAAGTATGCT | CCATGCAATG |
| 18181 | TTTGTGACAC  | ACACCAAAAA  | TATTACTTGT  | TGTTCACTCT  | AAATTCAAAT | TTAACTGGGT |
| 18241 | CTCCTGTATT  | TTATTTGGCC  | AACTTAGTTC  | CCAGGCCCAA  | AGAAAGAGGC | TTTTGAAATT |
| 18301 | TGCAAGAAAG  | CTGGTTGGAG  | CTGTGAGAAA  | GTGGACTTTG  | TAAACACAGT | ACCACCGAAC |
| 18361 | CAATTGTAAC  | TGTACTACCT  | CTAGACAAAA  | GAGAGGGCAG  | TCAGACAGTT | GTTCTGTATT |
| 18421 | TCTTCTTTCA  | ACAGTCAATTT | GAGCACTTAC  | TACAAAAACAG | AAGCTATGTG | TAAGGGTGGA |
| 18481 | GGCGTTAGCT  | GTTAATCAGG  | ACCTCCAGGC  | TAAGTTTCTG  | TATTAGTCCG | TTTTACGCT  |
| 18541 | GCTGATAAAG  | ACATACCCGA  | GACTGGGGAA  | TTTACAAAAG  | AAAGAGGTTT | AATTGGACTT |
| 18601 | ACAGTTCCAA  | GTGGCTGGGG  | AAGCCTCACA  | ATCATGGCAG  | AAGGCAAGGA | GGAGCAAGCC |
| 18661 | GATCTCTTAC  | TGGATGGCAG  | CAGACAGACA  | GGGAGAGAGA  | GCTTGTGCAG | GGGAACCTCT |
| 18721 | CTTTTTAAAA  | CCATCAGATC  | TCGTTAGACT  | TATTCACTAT  | CAAGAGAACA | GCACAGAAAA |
| 18781 | GACCTGCCCC  | CATGATTTCAG | TTACTTCCCA  | CCAGATCCCT  | CCCACAACAT | GTGGGAATTC |
| 18841 | AAGATGAGAT  | TTGTTACCAT  | ATCAGTTACC  | AACCCTTCCA  | GATAAATCAC | GTGAAATATC |
| 18901 | GCCATTAAACA | GAGTGAGCTC  | AGGTGGTTCT  | TCAGTGCATT  | TCTGATACCT | GAACCTTCCC |
| 18961 | TGGGAATTTT  | ACAGACCATC  | AGGCTCTCCA  | CCCTTTTGATA | GCAGGATAGC | AGGGCCAGG  |
| 19021 | TTCTGCAGGA  | GGAGATGTTA  | CCACAGGCCT  | GAAAGGGAGG  | GAGGGGCAGG | TGCTACAGGA |
| 19081 | AGATGCTGGC  | TCTGGATTCT  | CTGGAGGAGC  | TTTCAAGGGA  | AGTAGATACA | CACCTGCTCC |
| 19141 | ATCATTTTCA  | GTCCATCACA  | CTCTAAAAATG | CTTTGGACAA  | GAAAGCAATG | TTAAAGACAA |
| 19201 | ATGTGGCCCA  | TTTTCTGTGA  | CAAAGAGGGC  | TGCTCCCATG  | CCAGGCTATT | GGCACTGGTG |
| 19261 | GGCATGAGGC  | TTCTCTGCTG  | CCCTGGCCCG  | GGGGTTCTCT  | CACCTACCAT | TGGCTCTCTG |
| 19321 | ACACCTGGAG  | AGACCACCAC  | CCTTGGGCTT  | TCATGATGCT  | CACAGAAATC | ACACTGTTGG |
| 19381 | GACCTTTAAG  | AGCCTGGATC  | AACCTGGAACA | GGCAGGGAGT  | ACTAGGCAGC | CCAGCATTTG |
| 19441 | CCCCAAAATA  | TCCAGGCCTG  | ATAAAGAGAG  | AAAACAGGTA  | GCTCACAGGA | AAAGGATAAA |
| 19501 | AAAAGGAGGA  | GGGATTTAAC  | ATGAAAAGGT  | GCTTGATCTC  | CCTCATAATA | AAAAGACTGC |
| 19561 | TGATTCCCAT  | CAGGCAAGTG  | ACAGAAAAAA  | AAAAATTAAAT | TTAAAAAGAC | TGCTGATAAA |
| 19621 | ACCACAGCGA  | GACACTGCTG  | CTCAGGGATC  | TGAGGGTGTG  | GGCAGCCAGG | CTGCCACGCA |
| 19681 | TCATGGGTCT  | GAGAGGAAGA  | CCACACCCCT  | GGAGCAGAGG  | GCGGCTGATC | TGTGAGATGC |
| 19741 | CCTTTGACAG  | CACCTCAGCT  | TCCAAGAAAT  | AACCCTTTCT  | ATGTGAGCAG | AGGCATCCAT |
| 19801 | GGGGGGACAC  | ACTGGTGAAT  | CATCTGTTAT  | GTAGAAGTCT  | GAAAAACATC | AGGATGGAAC |
| 19861 | TGGTGAAATA  | AGTGTGGCCT  | CTGACGGAAT  | GGAGCGGTCC  | GTCTGCAGTG | CTGCGGGTGC |
| 19921 | CCCTCAGATC  | CTGTGGGTCA  | GTGAGAAAAAG | CAGTGAGGAA  | CAAGGCAGGT | ACTGTGTACT |
| 19981 | GTCTCTGCG   | TGCAAGGAAG  | GCCAGCGCAT  | GCAACAGAGT  | CCACACAGAC | ATAGCCTAAC |
| 20041 | TCTGGAAGGA  | AGAATGAGAA  | TGCAGTTTCA  | GTGGTGGCCT  | CTGGTGGGGA | GAAACTGGGT |
| 20101 | GAAGGGAGAT  | GTCAATTTCCA | TTTCTCTACT  | ATTAATTTTG  | TATTACCATG | CTTAAATGTT |
| 20161 | ACTTTTTTACC | TTTTTTTTTTT | TTTTTTGAGAC | AGGGTCTCTC  | TCTGTTGCC  | AGGCAGGAGT |
| 20221 | GCAGTGGTAC  | AATCATGGTT  | CACTGCAGCC  | TGAACCTCCC  | AGGCTCAAGC | AATCCTCCCA |
| 20281 | CCTCAGCCTC  | CTGAGTAGCT  | GGGACTATAG  | GCACGCATAC  | CACCGTCCCC | AGCTATTTTT |
| 20341 | TTTAATCAAG  | ATGGAGTTTT  | TCTATGTTGC  | CCAGGCTGGT  | CTCAAGCTCC | TGGACTCAAG |
| 20401 | CAATCCTCCT  | GCCTCAGCCT  | CCCAAAGGGC  | TGAGATTAAA  | ACGTGAGTCA | CCCTGCCCAG |
| 20461 | CCAATTGCTT  | TTTTAAAAAG  | ATTAATATGCA | TGTATACGCT  | CAGGCATCAG | CACACTTGGA |
| 20521 | AAGGATGAAA  | ATATCCGGAA  | GAAGGGTTCT  | TTTAAAAAGG  | TCCTCAAGTG | ATGCTGGCAG |
| 20581 | GCATGACGAA  | TGTCCCTGGT  | CACAAAAGCT  | CTGATCTGGC  | CTAACCTCTG | CATGTTAGAG |
| 20641 | ACTGGAGTGC  | GTGTGTGTGC  | GCGCAAAGTG  | TGGGGGGATG  | GGGGTGAGTG | TGTGTGGTGT |
| 20701 | GTAAGCATGA  | GTGTGTATGT  | GTGTGGTGTG  | GGGGTGTGTG  | CTGTGTGAGC | GTGTGTGAGT |
| 20761 | CTGTGTGTGT  | AGTGTGTGTG  | TGAAGTATGT  | GGTGTGTATG  | TGTGACGTGA | GGTGTGTGTG |
| 20821 | GTGTGTGAGT  | TGTGTATGGT  | GTGTGCATGA  | GCATGTGTGT  | GGGCATGTGA | TGTGTGTGTG |
| 20881 | GTGTGTAAGC  | ATGTGTGAGT  | GTGTATGTTT  | GAGCATGTGT  | GGTGTGTTGT | GATATGTGTG |
| 20941 | TGGTGTGTGA  | GCATGTGTGT  | GTGTGTGTGT  | TGTGTGTGGT  | GTGTGTGAGC | ATGTGTGTTG |
| 21001 | TGTGTGTGGT  | GCATGTGTGT  | GGCGTGTGAG  | CGTGTGTGTG  | CATTGTGTCT | GTGAGCATGT |

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SEQ ID NO Sequence

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| 21061 | GTGAGTGTGT | GTGTGTTTCAG | CATATATAAG  | GCATGTAAC   | GAACACAGCA  | CTTAGAGGG   |
| 21121 | CTCTCCTGGA | GTGAGAGGGG  | GTGGGTAGGA  | GGAGAAGGGA  | GGTGGGCTAG  | TGTGCTGAAG  |
| 21181 | TATCTACTCC | TTGTCATAGT  | CTGTGACAAC  | CCAGACTAGC  | CCATGAGCCA  | CCCTGTTCCC  |
| 21241 | TGCATTTCCT | ATGAGACCTC  | GGTGGACATG  | TTCCTTGAGG  | TGAGGCTGAC  | TGATGTCATT  |
| 21301 | TGACGATCTT | GATGCCAAAT  | CCTTTTATAT  | CAAAAACAAC  | CAGAACACTC  | TCTTTTCTCT  |
| 21361 | TAGTGCTTTC | ACCCAGATGA  | CCACATTTCA  | TCCTCCACGC  | CACCTCTGGG  | CAGGTGGCAC  |
| 21421 | TGCTGGTTTG | AAAGGGAGGT  | CTCCCTGGA   | GTAACCTCCG  | TGGGCGGATT  | CACACCTGCG  |
| 21481 | CCACAGTCTT | GTCCACGTCA  | GCCACCATG   | GTGGTCTCCG  | GTTCTCTCAG  | AATTCCTGCT  |
| 21541 | TTTCAGCTCA | TCCCCACATT  | CCCGGAGGGA  | CTGAGAGCGC  | AGCCCCAGGG  | CCCTGCTCTT  |
| 21601 | TGGGGGCGGT | CTCTACACCC  | AGAGAAGCAG  | CAAGGCATTC  | CTAGGTTTCT  | CTTTCAGATG  |
| 21661 | CAGAACTTCA | GTGTTTCAGAG | ATGTTCCAC   | TGGTCTCAGAG | AGGGCTCAGT  | TCAGCTTTAA  |
| 21721 | TGACTGCGCT | GTTGCGTGTG  | CTCTGCAGAG  | GGCGGGTGGC  | CCAGCGTGGC  | TGACTGCAGT  |
| 21781 | TTTCTGACG  | TGGAGCCCGA  | GCCTGCCCG   | CTGTTTATTA  | ATTAAGGATC  | ACTCTGCTTG  |
| 21841 | CAGAACCTCG | AACCTCCAG   | AACCTGTGAG  | TGGGAGAACC  | CCGAGAGGCC  | ACCTGGCCCC  |
| 21901 | ACTTCCACCC | TGCTGCCCAA  | ACCCCTCTCT  | TGCCTTCTGT  | ACAGTCACCC  | CAACTCCAG   |
| 21961 | TGATCCCCAT | CAACCATCTG  | ACAAGGGGAC  | TGAGAGGGAA  | GAGAAAGGAG  | GGGCCCCAAG  |
| 22021 | AGGAAGGTAA | AACGTGCGGG  | AACAGCCCCC  | AAATGTGTGA  | CAGCCTTCAG  | TGGAGTTGCC  |
| 22081 | CACCTTCCCT | TTTCTCTCTC  | CTGCAGGACC  | TCCCTTCTCT  | CCAGTCTCTC  | CCAACCTCTG  |
| 22141 | AGGTTTACAT | GAGAAAAGTC  | TGCAGAGAGG  | TGCCAGCATC  | ACAAGGTGTT  | AAGGACCACG  |
| 22201 | AGTTTGGCAT | TTTAACAGAT  | GCCAGAGCCA  | CTTGAGAAAT  | GTGGTAACCTA | AGCCCAGAGA  |
| 22261 | GGTACAGTTA | ACCTCCCCAG  | AGTCACACAG  | CAGGTTTCATG | GCAAAGCTGG  | ACTAGCACAG  |
| 22321 | GTGTCCTTCC | CCTGCAGATC  | CCCTTCTGTG  | CCCCACATCA  | CCTCCCTCCA  | GTGCTGGGGC  |
| 22381 | CACCTGGAGA | TGGGCCCTCA  | GACTCACCCG  | GCCAGAGGTG  | CCATCTCATG  | GGAGAGGTCT  |
| 22441 | GGCCAGGAAG | CATCGATATT  | TGAGATCCCA  | AGAAATGAAG  | ACTTGGCCTG  | TCAGATGACA  |
| 22501 | GACTTCGGTC | ATGGGAACAC  | GTGATCTGTT  | TTACACATGC  | GTCCCTCTCAG | CAGCAGCTTT  |
| 22561 | CCAGAACATT | CCCACTTTCT  | TCGTAGTGA   | GAAGAACTCT  | TTCCTTGACG  | CCTCTGCCCC  |
| 22621 | AACCTCTCTC | TCAGTGTCTT  | TGCTTCTAGT  | TCTTTGATAA  | ACCATTCTGC  | TTTGCAGAGT  |
| 22681 | GCGAGCTCTG | CCTTGCAGGG  | TTGCTCATCT  | CCTGTGCTGA  | GTAACCAACG  | CTAAGGTCTGA |
| 22741 | GTGGTCTGCT | ACCTCTCATA  | AGAGCTAGGG  | TGCTCTCATG  | CTGATGACTA  | GGACTTGCCC  |
| 22801 | TCAAGGAGAA | AAATAAATCA  | AAACAAAAGC  | AAAAACAGCA  | AACATGCATC  | TCTTAAAGAA  |
| 22861 | GGCTCTGAGT | CCAGGTAAAT  | TTCTTCCAC   | TGAAGCAGCC  | AGGCTGAATT  | CGAATTATCT  |
| 22921 | TGCCTCCCTG | TTAAAACTA   | ATGCAAAATT  | TCCTAGAGAA  | TATCCACTAA  | TTCTGGAGG   |
| 22981 | GGGCATGGGC | ATTCTCTGAT  | CCCATGAGAG  | GACCATTTGC  | TCTTCCCTCA  | GTATGCTAAA  |
| 23041 | TAACAGAAGC | GACATTGTGT  | GCTGGAAAAG  | ATCAGTGAAG  | TTAATAAGGT  | TTTTCTTGCC  |
| 23101 | CAGGGTGAGG | GAACAGTTCC  | CAATGACAAA  | TGCTGTATGG  | GAAGGGGTGG  | TAGAAGTGGC  |
| 23161 | AGCCCCCTTG | GTCCATCCGT  | AAAGTGAAC   | CTGTGGATCC  | TGGAGGATTC  | CAGCGTCTTT  |
| 23221 | TTTTTTTTTT | CTTTTTTTTT  | AAGACAGAGC  | CTTGCTGTCA  | CCCAGGCTGG  | AGTGCAGTGG  |
| 23281 | CACGATCTCA | GTTCACTGCA  | ACCTCCGCTC  | CCCGGGTTCA  | AGCGATTCTC  | ATGTCTCGGC  |
| 23341 | CTCCCGAGCA | GCAAGACTAC  | AGGTGCGCAC  | CACCATGCCC  | GACTAATTTT  | TGTATTATTA  |
| 23401 | GTAGAGACGG | GGGTTTCACT  | CTGTTGGCCA  | GGCTGGTCTC  | AAACTCCTGA  | CCTCAGGTGA  |
| 23461 | TAGCACCCTG | TCAGCCTCCC  | AAAGTGCTGG  | GATTACAGGC  | ATGAGCCACC  | ATGCCAGGCC  |
| 23521 | AGCATCTTTC | ATTTTCTCTG  | CTGCTTTGGC  | CCTTTCCTCT  | CTCACTGTCT  | TCCTTTTCCA  |
| 23581 | TTTCCAAAGT | CAGTCCATCT  | CACATATTAG  | ACAAAACTG   | CTAGAGCGCT  | TGTATTGTTG  |
| 23641 | CATCTCTCCC | TGCACCTGGC  | TGGTCTGTTT  | TTGGCCACTG  | AAGCGTTTCC  | CCCAGCTGTT  |
| 23701 | GCTTTAATCA | TTTTATTGTT  | ATTATGCCCT  | ACTTAAGAAA  | TGGATATGAG  | ATGCATTTAC  |
| 23761 | CTGTCCTTTC | CTGCCACTCT  | GCAGAGCCAG  | TAAGATGTGG  | TGGAAAGGGC  | CCAGGCTTTG  |
| 23821 | GAGGAGGGCT | GGCTGGGGTT  | GGATCTTGGC  | TGCCCTCTAC  | TAGCTGTGTG  | ACCTTGGGTA  |
| 23881 | AGTAGCTGGA | CCTCTCTGAG  | CCTGGTTTCG  | AATCATAGCA  | CCTCTCTTTC  | AGGGCTGCTG  |
| 23941 | TAAGGAATAG | CAGTGGGTGT  | TATAAAGCAG  | AGCGCACAGC  | CAGCAACTGG  | CCCCTAGCCA  |
| 24001 | CACTGCTGAG | CACCTACTGT  | GATAAGCTGC  | CATTGTGGTG  | TGTGAAGCAA  | AGGGGAAACA  |
| 24061 | TGCCCTGCTG | AGTGAGCTTC  | CTGTAGGGCA  | GGTTGTAGAA  | CCAGAGGTGG  | GTTCCAAGGT  |
| 24121 | TACAAAGGGA | CTCTTAGTGT  | ATTAGTCTGT  | TCTCACATTA  | CTATAAAGAC  | CTACCTGAGA  |
| 24181 | CTGGATCATT | TATAAAGAAA  | AGAGGTTTAA  | TTGGCTCACA  | TTGGCTGGGT  | GCGGTGGCTC  |
| 24241 | ACGCCTGTAA | TCCCAGCATT  | TTGGGAGGCC  | AAGGCCGGCG  | GATCACTTGA  | GGTCAGGAAT  |
| 24301 | TTGAGACCAG | CCTGGCCAAC  | ATGGTGAAAC  | CCTGTCTCTT  | CTAAAAATAA  | ATACAAAAAT  |
| 24361 | TAGCTGGCCA | TGGTGGTGTG  | CGCCTGGAAT  | CCCAGCTACT  | CAGGAGGCTG  | AGGTGGAAGA  |
| 24421 | ATTGCTTGAG | CCCGGGAGGT  | GGAGGTTGCA  | GTGAGCCAAG  | ATCGCCCCAC  | TGCACCTAG   |
| 24481 | CCTGGGCAGC | AGACTGAGAC  | TCTGTCTCAA  | TAAAAAATAA  | AAAAAAGAAA  | AGAAAAAGAA  |
| 24541 | TTGCAAGAAA | TAAATTATTG  | TTTATGAGCT  | ATATGGTCTG  | TGTTACCTTG  | TTGTGGGACT  |
| 24601 | GGGAGTCTTG | GCGTCTCCCT  | GACCTGCTCT  | GTTGCTGCAG  | CACCGCTCAG  | CCCTGCTGCT  |
| 24661 | TCCCTACCTG | CCTCCCTCTG  | GCCTCTCTCT  | CCTCCACCGG  | GCCCTTGGTG  | CCTCTCTTAG  |
| 24721 | AGACAGTCTC | CCTGGGACCG  | ATTGTGTTCT  | CACCTTACAG  | AGGCATCCAG  | GACTACAGAT  |
| 24781 | AACCAGAGGA | AGGGGCGCCC  | CCCCCGCCTG  | CCCTCTCTCT  | TGGCATCCTC  | ACGCTGCAGA  |
| 24841 | GGTCAGAGCC | TCATCCCAGC  | CCCTTACCTG  | CCCTTACTCT  | GTGGAGAAC   | GTGGTCAGTT  |
| 24901 | CGCCAGGCGG | GATCCACGAA  | CGGCCTTTGT  | GAGATGGTGT  | AGCTCACACC  | CAGAGCTGGC  |
| 24961 | TCCGATGACC | CTGTCTCTCT  | TACATGTTTC  | TACCTTCCCC  | TCCCTACCTT  | CCCCACTGCG  |
| 25021 | TGGGCGCAGA | GTGGAGGCAG  | ATGAGGTTTA  | AAGCTCAGAA  | GGGCTTAAAC  | GGGTTGGGGC  |
| 25081 | CAGTGGGCTC | ATGCTGTGTA  | TCCCGGCACT  | TTGGGAGGCC  | AAGGCAGAGG  | ATCACTTGAG  |
| 25141 | CCAGGAGGTT | CGAGACCAAC  | CTGAGCAACA  | TAGTGAGACC  | GCGTCTCTAC  | AAAAAATAAA  |
| 25201 | ATAAATAAAA | TTAGCTTTGC  | AGGGTGGCAT  | GCACCTGCAG  | TCCCTGCTAC  | TCAGAAGGCT  |
| 25261 | GAGGTGGGAG | GATCGCTTGT  | GCCCAGGAGT  | TTGAGGCTGC  | AGTGAGCTAT  | GCTGGCACCA  |
| 25321 | CAGCACTCCA | GCCTGAGTAA  | CAGAAATGAGA | TCTTGTCTCA  | AAACAACCAA  | ACAAACAAAC  |
| 25381 | AAAAGAAGGC | TTAAAGGGGG  | CTCCAGGTGG  | GCTTGGCAGC  | ACAAAGCTAT  | GAAGTTCTAT  |
| 25441 | CTTAGACACA | AGTTCTGTGA  | CTGGGCTTTT  | GCAGGCTGGC  | CTGGGTACCT  | GGCTGCCATA  |

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SEQ ID NO Sequence

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| 25501 | GACAGGGAAC  | CTTCCAGATG  | AGCTGCAGGC  | GTGGAGCACA  | GGAGCCAGGG  | TGCTCTTCCT  |
| 25561 | GGGCTCTGTC  | CACAGGCAGA  | ACGTACACAG  | TCTTTGTACA  | CGTCCGGCGG  | CTCTGGTGCC  |
| 25621 | TATTTTGTGT  | TGTGTTTTTC  | TTTTGTTTGG  | GGGGATGGAT  | TTGGTTTCCC  | CCGAGCCCTC  |
| 25681 | TGTCCTCTGT  | TCACCTGGCT  | GGTGCTCGGC  | AATGTTGACC  | AGCTGCCTGG  | CTGGAGTTGG  |
| 25741 | CAGTGGCTAA  | GGCTGTGACA  | GCTAACATGT  | TCCTGAGTCC  | TCTCATTCT   | TCACCATAAT  |
| 25801 | GCCCTGTTGA  | GTTTGAGAT   | ACTGTCTCTG  | TTTTTATCTC  | CCGGGGAAC   | TGAGGCTCAG  |
| 25861 | AGTGGCTAGG  | CCACCTTCCC  | ATGGTCCCTC  | AGCTCATGAG  | GGCCACACAG  | GGCATTGCGG  |
| 25921 | TGGCCTTCTC  | CTCAGCCTTG  | ACCCTCCGGC  | CCCAGCATTG  | CTGCCTCAAG  | GGGTCTCCTC  |
| 25981 | TGCTGAGCCG  | TGCACCTTCT  | GCCTGGCAGC  | TCCAACCTCTG | TGGCTGTGTT  | CAGTGGCTCA  |
| 26041 | GCACCTGCCC  | TTGACCTTCC  | CTGGCCTTCT  | GCGGATGCCA  | GACTGGAGCA  | CTCTGACAAG  |
| 26101 | GTCTGGGGTG  | GTTGTATGGG  | TCCTGTGACC  | TCTATACACC  | TCCCAGTGCC  | TGGGAATCCT  |
| 26161 | GCAGATACAC  | CCTCCTTAGC  | CGTCCCTAAC  | CATAGAGGAC  | ATTTCTGAGG  | TCCCCGAGAG  |
| 26221 | AGTGGGGCAC  | CCCCTGCAGG  | TCCAACCTGCT | GGGCCCCAGG  | AGGATAGCAG  | CAGCATGAGG  |
| 26281 | GGTTCCATTA  | GCCACAACT   | CACGGCATGG  | AACCTTCACC  | CACCTCGCCC  | CTCATCTGCT  |
| 26341 | GTTTAGCACC  | TGGCACGCCG  | TGTATACTTA  | CTGATTATTA  | CATTTTAATG  | GCAAAATTATA |
| 26401 | GTGGCAAACG  | TATGCATCTT  | TGCACAATTG  | TTGTACAGCA  | TGATGAACAA  | GTCATTAAATA |
| 26461 | GTAAAGAAATA | AATGTGAAAG  | TGAGAAAAAT  | CTGACTGCCA  | AAGTTTTTAC  | TCCTTCCTTC  |
| 26521 | CCTCCCCAGA  | CTTTTAAATG  | AAAGTTTAGG  | GATAATCCCT  | TAGTTGTCTC  | GCTAGTAGGA  |
| 26581 | CTTGCAATTA  | AAAGAAATGG  | GCCAAGAAC   | CTTCTACGCT  | TCTCCTTTTA  | GGTTTGGGTG  |
| 26641 | TAAATTCCGG  | GTATTTCTCA  | CTGATGAAAG  | CCTGGTGACG  | GGCAGACCGT  | GGGAAGCTTT  |
| 26701 | CATTTCCGGA  | ATGGACCATC  | AACATCCCTT  | GGAGAAGAAT  | TCTCTTCTCC  | AGACCCAGAC  |
| 26761 | CTGGTGTCCT  | GGCACCCATT  | GGGCAAGTGG  | GTCCTAGAAG  | ACAAACCTGG  | TCAGAGCCTG  |
| 26821 | GAGGCTGCTT  | AGCATTCCTC  | ACGCACATTA  | GCAGCTCGGA  | GAGCTCAGGA  | AGCCGAGGCC  |
| 26881 | CCTCCTTGCC  | TCACCAGCCT  | GGATCAGGAC  | AGCATCCCTT  | GGAAGACACA  | CAGGGCCTGG  |
| 26941 | CCTCTGATTA  | CCCAGCCTGG  | AGGAAAAGCT  | CAATCGAGCA  | TCATGTCAAC  | CGGTGCCCCC  |
| 27001 | ATGCAGGGTG  | GCACCTGGTG  | GACCCCAAG   | CCAATGATAC  | CACCTCACAG  | GAGTGCAGGC  |
| 27061 | CCATTGTGGC  | CAGATCATCT  | TGACTTTTCA  | AGATAAATCA  | GAAATCGTAT  | TTCCATGAGA  |
| 27121 | TATCCCTATT  | TGCAAGTGAT  | GGTGACTAAA  | TTAGAAGTTT  | TTGAATATTG  | TAACATGTTT  |
| 27181 | GTAGGCTGTT  | TGCTGTGTTT  | AAACTCTATC  | TGGAGGAATT  | CAAGCTAGAC  | TTCAGGAATA  |
| 27241 | ACTTCTTGAG  | GCAAGGATTT  | TGAGACCTTA  | GGGAAAGAAG  | GACGCTTTGG  | GGGTATTCTG  |
| 27301 | ACTGTTGTCC  | TCCTGGGAAG  | GAAGAACAGA  | GAACTAGAAG  | ACTGCCCTTA  | GCGAAGTTCA  |
| 27361 | AAGCACCTAA  | GCCCGGAGCC  | CTCAGCAAGT  | GTTCTTGAGT  | CACAGATTCT  | CCTTGAGGCG  |
| 27421 | CCTCTTTCTG  | GCTCCATAGA  | ATGGCTGATT  | CTGTAACCTG  | GTGAGTTTGC  | TTTTTTTTTT  |
| 27481 | TCCTCCATCA  | CCCAGGCTGG  | AGTGCACTGA  | AGCTGGAGTG  | CCGTGGAGCG  | ATCACTGCAA  |
| 27541 | CCTCTGTCTC  | CCAGGTTCAA  | GCAATTCTCC  | TTCTCTAGCC  | TCCCAAGTAG  | CTGGGATTAC  |
| 27601 | AAGCATGCAG  | CACCACACCT  | GGCTAATTTT  | TGTGTTTTTA  | ATAGAGACGG  | CCCGAAGTGC  |
| 27661 | TAGGATTACA  | GGCATGAGCC  | ACCGCGGCCA  | GCCATAAATC  | TGTGACTCTT  | GTTACAAAGG  |
| 27721 | GCTCTATATT  | TGCTCTTTGA  | GGGTGGTTTT  | GGTTTGATGC  | CTGTTGGTTG  | CCATCTTTTA  |
| 27781 | ACTAGGGATG  | TTTTATCAAA  | ATGCCCAGCC  | AAAGTGTTCA  | AACAAATTAT  | ACCTTAAAGT  |
| 27841 | TTGAAAATGT  | CTGGCACTTC  | TAATTCAATG  | CCTGTTGTGC  | CAGGCACCTG  | GCTGCTGAGG  |
| 27901 | TAGAGCAGAG  | GGCTAGCTAG  | AGAACACACA  | CACACACACA  | CACACACACA  | CACACACACA  |
| 27961 | CACACAGAGT  | GGCTTTACAA  | GTCAGTTTTA  | TATTCTACCT  | ATATGCAATA  | AAGGTATTAT  |
| 28021 | TATGTTGAGG  | TGCCTTGATA  | TAAAAATTTT  | TCTTAAAGGA  | GAGGATGCCT  | AAAACAGGCA  |
| 28081 | TTACCTGAAA  | CCTCTCTCTC  | CCAGCATTTG  | TTGTCTTCTG  | TCATGACTCA  | GGGTTTTTCA  |
| 28141 | TGAGAATGGG  | ATGGAATGT   | GGTCTAAAGA  | TAGGGCCAAT  | GTTGGGACTG  | GATCCCCCTC  |
| 28201 | GGGAAGTCAG  | ACCAGGCTAG  | GGCAGGTCTT  | TGAAGCCATC  | AGGAAAAGCC  | TCTGGAGCCA  |
| 28261 | GAAACAAAC   | AAAAAATAAA  | TGGTGTTAAC  | TAAACTCAGT  | CTCAATCTCT  | GAATAGGACT  |
| 28321 | CAAGTCAAGC  | AAAATAATTA  | AAGGAGTTAG  | CAAAGGGCAA  | GTCAGAGAGA  | CCGAGCAACA  |
| 28381 | CCAATGTCTT  | CCGGGAGCCC  | TGTGGCGAGT  | GACAGAGCCT  | GGACTCTGGA  | GTAGAACTCA  |
| 28441 | TCTTGTGTCT  | TCTTCTGCCA  | CTCGTTAGCT  | GGGTGACCTT  | GAGCCAAGCC  | CCTTAACCTC  |
| 28501 | TTGGACCCTA  | TGTTCTTATC  | TCTAAGTAGG  | GGCTGGTAAT  | ATCTTCCCCT  | TTGAGGAATG  |
| 28561 | CCCTCTAAGG  | GGTGTTGTGA  | AGATTTCGGT  | AGGTGGCAGG  | GGTAGGACTC  | CTGGCCAGAA  |
| 28621 | ACAGGCACAT  | AATAAATGCT  | AAGTCTCTCC  | TTCTCTCCAC  | CTGCTGGATG  | CTGTAGATAC  |
| 28681 | TAAGGATTTC  | GATGTGAATG  | AGACAAAACC  | CCTGCCTTCC  | AGGAGCCTTT  | GAGAATCAGA  |
| 28741 | GAACCTAGAC  | CATTTCACAG  | ACAAGGGGAT  | GCAAGGCTCTG | GATAAAGTTT  | TGGGGATCAA  |
| 28801 | TAGAGCAGAG  | GGCTCCCAGA  | GGATCCCAT   | GGGTTGACTC  | CTAACTCAAG  | GGCATGAGAC  |
| 28861 | AACCCCCAGG  | AAGGGCACCC  | TGGAAGGGGT  | CCGGCTGTCC  | CTGATTTACT  | TGTGGGCACT  |
| 28921 | GGGGGAATGC  | CCGGAGCCAT  | CCAGCCCTCA  | GGGCTCTGTG  | TGATTTCTGGG | TTCTCTCCAT  |
| 28981 | AAAAGATAAT  | CAGATTCTTT  | CACGTTAATG  | TCTTTCTCCA  | CCTCAATTGCA | CATCATGCAG  |
| 29041 | CTATTCAATTG | ACTCAGCAAG  | TATCAGCTTT  | GCATGCGACC  | TTGGCCTACC  | CACCTTAGCT  |
| 29101 | TTTAGTAATA  | GCTCCCTTCT  | TGAATAATAC  | AACCAAGTGG  | GAAACAGAAC  | CTAACTCTTA  |
| 29161 | CCTCTGGGAG  | GCTTATTTCG  | TTTGAGAAC   | TATGTCTCTG  | AGTTTGTGTC  | ATATGGCAGT  |
| 29221 | GAAGTTTTCG  | GCACACACTC  | TAGAGCCAGG  | CAGCCTGGGT  | TCAAAGCGCA  | GCTCTGCCAG  |
| 29281 | GTCTTAACCTG | CATGAATTTG  | GGCAAGTCGC  | TCAACCTCTC  | CATGCCTGAG  | TTTCTCATCT  |
| 29341 | TGTAAGATTG  | GAGCAATGGT  | AATACCTGCT  | TTTTAGGGTT  | GAGAAGAGAA  | TTAAATGAAT  |
| 29401 | TAAGATGGGT  | AAAGTGCTTA  | GAGTGGAGCT  | TTGCAAGTAG  | TAAGTGCTAT  | GTAAGTGTTT  |
| 29461 | GATTTAAAA   | GAAAGACCTT  | TAAATACATT  | CTTTGTTTAT  | TTTCAAGGCC  | CTTCAATTCA  |
| 29521 | CAACCTTACA  | TTTCAACAAC  | AAGCTCTGTC  | TCCCTTGGA   | TCCAGCCATA  | ACTCTGCTCA  |
| 29581 | CAAGTGTGAG  | ACAGGCCCCA  | GCAAGCTGCG  | ACGAAGAGGA  | GAGAAGGCAG  | CCCCCAGAC   |
| 29641 | TCCCAACCCC  | CTGTCCAAGA  | TGGCAAAACC  | AGAACACAGC  | CTCTGTACCA  | CCCCAGCAGG  |
| 29701 | TATTGAGAA   | CTGCAATCTC  | CAAAGCCAC   | TTCAATTGTA  | AATGTAGAGC  | CACGTGCGCT  |
| 29761 | TTAAGTCACC  | TGTCACCTCTG | GAGGCTCTTT  | TGCTCAGTTC  | CTCACCATTA  | GCAGGGATGA  |
| 29821 | CAGGAGTGCG  | AGGAGTGCGG  | TCGACTCCCA  | GATATTGGAG  | AGCGCTGGGC  | TAGCTGCCCA  |
| 29881 | TTCTCCCGGC  | CTCCACTCCT  | CTTTGCTGTC  | CAGCCATCAC  | TTGCTCTTTG  | AAGGCAACAA  |

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| 29941 | AAACAGAAAA  | CAGTGCCAAA | AGTATGGGAA | GAAAGCCAGC | TTCTCCCCTG  | GGGTGCCTGT  |
| 30001 | GATGCCATGC  | CCACCCCTCC | TGACCACGCA | GCCCCGTGG  | ACCCTCAGGG  | CCCCAAGCCC  |
| 30061 | CCATTTCCAT  | CACATGCGTA | CACCCATGTG | TGTCCATAGC | CGCCCATCTC  | AGTCAATAAG  |
| 30121 | GCTGCTCCTG  | CCCACCTGGA | ATAGTGGTGA | CAACCAGGAG | TGGCTTATGG  | GAACATATCC  |
| 30181 | AATGGCCTGA  | CAGCATGTCC | GCTGCAAAAC | GCTGAGGTAG | GACACTGCC   | TCATGTCTAG  |
| 30241 | CTGATCAGCA  | AGAGGCGCAG | TTGCTTTCTT | AGGTAACATT | GCTGCTGTGT  | CCTGGCCATT  |
| 30301 | GCTGGGGGGT  | GGCACTTAAT | CTACACCAGA | ATTTTCCCTC | CTGTATCTTC  | CAAGCTGCTT  |
| 30361 | GGATCTTGGT  | GCTGAATTAG | GTTGGACTTT | GTCTTGTGGG | GAAGGGAGGA  | CTATAGACCC  |
| 30421 | TCAACGTAAG  | CAATGGTCAG | ACTATTCTAA | GAAAACTCGC | CGAATTAAAG  | CATGAGGTAA  |
| 30481 | ATTTAGTTCT  | GACTTCTGTC | CACCCCACTG | CCACTGTCCC | CTTTTATCCC  | ATGATCCCTT  |
| 30541 | GCTTTTCTTT  | TCCTCCTCTC | TCCTATCTC  | TTGTGTTTGA | CGCATGATAG  | GAATTCAGAA  |
| 30601 | ATATATGTTT  | GTGGATTGTG | TTATTCACGT | AGCAAACCAT | TTCTTGAGTG  | CCTACCATGG  |
| 30661 | GCCAGGTAGA  | ATGGGCGGCC | CCGGGCTGCA | GTGGTTTCTT | CAGCCCTCT   | CCAGGGTTTA  |
| 30721 | CAGTGTGCAA  | GACGGTTTGT | GATGGGTCTT | CCCATCGAGG | ACCACACTCT  | TCTTTCTCTG  |
| 30781 | TGCCCCCTGG  | TCCTCAGTCT | CTGACCCAC  | TTCAAAGGCA | GCATTCACTC  | AGGGAAGCTC  |
| 30841 | CCATACAATG  | CTAGTCAGAG | TAAAAGTTTG | GACAAATTGC | CAGGAAGCAG  | CTTGTGAGTA  |
| 30901 | GATCAGTAAG  | GCCTTTAAAA | TATTACTACT | CTTTGACCCA | GAATTTTCACT | TCTAGGAATC  |
| 30961 | TGTCCTAAGG  | AAGTAGTCAC | ATGCAAAAGA | TTTATGTACC | AAGATGTTCA  | TCAAAGTGTT  |
| 31021 | GTTTTATAAC  | AGGAAGTCTC | AGAAGCTGGA | TAAATATCCA | ACCTCTGGAA  | ATGGTTAGAT  |
| 31081 | AGAAATAGTAT | GTAGCCATTA | GAAAATTATG | TCTATGGGGT | TTAAATGTGC  | ATGGGAAAAAC |
| 31141 | ACTTCTGACA  | TAAAAGAGCA | TGAGAAGTGT | ATATTTAGCA | TAATCTTAAC  | TATGTTTTAG  |
| 31201 | AATGCACAGG  | AAAAAAATGT | ACAAACATAT | TCATAGTGAT | GTCTCTGGTG  | GTAGGATTAT  |
| 31261 | GATCAGTAAG  | TACTTCTGTC | TCTTCATATT | TTCCTGTATT | TGATAATACA  | TGCATATGTT  |
| 31321 | GTTTTTAAAA  | TAAGAAAAAT | TTTAAGTTTA | AAATTGGAGC | TGAAAAGTGT  | TTTtaggtca  |
| 31381 | GGCGAGGTGG  | CTCACACCTG | TAATAGCACC | ACTTTGGGAG | GCTGAGGCAG  | TCAGATCACT  |
| 31441 | TGAGCCCGAG  | AGTTCGAGAC | CAGCCTGGCC | AACATGGTGA | AACCCCATCT  | CTACTAAAAA  |
| 31501 | TAAAAAAATT  | AGCCATGTGT | GGTGGCACAC | ATCTGTAATC | CCAGCTACTT  | GGGAGGCTGA  |
| 31561 | GGCATGAGAA  | TTGCTTGAAC | CCAGGAGGTG | GAGGTTCGAG | TGAGCCAAGA  | TCGTGCCACT  |
| 31621 | GCACTCTAGT  | CTGGGCAACA | GAGTAAGACT | CTATGTCAAA | GAAAAAATAA  | AAAGAAAAAG  |
| 31681 | CTTTTAAAAA  | AGTAGCAGAC | ATAACTATAT | AATCCTTACT | AAGCTGTCCG  | TCAAATTTTT  |
| 31741 | ATTTATATAT  | TTATTTTATT | CATTTATTAT | TTTtagacag | GGTCTCACTC  | TGTTGCCCGA  |
| 31801 | GCTGGAGTAC  | AGTGGCGTGA | TCATGGCTCT | CTTCAAACCT | GACCTCCCGG  | GCTCAAGTGA  |
| 31861 | TCCTCCCATC  | TTAGCCTCCC | AAGTAGATGG | GACCCACAGT | GCATACCACC  | ACACCTGGCT  |
| 31921 | AATTTTTTTT  | ATTTTTTATT | TTTAGAGATG | GTGTTTACTA | TGTTGCCCGA  | GCTAGTCTCA  |
| 31981 | AACCTCTGGG  | CTCAAGCTAT | CCTCCACCTT | CGGCCTCCCG | AAGTGCTGGG  | GTTACCAGCA  |
| 32041 | TGAGCCACTG  | TACCCAGCCC | TCAAAATTTT | AAAAATCTAT | AAGAGACATT  | ATTGGACAAT  |
| 32101 | TAGAGAAATT  | CACATATGGA | CTTATAATAG | TATCAGAGTG | TGTGGTGTGA  | TGGTTCTGGA  |
| 32161 | GGGAATGAGC  | TTTTTCTTTG | GAGACAGGCT | TTTCTATGCC | CACCCTTTTA  | TCTTGCTAAC  |
| 32221 | TTATCATCAT  | CCAGGTTCCA | GCAGAAACAT | TACTTCCCCC | AGGAAATTTT  | TTAAGGGTGC  |
| 32281 | AGTATCATGA  | TGTCTGCAGC | AAATTCCTCA | ATAGCTCAGG | AAAAAAGTAC  | GTGTGTGGTA  |
| 32341 | TGAGTGTGTG  | TATGTATGTG | TGTATATATA | TACACATATA | TACACATATA  | TATACATATA  |
| 32401 | TGTGTATATA  | TATACATATA | TGTGTATATA | TATACACACA | CATACACATA  | TATATACACA  |
| 32461 | CACACATACA  | TACATGTATT | TTTATATAAT | TATATATGCA | GAGAGTGCAA  | ATGTTGCCAA  |
| 32521 | GTTAAAGATT  | GGTGAGTCTA | GGTGAAGGGA | ATATGGTATT | TATTGTATTA  | TTTGTGCAAC  |
| 32581 | TTTTCTTAAG  | TTTGAAAATT | TTCAAAACAA | AAAATTGGAG | GAAGAAGGCA  | TGCCAGTCTA  |
| 32641 | CCCCAAGCCC  | TCCATTGGAA | TGCTGAAAAT | CTAAACAATG | TGATTTGGCA  | ATTTCAATTC  |
| 32701 | TTTTCTGTGT  | TGGGCCAGTA | GTCTTAGAT  | GTGGGGGAG  | GGGGTAGTCG  | CTGAGGTGTG  |
| 32761 | GTTGACTTAG  | GATGGAAGAA | GCAGAAGTCA | AGACTCCCAG | GGTCAAAGTG  | GTTTGTCTCT  |
| 32821 | CTGACCCAAG  | TGTGGGAGGC | CCAGAGTCAG | CGTTTCAGGT | GTGCTAAATC  | AGCATGGTTC  |
| 32881 | TATTCACGGC  | CAAAGTCCAC | CCTGGGCACC | TCTCTGGCAG | CAATCTTGGG  | TGACTCTACT  |
| 32941 | AAGGCCAGGC  | CTCCATGACC | CTATGTCTGG | ATCCCATATC | TCCACCTCTC  | CCACTGTCTC  |
| 33001 | AGGAACGGTG  | CTTAGCTTTT | TCTTTTCCCT | CTCCTGTCTT | CTTTGCCAGC  | ATGTAGAAAG  |
| 33061 | TTTAAATAAT  | TCCCCTCTTT | ACAACAAAAC | AAAACATACC | CCCTTCAGTC  | AACCAACCCTA |
| 33121 | GCTCTCTTCT  | CCTTTTCCCA | GCCAGATTTT | TTTAAAAGCA | TCCTAGGCCA  | GGCGCGGTGA  |
| 33181 | CTCACGCTGT  | TAATTCAGC  | ACTTTGGGAG | GCCAAGGTGG | GTGGATCACA  | AGGTCAGGAG  |
| 33241 | ATCGAGACCA  | TCCTGGCTAA | CATGGTGAAA | CCCCATCTCT | ACTAAAAATA  | CAAAAAAGTA  |
| 33301 | GCCGGGAGTG  | GTGGCAGGTG | CCTGTAGTCC | CAGCTACTCG | GGAGGCTGAG  | GCAGGAGAAT  |
| 33361 | GGCGTGAACC  | TGGTAGGCGG | AGGTTGCAGT | GAGCCGAGAT | GGCGCCACTG  | CACCTCCAGC  |
| 33421 | TGGGTGACAG  | AGTGAGACTC | CGTCTCAGGA | AAAAAATAAA | AAAAAATAAA  | AAAAGCATCC  |
| 33481 | TCAGCACTTT  | GGCAACTCCA | TCTCTCCCA  | ACATGTCCCT | GTTACTGGAA  | TCCAGCCAGG  |
| 33541 | ACTCAGCCCC  | GATCTTCTTA | CTCTAACCCG | TGTCTCTAGT | TAACAAGGAC  | AGGTTTATGC  |
| 33601 | TGCAGTGACA  | AACAAGATCC | CAAATTCTTG | TGGCTTCACA | CATCTGGCAC  | CACCTCATCT  |
| 33661 | TCCAGCCCTTA | GGAGTCTACT | TTTAGTTTCT | TGAAAACCTT | TTACAGTTTT  | CTGTTGGGGC  |
| 33721 | CTTGTCATAT  | ACTATTCCCC | TGGAATGTTT | TTTCTTATCC | CCTCCCTTTC  | ACCTTGTCTAA |
| 33781 | CTTGTCGCCA  | TCCTTCAGGT | CTCAGCAGAA | ACATCACTTC | CTTGGGGAAG  | TTTTCTCCAA  |
| 33841 | CACCCACACT  | ACACAGGTGT | CCCATCTACA | CTCCTATGAC | TTTGTGGTAC  | TTGTCTCACT  |
| 33901 | TCATTTTCCA  | CTGCCCTCCC | CACAAGGCAC | CTGCACAAGG | GCAAGGACCG  | TACCACTGTA  |
| 33961 | CCTATGTAC   | TCATTGCTGT | GGTCACCTGC | ACTCTGGCTG | CCTACCTTAA  | CTACACATTA  |
| 34021 | GAATCAGCTG  | AGGAGCTTTT | AAAGCCACAA | TGCAAGACTC | CACCCTAGGC  | CAATTGGATC  |
| 34081 | CAAAATCCCTG | GGGTAGGGCC | AGACATCAGT | GGAGTTATAT | ATACATATAT  | ATATTTTGTT  |
| 34141 | TGTTTGTTTT  | TTTGTTTTTT | GAGACAGAGT | TTTGCTCTGT | CACCCAGGCT  | GGAGTGCAGT  |
| 34201 | GGCGCATCT   | TGGCTCACTG | CAAGCTCCGC | CTCTCGGGTT | CACACCATTC  | TCCTGCCTCA  |
| 34261 | GCCTCCTGAG  | TGGCTGGAAC | TACAAGTGCT | CGCCACCACG | CCAGCTAAT   | TTTTTTGTGT  |
| 34321 | TTTTAGTAGA  | GATGGGGTTT | CACCGTGTTA | GCCAGGATGG | TCTCGATCTC  | CTGACCTCAT  |



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SEQ ID NO Sequence

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| 34381 | GATCTGCCTG  | CCTCATCAGC  | CTCCCAGAGT | GCTGGGATTA  | CAGGCATGAG  | CCACTGCACC  |
| 34441 | CGGCCATCAG  | TGGATATATT  | TTTAAAGCAC | TGCAGAGAAT  | TCTGTTGCAT  | CAGCTTGAGA  |
| 34501 | ACCACTGATC  | TGCCTTGTGC  | TTCACATTTA | AAACTTTTTT  | TTAATGAATA  | AATAAACCCC  |
| 34561 | AAAAAATTAA  | TCTCCCTAAG  | CCTCCCTAGA | AGATAGGATG  | GTAAGGATAT  | TTTCCTAGGT  |
| 34621 | AAAAATATGT  | TAATTTTCATA | TTTCATGAAA | TTTCATGTTT  | CATTTCATC   | AAGCTCTGTC  |
| 34681 | ATACACCTTA  | CATGGGGCAA  | GCCCAAGTGC | TGGGCAGGGT  | GTAATTATAC  | TCATTACACA  |
| 34741 | GGCAAGGAAA  | AGTCACATTA  | GGTGATGGAG | CACAAATAGG  | CAGTTAATGG  | TTTCAGGGCT  |
| 34801 | AGTTAGGATA  | TGTTTGCTTT  | TCAATTGCAA | GTAATAGAAG  | CCCAAGAAA   | TTGGTTATTT  |
| 34861 | ATATAATATA  | ATTGATTGGT  | TCCCAAATTT | GAAAAATTCA  | GGAATAGACC  | CAGCTTAGGT  |
| 34921 | ACAGCTGGAT  | CCAGTCACTC  | AAACAATGTC | ACAAAAGAAC  | CTTGACAGG   | AATGTATCCT  |
| 34981 | GTGTTGACTC  | TACTTTGCTC  | TGAGTAGTCT | TTCCCCAGGT  | GATGATAAAA  | ATGGTCATCA  |
| 35041 | TCGCCAGGCT  | TGTGTCCTGT  | TTAGTAGGAA | TATACAAGAA  | GAGCTCAGTA  | AATGCTGGCC  |
| 35101 | CCACCCTAA   | GCAAAAACAA  | AACTTTTGTT | GTTGTTATTG  | TTGTTTTAAA  | TAACAGCTTA  |
| 35161 | GACCTTTGCT  | CTTTCCTTGT  | TATTCTCTTT | CATCTGTAAT  | CCAGTTTTCT  | ACTTCTGAAG  |
| 35221 | TATAGAATGT  | TCTGATGATT  | TATTCTTCAT | TACCCACAAC  | TTGCACATGT  | TTATTTAAAA  |
| 35281 | ATGCCAGAGT  | TGCCCTGGCCG | TTGTGTGCTG | TTAACCTTTG  | TTTGCTGTTA  | GTGGATCCCT  |
| 35341 | GAAGTTTCAGG | CTCCCAGGGG  | AGCAGATAAT | GGGTATCCAG  | TTCTGCAAT   | ATCCACCCTC  |
| 35401 | TGGCAAGCCA  | AGTTCCTTCC  | TGGGTAAGGT | TTTGCCCTACC | TGCATTCCCTA | GGGAAGTTTC  |
| 35461 | TGGGCTTGAC  | CACCAAGCCA  | GCTCTGAGAA | GGGGTGCCATA | AGCCCCACCA  | TGCTTTGGCT  |
| 35521 | ATGTCCTTAT  | AGAATATTTT  | ATGTTGTTAC | TGAAAACATA  | AGGAAGATGG  | GTGCGGTGGC  |
| 35581 | TCATGCCTGT  | AATCCCAGCA  | CTTTGGGAGG | CCAAGACAGA  | TTGATCACTC  | GATGCCAGGA  |
| 35641 | GTTCAAGACC  | AGCCTGGCCA  | ACATGGTGAA | ACCTTGTCTC  | TACAAAAACA  | AAACAAAAACA |
| 35701 | AAAAATTAGCC | GGGTATGGTG  | GCATGCACCT | GTGGTACCAG  | CTACTCAAGA  | GGCTGAGGCA  |
| 35761 | CAAGAATCTC  | TTGAACCTGG  | GAGGTAGAGG | TTGCAGTGAG  | CCGAGATCGC  | ACTACTGCAT  |
| 35821 | TCCAGCCTGG  | GTGACAGAGC  | AAGATTCTGT | CTCCAAAAAA  | AAAAAAAATA  | AGAAAAAGGA  |
| 35881 | AGCTAAAGGA  | GAGAGACTAA  | AATGATATCA | GGTTCTTGGA  | GACCAACAG   | ACATGATTTT  |
| 35941 | GCTTCATGGC  | AGGACAGCCG  | GAAGAAGTGG | GATTATATCC  | TCACATTACA  | AATAAGAAAA  |
| 36001 | CTGAGACTCA  | GAATGGTTAA  | GTCACCTTGT | CCAGGCCACA  | CAGCCAGTAA  | ATTACAGAAA  |
| 36061 | CAGAATTGTA  | ACCCAATCT   | TCCAGCTCCA | AAGCTTGTGT  | TCTTTTCACT  | ACCTCCTGCT  |
| 36121 | TAATTTTTTA  | ATTCTAAGA   | TTAGACCCCT | CATCTATCCA  | TGACACCTGC  | CTGTCATCCC  |
| 36181 | CTGAAAAAAG  | GTGAACGCCG  | TTCAGAAAAT | TTTCTAGCCT  | GAGCTCACTC  | CCAGTTCACT  |
| 36241 | TTGTTTTGCT  | TTGTCATGGC  | TGCCCAGTCC | CCACTTGTAG  | ACCAGGAATA  | GGTCATGGGT  |
| 36301 | GCGGGGACTA  | CACGCTGTGC  | CTGCTGCAAG | GGCCGGCCTC  | TGTTTCCGGG  | GCTGAGTGGG  |
| 36361 | GGCCAGACCT  | GCCAGGAGCA  | CCATCTTCTG | TGGGTCCTGC  | CTGGATGTCA  | CATCCCGGCC  |
| 36421 | CCAAGAAAGTC | ACTGCAAAAC  | TTGCTATTAT | TGAGCTTCAC  | ATCCTAGAA   | TTGCTGTAC   |
| 36481 | TGTGGCTGCT  | GCATGAAGTT  | GTCTGAGAG  | AAACGGGCAT  | TGTCATTAA   | AGGGAAATTG  |
| 36541 | ATGGTCTGGG  | GGAAAAGTCA  | TCCTCATTCT | CTTGCGATC   | TATGGGTGAT  | TGAGACTGGC  |
| 36601 | GCATGTTGAA  | GGGGTTTCTC  | AGCCATCGTG | TGCCATGTTA  | TGGAACAGTG  | GTGTAGCCAG  |
| 36661 | CCATTTGACA  | CCCAGCGCTG  | ACCTTTGTTT | AACAACCTCA  | CCTATATATG  | ACAAAATGAT  |
| 36721 | TGTCAGAAAT  | AATCGTGTA   | TGAAATGACT | GTAATAATGG  | CCAGAAAAGA  | AACGCAGATA  |
| 36781 | GTAAATATGT  | TCTCTTGTG   | AACTCTGTAC | ATATAATTGC  | ACCAGGATTT  | TTTTCAAATA  |
| 36841 | AAAAGTAAAT  | ATTATACTAC  | AAAAAAGGGA | AAAAGCACAA  | GCATTTATTA  | AATAGCTTTC  |
| 36901 | TATATCTTTT  | TGAGTTTGTG  | TCCTTTGATT | GCAGACTGAT  | GTAATATTTT  | ATGTAATCA   |
| 36961 | TTGCTTTGTT  | ACTAAGTGAA  | CTTTAAGAAA | AGTGAGACGT  | CTGCAGAGT   | TGCCATAAT   |
| 37021 | TTAGCAGCTA  | CTGTATTGTA  | CCATTGATGT | ACGGCTTTAT  | TTTCTTGATT  | AATTATTTAA  |
| 37081 | ACAAATATAAT | TCACAATTTT  | AAAATAATAA | ATTTCCACTT  | AAAATGGTAT  | TTAAACTCAG  |
| 37141 | CAAAAATATAT | CATCTATGAG  | TAAAATTTGT | ATTTACCAG   | CAAAAATATT  | ACAGTTTGTG  |
| 37201 | GTTACATATC  | TGCTCACTG   | TTTTAAATTT | TAAATACAAA  | AACTCCAAGT  | AGGCTGGGTG  |
| 37261 | TGGTGGCTCA  | CACCTGTAAT  | CCCAGTACTT | TGGGAGGCTG  | AGGCAGGCAT  | ATCGCTTGAG  |
| 37321 | TTCAGGAGTT  | CAAGATTGTC  | CTGGGCAACA | TAGTGAGATC  | CTGCTCTAC   | TGAAAACAAT  |
| 37381 | TAGCTGGGTG  | TGGTGGCACA  | TGCCCTCGGT | CCCAGCTACT  | CAGGAGGCTG  | AGATAGGAGG  |
| 37441 | ATCACTTGAA  | CCCTGGGGGA  | CAGAGGTTGC | AGTGAGGCAA  | GATTGCACCA  | CTGCACTCCA  |
| 37501 | GCCTGGGTGA  | CAGATTGAGA  | CCCTGTCTCA | AAAAAAGAAA  | AAAAAAAAG   | AAACACAAAA  |
| 37561 | ACTCCAGGTG  | GTGCGACAGA  | ATGACAGGAC | TGAAGTAACT  | TAGCTCCAAT  | TTCTGTCTTC  |
| 37621 | ATAATCACTG  | TCCTACCAT   | GTCTGTGCTT | AGAATCTACT  | TGCTTAATGC  | AGGAACATGT  |
| 37681 | GTCTCTACAG  | AGATGGAAAA  | TGCAAAATGG | GCCAGAAGCA  | AGCTGGAAAT  | TCTGAACCAT  |
| 37741 | TAAGAATTTA  | CTCTCTGCCA  | GGCACGGTGG | CTCACGCCCTG | TAATCCCAGG  | ACTTTGGGAG  |
| 37801 | GCTGAGGCAG  | GCAGATCATC  | TGAGGTGAGG | AGTTCAAGAC  | CAGCCTGGCC  | AACATGGTGA  |
| 37861 | AACTTCACTC  | CTACAAAAAT  | ACAAAAATTA | GCCAGGCATG  | ATGGTGGGTG  | CCTGTAATCC  |
| 37921 | CAGCTACTCG  | GGAGGCTGAG  | GCAGGAGAA  | CGCTTGCACC  | TGAGAGGTGG  | AGGTTGCAGT  |
| 37981 | GAGCCGAGAT  | CTATCTGCAC  | CATTGCACTT | CAGCCTGGGA  | GACAGAGTAA  | GACTCCTACT  |
| 38041 | CAAAAAAATA  | AAAAAATAAA  | AAGAACTTAC | TCTCAAAATA  | AATACGTGTG  | GCTGACTCCA  |
| 38101 | CATATGGTAG  | GGCCAACTGT  | ATAACTAGAA | GTTCTCCAAA  | TAACCTCTGT  | GGAGAAAAAA  |
| 38161 | AAGTTTATTA  | AAGGTTAACT  | TTTTTAAAGT | GCTAACTAGA  | ACCTTACTAA  | CACTGAGATC  |
| 38221 | GCACCAATTG  | TTTATAACTT  | AGACAGGGCC | GGGTGCAGTG  | GCTCATGCCT  | ATAATCCCAA  |
| 38281 | CACCTTTGGGA | GGCCGAGGCA  | GGTGGATCAC | TTGATGTGAG  | GAGTTCGAGA  | CCAGCCTAAC  |
| 38341 | CAACATGATG  | AAACCCCATC  | TCTACTAAAA | ATACAAAAAT  | TAGCCAGGCA  | CGGTGGTACA  |
| 38401 | CGCCTGTAAT  | CCCAGCTACT  | GGGAGGGGTG | AGGCAGGAGA  | ATCTCTTGAA  | CCCAGGAGGC  |
| 38461 | GGAGATTGCA  | GTGGGCCAAG  | ATCGCACCAT | TGCACCTAG   | CCCCAGCAAC  | AAGAGTGAAA  |
| 38521 | CTCTGTTTCA  | AACAAACAAA  | CAAAAAAATA | AACCTCTTGG  | ACCAGGAAAA  | TATTTTTTAA  |
| 38581 | GGGAGGAGTA  | TTTTATCACT  | GGCATTGTTT | AGGATTGAGC  | GCACATGATG  | CTAATGAAAA  |
| 38641 | GCAGACTAAC  | TATTAGTTGG  | TTTTATTACT | GTTTTTGAAC  | TCTCTCTCTC  | CCTTTTTTTT  |
| 38701 | TTTTTTGAGA  | CAGAGTCTCT  | CTCTCTGTCA | CCCAGGCTGG  | AATGCAGTGA  | CTGCAGTCTC  |
| 38761 | AGCTCACTAC  | ATCCTCTGCC  | TCCTCAGTTC | AAGTGATTCT  | CGTGCCCTAG  | CCTCCCGAGT  |

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SEQ ID NO Sequence

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| 38821 | AGCTGGGATT  | ACAGGGGCACC | ACACCAGGCT | AAGTTTTTGT  | ATTTTATAGTA | GAGGCAGGGT  |
| 38881 | TTACACCATGT | TGCCCAGGCT  | GGTCTCAAAC | TCCTGGCCTC  | AAGCGATCTG  | CCCATCTTGA  |
| 38941 | CCTCCCAAAG  | TGTTGGGATT  | ACAGGCGTGA | GCCACCGTGC  | CTAGCCCTGT  | TTTGAACCTC  |
| 39001 | TCTAGAGACA  | GTCCAGCCCC  | TTATTACTTG | TCCTGAGGCA  | GCTGCTCCCT  | TCACCTGGCC  |
| 39061 | CCCCGCATTG  | TGTTCCGGAC  | CCTTGCTCTG | GTGGTGCTAA  | AGAATATCTC  | TGTCGATCCT  |
| 39121 | TTGGGGACTG  | GGGAAACTGA  | GGCCCACTGC | CACGCGATGC  | CATTGTGTCA  | GGGAAGATTA  |
| 39181 | GGTCATCTGC  | TAGGTCCCCA  | GTCACTTGAC | CTTCTTCCCA  | GACAGGAAGA  | AGCTGCTCTG  |
| 39241 | GGTCTCTCAG  | TGCTCCACGT  | GTCTTTGCAC | ATTGAAATGT  | TTTCTGATTT  | TTTTTTTTTT  |
| 39301 | TTTTTGCTGT  | TACATTTACT  | TTTAAAAAAT | AACAAGCAAT  | AAAATGTTAC  | ATTTGAGAAG  |
| 39361 | GTGAAATGA   | GAATTGATTT  | GAGTTAAAT  | CTAGCAGATT  | TTTCTTAGAA  | GAATGATATC  |
| 39421 | ATCATCTCCA  | GCTACCTGCA  | ATTGATCTAC | TCTGAATTAA  | GAAAGAGACT  | TCCATTGTGT  |
| 39481 | GTTTATATTT  | TGCACTCTTG  | ATGTGTTTCT | TTAAATTATG  | GTGATGGGCC  | AGGTGTAGGA  |
| 39541 | GCTCACACCT  | GTAATCCCG   | CACCTTGGGA | CTCTGAGGAG  | GGAGGATCAC  | TGGAGGCCAG  |
| 39601 | GAGTTCAAGA  | CCTGCTCTGT  | ACAGTAAAT  | TTAAAAATTA  | GCCAGGCATG  | GTAGCATTCA  |
| 39661 | CCTGTAGTCT  | TAGCTACTTG  | GGAGGCTGAG | ATGGGAGGAT  | TGCTTGAGCC  | AGAACTTTGA  |
| 39721 | GGCTACAGTG  | AGTTATTTTC  | ACGCCACTGC | CCTCTAGCCT  | GGCTGACAGA  | GCAAGACCTG  |
| 39781 | CCTCAAAAAA  | ATAAGTAAAA  | AATAAATTAA | ATTTCAATCA  | TTAGCAGTCA  | TTAGGATATT  |
| 39841 | TAAATACAGT  | ATGTTGAATC  | AAAGTTACGC | ATGTGTGTAT  | TTTTTTTTTC  | AGAGAGTTGT  |
| 39901 | TTATCATGTG  | GGTTTTAATT  | TAACTTTAAA | AAAAATGTTG  | CTGGACAGTT  | GCCCCAAATG  |
| 39961 | GTCATCTCAG  | CATTGTGTTG  | AGAACGTATG | TCCTGCGGGC  | TCCTCTGTCT  | CTGGAGTTTT  |
| 40021 | GCTAGCTGAC  | AGCCACTGGC  | TAGTTAGAGA | CTGCAGTCAG  | CACAGATGCA  | GGCGTGGACT  |
| 40081 | TGCGCACGTA  | ACCATGTCAA  | TGCAAGGCCA | TCACCTTCTA  | AAAATTCCTG  | ACCCTGCTGT  |
| 40141 | CTGAGATGGT  | GGTGCAGCGG  | ATAGAACTCT | GCTCTAAGAG  | GCAGTAGCTA  | ATTCATGTGC  |
| 40201 | TTCTTTGCCC  | TTGACTAGCT  | GAGTGACTTT | GCACATGGGG  | CTTGCCCTCT  | TGTGCTTGTG  |
| 40261 | TCTGCAAAAG  | GGAAATCATCT | TTTCTTGCT  | AGACAGAAGG  | TGGACCCCTG  | ACCTATGGCC  |
| 40321 | TTTTTGAAGT  | TTCCCCCGCG  | TTCTTAGAAG | GACCTCTGAT  | CCTACTGAGT  | TTAATACCCA  |
| 40381 | CGGGTTAATA  | ATTGGGAAAA  | GCAAAGGAAG | CGCTTCTGTT  | TAGGTAATTA  | TATGCATGTT  |
| 40441 | TTTGTCTTTT  | TCTGGCTGGA  | AAGATATCCA | AGCCACTGGG  | AAGGTCCGTG  | GCTACCCAGG  |
| 40501 | GTAGCCCTCT  | CTGGGGAGGG  | CTGCTATATC | CAAGAGCCCC  | TCATGAGAA   | TTGAAATCG   |
| 40561 | ACCATGGTAG  | GGCCTGCTGA  | CTTTTGACAG | CTAATGGTGT  | GCTGAGAA    | GTCCCTCCAA  |
| 40621 | AGATGCCCTT  | CCATTCCCTC  | GGGAGAGTCT | GGGCGAGCCC  | TACTGGGGGC  | TGGGATGCTG  |
| 40681 | GCTCTTCCCT  | CAGCCTCCAC  | CCCAACTGCT | CTCTTCCCTC  | CTCCCTCCCT  | CAGCCCCCTA  |
| 40741 | ATTTCTCTCA  | CAAGGCTTTG  | TTCTGCAGCA | ACCTTTCCCTA | ATGCAGTCTC  | GGCCTCTTCG  |
| 40801 | CAGCTTCATT  | ACATAACCTT  | CCGTGGACTC | CTGGTCCAA   | GATCACCCCA  | GAAAGCCAGT  |
| 40861 | CAGAGGTAGG  | CACGCAGCTG  | GGGTCCATT  | ACTTACCTTC  | CCCACCCCT   | CGGAACCTAG  |
| 40921 | AGGTGGTGCA  | GGAATTTGGA  | CTCCAAGAAT | TAACAGCTCC  | ACCACCATCA  | CCAGAGCCAA  |
| 40981 | AACTCAGGAT  | GCATGTGCTT  | CATCTGCTGC | TTATTTCCAG  | CTGAGAGCCA  | GTGGTGCCAT  |
| 41041 | GGTTCTCTAG  | GGAGCCGGTC  | CCCTGATGCC | GGCTCCTGGC  | CCCAATCTC   | CTGATCCCG   |
| 41101 | GCTCTTCCAG  | AATGTCTTGT  | CTCCACCATC | GCCTTTGACC  | AATGGTGTCC  | CTTGCCCTGG  |
| 41161 | TAATGTCCCC  | TTTGCCTGAT  | GATGGCCCTG | TCACCTCTCT  | CTTTAGCACA  | GAGGAGGCTG  |
| 41221 | TTTCATCCCT  | TCAAGCCTGC  | CTCCCTTCA  | AGTCTTAGCT  | CAAGTTCACT  | TTCTCCGAG   |
| 41281 | AGCCTTCTCC  | AATCTTCTTG  | ACTACGTCTC | CTCTCAGCTC  | CAGCAACCTC  | TGTCTCTGGC  |
| 41341 | ACTGATTCCT  | TACTTAGCTA  | AGAGAAATCA | AGACACTTGG  | GGCTCAGGAC  | AATCTGCTTT  |
| 41401 | CTCTCTTCTT  | ACCCATGGCC  | TTGGACTGTG | TGTACCTCTT  | TGTCTCCACT  | CCCAACCCCA  |
| 41461 | ACCCCCAGAG  | GGCAGAGAGC  | ATGTTGTCTG | TCCCTTTGCT  | CAGCATGAAG  | CCATGCGTGT  |
| 41521 | GGTAGATCGG  | CAGAGTTCCA  | TAACCTGTGT | TGACCGAGGG  | GTCACTTTGC  | TCTGAAATTA  |
| 41581 | CCCCTGTGTC  | CTTCAGTATT  | TGCACAGATA | GCTTCTTGCC  | CAGACCGAAT  | ATATCCAAGG  |
| 41641 | GCATGGCCCA  | CCTCTGCTCC  | TGTTTCCAGG | TCCCTGGTGG  | GGGTTAGTTC  | ATGCCTTCTC  |
| 41701 | CATAATCTGC  | CCACTGGCCT  | GGTCTCAAG  | GTCTTCCCAA  | CTGCTCAGCC  | AGAGTTGAGA  |
| 41761 | AAATGGGTCT  | CTCCATCTCT  | TTGTGTCTGT | TCTCTCCTTC  | CTGGCCCACT  | CTCCTGCCCA  |
| 41821 | CAGGTATCCA  | GGGCTGCCT   | GTAGCATTAG | AGGACATACA  | TGCACATGCG  | TGGGCATGGG  |
| 41881 | ACACTCAGCT  | AGCCTCCAAG  | CACAGCATCA | ATAATGCATT  | CTGTGCTTTA  | TAGCATGGAA  |
| 41941 | AGCTGCTCTA  | AACTTTATTA  | CACAGTGGAC | ATGTCTGAAG  | CAGCTCCCAA  | ATCCACCCCT  |
| 42001 | GAGTGTGTTG  | GAATTGGCAA  | GCCTATCACT | TGGGAGTCTA  | GTTTTTTTGT  | TCGTTAATAA  |
| 42061 | TAGATGCTTC  | CTGTGGCCCC  | AGCTTGGCAA | TTTTGATTTA  | AAGTGATCTT  | AACTGAAGAG  |
| 42121 | ACTAATGGAG  | GGGTCTGAAT  | TTGTGCCTTT | TAAGCACAAA  | GTATTGCTCT  | TAATTAACCTG |
| 42181 | GATTCTATCC  | TTTGAGCAGG  | CAGAGGCCCT | CCCCCAAGGG  | CGTCATTAAC  | GATCCACATC  |
| 42241 | TGGACATCTT  | CCAAAGCCTT  | CTTCTGTTTC | AGGCCAACCG  | CAGGTGTGTT  | CCTGAACACC  |
| 42301 | CAGGAGGCTA  | TGAGAGCCAC  | ATATGCCTCC | CAAAATACACA | CAGTGTGCAT  | GCCAGGGGAC  |
| 42361 | ATAGAGCAGT  | GTGCAAGATC  | CCATTCCATC | TCTCTCCACC  | TGGGAGAGGA  | TGGCTCTTCT  |
| 42421 | GTCTGATTCA  | TGGCTCAAAG  | TGGTAAAGGA | GCTCCCCACT  | CCCCGTCCCA  | CGCCTACTCA  |
| 42481 | GAGTCTGCAA  | ATATGTATGC  | GATATGAGAG | CTCGTCAGTT  | AGTGTCTCTT  | AGTGTGGCGC  |
| 42541 | ACATTTGAGG  | AGTCTGACTC  | CCCTCCAGCA | CAGGCCAATG  | TGCACTGCTC  | TCCTATCTTT  |
| 42601 | GTACCCCCAC  | TGTTGCACTG  | TGCAGAGGTT | GGAGCCATAG  | AAGTACCAGA  | GCTGTGAAAG  |
| 42661 | GAGAGGCCCC  | CTCTCACCTC  | TGCCCTGGTC | TCCATCCCCA  | CTTTCTCTAG  | GAAGCTAGTA  |
| 42721 | GGTGCTGACA  | GGGAGAGAG   | GGGTCCAGAA | ACAGTGGCTG  | ATGCTTGCAA  | ATGCTTGCAA  |
| 42781 | TCCTAGCACT  | TTGGGAGGCT  | GAGGCAGGAG | GATCATTGTA  | GGTCAGGAGT  | TTGAGACCAG  |
| 42841 | CCTGGGCAAT  | GTAGCAAGAC  | CCTATCTCTA | CAAAAAGAAA  | AAATGTAATT  | AGCTGGGTGT  |
| 42901 | GGTGGTGGGC  | ACCTGTAGTC  | CTAGCTACTT | GGGAGGATGA  | GGTGGGAGGA  | TTGCTTGAGC  |
| 42961 | CCAAGAGTTT  | GAGGTTACAG  | TAAGCTGTGA | TTGCACCACT  | GCACTCCAGC  | CTGGGCAACA  |
| 43021 | GAGCTGAGAC  | CCTATCTCAA  | AAAAAGAAAA | AAAAAAGAAA  | AGGAGAGAGA  | GAGAAAGAAA  |
| 43081 | AGAAAAGAAA  | AAAAAAAAG   | AAGGGAAGGG | AAAGCCCAGA  | AGAGTGTGGG  | GAGAGGAGGC  |
| 43141 | GGCCGTCAAT  | CTGGGGCCCT  | CAGTGTGCAC | AACCAGATAA  | CACATGCTCT  | GTGGGCTTTT  |
| 43201 | GTACCATTTT  | GCTTGAGCAT  | AAAGAAAGGA | AGGCTGCCCC  | TAAATAGAAA  | GCACTCTGGA  |

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SEQ ID NO Sequence

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| 43261 | GGCAAACAAA  | TCTGACTCCA | ATCCTGGCCC | TGCCACTTTC  | CCAGCTGAGG | ACTTAGACAA  |
| 43321 | GCACCTAGC   | CTCTTGGACA | TTCTCAGAGC | CATCTGCTGC  | AAGTGGGTGC | TGCCATACCC  |
| 43381 | ACCTTACTGG  | GCAGGCTTGG | GGGACCAAGG | GTGGTAAATG  | GCTCAGTCTT | TCATGATGCG  |
| 43441 | GCCACACAGC  | AGGTGCGCCA | TCCAGGTCCA | TTTCTTTCCT  | TCCTTTCCCC | CAAATCAAGT  |
| 43501 | TGTCATTAAA  | GTACTAGTCC | ACATTAATGA | AATCAACTGT  | ATTAATTTTC | TATTTGCTGC  |
| 43561 | TATAATAAAT  | CATCAGAAAT | TTAGTGGCTT | AAACCAACAC  | AAATGTATTA | CCTTACAGTT  |
| 43621 | CTGGAGGCCA  | GAAGCCCTCC | ATAGGTGTCA | CTGGGCTGAA  | ATCAAGGTTT | TGGCAAGGTT  |
| 43681 | GCGGTCACTT  | CTGGAGGGTC | CAGGGGAGAA | TCCATTTTCT  | TCCTTTTTC  | AGCTTCTAAA  |
| 43741 | GGTTTCATGC  | ATTCTTGGC  | TCATGATCTT | CTATAGCTAT  | AGTCAGAAAA | ATTTTCCATC  |
| 43801 | AATCATCTTC  | AAAGCCAGCA | ATGGCAGGAT | GAGTCCTCAC  | ATCACCTTGC | TCTGACACCA  |
| 43861 | GTTCTCTGCC  | TCCCTCTTCC | ACATGTCAGG | ACCCTCATGA  | TTACTTTGGG | CTCACTCTGA  |
| 43921 | TAATCTGGGA  | TGATCTCTCT | ATTTTAGAGT | CAGCTGACTG  | GGAACCTTAA | TTCCATCTAC  |
| 43981 | AACCCCAATT  | CCTCTTTGCC | ATGTACAGTG | ACATATTCAC  | AGGTTCTGGG | GATTAGGACG  |
| 44041 | AGCCTGTCTC  | TGAAAGGCTA | CTTTACATGA | AAATTCATTT  | TTTTAATTA  | GATTTTTTTT  |
| 44101 | TCCTCTTGAG  | ACAAGGTCTC | ACTCTATGGT | TCAGGCTGGA  | GTGCAGTGGT | ATGATCACAG  |
| 44161 | CTCACTGCAG  | CCTCGACGTC | TCTGGGCTCA | GGTGATCCTC  | CCACCTCAGC | TTCCCTAGTA  |
| 44221 | TGTGGAAGTA  | CAGGGGTGAG | CCCCATGCC  | CAGCTAATTT  | TTTTTTTTTT | TTTTTTTTGA  |
| 44281 | GACAGAGTCT  | CACTCAGTCA | CCCAGGCTGG | TGTGCAGTGG  | TGCAATCTCA | GCTCACAGCA  |
| 44341 | ACCTCCGCCT  | CCTGGGTTC  | AGTGATTCTT | GTGCCTCAGC  | CTCCCAAGGA | GCTGGGACTA  |
| 44401 | CAGGTGTGCA  | CCACCACGCC | CGACTAATTT | TTGTATTTTT  | AGTAAAGATG | GGGTTTCACC  |
| 44461 | ATGTTGGCCA  | GGCTGGTCTC | AAACTCCTGA | TCTCAAGTGA  | TCCACCAACC | TCAGCCTCTC  |
| 44521 | AAAGTGCTGG  | GATTACAGGT | GTAAGCCAAC | ATGCCCCGCC  | CCAGCTAATT | TTTAAATATT  |
| 44581 | TTTTTTGTAG  | AGATGGGGTT | TTACCATTTT | GTCTAGGCTG  | GTCTTGAAGT | CCTGGGCTCA  |
| 44641 | AGCAAACCTC  | CCACCTTGGT | CTCCCAAAGT | GCTGGGATTA  | CAGCATGAGC | CACTGCACTC  |
| 44701 | GGCCTTAAGA  | GAAGATTAA  | TAATTAATAC | TTTACAACAA  | GATCTGGAAG | AGGTGGGATG  |
| 44761 | AGTAACATAA  | TGAGGATACA | AGTAACCCGG | GTCATATTTG  | CTAATACCCT | TGGTCACATT  |
| 44821 | GAACCTGATA  | TCTTATCAGA | TTTTCTTAAT | CAGCTCCTTT  | AGCAGCAGTG | TTGCAGCATC  |
| 44881 | TTATCTCATT  | TTGTTTTTTG | TTTTTTTGCC | TAGCACATGC  | CTGTAAATCA | CTGGATTGAG  |
| 44941 | GATGTTAGAT  | GTTGTTGTGC | CTTTGGATGC | TTCTTATAAA  | TCCATATTTT | ATGGCTCCCT  |
| 45001 | GGAAAGTGCT  | ATGCAAATGA | TAAGCTGCAA | GGATGGAAAG  | GAAATTGCAG | TGCTCCTGAA  |
| 45061 | TTGTAATATG  | GCTTTTACGA | GGAGGTTTCT | AATTACTCGC  | TCTTTCTCTT | GAACTGAGGA  |
| 45121 | GTTGAAGTGT  | AGGTGGCAGA | TCCATAACAG | ATAATCATGT  | GTGTGATGTG | ACTTCAGCCT  |
| 45181 | GAGCGTCGAG  | GACCAAGTCA | CAGAGCAGGA | ACAGCCACTC  | TCCAGTGTCC | TTGGGGCTAC  |
| 45241 | GTCTGAGGAG  | AACCTGGGAT | TTCATATATG | ACCTGCACGT  | GCTGGGGGGC | TCTCTTGACG  |
| 45301 | GATGTTCTGC  | CCTCTGAGCA | TGTTACAGAT | TCTGACATTC  | TTATGTTTCT | TCTGTGGAGA  |
| 45361 | GACATGTACT  | TAGTGACCTA | ACTCACTTTA | GCATATTTTT  | GCTCATCGTT | TGTGTAGCTT  |
| 45421 | AAAGGAATCA  | GATAATTACC | CCCTCCCCAC | TACTTTCGGA  | AGCACAATG  | CAATGCCCTA  |
| 45481 | GATGTTACT   | GGGACTCAA  | AAAGAAAAGA | GAGTAGTAAA  | ATCTATTAAA | GGGGACAAAG  |
| 45541 | ACAGCCTATA  | TACTACAAGC | TTTCTATTTT | TATGGCAGAG  | AATGCCATTT | TCTAAGTAAA  |
| 45601 | CAGAGAACTG  | CATTTGACCT | GCAATATCAA | ATGCATGGAT  | TTGATGCTTT | GGAAAGCAAC  |
| 45661 | TGTTTTCTCG  | GTTAATCTGG | GTGTCTTCCG | TGAAATGTCC  | TCCTGCCTTT | GGCTTAAACA  |
| 45721 | CTAGCTTTGT  | CTACAGCCAT | TCCATCCTGA | ACCTGCCCAA  | TCTTGTCTGA | ATCCTGGTTT  |
| 45781 | CACCACTGAC  | AAGCTGTGTG | TCCTTGGGCA | AGTTACTTCA  | CCTGTCTGTG | CTTCAGAGTC  |
| 45841 | CTCATCTGTG  | AGTTGGGGAA | CTGTGACAGA | ATCTACCCCA  | TAGGGCGTAG | TGAGGATGTG  |
| 45901 | TTGAATTATC  | CCAAGTGGCT | ACACAGAGTA | AGCACTCAA   | TGATGTCTAT | GTTGTCTATGA |
| 45961 | TTGCTGTGTAC | CAGAGCCTAG | AGTTCACTCT | GATACTCGAG  | TCTGTGGCCC | ATCCAGCCCA  |
| 46021 | GGTAAGGAAT  | AGTTGGAGGA | GTTGGGCATG | TTCAAGCTTGA | AGAGGAGACG | ACAGGGGATA  |
| 46081 | TGGGATAGTT  | GAATCTGTGA | AGGGCCCCCT | GGGATGAAGA  | ACTGGCATGT | TCTGTGTGGC  |
| 46141 | TCCAGGGCAC  | TGAGCAGGAC | CCATTTGCCA | AAGTCTCAGG  | GACACAGTTT | CTAGCTATAG  |
| 46201 | ACAGAAAAAT  | TTTTCTGTAC | TCAGAGGATG | AAAATAGAAT  | GAGCCCCCTT | AAGAGGTAAT  |
| 46261 | GAGCTCCCTG  | TCATTGGAAG | GATTCCAGAA | GAGCTAGGTA  | ACCACCTTAG | GTGCTATCAA  |
| 46321 | GGGGCTTTTT  | TCTTTAAAGT | CCTTTCCAAA | AGCTTCTGAG  | ATTGCATAAA | CAATAGGAAG  |
| 46381 | CCATCTTGGT  | GCTTTAACAC | AAACTCTCCC | CAGTGATGAG  | GGTTGAGCCA | AAGCCAGATT  |
| 46441 | GGCAAGCAGA  | GAGGAGACTT | GTGTACAAGG | AGTTCTCTGA  | GTCAATTGCT | TTTTCTTGT   |
| 46501 | TCTAGCCAGC  | CAGAGGGCTC | CTGTTGGAAG | ACAGAGAGCC  | GGAGAGGCTG | AGGCCCTGACC |
| 46561 | AAACCAGCTT  | CTGCAGGCCA | GCTGGGAGGC | CACAACTCCT  | ACCTACGGGA | AAACTGAAGG  |
| 46621 | GCATCTCTAT  | TTTTAGATTA | GCAAAAGAAA | ATAAATTTAA  | GTTTGAGTCT | CCTTTGCAAC  |
| 46681 | TTTTAAAAGA  | CATCTTTTAT | GAGATGATCA | TTCAATTTCT  | ATAAAATTC  | CCCCTTTGA   |
| 46741 | GTACAAATCT  | AGTGGTTTTA | GTCTTCTTGT | ATGATTTTGA  | TGTTCTTTTC | TTAAGGCTCT  |
| 46801 | TGGAAGACCC  | AGAAGCCTCT | CAGACACAGG | TGGGTGTGGA  | GGGCGTAGCA | CAGAGGCAGA  |
| 46861 | CTTCTCATTT  | CCTGGGCTCT | CCCTTTAATG | ACTCTCAGAG  | ACCCCTCCTT | CCCCCTGCC   |
| 46921 | CTGGCTTCTA  | CCCCAGGGGT | GTAGAGTTTT | GCCATTTTCC  | AAGCAGAACT | TCATTTCTCT  |
| 46981 | TTCTGTGTCT  | ACACTCTTTG | TGCTTCTTTC | TTGCCAGCTT  | TTTCTCCTTT | GCCCGCCCTT  |
| 47041 | CCTTCTTCTC  | TTCCCTCCCT | CCCCCTTCTC | CTCCTTCCCT  | CTTTCCTCTC | TTCCCTCCCT  |
| 47101 | CCACCTTCTC  | CCCCTTCCCC | CCTTCCCTCT | TTCTTCTCTT  | CCTTCTCTGC | TGCTTCTCTT  |
| 47161 | CCTTCTTCTC  | TTCTCTTCTT | CCTTCTTCTC | TTCTTCTCTT  | CCTGGTATGT | GACTAATTTT  |
| 47221 | TGTTTCAGGA  | CATAAATGTT | GTCCAGGCTG | TTCTTTGGTC  | TTTCTGTGTT | ATAATGGACA  |
| 47281 | TTTGCCATTT  | AGAGAGGCTG | CTTTTCTCTG | AATCATGTTC  | TTGGGGCCCA | GAACCTAGGT  |
| 47341 | GTGTGCTTCT  | GACTTTGTTT | TCTTCTGTAT | CCAAATCTGT  | ATATGTCCAT | TTAAATTTAT  |
| 47401 | CTAGACCCAC  | AGGGCACTGT | GGGACAGATC | CTCAGTGGAA  | CATGACTCTG | TAACGAGAGC  |
| 47461 | ATTTTGTTTT  | GTCAAAATGA | GAACATATTA | TTGCCCTTCA  | TCTGATTGTA | AACATAATAC  |
| 47521 | ATGTTTATAA  | AACAGTATAA | TGAGACAAAA | ATGTAGACAC  | TAATAAGGGA | AAATCTCCCT  |
| 47581 | AATGTATATT  | CTCTTCACAG | AGAAAGCCCC | TGTTGGGCAT  | ATATACTCTA | GTTTGTTTAT  |
| 47641 | TTGTTTGACT  | ACACATATAT | GTATTTCTTT | CTTATGTATA  | AAAATTTCTG | ACATGCACAT  |

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SEQ ID NO Sequence

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| 47701 | TTCTGCAACT  | ACTGTTTTCA  | CTTGATGATG | CATGGACCTC  | TCTAGAGTGT | ACGTTTCTTC  |
| 47761 | TTCCTTACAA  | AGCAGTTGGC  | TTGCGCCAGG | GTGCACCAGG  | ACACGGTTTT | GGCTCTGTCC  |
| 47821 | CCAGGGTGTC  | ACGGGACCAG  | GGGATGATCT | CACAGGGTCT  | GCCATCTGCC | CTGCCTGGCC  |
| 47881 | GGAGGCTGCA  | TCGAGAGGGC  | CAAGGGGCAC | CACGTGTCTG  | GGGTACTGTC | AAACAAGAGC  |
| 47941 | CTTCAGAGCC  | TTCCACAGTC  | TTTCTTTTGC | TTCCACGATC  | TGCTTCCCCG | CTGGTGGACT  |
| 48001 | CTGAATCTAG  | AACTAGCTCC  | AGGCGCCTCT | CCAAATTTCAG | ACGGGAGCTG | GGGCACTATT  |
| 48061 | ATAATGCAAA  | TCTAGGCAAA  | GCCTCCCAA  | TACCAGGATC  | CAGAATGGGG | TGGGGCCCTT  |
| 48121 | TGCCCTAGAA  | AGCTGTTTAG  | TTTGAAAATA | CAAACAGGAG  | ACAGAAAAGT | TTGGCTAAAT  |
| 48181 | TAATGGATAA  | AGTTTTAACG  | ATGGTAACCA | TAGTAGGGTT  | CATCGACAGC | CAGCGATGGT  |
| 48241 | TCTGAACACT  | TGACATGAT   | TAACCTACCT | AATCCCCACA  | TTTACAGAC  | AATGCAAAGG  |
| 48301 | AGGCTCTGGG  | AGGTGAGTG   | ACTTGCCCCA | AAGTCGCACA  | GCTCCTAAGT | GAAGGATTCTG |
| 48361 | GAGTGGACTC  | CAGGCAGCCT  | GGTCTGACTC | CCTGCACTGC  | GCTGTGCTTA | TCTCTGGCCC  |
| 48421 | CAATGCCGCC  | ATGCAGAAAGT | GTCTGGGGGC | ACTTTGTCTC  | TGTCAGACAG | AATTCGGAGA  |
| 48481 | TCTGTATGCT  | TGCCCTGGTA  | TGGCACTTCT | CTTTTTTTGA  | GACAGAATCT | CACCTGTGCA  |
| 48541 | CCCTGGCTGG  | AGTGCAGTGG  | CATGATCTCA | GCTCACTGCA  | ACCTCCGCCT | CCCAGGTTCA  |
| 48601 | AGCAATTCTT  | GTGCCTCAGC  | CTCCCAGTA  | GCTGGGATTA  | TAGATGTGCA | CCATCGTGCC  |
| 48661 | GTGCCACCTT  | TTTGTACTTT  | TAGTAAAGAT | GTTGTTTTGC  | TGTGTGGCC  | AAGCTGATCT  |
| 48721 | CGAATCTTTG  | GCCTCAAGTG  | ATCTGCCTAC | CTCAGCCTCC  | CAAAGTGCTG | GGATTACAGG  |
| 48781 | CATGAGCCAC  | CATGCCTGGC  | AGTGTGGCAC | TTCTTACGTG  | TGTTACAGCG | ACACTGTTTA  |
| 48841 | CTTTCTGCTC  | CTCCAAGACG  | GTGCTGAGCT | CAGGTCGTTC  | ATTACTGGCA | GACAACTGCT  |
| 48901 | GATTTCCAAC  | AGAATTGCCA  | TCCTCTTCTC | CCCTGCGACT  | TTCAGAGTGT | GACCTCAGAC  |
| 48961 | TCAAAAAATTA | GAAGTGAAAA  | CATCTTAAAA | ACTATCACCT  | TTTCTTCCTA | ATCCTCCTCT  |
| 49021 | CCCTCCCTGG  | TCTTCTTGT   | TGTCCTTCT  | TAATGAACTA  | TCATGGCAAA | AAGAGCCCAT  |
| 49081 | TTCTGGTCAT  | TTTCTGTGGC  | CTTTCAAAC  | CCCACCTACC  | CCACTGCTCC | TGGGTGCATT  |
| 49141 | ACCCGAAAGC  | TGAGACTTCA  | GTGCAGAAAG | TGCCAGGGCC  | TCTGTCCCC  | CAGATCGCCT  |
| 49201 | TCCTTGTCTT  | CCCTGTGCTT  | GCCTGTCACA | TTGTGTGGGT  | TCCAGCGCTG | GAAGGAATGA  |
| 49261 | GGAACAGATT  | CTCTGGTTCT  | CCTTTTGAAG | TTTACCTTCG  | CTCCACCACT | TCTGAGACCT  |
| 49321 | TCCCGGAAGT  | TGCCCTTGT   | TTCTCTCCTC | TCCAGGGCTG  | CCCCAGAGCT | GCCTCTCACC  |
| 49381 | TCTTCTCTCT  | GTCAACCCAC  | CACCATCAGG | GCAGAAGTTG  | GGACAAAGCC | TCTCCTACTG  |
| 49441 | GCTCTGCTT   | TTCTCCCTTA  | GGTCCAGCCT | CCTCTTCTCC  | ATCTTCAGGA | GTCTCCTTCT  |
| 49501 | CCACTCACAC  | GTCACTGACT  | CAGCACCTCG | CATCAGTCCA  | GAATATGACT | GCTTGTTCAA  |
| 49561 | GTGCCACCTT  | TCTCATGCAT  | TTTTTTCTAG | TGACAATCAC  | AGCCACCTCT | TGGGGCAGGA  |
| 49621 | GTGTCATCAT  | CCCCATGTTT  | CAAATGAAGA | ATTGCAGTTC  | AGAGAGGGCA | AGTACTGGC   |
| 49681 | CCAGCCTCAA  | CAGCTAGCCA  | GTGGACCCCA | CCAGGGCTTC  | TGACTCCAGT | CCGGGTTCCT  |
| 49741 | TTTCCACCCA  | AATCCATGGA  | GGGAGCTGAG | CCGAGAACAG  | GTGTCCTTCA | GGAAGACGTG  |
| 49801 | AAGCCAAAGC  | CTCCACCTCC  | AAACTCAGGG | GCCCAGGGAG  | TCCAGGCACC | CATCCACTCA  |
| 49861 | CAAGGCTGGA  | TATGGTGCAT  | TCCAGGAGAG | GGGTTGGGGG  | CGAGTGGCCT | CTCTGTGTAC  |
| 49921 | CCGTGGGGAT  | AGATGCGCAA  | GTGGCATCGC | CACATCTGTA  | GTCCTGGCTT | CATGGGTGAG  |
| 49981 | CTCCAGGTCC  | AACGAGAAGC  | CAAGCAGGGG | GCCCTTCAAG  | CTCAGCTTTG | GGCCCGGGTC  |
| 50041 | GGGGTACAGG  | GTAGAGCGGG  | CCTCCCCAGC | CCCTGCCATG  | AGGCCAAGGC | AGTGCATCGT  |
| 50101 | TGCGACCGTT  | CATTACAGAA  | CCAAAGCCTA | GGAGCTGGTT  | ATCATTCCGG | TTTACAGCTG  |
| 50161 | ATGGAAGAGC  | AGGTGCTTCC  | GAGAACCCAC | AGTGTCTTTT  | GGCCAGTGAC | CCAAGGGTGC  |
| 50221 | CTCTGAGAGG  | CCTCGCAGCA  | CCCGGAGGTG | CTGCTGAGGC  | AACGCCCTGA | CTGTAAGAAG  |
| 50281 | GACCATTTCT  | CCTCAGAGAG  | TGGCGTGAT  | GCTGCTGCGA  | CAGTCCCAAC | ATCCCTCCCG  |
| 50341 | ACTCTCACTC  | CCAACAGACT  | TCCCACTGTA | AAGCTGAACT  | CTCCAGCAAA | TCACCTCTCG  |
| 50401 | CCAGACTCTC  | TCCTCACTCT  | CTCTGGGTCC | ACTAGAGGTT  | CCTCAGCCTC | TCTTTGCCTT  |
| 50461 | GGTTTTTTCCA | GCTGTAAAT   | GGAGCAAGA  | GGGCCTATGT  | ACCCACAAAG | GTGTGGTTGG  |
| 50521 | AGCGACTCCT  | CCTACATTAG  | GGCCTCGAGT | GGGGCTTCAT  | GATTGGTTGG | TGGAGGTCTC  |
| 50581 | CAAAACCCAC  | CAGTGCCACC  | GAAGGCTGAG | ACTGCAGATG  | CAATGCCACA | GGTGTCTTTC  |
| 50641 | CTCAGCCTGG  | GCAGCTGAAC  | ATCATGTGTA | AAACGGGGAT  | AATAAGATAA | TAACAGCCCC  |
| 50701 | TTGCACCTAT  | GTGGCTGTGA  | GGATTAAACA | AGATAAATGT  | GTAACAGTGC | CTGGCTATAG  |
| 50761 | AAATATTTAC  | TCTTGTATT   | AAGGGAAGAA | TATGTGTGGC  | TAAAAAGGGA | TCGAAGATGT  |
| 50821 | AAAAGCCAAT  | CCCTCCCCCT  | CTAGCATATT | TAAGGGTAAT  | GTTGAGTTGG | TTTGTGGACC  |
| 50881 | ATTTGCTGCC  | TGTTAGAGCT  | GGAAGGTAGG | GACCCCTCT   | CAACAGCGAT | GCTACAAATT  |
| 50941 | ATACCCATTG  | GAGGTCAACC  | AAAAGACAAA | GCTTATTGGC  | TGGACATGGT | GGCTCACACC  |
| 51001 | TGTAATCCTA  | GCACTTTGGG  | AGGCCAAGGC | AGGCGGATCA  | CTTGAGATCA | GGAGTTCGAG  |
| 51061 | ACCAGCCTGG  | CCAACATGGT  | GAAACCCCAT | CCCTACTAAA  | AATACAAAAA | TTAGCTGGGC  |
| 51121 | GTGGTGGTGC  | ACACCTGTAA  | TCCCAGCTAC | TCAGGAGGCT  | GAGGCAGGAG | AATCACTAGA  |
| 51181 | ACCCAGGAGG  | TGAAGGTTGC  | AGTGAGCCGA | GATCGCACCA  | CTGTACTCAA | ACCGAGGCCA  |
| 51241 | CAGAGGGAGA  | CGCAATCTCA  | AAAAAAAAGA | AAAAAAGACA  | AAGCTTGTTA | ATACCAGCAT  |
| 51301 | ATTGTTAAGG  | GAATAAAGTA  | GGCTGCAGAA | CAACTGGTGT  | AATATGGTGC | CATGTAGGGA  |
| 51361 | AAATTACATG  | TGTGCATAGG  | AGAGGGGTCT | GCAAGGTTGT  | GCCCTAAGAT | GTTAGAGTGG  |
| 51421 | TTCTTTTGCT  | TTTCTCTTTT  | ATAATTTTGT | ATTTGACTTT  | TAAATAAGGA | CCATAAATCA  |
| 51481 | CTTTTATAAA  | ATACATTCTC  | TCCAGCCCCC | ACTACTCTCT  | TAAAGAATAA | GAGTGGTTTG  |
| 51541 | CCCAAGAAAG  | ACAGTTTTTT  | TTGCTCTGGT | TTTCTTGATT  | CTGACATCAG | AGGAAACTGC  |
| 51601 | TTCTCATCCA  | CTTGGGGCTC  | TGGGTTACAG | GGATTCAATT  | CAGGCAGATT | AAAGTGGTGA  |
| 51661 | CCAGGGGCAT  | TCGTGGACAC  | AGGGAGGGAC | AGGAGCACCA  | TCAGTTTGTC | TCACACAACC  |
| 51721 | ACTGTCACTC  | TCACTGAAGG  | CTGTTCCTTG | ATCAAAAAA   | GTATTGGGCC | AGGCACGGTG  |
| 51781 | GCTCACACCT  | GTAATACCAC  | CACCTTGGGA | GGCTGAGGTG  | AGTGGATCAC | TTGAGGTCAG  |
| 51841 | GAGTTCGAGA  | TCAACCTGGC  | CAACATGGTG | AAACCTTGTC  | TCTACTAAAA | GTTCAAAAAAT |
| 51901 | TAGCCAGGCG  | TGGTGGGTGC  | CTGTAGTCCC | AGCTACTTGG  | GAGGCTGAGG | CAGGAGAATT  |
| 51961 | GCTTGAACCC  | GAGAGGTAGA  | GGTTCAGTGC | AGCCGAGATG  | GCACCACCAC | ACTCCAGCCT  |
| 52021 | GGGCGACCGA  | GGGGGACTCT  | GTCTTAAAAA | AAAAAATAAT  | AAATATATAT | ATATATATAT  |
| 52081 | ATATATGTCA  | AAAATGGGGT  | AGTTTTTAGA | TCTATAGTAG  | TTCTAAAAAC | AAAGGCCATC  |

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SEQ ID NO Sequence

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| 52141 | CAAGCATGAC  | AGATTTACAA  | GCACTATTGG  | CTATTCAGT  | AGTTACAATG  | GAGGAGAGAA |
| 52201 | GCTTTTAGTT  | AAAACAACA   | AACAACACAA  | CAAACCCAGA | AACCTTAGGT  | CAAAACCAAA |
| 52261 | ATTGTCCTCT  | CAGACACAAT  | CTGGGAATTT  | TCTCATGACA | GTGGGCATTA  | GCCAACTGAC |
| 52321 | ATCAGCAGCA  | ACCATCCGTG  | TGCACACAGT  | GGCACCACCT | CCTCCCAAAA  | AGCAGCCTTC |
| 52381 | ATCTATGCC   | TCATACAATC  | GTTGATTATT  | CTCTTTGGAT | TGAGGCCCGG  | AATTATTTAA |
| 52441 | GTTTCTTCTT  | GCCAGCATGA  | GTCTTTCCTT  | TCTGTATGCT | CCTTATCTTC  | TCTCTTTAAT |
| 52501 | TTGGCAGTTC  | TGCTTGAAAT  | CTGGGTCTTT  | CATTAGTAGT | AGTTCAATTT  | GGTTCAGAA  |
| 52561 | CATTGCTGG   | TGTGATGCAA  | TGTGACCAGA  | GCTCACACTT | CAGAGCTCTT  | CAAGGGCCAG |
| 52621 | TCTTACTGAG  | CACCTCCAG   | TGGCTGCCTG  | TGTGCTGGGC | GCCACTTGTG  | GTGGGCAGGA |
| 52681 | GAGAGGAGGG  | GACACAAAAG  | GAGACACAGC  | TCCTTCTTAG | AAGCTCAAA   | TTGGGGACCA |
| 52741 | GCTGCCACAG  | AAGAGTATGT  | TTAGCATCTG  | AGACACCAAG | ATCCAGCGTC  | ACAAGGGTGT |
| 52801 | TTATTAAAGCC | TCCTCATCTC  | TTTCTTTTTC  | TTTTTTTTTT | TTTTTTTCTT  | CAGGCAGTCT |
| 52861 | TACTCTGTCA  | CCCAGGCTGG  | AGTGACAGTG  | CATGATCTCG | GCTCACTGCA  | TGCAACCACC |
| 52921 | ACCTCCCGGG  | TTTAAGCAAT  | TCTCTGCCT   | CAGCCTCCCC | AGTAGCTGGG  | ATTACAGGTG |
| 52981 | CCCACCACCA  | CACCCAGCTA  | ATTTTGTGT   | TTTTAGTAGA | GACAGGGTTT  | CACCATGTTG |
| 53041 | GTCAAGGCTGG | TCTCGAACTC  | CTGACCTCAG  | ATGATTACCC | CACCTCGGCC  | TCCCAGTGTG |
| 53101 | CTGGGATTAC  | AGGTGTGAGC  | CACCGCGCCT  | GGCCTTGCTG | TTGATTATC   | TATAGTATGT |
| 53161 | TTGACTTGAT  | GACCTCCAGT  | TACCTTAGAC  | AGAGGTTCTC | ATCTAAGCTC  | CAACTTTCCA |
| 53221 | TTTCCTTTGT  | CCTCGTCTTT  | CCCTTAAACC  | CCTCCACATT | TCTCTCAAAA  | TCACCCCACT |
| 53281 | TCTAAAAAAT  | ACTGTTTATT  | TTTCTTTTAA  | ATTTCAAATT | ATCTATACT   | ATGAAATAA  |
| 53341 | ATCAAAATAG  | CATGGAATAA  | GCGAAAAAAA  | TGGATCCCAC | CCTTCCCCAC  | TCCCATTCCC |
| 53401 | TAGGGCTAAC  | CATAGTTAAC  | CATTTAATGA  | CTAGGTTTTT | TTGTTGTTGT  | TATTTTTTAT |
| 53461 | TTATTTTATT  | TGAGACAGAG  | TCTTACTCTG  | TCACCCAGGC | TGGAGTGCAG  | TGGTGTGATC |
| 53521 | TCGGCTCACT  | GCAACCTCTG  | CCTCCAGGT   | TCAAGCATTC | TCCTGCCTCT  | GCCTCCTGAG |
| 53581 | TAGCTGGGAT  | TACAGGTGCC  | TGCCACCACA  | CCTGGCTAAT | TTTTGTACTT  | TTGGTAGAGA |
| 53641 | CAGGGTTTCT  | CAATGTTAGC  | CAGGCTGGTC  | TCGAACCTCT | GGCCTCAAGT  | GATCTGCCCA |
| 53701 | CCTTGGCCTT  | CCAAAATACT  | GGGATTAAGG  | TATGAGCCAC | CGCACCAGC   | CCTCTGGGC  |
| 53761 | TCTTTTCTTT  | TAGTTGCACT  | CGCTCCCCGC  | TCCTGGAGTA | GAGGGATTTC  | CGAGAGACTG |
| 53821 | TGGGCTCCAG  | CCTTCACTA   | GGCCAGGAC   | TAGGATGCCT | GCCCTAACAT  | TTATCTTTAT |
| 53881 | ACCTTAAAGC  | AAAACAGCTG  | GACCATAAGC  | ATTCAAGAAC | AAACTGTGAA  | TAAGGAGAAA |
| 53941 | GTTCTCCAG   | GAAACAAGAG  | CTTTAGTTCT  | GTTGGGCCAG | CCCTTATATT  | CCTTAGCTGT |
| 54001 | TACCACTCAC  | TGCTTGATTT  | AATCTCGGCT  | ATCACTTGCC | CTGACAGGTC  | TGCTGCTGGT |
| 54061 | GCCAGGATGT  | CTGGGTTTTG  | AAGCCTGGCT  | CCATTACATA | CTTCTGTGT   | GACCTTGGGC |
| 54121 | AACTTACTCA  | ACCTGTCTGT  | TCCTCAGTTT  | CCCCAGCTGT | ATTATGTCAG  | CATAATAGTT |
| 54181 | TGTTTGTGTA  | ATTAATGAG   | GTAATAACTG  | GAAATGTTC  | AAACATGGTT  | CCTATCATGA |
| 54241 | GAAATCTGTC  | TTTCCGCCTA  | AATGTGCTGG  | AAAATTCCTG | GTGGTGCAGA  | ACAGGAGACC |
| 54301 | AGAGCAAAGG  | AAAGACAGGG  | TGCAGAAGCC  | AAAAATTACC | TTGGAGAAAC  | AAGCGCATGT |
| 54361 | TAGGCTTCT   | TTTGGATTCT  | AGGTTTATCT  | CTGCTTGGTC | TTCACTTACC  | TGCAAGAGAT |
| 54421 | CCATTTAGGG  | GATTTTGTGT  | TGTTTTTAAC  | GATAGCTTTA | TTGAGATATA  | ATTATATATG |
| 54481 | CATAAAGTCT  | ACTCTTTTAA  | AATGTTTCCG  | GTATATTAC  | AAGGCTGTGC  | AGCCTTCCCT |
| 54541 | GTCCTTGTAT  | CCAGTCTGAG  | TTTTTAACTG  | AAGGGATAAG | GAGGACCAAC  | CTTCCCCAG  |
| 54601 | ACCAGAACCG  | CGGGCCAGGG  | GGCGATTCTG  | CTGAGTCACC | GCGGGCGCCT  | GGTGCAGGCG |
| 54661 | GGCGGAGCCC  | GGGACCTTCC  | TTGGCTGCCC  | CCTAGCGAGG | GCCGCAGCGC  | AGCCTGAGAC |
| 54721 | ACCCGCGGGG  | GCCGCTCCAC  | GGCCGTCGGA  | TTTAGACTGG | AAGCTCGGTC  | CAGGTCCCCA |
| 54781 | GCTTGATGCG  | CCCGCGGTGT  | AGGAGACCAG  | CCCGACTCGG | GCTTCCCCCT  | AGCCCTTGGA |
| 54841 | CTCTTGATCT  | CAGCAGGGCC  | TGGGTAATGA  | ACGTGAGTCT | CCCTTTCCCA  | AAGGGGTTGC |
| 54901 | TTCTGTTGGA  | AGGCACCCGT  | TTGATACAGT  | AGCATAGAGA | TGGGTTTATG  | CATCAAAATA |
| 54961 | TCAGAATTCA  | AGCCTTGCTC  | TCTGCTTACT  | AGCTGTGTGA | CCCTAAAAAG  | GTTTCTGAAC |
| 55021 | GTCTCTGAGC  | TTCACTTTCC  | TCATCATTC   | TTCTCACGGG | GTGGTTGTGA  | GCATTACAGA |
| 55081 | GATCCTCTCT  | GTGAAGCCCC  | TGTGAGTGGC  | TCATCCTGAG | GGCTGAAATA  | AACATGTTAT |
| 55141 | TAATAATCCA  | AAACTGGCAA  | GGGATGTTGA  | CTGGTCCCCC | TCCCTTGCCC  | AAGGAGCTTT |
| 55201 | CTAGAACCTG  | AGTTATCATT  | ACCAAACCTG  | ACTGCCTTGA | GTAAGAAAGT  | TAGAAGGAAT |
| 55261 | GAGGAAGGATG | GTGGCAGGTG  | GAGGAAGGCG  | GATTGGTCAT | CACCTCCTTG  | CAGCAAGAAA |
| 55321 | CAGCCCCAGA  | TCGTGGGAAA  | CCTACAGACC  | TGCTAGACAG | ACTAGGAGCA  | AAAGCTGGGG |
| 55381 | CTTTAAGAAAT | CCCCAGGGAG  | GTCTCTCTGA  | GAGAGTAGCC | AGTTGGATTT  | TGTAAGCAGA |
| 55441 | GATTTGTGTT  | GGGAGGAGGT  | GACAACGTAG  | GGAGCAGAGG | GGCAAAGCTG  | TCGGGAATCC |
| 55501 | TGCCCTGAGG  | GCAGGGATGT  | GTGTTGGGGG  | GAGTTGGGTC | ACTGGGGCTC  | GGTGGCCTTG |
| 55561 | GGCAAGTTTC  | TACCTCTCAG  | GTCCCTTACC  | CACCTAGGGT | CGCCATCCTG  | CCCACCTCAC |
| 55621 | AGGTTACAGT  | GAGCCTGGAT  | GCACTGTCAT  | GGGCAGGTGC | CCAGGAAAAAT | GGCAGACATG |
| 55681 | TTCCAAACAG  | CACGCAGCAT  | TCCCCAGTGA  | TGCCCAGGGT | CACCTTGGAG  | GTGGGCGAGA |
| 55741 | TGCCCTGGGGT | TTCTCGTCCA  | CCCCACAACA  | CCTCAGGGGA | CAGCCAAAGC  | TGTCCCTTCA |
| 55801 | GGTAAGCTGC  | ACAGAAGATG  | TGAACCTGTC  | TGCAAGAGCT | CTATTCTTTT  | GGAGCAAAAG |
| 55861 | GGACCCAGGG  | TCTCACCTGC  | ACATCCCTGT  | CCCTGAGGGC | CTAGGGGTTT  | TTGAGGGCCC |
| 55921 | CAGCCTTGGC  | AAAATGAGGA  | AGAAGGTGAA  | GGTTGTCTGG | GCCCTTGCCA  | GGCTCTCTCC |
| 55981 | TCGGCCACGC  | ACTCCCCCTC  | CTGCACACAC  | ACCCCTTCTC | CTCCACCCCA  | TCTCCATTGT |
| 56041 | TGTCAGAAAA  | GTCACAATAA  | AAAGGTCCAT  | ATTGTCTAGT | TCCCATACTT  | TTAATTTTTA |
| 56101 | AAATTTTATT  | TATTTATTTA  | TTTATGTATT  | TTTTGAGACA | GAGTCTTAAC  | CCAGGCTGGA |
| 56161 | GTTCAGTGGC  | ATGATCTAGG  | CTCACTGCAA  | CCTCTCCCTC | CTGGGTTCAA  | GTGATTCTCA |
| 56221 | TGCCCTCAGC  | TCCCAGTAG   | CTGAGATTAC  | AGATATGTGC | CACATAGCCC  | AGCTAATTTT |
| 56281 | TGTATTTTTA  | GTAGAGACGG  | GGTTTACCCA  | TGTTGGCCAG | GCTGGTCTCG  | AACTCCTGGC |
| 56341 | CTCAAGTGAT  | CTGCCTGCCT  | GAGCCTCCGG  | AAGTGCTGGG | ATTTCAGGTG  | TGAGCCACCG |
| 56401 | CACCTCGGCTC | CACACTTTTC  | ACTTATTAATA | AGACTGTGGT | GTCCATCAAT  | GGATGAATGA |
| 56461 | ATAAACCAAT  | GTGAGACTATC | CCTCCCATTA  | CCCAAGGAAT | GAAGCACGGA  | GCCGTGCCAA |
| 56521 | GATCTGGATT  | CACAGTGAAG  | GAAGCCAGTC  | ACCAAAAGCC | ACGTGCTGTG  | TGACTTCCCT |

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SEQ ID NO Sequence

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| 56581 | TATACGAAAT  | ATCCAGAAGA | GATACATCCA  | TGGTGACAGA  | AAGTAGATGA  | GCAGCTGGGG |
| 56641 | ACTGGCGAAG  | GGGAGAAGGG | GGAGCAGCTG  | TCTATGAGGT  | CCAGCCTTTC  | TTCTGGGTTT |
| 56701 | GGTGAGAATG  | TTTTGGAAC  | AGATAGAGGT  | GATAGTTGTA  | CAACATTGTG  | AATGTACTAA |
| 56761 | ATGCCACTGA  | ATCATTCAAT | TTAAATCGTT  | CTTTACGTTG  | CATGAATTTT  | AAGTCAATCA |
| 56821 | AAAACAGTTG  | TTTGAAAAGA | GAAAAGCCTA  | TGGGTAGCGG  | CAGCAGTGAT  | TGGATTATG  |
| 56881 | ATTCGATTCC  | ATGGCTCATC | CCTCCCTGC   | CTCACCCCTC  | CGCCCTCCGA  | CGTCTTCTTC |
| 56941 | TTTTACTCTG  | AACTGTTATC | TTGTGTTCTA  | TCTCTCTCTC  | TCTCTCTCAA  | CCCTGCAGAC |
| 57001 | AGTTTTCCCT  | TTCTTTGTCT | GCCCCCACC   | TCCAGATTTC  | CGTGTCTCCA  | GTGTCTCCCT |
| 57061 | ACGAGGCATG  | AATTGAGACT | GGGAGGGTGT  | GATTCTGAAG  | AAGGCACCAA  | CAGTGACTCA |
| 57121 | GCTAGCCCTT  | TCCCCCACC  | CGCCCCCCGG  | GCCTCAATTT  | AGCTAAAAAA  | CCACAGGGAC |
| 57181 | GGACTCGAGA  | GGCAATACCT | TTCCAAGGTT  | CCCTAAAAAA  | TGTCCCATTT  | TAGTGTCCAG |
| 57241 | GTTTCACTCA  | ACTTTAGTGC | CTCCCCTAAA  | ATGTGTTTCT  | TACCTCCAC   | CCCACTGCAT |
| 57301 | CTAAGTCACT  | GCCTGAGAAA | ACAGGATTGA  | GGAAAGGAGA  | AAGGAAGAGA  | GAGAGAGAGG |
| 57361 | AGGAGAGAGA  | GAGAGAGGGA | GGAAAGGCTGA | TGGATTTAGA  | AAAGAAGAAA  | ACAAGTGGTC |
| 57421 | TGAGGAAAAC  | AGCCTTGGTG | TGTTTTATTTT | CCTGTCTGTG  | TATCGCTTCT  | CGGCCTTTTG |
| 57481 | GCTAAGATCA  | AGTGATTTTT | CCTGTCTGTG  | TGTCTCGCTT  | AGATTACAGG  | GATCTGTGGG |
| 57541 | TCATTGACACG | TCTGGTCCAG | GCTGCGTAGT  | CACCTCAAGG  | GCATGCTTAT  | TGATGTGTTT |
| 57601 | TTCAATTAC   | TATCTTTGCA | TGGGAGTCCC  | AGGCCAAGAG  | GCACAGCTGC  | GCCATTTGTC |
| 57661 | TGTTGGTTTA  | GATATCCTTT | ATCCAGTTCT  | TCCAGAGAAA  | TCATCTCGCC  | CTTCTGGAGG |
| 57721 | AGGTGGGCGA  | CAGGGGTCAG | AGATGGGAGG  | GAAAGGAAGG  | AGCCAGGTCC  | TTGGCTAGGA |
| 57781 | TGCCAGGGTC  | CCCTGCCTCT | CACCTGGCCT  | GGGCTGGAGG  | CCTCCTGCTG  | TCCTGTCACT |
| 57841 | GATCACTACC  | CCGCCCCAGC | CTCCTGAGTT  | AGAAGACACA  | GGCTAAAGTA  | GAGTATTTCT |
| 57901 | TCATTGAAAA  | ACCCATACAA | AATAAAGGTT  | CATAAAAAAT  | AAAAATTAG   | ACTGGGTGCT |
| 57961 | GTGGCTCACA  | CCTGTGATCC | CAGCACTTTG  | GGAGGCCAAG  | GCAGGTGGAT  | CGCTTGAGCC |
| 58021 | CTGGGGTTCA  | TGACCAGCCT | GGGCAACATA  | GTGAAACCCC  | ATCTCTACAA  | AAAATACAAA |
| 58081 | AAATTAGCCA  | GCATGGTGG  | TGCATACCTG  | TGGTCCAGC   | TTCTCAGCCT  | ATGGACCCAC |
| 58141 | ATAGAATACA  | ATGTCAGCAT | AAGAAGGGAG  | CCCTGGGGTC  | ACCAAAATGGT | TTGGGCGGCA |
| 58201 | AAGAACCCTGA | AGGTGTAGAG | AAGTGGCTTG  | GTTACCCAGC  | TGTTGGATGT  | GAGACCTGGC |
| 58261 | CACTGCTTCT  | TCCATACCCT | AGACCTGCAC  | CCTGACATCT  | CAAGTAAAAA  | GTGGGGGAT  |
| 58321 | GTTTTATGGT  | CCAGGATGAA | GGAAGGGCAG  | TGAGGGGCGA  | CGGAGCATCA  | CTTTGCATTT |
| 58381 | CTGTCGCTCT  | CTTACTGGCT | GTGTGACCTG  | GGGCAGGTAA  | CTTCCCAGAC  | TCCTGGGAAT |
| 58441 | CATAACACCT  | ATGATGATGA | TGATGATGAT  | GATGATGATG  | ACACCTACCT  | CAAGGATTGC |
| 58501 | CCTGAAGGGT  | CACAGAGATG | CCTGCAAGGC  | ACCTGCATGG  | AGCAAGCGCC  | CCTTCTCTGG |
| 58561 | CAGGTGCTGG  | GTGAGCACTA | CCTGCTGCCA  | GGCCCTGGGG  | CTATGGCACT  | GCCTGACCTT |
| 58621 | GCAAGTCCTA  | CCTGGCGAAG | CTGTCTGTTT  | TGTGCTCAGT  | CAGTGTGGT   | TGTAAGACTG |
| 58681 | AGAAGAGTCA  | CTTCAATTTG | CTCTCCAGGG  | ACATCTTTCT  | GGGTCTCTAT  | TTCTGCCTAT |
| 58741 | GTCAAGTAGC  | GCCTCAAGGA | TGCTCTTGAA  | AATGGGCTTG  | TCTTTCTTAA  | CATGGCAGGT |
| 58801 | AGTCCCAAA   | GCATTAGCAT | GGGGCAGCTG  | ACCTAGCCCCA | GCCAAATGCAG | TGCAGTGACT |
| 58861 | CTTGCAACCG  | AGTCTAATCA | GAAGGTCCAT  | GAACCTACGA  | GCATTTCTCT  | TCCCAGGATC |
| 58921 | AGGGTGGAGG  | CTGAGCCTCC | CTGCTTAGAG  | ATTCTTCCCA  | TGCATTCCAC  | TTTTTTCCCC |
| 58981 | AAAAGAAAAAT | ATTGACCTTT | GAGAGGCACA  | CAGTTTATTT  | ATTTTGCATA  | GTAAATAGTA |
| 59041 | GCCTGTATTT  | TAAGGATGAG | TTGATTTCTG  | CATCAGCCCC  | TGTAGGTCAT  | CAGCCTTCTA |
| 59101 | TTGGTGCATC  | TGACTCTCTC | TAGCCCTGCA  | GGGATGGTGG  | AGGGGGAGGG  | GAAGGAGGGA |
| 59161 | TCTTTATTGG  | AAACCAGGAC | AGTGAGACTC  | ATTGCCCTGT  | CATCTGCTCT  | GTGGTGTGTA |
| 59221 | ATGAGGCAGC  | CCAACAGAGA | AATACCCTGA  | GCGAGCATCC  | CCAGCCTCCA  | AAACAGTGGC |
| 59281 | GCATTGCCCT  | GAGTCCCTGG | AATGACCTTT  | GATTCTCTCT  | CTCTGACTT   | GGAACCCATG |
| 59341 | GAAACCTCTA  | GAAGCAGCTG | AGGAAAACCC  | AACATGAAAA  | GCAGAATCC   | ACACTGAGAA |
| 59401 | TATAGGAGGT  | GATCGGAACA | TACAATGATT  | CTTGCTAAGA  | CCGATTACCA  | GTTTTTCTTT |
| 59461 | TTTTTCGATC  | GAAGAAATAC | TGGAGAAGCC  | TAAAGAAGGA  | GTCTAAAAAC  | TCTGGCACGT |
| 59521 | GGGCCAAAAAC | TGCTCTTGAG | CTAAGAATGA  | TTTTCACATT  | TTTAAAGTGGT | TGAAAAATGA |
| 59581 | AATAAAATAA  | GATGATGTTT | TGTGACACAT  | GAAAGCTATG  | GGAAATTCAA  | ATTCTAATAT |
| 59641 | CTATAAATAG  | TGTTTTATCA | GAACACAGTC  | ATGCTCATTT  | ATTTATGCTC  | GATGGCTGCT |
| 59701 | TTCCCGCTAC  | AATTACGTTG | AGCAGTTACA  | ACAGAGACCA  | CGTGGCCAC   | AAAGCCTTAC |
| 59761 | AATATTTACT  | ATCTGGCCCT | TTCCAGAAAA  | AAATGTGCCG  | ACTCTTGACC  | TTAACCTCAG |
| 59821 | CAATTTGGGA  | GGCCGAGGCA | GGCGGATCGC  | TTGAGCTCTG  | GAGTTTATGA  | CCAGCCTGGG |
| 59881 | CAACATAGTA  | AGACTCCATC | TCTACAAAAA  | ATACAAAAA   | TTAGCCAGGC  | ATGGTGGTGC |
| 59941 | ACACCTGTGG  | TCCTAGCCAC | TCGGGAGACT  | GAGGTGGGAG  | GATCGCCTGA  | GCCCAGGAAG |
| 60001 | TCGAGGCTGC  | AGTGAGCTGT | GATGGCACCA  | CTGCACCTCA  | GCCTGGGCGA  | CAGAGCAAGA |
| 60061 | CCTTGTCTTC  | AAATAAATAA | ATAATGCAAA  | GTAATAATAA  | TAAAAACATA  | TAAAAAGGAA |
| 60121 | TCAATTTAAA  | ATTATAATGA | AAGCTGGCCG  | GGCATGGTGG  | CTCACGCCCT  | TAATCCCAGC |
| 60181 | ACTTTGGGAG  | GCTGAGGTGG | GTGGATCAG   | AGGCCAGGAG  | ATCGAGACCA  | TCTTGGCTAA |
| 60241 | CACGTGAAA   | CCCCGTCTCT | ACTAAAAATA  | CAAAAAAATA  | ATTAGCCGGG  | CACAGTGGCG |
| 60301 | GGCGCTTGTA  | GTCCAGCTA  | CTCGGAGGCG  | TGAGGCAGGA  | GAATGTCTTG  | AACCCGGGAG |
| 60361 | GTGGAGCTTG  | CAGTGAGCCG | AGATCGTGCC  | ACTTGCAGTC  | CAGCCTGGGC  | GAAAGAGCGA |
| 60421 | GACTCCGTCT  | CAAAAAACAA | AACAAAAACA  | AAAAACAAAA  | AAAAATTATA  | TGAAAGCCAA |
| 60481 | GGGGCATAGT  | AGACAAATTT | TTCTAGAGCT  | CATTAAAGTC  | AATGAGTCAC  | CAGTTAGTAA |
| 60541 | AACGCAGTCA  | CGGGGAAGAG | AGGGCAGGAT  | TCTTTGAAGC  | AGCGGCTCTC  | CTAAAAACAA |
| 60601 | CCCACCCCTG  | TCCAGCTGCC | TTCCCTCTCT  | AGGGTGTTC   | CTTTGACTGT  | GTGACCCCCA |
| 60661 | TCCCTATTTT  | CCCAACCGTC | CAAGCCAC    | TCTAGCATAA  | TACGAGCTTT  | TAATCCCTCT |
| 60721 | CCCTGACCCC  | AACCCGATTT | TGAAGCCCAG  | TCTAGTATTT  | TCTCAAATAC  | ACTTCTTGGC |
| 60781 | TCCATTCTCT  | CCTTTCCATC | ACCTCTGCCT  | TTTCACTGCA  | TGCTTGGACC  | ACTGCAGTCA |
| 60841 | GCTCCCTATG  | AACAGTTGCT | CTCTACCCAT  | CCAATCGGCC  | CCGCTGCTGT  | CTGCCAAATT |
| 60901 | CACCGAGGCG  | ACCTCTGTGG | TGCTGCCTGT  | GGACAAAGTC  | CAAGCCAGCG  | ACCTCACCCA |
| 60961 | CCTACAGGTG  | AGTGGGGAGC | AGCCAGCGTG  | TCCAGTGGTT  | TACCCCATCG  | CCACAGACTT |

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SEQ ID NO Sequence

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| 61021 | GGTGATGTGT  | CGATGTGCAG  | AGAAGGGGTG  | TTGGCAGCCA  | CAACACAAGC | AACCCCGCCC  |
| 61081 | CATGTGAGAT  | CTAAGATGGG  | CGTGCTGGGA  | GCCACCTCTG  | AGAATCCAAC | AGAAGGCAGA  |
| 61141 | GGGGAGAACG  | GCTCACACGG  | CACAAACACT  | CCTTCCTTTT  | TTTTTTTTCT | TTTTCCTTTT  |
| 61201 | TGAAAGGAGT  | CTCACTCTAT  | TGCCCAGGCA  | GGAGTGCAGT  | GGTGCATCT  | CAGCTCACTG  |
| 61261 | CAACCTCCGC  | CTCCTAGGTT  | CAAGCGATTG  | TCCAGCCTCA  | GCTTCCCAAG | TAGCTGGGAT  |
| 61321 | TACAGGTACA  | CTCCACCATG  | CCCGGCTAAT  | TTTTGTGTTT  | TTAGTAGAGA | CGGGGTTTCC  |
| 61381 | CTATGTTGGC  | CAGGCTGGTC  | TTGAGCTCCT  | GACCTCAGGT  | GATCTGCCTG | CCTTGGCCTC  |
| 61441 | CCAAAGTGCT  | GGGATTACAG  | GTGTGAGCCA  | TGGGGCCTAG  | CCTCCTTCCA | TTTAAATGTA  |
| 61501 | TGCCATAATTT | GCCCAATTGAG | AACGGCTGAG  | ACGCATTTTA  | AGTGGCCAGG | GTCTACTTAG  |
| 61561 | AGTTAGTGCT  | CATGACCAGG  | CCCAGGTCAA  | GCCTGGCTGG  | CCAGATGGTG | CCTTTGACCT  |
| 61621 | GCTCTGTCTC  | TGTGCAAAAG  | AATGAGCTGA  | AGGATGGGGG  | TGCAGTGTGT | GGGCAGTGGG  |
| 61681 | CTGGGGCTGG  | CAGGACTCAG  | TGACTAAGGG  | AAGAGAACTT  | TCCTCACTAC | CAGCCTGTCT  |
| 61741 | TTTCAGGGCA  | CCGCGGGGGG  | CTTTGGGACT  | TGGTGATGAA  | CACAGCACAG | AGAGCTGTCC  |
| 61801 | AGCATGCGGG  | TCCCTGGCTT  | CTCACACTTC  | CCAGGCTCCT  | TCAGAGGCTC | TCCTCAAAGG  |
| 61861 | GAGCTGCTCT  | CTCTAGAACC  | CATGAATTTG  | GAATATAGGC  | AACCACTGCA | TTGGGGACCA  |
| 61921 | CTGACCTCAA  | ACATAGAGAC  | CAGAGCAAAT  | GGGGCTCATC  | ACGTGAAACT | CATCTGGAAC  |
| 61981 | TCTAGCATGCT | TCTTTTATAT  | ATATATATAT  | ATATATATAT  | ATATATATAT | ATATATATAT  |
| 62041 | ATATATATTT  | TTTATTATTA  | TACTTTAAGT  | TCTAGGGTAC  | ATGTGCACAA | CATGCAGGTT  |
| 62101 | TGTTACATAT  | GTATACATGT  | GCCATGTTGG  | TGTGCTGCAC  | CCATTAATTC | ATCATTTACA  |
| 62161 | TTAGGTATAT  | CTCCTAATGC  | TATCCCTCCC  | CACTCCCCCC  | ACCCACAAAC | AGGCCCCAGT  |
| 62221 | GTGTGATGTT  | CCCCTTCCTG  | TGTCCAAGTG  | TTCTCATTGT  | TCAATTCCCA | CCTACGAGTG  |
| 62281 | AGAACATGCT  | GTGTTTGGTT  | TTTTTGTCCT  | TGCGATAGTT  | TGCTGAGAA  | GATGGTTTCC  |
| 62341 | AGCTTCATCT  | ATGTCCTTAC  | AAAGGACATG  | AACATCATAT  | TTTTTATGGC | TGCATAGTAT  |
| 62401 | TCCATGGTGT  | ATATGTGCCA  | CATTTTCTTA  | ATCCAGTCTA  | TCATTGTTGG | ACATTGGGGT  |
| 62461 | TGGTTCCCAAG | TCTTTGCTAT  | TGTGAATAGT  | GCCGCAATAA  | ACATACGTGT | GCATGTGTCT  |
| 62521 | TTATAACAGC  | ATGATTTATA  | TTCCCTTGGT  | TATATACCCA  | GTAATGAGAT | GGCTGGGTCA  |
| 62581 | AATGGTATTT  | CTAGTTCTAG  | ATCCCTGAGG  | AATCGCCACA  | CTGTCTTCCA | CAATGGTTGA  |
| 62641 | ACTAGTTTAC  | AGTCCTACCA  | ACAGTGATAA  | AGTGTTCTTA  | TTTCTCCACA | TCCTCTCCAG  |
| 62701 | CAGCTGTTTG  | TTCTTGACTT  | TTTAATGATC  | GCCATTCTAA  | CTGGTGTGAG | ATGTTATCTC  |
| 62761 | ATGGTGGTTT  | TGATTTGCAT  | TTCTCTGATG  | GCCAGTGATG  | ATGAGCATTT | TTTACGTTGT  |
| 62821 | CTGTTGGCGA  | ACTCTAGCAG  | CTTCTTTTCA  | CAAGTTCATG  | GAGAGAGGTT | TCCCCTGAG   |
| 62881 | GGAATCACAT  | CTGTCTGATC  | AAAAGAGGCT  | TGGGAAATGG  | CTCTCTGTTT | CATTCCCTGA  |
| 62941 | AAACCTCTGA  | TGGAACCACT  | GCCACTGTGG  | CAGCCCCAGC  | ACTGGCACCC | CAGCCATGAT  |
| 63001 | TGGTGCCCA   | GCCACATCTC  | TGCTGTGAGC  | CCCAGAGCCC  | TGGTTAATTA | ATCATCCACG  |
| 63061 | TGTTGATGGG  | GAGAGGCCCA  | TTCACAAAAG  | CGACATAAAG  | CCCAGGGAGA | CGTGGCCGTG  |
| 63121 | GCAAGAAGGG  | TGTGGGACTA  | CATTCCGCCC  | CCAACCTGAGA | GATTCAAGAA | CCAGAAAAAA  |
| 63181 | ATGGAATAAC  | ATACTGTGCT  | CTTGGGTGGG  | AAAACATAAT  | ATCATGAAGG | GAGCAATTTT  |
| 63241 | TATAGTTTGT  | GCCTATAATA  | CAATTCCAGC  | CGAAATCCCA  | GTGGAACTTT | GAGAATTTGC  |
| 63301 | AGGAAAAAAA  | AAAATGTCTA  | AAGTACATCT  | GGAAGACAAA  | CTTACAAGAA | GGTCAAATAA  |
| 63361 | TTTTGAAAAA  | GAAATGATA   | TCTAAGCCCA  | CCTAGAGAAT  | AAGACTTGAG | ATCCAAAGCT  |
| 63421 | AAATCAGGAG  | GCTCTAGCAA  | AATTGACAGA  | TAAGCAGGAC  | AGAGTGATG  | GTGCATTCAC  |
| 63481 | CTGGGAAGA   | GGGCAGATTG  | GTCTACAAAT  | AGGCCTGGGT  | CCACTGACTT | TAGCTGTTAT  |
| 63541 | ATTTGGGGAG  | AAACTTTTCA  | ACCTCACTCC  | ATCTTAAACC  | TAAAAATATT | CCAGATGAAT  |
| 63601 | TAATAAATAT  | AAAAAATTAG  | ACCACTAAAA  | ATGTAGAAGA  | AAATGGATGA | TCTTTCTATA  |
| 63661 | CCATAGAGCA  | ATGGAATAAA  | TCACAAAGGA  | AAACAGATTT  | GACTATATAA | AACTTAAACC  |
| 63721 | CTGCCCATCA  | AAAACCATCA  | GAAACCAAAA  | TAAAGGCAA   | CCAACCTGGG | AAGATAGTTG  |
| 63781 | CCACAAATAT  | GATCAAGGGT  | TAAATGTTAT  | CATAAATTAA  | GAGCCACAC  | AAGTCATTAG  |
| 63841 | AATAAGCACT  | GAGACCTGAA  | CAGACAAGCA  | AAAAGAATGA  | GAGTGGGTCG | GCGCGCGGCG  |
| 63901 | TCATGCCTGT  | AATCCCAGCA  | CTTTGGAAAG  | CTGAAGCAGG  | CGGATCACTT | GATCCCAGGA  |
| 63961 | GTTCCAAACAC | CAGCCTGAGC  | AACATGGTGA  | AACCCTGCCT  | CTACAAAAGT | CATAAATATT  |
| 64021 | AGCCGGGTGT  | GATGGCACAC  | GCCTGTAGTC  | CCAGCTACTC  | AGGAGGCTGA | GGTGGGTGGA  |
| 64081 | TCACCTTGAGC | CCGGGAGGTA  | GAGTCTGCAG  | TGAGCCAAGA  | TCACACCGCT | GCACTCCAGC  |
| 64141 | TGGAGCAACA  | GAGTGAGACC  | CTGACTTAAA  | AGAAAAAATA  | AAAAAAGAG  | GAGAAAAATG  |
| 64201 | CTGATCTCAC  | TAGTAATTAA  | AACATCAGGC  | CAGGCGCAGT  | GGCTCACACC | TTTAATCCCA  |
| 64261 | GCACTCTGGG  | AGGCTGAGGC  | AGGCAGATCA  | CTTGAGATCA  | GGAGTTCTAG | ACCAGCTTGG  |
| 64321 | CCAACTAGGT  | GAAATCCCGT  | CTCTACAAAA  | AATACAAAAA  | TTCGCCAAGC | GTGGTGGCAC  |
| 64381 | ATGCCTGTGA  | TCCCAGCTAC  | TCGGGAGGCT  | GAGACAGGAG  | AATTGCTTGA | ACACGGGAGG  |
| 64441 | CAGAGGTTGC  | AGTAAGCTGA  | GATCGTACCA  | TTCCAGTCCA  | GCCTGGGCTA | CAGAGCGAGA  |
| 64501 | CTCTGTCCCA  | GAAAAAATTA  | AAACATCACA  | TATTTAAACA  | ACTCTAGGAT | ATCATTTAAA  |
| 64561 | AAAACATTAA  | TAGACTGTTT  | TTTAGAGCAC  | TTTTAGGTTT  | ACAGTGAAC  | TGAGTGAAG   |
| 64621 | GTACAGAGAC  | TTCCCGTAGT  | TTCCCTGCCC  | TCCACGTACA  | GCCTCCCCCA | CTGCCAACGT  |
| 64681 | CCTGCACCAG  | AGTGGTACAC  | TTGTTACAAC  | CAATGAATCC  | TCATTAACAT | ATCATTATCA  |
| 64741 | CCCAAGTTCA  | TAGTTTACAT  | TAGTAAAAACA | TCATCTTTCA  | TCTATAAGCA | CAAAAAATTTT |
| 64801 | TTGGCATTTA  | TTTAGGTGTA  | TGATTAACCT  | AGTGTGTACA  | AGACTCACAC | TTCATACCCA  |
| 64861 | CTTGCACTGC  | ATCTGAGAAG  | CAATTGGTGT  | CTACAGCCGC  | TACACCTCTA | ACAAGCCCGA  |
| 64921 | TCTTGTGTTGA | AAAGCAATTG  | GTGATGCTTC  | TCAAAATTCT  | ATGGACAAA  | TCAGCCGGGC  |
| 64981 | ATGGTGGCTC  | ATGCCTGTAA  | TCCCTAAACT  | TTGGGAGGCC  | GAGGCAGGCA | GATCACCTGA  |
| 65041 | GGTCTGGTGA  | AACCTGTCT   | CTACTAAAAA  | TGCAAAAAAT  | ACCCAGGCAT | GGTGGCTGGG  |
| 65101 | GCCTGTAATC  | CCAGCTACTC  | GGGAGGCTGA  | GGCAGGAGAA  | TCGCTTGAAG | CAAGGAGGCG  |
| 65161 | GAGGTTTCAG  | TGAGCCAAGA  | TTGCACCACT  | GCACTCCAGC  | CTGGGTGACA | AGAGTGAAAC  |
| 65221 | TCCATCTAAA  | AAAAAAAAT   | TATGGACAAA  | GTTTTTCAAA  | AAGATATTTA | ATGCAACTTT  |
| 65281 | ATTTGTAATA  | TTGGAACATC  | TGAGGCCATT  | TCAGTGTCAA  | CTATTAGGGG | ATGGTTAGGA  |
| 65341 | AAATATGGTA  | CATATGTGGA  | AAGGAACATT  | TGGTAGTTAG  | TGCCCTGAT  | GTTTACAAAG  |
| 65401 | GCTTTTAGTG  | ACCAACAAAT  | GCTCATGCTA  | TAATCTTATG  | TGAAAAAAGC | AAGTAGCATA  |

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SEQ ID NO Sequence

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| 65461 | ATTGCAACTA  | TATTTTAAAT  | GCATAGAATA  | AAAGGCTAGA  | AGGAAATATC | ACAGATCCTT |
| 65521 | GACATACATT  | CCCAAACCTT  | TGTAAATCCG  | CGGATTTCATG | AAAACAGACA | CATTTGCACA |
| 65581 | AGTGCCTGAT  | CTTTTCTGTT  | ATACATTTCAT | TAGAAGTCAA  | GCCCTGGTGC | CACAAAGTAT |
| 65641 | CTGCCTTTTC  | AAATGTGATC  | AGAATGTTCT  | CTTTTGCTTC  | AAGGCCATTT | TTCACGAAGC |
| 65701 | ACGTGGCCTG  | TTGCCCTCTC  | ATCAGAGTCA  | CCGTGTGCC   | TGGAGGACTG | AGAACAGCAG |
| 65761 | AGCCGTTTTA  | GGATGGGACA  | GGGCAGCCAG  | GAGGATTGGG  | CTCACTCCCT | ACTGAGTGCC |
| 65821 | TCACTCCCGT  | ACAGCCCCCA  | TAGAGGAAGA  | GGGGTTCAAA  | TTTATTCCTC | AGCCAGATGG |
| 65881 | GATGTGCCGC  | CTGTCTTGA   | ATTTCACATC  | ACTTATGATG  | GACCAAAATT | CCAAAAGCTG |
| 65941 | AATCCATGAT  | TGTCAAAGTC  | TGGTATGGCA  | GGATGTCAAC  | AGTAATCGTT | TCTGGGCAGA |
| 66001 | GGGATGATTT  | TCTCTTCCCA  | TCTTGCTTTG  | TATAAATACA  | TTTCTCTATA | TAAGGTTGTA |
| 66061 | TTACTTTTTCT | CATCAAGAAA  | TAGCAAAAGTA | CTGTTTTACT  | CAAAATATGA | ATAGAGCCAG |
| 66121 | GCATGGTGGC  | AGCTTATGCC  | TGTAATCCCA  | ACACTTTGAG  | AGGCGGATAT | GGGAGGATCA |
| 66181 | CTTTAGCCCA  | GGAGTTTGAG  | ACCAGCCTGG  | GCAACATAGT  | GAGACCCCTG | TCCCCACTCC |
| 66241 | CCCAAAGAAA  | ACCCACAAAG  | CATTTATCCT  | GGATTATTCA  | CAGGGGCCAA | AAAAAAAAAA |
| 66301 | AAAAATTCAG  | GCCTCCTATA  | GCCATGAGCT  | ACGAATATGA  | AAATATGCAA | ATGTGTAAGA |
| 66361 | AAAGCCAGCA  | CATCCGATTT  | TTACTTTTAC  | TTTCACACCT  | CTGTCCACCA | TGTTCCAAGA |
| 66421 | GAGGAAACTT  | GGTCATTGAA  | AGGAATAGAT  | CAAATCCAAA  | GAACAAAACC | ACTGTGCTCA |
| 66481 | TTAAACTTCT  | TAGTGTTCAC  | AAAGCTTTAG  | CTGCAGGTTG  | AATGGGGCAA | CCCGAATTGG |
| 66541 | CTGGCTCACC  | TGGGCTGCAG  | GGAGCAGAGA  | TCGCACACT   | GCATCCAGC  | CTGGGCAACA |
| 66601 | AAGCGAGACT  | CTATCTCAA   | AAAAAAAAAG  | TTCAATAATT  | CAAGTTATG  | AATTATTTTT |
| 66661 | AAAATAATAA  | TAATTTACAA  | TAAAGATGAG  | GACAAAGTGT  | GAGTAAATGG | TGGTTTCTAT |
| 66721 | CCAGCTCTGT  | TGAGCTGAAG  | TGGCATCTCC  | CTGCTGGGGC  | TTTGGGGGAA | GAAGGGTGTG |
| 66781 | TGTTGCTCTT  | CAGATCCCAA  | GCCTCATGCC  | CCTACTGGGC  | CCTGTGGGGT | GCTTCTCAGC |
| 66841 | CCACCAGGAG  | AGCCACCCTG  | GGAACACACA  | CGTGGGGGAC  | CTGTTGGGGT | CCGGTGTGGT |
| 66901 | GAATGGGGGC  | CACAGCCTGA  | CTCCAGGAAG  | CCAGCAAAC   | CGGAGCTGGA | GGAGTCAGGA |
| 66961 | CACCCCCGAT  | GAGTCAAGAG  | TTGGTTTTCG  | TGCCAGTTGA  | CATCTGATTG | AACCATCTCT |
| 67021 | TCACTTCTCC  | GTGCCCTCACT | TTCCCTTACCA | GACAGGCTCT  | GCTGATGCTG | TCCCTCTCCT |
| 67081 | GTTTCAGTCG  | GCCCTCACCG  | TAAAGAGAGAA | AGAGCAAAC   | GCTGGGGCAG | AGCATTGATT |
| 67141 | TTTTTAATGA  | AGTGGAAAGA  | GAGCTGGGAA  | TAAACAAGTCG | GGCCCACTC  | ACCTGCCTCA |
| 67201 | CCTGTGGGGT  | TTATTTGTTT  | TGTTTTTTTT  | TTTTTGTTTT  | GAGACAGAGT | TTCACCTGT  |
| 67261 | CACCAGGCT   | GGAGTGCACT  | GGTGTAATCT  | CAGCTCACTG  | CAACCTCCAC | CTGCCAGGTT |
| 67321 | CAATTGATTC  | TCCTGCCTCA  | GCCTCCCCAG  | TAGCTGGGAT  | TACAGGCACC | TGCCACATGC |
| 67381 | CTGGCTAATT  | ATTGTATTTT  | TAGTAGAGAT  | GGGGTTTTAC  | CATGTTGGCC | AGGCTGGTCT |
| 67441 | CGATCCCCCT  | ACCTCAGGTG  | ATCCACCCAC  | CTCGGCCTCC  | CAAAGTGCTG | AGATCACAGG |
| 67501 | CGTGAGCCAC  | CATGCCCTGG  | CGTCACTGG   | TGGTGTGAA   | TATGAAGTGC | TGCGGTGTTG |
| 67561 | GTAATTAAG   | CAAGCAGATA  | GATGTAATA   | ACGCTTGGGC  | AGGAATATGG | AGCACGGGAT |
| 67621 | GAGGATGGGC  | GGCCAACTGT  | TAGAGAGGGT  | AGCAGGGAGG  | CTGAGATCTG | CCTGCCATGA |
| 67681 | ACTGGGAGGA  | GAGGCTCCTC  | TCTCTCTTCA  | CCCCCACTCT  | GCCCCCAAC  | ACTCCTCAGA |
| 67741 | ACTTATCCTC  | TCTCTTCTT   | TCCCCAGGTG  | AACCTTGAAC  | CAGGATGGCT | GAGCCCCGCC |
| 67801 | AGGAGTTCGA  | AGTGATGGAA  | GATCACGCTG  | GGACGTACGG  | GTTGGGGGAC | AGGAAAGATC |
| 67861 | AGGGGGGCTA  | CACCATGCAC  | CAAGACCAAG  | AGGGTGACAC  | GGACGCTGGC | CTGAAAGGTT |
| 67921 | AGTGGACAGC  | CATGCACAGC  | AGGCCCAGAT  | CAGTCAAGC   | CAAGGGGTGG | CGGGAACAGT |
| 67981 | TTGCATCCAG  | AAATGCAAG   | AAATTTTAAA  | TACATTATTG  | TCTTAGACTG | TCAGTAAAGT |
| 68041 | AAAGCCTCAT  | TAATTTGAGT  | GGGCCAAGAT  | AACTCAAGCA  | GTGAGATAAT | GGCCAGACAC |
| 68101 | GGTGGCTCAC  | GCCTGTAATC  | CCAGCACTTT  | GGAAGGCCCA  | GGCAGGAGGA | TCCCTTGAGG |
| 68161 | CCAGGAATTT  | GAGACCGGCC  | TGGGCAACAT  | AGCAAGACCC  | CGTCTCTAAA | ATAATTTAAA |
| 68221 | AATTAGCCAG  | GTGTTGTGGT  | GCATGCTAT   | AGTCCTAGCT  | ACTCAGGATG | CTGAGGCAGA |
| 68281 | AGGATCACTT  | GAGCCAGGGA  | GTTCAAGGTT  | GCAGTAAGCT  | GTGATTATAA | AACTGCACTC |
| 68341 | CAGCCTGAGC  | AACAGAGCAA  | GACCTGTGTA  | AAAAAAAAG   | AAAAAGAAAA | AGAAAGAAAA |
| 68401 | AAATTTACTT  | TGAGTTACCC  | ACATGAGTGA  | ATGTAGGGAC  | AGAGATTTTA | GGCCCTTAAC |
| 68461 | AATCTCTCAA  | ATACAGGGTA  | CTTTTTGAGG  | CATTAGCCAC  | ACCTGTTAGC | TTATAAATCA |
| 68521 | GTGGTATTGA  | TTAGCATGTA  | AAATATGTGA  | CTTTAAACAT  | TGCTTTTTAT | CTCTTACTTA |
| 68581 | GATCAGGCCT  | GAGTGGCCTC  | TCTTTAGCAA  | GAGTTGGTTA  | GCCCTGGGAT | TCTTACTGTA |
| 68641 | GCCACATTAA  | TAAACAACAT  | CGACTTCTAA  | ACATTCTATA  | ATACCATCTT | TTGGCCAAAT |
| 68701 | TGACTTCGCC  | TCTTCCTCTC  | TCTTTCCAAA  | TGAAATGTGT  | TTCATTTTAC | TGTACAGACA |
| 68761 | CATGGTTGGG  | GACCCACAG   | AGCACACAGC  | CCTCCCTCTG  | CCTTCCCATG | CTGGCCCTTC |
| 68821 | ACCCACTGCT  | GGAGTGCCAG  | GTTGGTCCAA  | GGGTGGGACC  | AAGTTGTCTG | AGGTTGTCTC |
| 68881 | AAGGTTGGTC  | GAGGCTGTCT  | CCGCGCTGGG  | TTGTGCTACA  | AGGAGCCCTT | CTTTCATGG  |
| 68941 | GTGTGGCTGG  | CAGTGAGTGC  | TCACAGCAAC  | AGCCACACAGT | GCAGCCGAG  | GGCAGGATGG |
| 69001 | ACTCAGTCCC  | TGCCCTCCATA | CCCATTTCTA  | AGGAGGCCAA  | ATGGCAAACA | CTCTACTTTT |
| 69061 | CTCTTTTAAAT | GCTAAAAATA  | AGAAAAACCC  | TTGCAGCCCA  | GGGTATGGGT | AGTGCATGGA |
| 69121 | TGCCGTGGAG  | TTGTGAGGTG  | GGAAGTGACC  | TCTGTGGAT   | ATGCTTATTC | AGGAAGATTG |
| 69181 | CTGGAGTGGG  | TGGGGTCTCT  | GGGAGGTCCC  | CTGAGTGTGG  | GAAGCTGGGA | CCACCAGCTT |
| 69241 | TCTCGCACAG  | GGAGTGGCCA  | TCCCAGCTTG  | GAGAGGTTCC  | AGGACTGGTT | GGGAGGCACG |
| 69301 | TTTCAGATTT  | CTATCTGTTG  | AATCAGCGAA  | GATATTGGAT  | TATGAGGAAT | TTGGGAATTA |
| 69361 | GGAAAGTGGG  | TGCAGGTGGG  | TTGGGGGTAG  | GTGAAGGAAG  | ACATGGGCGT | ATTGGGGGAG |
| 69421 | CAGGGGCTGC  | TCAGAGGTGT  | TCCAGAAGCT  | CTGGGTGAGG  | AGGTGAGAGG | GACCGGGGAA |
| 69481 | TGCAGCTCGG  | CCCAGCCTCC  | CTGCCTGAGG  | TCAGCCATCA  | CGTGGTGATG | GCAAGATGGA |
| 69541 | AATGTGCTTT  | CTGACTGCTC  | CAGCCAGTGC  | TGCCAGATTTC | AGCTCCCCAG | GGAGGGCACC |
| 69601 | TGAGAGGCTC  | CAAGCCAGGA  | GATCTGTTTT  | CTCCTTTGTT  | TTGTTTTTTT | TTGTTTTGTT |
| 69661 | TTGTTTTATT  | ATACTTTAAG  | TTCTAGGGTA  | CATGTGCACA  | ACGTGCAGGT | TTGTTACATA |
| 69721 | TGTATACATG  | TGCCATGTTG  | GTGTGCTGCA  | CCCATCAACT  | TGTCAATTAC | ATTAGGTATA |
| 69781 | TCTCTAATG   | CTATCCCTCC  | CCCCCTCCCC  | CACCCCTGTT  | TTTCTCTTTT | GAATCCTTCT |
| 69841 | TAGAGGCCGG  | GCGCGGTGGC  | TCACGCCTGT  | AATCCCAGCA  | CTTTGGGAGG | CTGCGGCAGG |



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| 69901 | AGGATTGCTT  | GAGCCCAGGA  | GTTCCAGACC  | AGCCTGGGCA  | ACATAGTGAG  | ACCTCGTCTC  |
| 69961 | TACAGATAAT  | AATTTTAAAA  | ATTATCCGGG  | CATAGTGGCA  | TGCACCTATA  | GTCCCAGCTA  |
| 70021 | CTCAAGAGGC  | AGAGGCAGGA  | GGATCACTTG  | AGCCCAGGAG  | GCGGAGGTTG  | CCGTGAGCCA  |
| 70081 | AGATCCCACC  | ACTGCACTCC  | AGCCTGGGCG  | ACAGAGACCC  | CCATGTCAAA  | TAATAATAAT  |
| 70141 | AATAAATAAA  | TCCTTCTCAG  | TCCCTTCCTC  | ACTGTGTCCC  | CCTCCACTGA  | ATTTTCCAC   |
| 70201 | CTCCTCTCCC  | ACTTCCCCCA  | CTCCCCTTTT  | CCCTCTCCTT  | CTCTCCCCAC  | TCCATCTTTT  |
| 70261 | TCTTTCTCTG  | CTGTTTCTCG  | TCCCTCCCTC  | CTCTCCATCC  | CACAACACTG  | CCTACCCCTG  |
| 70321 | CCCTGCCCCA  | CCCTGGTGCT  | CAGGATGTGT  | GAAGTGAGGG  | GTGGTAGCCC  | CCAAGACCTC  |
| 70381 | AACCCCGAAG  | GTTAGCCTGT  | TGAAACCACT  | TTCTCCCAGC  | TGCCCCCTGT  | GCAGTTGGTG  |
| 70441 | CTGCTGGGGG  | AAACTGGGAT  | TGGGGGCCAG  | ATTTTGCCCT  | TTTTCCTGAC  | AAAGAGAGAT  |
| 70501 | GAAGAATTCT  | CTCACCAGGT  | GCCTGGGACT  | GGGGTGTGGG  | TGTCCACAGC  | TATCCAGCG   |
| 70561 | CATCTGTTCT  | GCATCATGAT  | TAATAGTGCT  | GCTTTCAGCC  | GGGCGCGGTG  | GCTCACACCT  |
| 70621 | GTAATCCCAG  | CACCTTTGGG  | GGCTAAGGTG  | GGCAGATCAC  | AAGGTCAAGG  | GTTCGAGACC  |
| 70681 | AGCCTGGCCA  | ACATGGTGAA  | ACCTCGTCTC  | TACTAAAAAT  | ACAAAAATTA  | ACCAGGTGTG  |
| 70741 | GTGGTGGGTG  | CCTGTAGTCC  | CAGCTACTTG  | GGAGGCTGAG  | GCAGGAGAAT  | CACCTGAATC  |
| 70801 | TGGGAAGCAG  | AGGTTGCAGT  | GAGCCAAGAT  | CGTGCCACTG  | CACCTCCAGC  | TGGGTGACAG  |
| 70861 | ATAGTAGACTC | CGTCTAAAA   | AAAAAGGAGT  | TTTGCTCTGT  | CGCCAGGCT   | GGAGTGATGT  |
| 70921 | GGCGCCATCT  | CGGCTCACCG  | CAACCTGCGC  | CTCCCGGGTG  | CAAGCGATTG  | TCCTGCCTCA  |
| 70981 | GCCTCCCAG   | TAGCTAGGAT  | TACAGGCGCC  | TACCACCACG  | CCTGGCCAGT  | TCTTGTAATT  |
| 71041 | TAGGAAGAGA  | CGGGGTTTCA  | CCCTGTTGGC  | CAGGCTCGTC  | TGGGACTCCT  | GACCTCAGGT  |
| 71101 | AATCCGCCCA  | CCTCAGCCTC  | CCAAAGTGCT  | GGGATTGCAG  | GCATGAGCCA  | CCGTGCCCAG  |
| 71161 | TCAATCCCTT  | CTCAAAAAAA  | AAAAAATAGT  | GCTGCTTTCT  | CTTTCAGATG  | TCCTGATTTG  |
| 71221 | GGTGATAGTA  | AATGCCACTC  | TACTTATAAG  | GGATCTACCT  | CAGAAATGCT  | ATTTGGGACAT |
| 71281 | TTTTGTAGCA  | CTCTACTGTT  | GGCAGCAGGT  | GATGCTCACA  | ACAGCCCGTG  | AGGGTGGATG  |
| 71341 | ACGTCCTGTT  | CACAGATGAC  | AAAGGAGCCT  | CATGCTCAGA  | CCGTGGGCTG  | CCAGAGCAGG  |
| 71401 | TCCATGGGTG  | CAGCCCCACA  | TGGACCATAT  | TTCCCCCTTG  | TCACTCTTTC  | CACCAAGCTC  |
| 71461 | CCTTGAAGCT  | TCAGTTATTA  | AGCTCTCTTG  | GGTGAATATC  | AAGTTAGAAT  | CACAACATGT  |
| 71521 | GCCTCATATG  | GATTGTGCCA  | GTGAAAAATG  | ACATTCTATT  | TAGAGGCAGG  | GCAGCCTGGC  |
| 71581 | TATAGTATGT  | TTTAAATAT   | GTATTATGCT  | GCAACAAATG  | TACCATGATC  | CTGTAAGATG  |
| 71641 | TTCACAACAA  | GGGAAC TGGA | TGTGGGGTAT  | ACTGTCTGTA  | CTAACTTCAC  | AAGTTTTCTG  |
| 71701 | TAAATCTAAA  | ACTGTTCCAA  | AATAACAAGT  | TCGTTTAAAA  | TAACTCCAG   | GAGACCAGGT  |
| 71761 | ACGTGATAGT  | ATGCCTATAA  | TCCCAGCACT  | TCGGAAGGCT  | GAGGCAGGTG  | GATTGCTTGA  |
| 71821 | GCCCAGGAGT  | TTGAGACAAG  | CCTGGGCAAC  | ATGGTGAAAT  | CCTGTCTCTA  | AAAAAAATCA  |
| 71881 | CAAAAATTAG  | CCAGGTGTGG  | TGGCGCATTC  | CTGTAGTCCC  | AGCTACTTGC  | GGGGCTGAGG  |
| 71941 | TGGGAGAAAT  | ATCTGAGCCC  | AGGAGTTTGA  | GGCTGCAGTG  | AGCTGTGATG  | GTACCACTGC  |
| 72001 | ACTCCAACCT  | GGGCAACAGA  | GCAAGACCCT  | GTCTCAAAAA  | ACAAAAATGA  | AATAAAGTCC  |
| 72061 | AGGAAGAAG   | TAGGTTTAC   | CACCTCTATT  | TTCTGAAGAG  | AAAACATAAT  | TTAATGTGTA  |
| 72121 | TAGTGAGTGT  | AAGTTTACCA  | AGTTAGTGTT  | TGAGTTGCCT  | AAAATATGTT  | TGCTAAAACT  |
| 72181 | ATTCAAAGCT  | TTACACATAA  | ACATGATCAG  | AAGTCTCTAT  | CCAAAACATA  | TGTGTGTGTA  |
| 72241 | TATATATATG  | CACATATAT   | ACTGTATATA  | AAAAATGCAA  | ATCTAAATTG  | CCAACCTTTT  |
| 72301 | AGAAATTGCT  | CTGAAAGGAA  | AGCATTTCAA  | GATAATTTGC  | TTACCCAAAG  | AATATACTTT  |
| 72361 | CCAAGAAAGC  | AAGTAATACT  | TAAGGTGTTT  | ATAATCTCTA  | TCAAATTAAT  | TCTTGCTACT  |
| 72421 | GAAAGCTTAC  | AAGGAGCTGT  | TTTGATGTCT  | GGTGTGACAG  | GTTTGACTTG  | GCAGAAAGGTG |
| 72481 | TACCTTTTAC  | AACAACATTT  | TAAATAAGTG  | ACAGAAGACA  | AGAAACTACA  | CGTTAAATGC  |
| 72541 | CAGAACAAAG  | AGTGTCTAAG  | TGGATGCTAA  | GAGTTGAAAT  | ATGGCTGGAT  | ACCTGCCCCA  |
| 72601 | GAGAGCTGAA  | AAGTAGATGA  | AAGTTGGTTA  | CCTATAAACT  | AGTGACCCCT  | AATGAATTA   |
| 72661 | AAGGTGTTTA  | TGAGTTAACT  | TGTTATGCCT  | TCCAGATAAG  | ACATGCAAA   | GGGGCTTCTT  |
| 72721 | CCTCCTTCAC  | TACTTCCAAG  | GGATTTAACA  | AGGAGACCAA  | TGCAAAATGAT | AAGGACTGTA  |
| 72781 | GGGCTCAAGC  | TGGGGACAGA  | TTGGGGAAAG  | GGGGACCATC  | ATGCCCATAT  | AGATGTCCCT  |
| 72841 | GTGCCCTGGC  | AGTCAAGGCT  | GCTGAAAAAT  | AACAAAAACC  | AGAAGTCTGC  | GTGATGCTGC  |
| 72901 | CTCTCCATTT  | GTCCAAAGCC  | TTCTTGCGGC  | AGTTTGCAGG  | CTTTTGCAAA  | AGCTCCAGGA  |
| 72961 | CCAAGGAGCT  | ATGTTTCATG  | TGGAAGCTTG  | TTCAAGGATTA | GCTGTTCTTT  | GTGGGATGGG  |
| 73021 | TGCAGCCAGG  | GCCAGGTGTC  | CAGGGACAGT  | GTTTAAACAA  | AGGGCATGAG  | GTGCTGTATC  |
| 73081 | TCACAGTGGA  | ACTCCACTTG  | CCTTTTTTTC  | ATCTTCTCAT  | TCTGCTTCAT  | GCACAGAACC  |
| 73141 | AGCCCCATCC  | TGAAACTGAC  | TCTAAATTAC  | TCCCGCCCCA  | GGTGGAGTGC  | CTTCTCGGA   |
| 73201 | GTTCAACAGA  | GCCTTCTGT   | CGCCCAAGGG  | ACAACCTCAC  | TGAATGCCCA  | AGCCACACCC  |
| 73261 | AAAACCTAAC  | AAGTAAAAAC  | CAAAATCTGT  | GCTCCCCCAT  | CCTGGGCCAT  | TCCTGTTTTT  |
| 73321 | TCTACTGCTG  | TTGGTGATAC  | CACCATCAGC  | TTGTCCATCA  | TGACCTTGGC  | CAGTTCCTCC  |
| 73381 | CACAACCTTC  | CACAGACCCC  | AGGGACCTCA  | CCTCCATTCC  | ATCCGACACA  | GATCTCTCTA  |
| 73441 | CCACAACCTT  | TGGTTTTGCA  | ACAGCAGCCA  | TGAGACCTTT  | ACACCCCTCC  | CCCTTCATCC  |
| 73501 | TGTCCCCCAC  | TGAGGCCCCA  | GAGCCATTCC  | TAAAGCAGC   | GCGCCACAAA  | CTATAACCCA  |
| 73561 | CAAGCCCAAT  | CTGGTACCCA  | GCCTGTTTTG  | CACAGCCAGT  | GAAGTGACAA  | TGATCTTTTC  |
| 73621 | ATACAGCCAG  | AAAAACAAAA  | CAAAACAAAA  | AAAAACAAAA  | AAAAACCCCC  | CCATTCTGAG  |
| 73681 | CATGTGACTT  | CCATGTTCAA  | GATGCTCTAT  | GTTTCAAGAA  | GCCCTTGAA   | AAGGAGGAAG  |
| 73741 | GGGAGCTGGG  | CACAAAGGGA  | GACCTCTCTA  | GCTGAGCTCC  | TCCCATCCAG  | ACATTTTCCT  |
| 73801 | GGACTTCCTA  | TCCAATGACT  | TCCCTTAGCT  | TCTTATCAGC  | CACCCCTGTC  | TGCCAGGAG   |
| 73861 | GCTGGAAGAT  | GTGGCCTTTT  | AACTGGGCAC  | AGCTCTGTCC  | TCTATCATAT  | CAGGGCTCTG  |
| 73921 | TTCCCAAGGA  | GGGTAGAGAG  | AATGGACACC  | AGGTGGACCC  | TCAGCAGTCT  | GTGCCACAGA  |
| 73981 | GGGAGTGTTT  | GCAATTTCCA  | GACTAAAAAGT | CCCCATGTGC  | TTGACGGGGT  | ATGTGACTAC  |
| 74041 | AACGTGATGC  | TTGACTTTTC  | CTCATATGAC  | CAGAGCCACT  | TTGTCCATCT  | GGTACAATGT  |
| 74101 | CAGCTATCTG  | CTAGGGGCCC  | TCCAGGATTC  | CCAGTCAATT  | CCATATCTGC  | ATCACCACCA  |
| 74161 | TTGGCACTAA  | ATAAAATAAA  | ATACTCAAGT  | TCCTGCTGGT  | GAGCATGAGC  | AGTGCTACAC  |
| 74221 | TGGGCCCTTC  | AACCAAGGTG  | ACATGATAAT  | GACTGAAAA   | AATCACTGCC  | ACTTATTGGG  |
| 74281 | GACGTCTCAT  | CTGCCAGGCA  | TGGTACAAAG  | TGCTTTAAAT  | AAGCATTCAA  | CAATTTTCATG |

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| 74341 | CTGACAGAAG  | CCCTGTGAGC  | CAGTGGAGCT  | ACTACTATGC  | CCATTATACA  | GGGGAGAAAA  |
| 74401 | CTGAGGCAGA  | GAGAGGTTAG  | GTAATTGCT   | CAGCCTCACA  | CAACCAATAG  | GTGGTGGAGC  |
| 74461 | CAGGATTTGG  | GCCCCATCTG  | CCTGACTCTC  | TAGAGGCTCT  | ATCTTCCAGT  | CTTCCAGAGT  |
| 74521 | TGAGTCTAAG  | CCATGAATAG  | GACAATTAGA  | CAGCAGAGGA  | AACCCATTCA  | GCCACCATGT  |
| 74581 | GCATGAAGAG  | TAAGGAATTT  | CTGTCATACA  | GAGGGGAGTG  | AATTCACTGA  | GCTGAGAGCT  |
| 74641 | GAGGAACCAT  | TGATCTGATG  | GCTGAGACAC  | CAC TGGGAAG | ACTGGAGAGG  | CTTTTCTGGG  |
| 74701 | CATGCAGTGC  | CAGGCACAGG  | AGGAGCTGAG  | GGAAGATGAC  | TAAGAGGTAC  | TGGCAAAGAA  |
| 74761 | TTCAGAAATT  | CTGATGGAAG  | CTTTACATGT  | TACCATCACA  | TCCATCCATC  | TATCCACCCA  |
| 74821 | TCCATCCACC  | CATATCTTCC  | TCCCTCCACC  | CAATCATGCA  | TACATCCAGT  | CATCTATACA  |
| 74881 | CCACCCACCC  | ACCCATCCAT  | CCATCCATCC  | ATCCCTTCAT  | CCATCCCATC  | ATCCATCCAA  |
| 74941 | TTATACATAC  | ATCCAATCAT  | ATATCTGTAC  | ATAATCCATT  | CTTCCCTCGG  | TTCATCCATC  |
| 75001 | CATCCATTCA  | TCCATCCATC  | CACCCATCCC  | TTCCTTCATC  | CTTCCCTATCA | TCCATCCAAAT |
| 75061 | CATATATCTG  | TACATAATCC  | ATCTTCCCT   | CGGTTTCATC  | ATCCATCCAT  | TCCATCCATC  |
| 75121 | ATCCACCCAT  | CCCTTCCCTC  | ATCCTTCCCT  | TCATCCATCC  | AATCATACAT  | ATATCCCAATC |
| 75181 | ATACATCTGC  | ACATCACCAG  | CTCATCCATC  | TATCCATTTA  | TCCATCCATC  | CTTCCCTTCCA |
| 75241 | TCCATCATTC  | ATCCATCATA  | CATACATCTA  | ACCATACATC  | TCTACATCAT  | TCAATCTTCC  |
| 75301 | CTACGATTTC  | CAATTATCC   | ATCATTCCCT  | CCTCCATCCA  | TCCCATTAATC | CATTTGATCA  |
| 75361 | TACATATATC  | ATCTATACAT  | CATCCATTCA  | TCCATCCATC  | CATCCATCCA  | CCCATATCTT  |
| 75421 | CATCCAATCA  | ATCATACATA  | CATCGAATCA  | TCTACACATC  | ACCCATCCAT  | CCATCCATCC  |
| 75481 | ATTCATCTAT  | CCACCCATCC  | ATCCATCCAT  | CCATCCATCC  | ATCTATCCAC  | CCATCCATCC  |
| 75541 | ATCCATCCAT  | CCATCCATCC  | ATGTAACCAT  | CCAGTCATAT  | ATCCAATTAC  | ACATCCATCC  |
| 75601 | AGTTATACAT  | TCATACATGC  | ATCTAATCAT  | TCAATTATAC  | ATACACACAT  | CCATATAAAT  |
| 75661 | CTACCATCCAA | TTATACCTCC  | ATCCAATTAC  | ACATTCATAC  | ACCCACCTAA  | TAAATTATTA  |
| 75721 | ATTCATATAT  | CCATCCATAT  | AATTATACAT  | CAATTATACA  | TCCATCTAAT  | CATTCAGTAA  |
| 75781 | TTACCCACCC  | ATCCAGTCAT  | CTATCCAATA  | ATACATTTCAT | CCAATCATCC  | ATCCATCCAT  |
| 75841 | CCACCCATTC  | ATCCATCCAT  | CCGTCCGTCC  | ACCCATCATG  | GTATGAGCCA  | TGATTTACCA  |
| 75901 | CGATGGTCCC  | CTGTGGACAG  | CCCAGGTGGG  | GCAGAACTGA  | AGGGAAGCCC  | AGGGCTGCCC  |
| 75961 | CCATAAAATC  | TTGCCTCCTT  | TACATGGATG  | AGAAGTAGAT  | CCACATGTAT  | AAATCCTCAT  |
| 76021 | GATTTGAAGG  | TGCTTTTACC  | AACATTCAT   | CATGGGATTC  | TCCCAGGAGC  | TCTAGGAGGA  |
| 76081 | GGCAGGTAGA  | GTTGAGGTCA  | TCTCACGCAT  | TTTACAGATG  | AGGAAACGGA  | GGCCCTGAGA  |
| 76141 | GGCAGGTCCA  | AGGCCACCTG  | ACCAGAAAGA  | AGTGGAACCTG | GGACTTGAAC  | CCAGCCATCT  |
| 76201 | TGCCCTCTGG  | TCCATGCTC   | TCTAGCCTGT  | AACTCCTGCT  | TCCCTGGTGGG | GCATCTCCAG  |
| 76261 | GAGGACCCTA  | TGGGCTGGCC  | ATGGGCTGCT  | CCTGGAGTCT  | TTTGCTCTGT  | GTGGCCATCC  |
| 76321 | TTCCCTCCCTC | AGGAGAGTGT  | GTGCTCCCAG  | AGCAGAGGCT  | GTATCTTCTG  | AGCATTTTGT  |
| 76381 | CCCTTCCCTAG | TACCTAGCAC  | TCAGCTCTGT  | ATACATTGGG  | CTCTCAAGAA  | TTCTCAACCT  |
| 76441 | TCCAGAGTGT  | AAGGCCCTGA  | CCTGCTCAGC  | CCTGGATACT  | GCATGATGCA  | TTGATAAGCC  |
| 76501 | CATAAAATAA  | CCAGGGCAGA  | TTGACTCCCA  | GTGGCCAAAG  | TGCCACAGGG  | AAGGGACAAT  |
| 76561 | TAGGCCCTTC  | TAGGAGGAGG  | AGGAGGTAGT  | TTTCTCATTT  | CTATTAAGGC  | AACAAAGACT  |
| 76621 | GCCTTACTAA  | GGACATTCTT  | GGTGGAGGGC  | GTGACTGTCA  | ACCAGTGTGA  | TCATTTGGGC  |
| 76681 | CTCTCTTGCC  | CAGGCTTCCC  | ATTCTGAAAG  | GACAGTTTTA  | TTGTAGGTAC  | ACATGGCTGC  |
| 76741 | CTTTTCAAAAT | GTAATCACA   | GCTTGTCCAT  | CAGTCTTTGG  | AGGTCTTTCT  | ATGAAAGGAG  |
| 76801 | CTTGGTGGCG  | TCCAAACACC  | ACCCAAATGTC | CACCTAGAAG  | TAAGCACCGT  | GTCTGCCCTG  |
| 76861 | AGCTGACTCC  | TTTTCCAAGG  | AAGGGGTTGG  | ATCGCTGAGT  | GTTTTTCCAG  | GTGTCTACTT  |
| 76921 | GTTGTTAATT  | AATAGCAATG  | ACAAAGCAGA  | AGGTTTCATGC | GTAGCTCGGC  | TTTCTGGTAT  |
| 76981 | TTGCTGCCCG  | TTGACCAATG  | GAAGATAAAC  | CTTTGCCCTCA | GGTGGCACCA  | CTAGCTGGTT  |
| 77041 | AAGAGGCAT   | TTGTCTTTTC  | ACCCAGGAGC  | AAACGCACAT  | CACCTGTGTC  | CTCATCTGAT  |
| 77101 | GGCCCTCTGG  | TGGGGCACAG  | TCGTGTTGGC  | AGGGAGGGAG  | GTGGGGTTGG  | TCCCTTTTGT  |
| 77161 | GGGTTTGTG   | CGAGGCCGTG  | TTCCAGCTGT  | TTCCACAGGG  | AGCGATTTTC  | AGCTCCACAG  |
| 77221 | GACACTGCTC  | CCCAGTTCCCT | CCTGAGAAAC  | AAAGGGGGCG  | CTGGGGAGAG  | GCCACCGTTC  |
| 77281 | TGAGGGCTCA  | CTGTATGTGT  | TCCAGAATCT  | CCCTGTCAGA  | CCCCACTGA   | GGACGGATCT  |
| 77341 | GAGGAACCGG  | GCTCTGAAAC  | CTCTGATGCT  | AAGAGCACTC  | CAACAGCGGA  | AGGTGGGCCC  |
| 77401 | CCCTTCAGAC  | GCCCCCTCCA  | TGCCCTCCAGC | CTGTGCTTAG  | CCGTGCTTTG  | AGCCTCCCTC  |
| 77461 | CTGGCTGCAT  | CTGCTGCTCC  | CCCTGGCTGA  | GAGATGTGCT  | CACCTCTTCG  | GTGCTTTGCA  |
| 77521 | GGACAGCGTG  | GTGGGAGCTG  | AGCCTTGCGT  | CGATGCCTTG  | CTTGCTGGTG  | CTGAGTGTGG  |
| 77581 | GCACCTTCAT  | CCCGTGTGTG  | CTCTGGAGGC  | AGCCACCCCT  | GGACAGTCCC  | GCGCACAGCT  |
| 77641 | CCACAAAGCC  | CCGCTCCATA  | CGATTGTCCT  | CCCACACCCC  | CTTCAAAAGC  | CCCCTCCTCT  |
| 77701 | CTCTTTCTTC  | AGGGGCCAGT  | AGGTCCCAGA  | GCAGCCATTT  | GGCTGAGGGA  | AGGGGCAGGT  |
| 77761 | CAGTGGACAT  | CTGATCTTGG  | TTTAGTATCC  | TTTATTTTGG  | GGGCTCTGGG  | TGTGGCCTGG  |
| 77821 | GCCTCTGGAC  | TTTGGCCACG  | GTGTTTGTTC  | CAGCCCTTCT  | CCTAACCTGT  | CCTTTCCAGA  |
| 77881 | CACCTCGCAT  | CTAGGTTATT  | AGCACCTCGC  | ATACTTTCTG  | ACATGCTCCT  | CAGTCTTGAT  |
| 77941 | TTTGACCATC  | TTCTCTTGCT  | TCCCATCTGT  | GTCAGTCAAG  | ACTGCATTTC  | GCTGTAAGAA  |
| 78001 | ACAGAAACCC  | CACTAATCTG  | TGGCATTTC   | ATGAAGAGGT  | TTACTTTTCT  | CACATAATCA  |
| 78061 | GATGCTTAGA  | CTTGGCCAGC  | ACCTCAAGGG  | TCATTGATGC  | TCTCTGTCT   | TTATTTTCTG  |
| 78121 | TCATCTTTAG  | TGGTTGGATT  | GTGCTTCAT   | GGTTACAAAG  | TGGCTGTGTC  | ACTTCCAGGC  |
| 78181 | ATCACATCTG  | CCTTTGAAGC  | AGGAACAAGT  | TGCAAGATAA  | AGTGGCCAAA  | AGGGCCCTGA  |
| 78241 | AACTAAATGT  | GTCCCTTAG   | GAAAGCAGGA  | GTTTCTTTCG  | AAGTGGCAAT  | CTTCTGCTTA  |
| 78301 | TGTCTCATTC  | GCCAGAGCTG  | GGTCTTACGG  | CCACCCCTTG  | CTGCGAGCAA  | GGCTGGGACA  |
| 78361 | TTGAGCATTT  | TGCCGTCCAA  | CCTCTTTAGC  | AGAATAAACC  | AAGGGGGAAG  | AACGTTAATA  |
| 78421 | GTGGCTTTTC  | AGTCACTAGT  | TGGCAGTATC  | TGCCCTCTTA  | TCTTTCCATC  | CTCCCATGG   |
| 78481 | AGTTTCAAGG  | TTCTTTTCTC  | AGTACTTCTT  | CAGGCTCTGC  | ACGTTTCATT  | GGATCTTGTG  |
| 78541 | TCTTGGGGTG  | AAAACATGGC  | CCAAGTGTCT  | CCCCAAGCAT  | CCACCTTTGG  | ATTAATTTGG  |
| 78601 | AAAATGGCTG  | TCAAGTGCCC  | GCCTCTTGCT  | TGGTATAATG  | CTACAGCTTT  | AGAGGACGCA  |
| 78661 | GCAGCATGG   | GCCTTGCCGC  | TGAGGTTCTT  | AGCCTCATGA  | GAATATCCAG  | ATCAGATTCT  |
| 78721 | CTTGGCTCCT  | TCTTAGAGCC  | AGTGATGCAA  | GACACTTCCT  | GCTCATCTTG  | TCGGGACGGT  |

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SEQ ID NO Sequence

|       |             |            |             |            |             |            |
|-------|-------------|------------|-------------|------------|-------------|------------|
| 78781 | TTTACAAGTT  | GCCTGCCATC | CTGAGAAAAGT | CTACAAAACG | ATGCCAGACC  | TCATGCCAGC |
| 78841 | TTCCCAAGCC  | TTGACTCTCA | GTGCTCCCTC  | AACAGGATTC | TGGAAGAATC  | TCCCAACAA  |
| 78901 | GTCGCAATGC  | CCTCTGGACC | CTGTGCAGGC  | ATGAGACTCA | AGAGCATTGG  | CTCCCAACCC |
| 78961 | TGGTGGAGGG  | AACACTGCTG | GGGCTGGGAT  | CTTGCTGGT  | TGCTCCGCCT  | GCACCCAAGA |
| 79021 | CAACCATAAAT | TAAATGTCC  | TTCATTGAAC  | TTGGAAGCC  | TTCAAAGCTG  | ACAACCTCTT |
| 79081 | ATGTGTACCC  | GGAAGGCCT  | GGGAGTGTGC  | CAGGGCATTG | CTCGGGAGGG  | ACGCTGATTT |
| 79141 | GGAAGCATTT  | ACCTGATGAG | AGACTGACAG  | CAGCTCCTGG | TAGCCGAGCT  | TTCCCTCCTG |
| 79201 | CCTCTGCTGT  | GAAGGTGGAC | CCATCCAACA  | GTCAAATGCC | TGACTCTGGA  | CAGGAGCGGA |
| 79261 | CCTATTTTATT | GCCATGCAAG | GGACTCTGCA  | CTTTTGAATT | GTGGGTCTATG | GGCTTGGATT |
| 79321 | TAGGGGTTAG  | AGCTGGGAGA | AGTCTTGGAA  | GTCACCTAGA | GATGACACTG  | CCATTTTGCA |
| 79381 | GATGAGGAAA  | CCGTCCAATC | AAAATGGACC  | AAGGACTTGC | CCAAAGCCTC  | ACAGCAAAAC |
| 79441 | CATAGGCCCC  | CGCACTAACC | CCAGAGTCCC  | TGTGCTGTCT | TAAGGATCAT  | ATAGTTGTAA |
| 79501 | GCAATCATCT  | GGTTTTCACT | ATTTCTTCTT  | TAAAAATGCC | TGGGGCCATG  | CCCAGCAGTC |
| 79561 | GTTTTCACTG  | CAGCGTTTAC | ACAGGGCTGC  | CGGGCTTTCC | TGGTGGATGA  | GCTGGGCGGT |
| 79621 | TCATGAGCCA  | GAACCACTCA | GCAGCATGTC  | AGTGTGCTTC | CTGGGGAGCT  | GGTAGCAGGG |
| 79681 | GCTCCGGGCC  | CTACTTCAGG | GCTGCTTCT   | GGCATATGGC | TGATCCCTC   | CTCACTCCTC |
| 79741 | CCTCTGCTGT  | TGCTCCTGCG | CAAGAAGCAA  | AGGTGAGGGG | CTGGGTATGG  | CTCGTCTGG  |
| 79801 | CCCCCTCTAAG | GTGGATCTCG | GTGGTTTCTA  | GATGTGACAG | CACCCCTTAGT | GGATGAGGGA |
| 79861 | GCTCCCGGCA  | AGCAGGCTGC | CGCGCAGCCC  | CACACGGAGA | TCCCAGAAGG  | AACCAAGGTT |
| 79921 | GAGGCTAAGC  | CCCAGAGACC | CCCAGGCAGT  | CAAGGCCCTG | CTGGGTGCTC  | CAGCTGACCT |
| 79981 | GTGACAGAAG  | TGAGGGAGCT | TTGCGTGTTC  | ATCCTCCTGT | GGGGCAGGAA  | CATGGGTGGA |
| 80041 | TTCTGGCTCC  | TGGGAATCTT | GGGTGTGAG   | TAGTCTGATG | CCTTGGTGCT  | CAGTTACCTC |
| 80101 | CCTGGCTGCC  | TGCCAGCCTC | TCAGAGCATT  | TAGGGCCTTC | TGGACTTCTA  | GATGCTCCTC |
| 80161 | ATCTTGCCTC  | AGTCAGCGCG | TCAGTTCCAG  | AGACTTCTCT | GCAGGGTTTT  | CTGGGGCAGG |
| 80221 | TGGTGGCAGA  | CCCGTGCCTT | CTTGACACCT  | GAGGTGAGTC | CACCCCTCCTG | CTCAGACTGC |
| 80281 | CCAGCACAGG  | GTCACTCTCC | AAGGGGTGGA  | CCCCAAGATC | ACCTGAGCGC  | ACAGAGGGTG |
| 80341 | CAGATGACTG  | GACCACACCT | TTTGGTGATC  | TTAATGAGGT | GGTCCCAGAG  | GAGCTCAGAC |
| 80401 | ATGCAATCTA  | GCATCCAGTT | CTGGGACTCT  | GTCTCCTTTT | CAACAGTATT  | CATGTAGAAC |
| 80461 | AGGCATGACG  | AGAAATGCCT | GTCAACATGG  | GTGATGGGGA | ATCAATCAGA  | CAGGGCGCCG |
| 80521 | GGCTCAAGGC  | TGCAGTCACC | CAAGAGTGGC  | TCAGCCCACC | AGGCCCTAGG  | AAACGCCTGC |
| 80581 | ACAGCCTGGA  | GCTCCTGGAG | TCATTTCTCT  | CATGTCTTCT | TCACTGCACT  | TACGTAAAGA |
| 80641 | TGCCAGCCAT  | TGGTTTGGTG | ATTGGAGGG   | TGCCCAGTTG | CCCACAAGA   | AATGCAGAAG |
| 80701 | AGGCCTAGCC  | AGGATTTCAC | CAGCAGTGGA  | GAGTAGAGAA | GATGTGGCCA  | GAAAAGAGTT |
| 80761 | TCCTTTCTCT  | CCTAAAGATG | GTACTCCCTG  | CAGTACTGGG | GGAAGCCTGC  | AGCATTTCTC |
| 80821 | AGGGCTCTGT  | GTGTTGAGAG | CAGCCCCACC  | CTGGCCCTTC | CTGAGTGCAT  | TTCTGCTTTG |
| 80881 | TGACTTGATC  | CGTGAAGTCC | CCTGAGATGG  | GCAGAGGGGA | TGCTCTCGAA  | GCTGGGGCAG |
| 80941 | AGCCTCATCC  | TTGAACGTGA | AGGACGTTTG  | AAGACTGTGG | CATGATCACA  | GGATGAGATC |
| 81001 | ACAGGGAACT  | TGAGTTTCTC | TCCTCCTCTC  | CCTTCACAGT | TATTTCACTG  | AGGGAAATCC |
| 81061 | CTCCCCTGCC  | CAGAATGAAA | ACTCTAGCCA  | ACTCTTGACT | TTTCCATCAC  | TCCAAAGTAG |
| 81121 | TTGAAAGTAC  | ATTAGTCTCC | ACAGTGGCAA  | AACAGTGTGC | AAAAGCTAAA  | TAATTAGAAC |
| 81181 | AGCCAGTCCC  | ATGTGACAGT | CAAAGCTTCT  | AACTCCATTC | AAAGTTGCAG  | CCATTTCCCT |
| 81241 | CGAGGGCTGG  | CAGGGAGGGG | AGGGGTAAGA  | GAAACAGGAA | GGTTCTTACT  | GAGTTGGTCC |
| 81301 | TGGTGTGAGC  | TGCGTACAC  | TCCTCGACA   | GGTTTCAAGG | AGACTCTCTC  | TCTCTCTGTC |
| 81361 | CTCATGGGGA  | CCTTATTGTA | ATTCTTCTAC  | TCTTACCCCA | GCCTGCCATC  | TCCAGCTATC |
| 81421 | CTCCCCTGAA  | GAGCCCTTCT | GCTGCGCTGG  | ATTCTGGTGG | CCATGTCTATC | TCCTCGGCC  |
| 81481 | CGTGGGAGTC  | TGAAGATCTG | GCTGCAGCCT  | CACCTCTGAG | GTCTCTGCTG  | TTGCCACCTC |
| 81541 | TTAAACATGA  | TCTGAGGCTC | CCATGCACCT  | TGACCTGTGC | CCACATGGGG  | CCCACGGGAA |
| 81601 | ACACGCTGGC  | AAGCAAACCT | TGGGTGTGCA  | GACGTTTCTC | AGGGCTGCAG  | CACCTGTCTC |
| 81661 | TTGCTCTGCC  | CCCAAAGCAA | GGCCAGCCCA  | TCTTCCATCC | TCTAGTGTTC  | CTTGGTGGGG |
| 81721 | CCTGACCACG  | AGTCCACCAG | GTCCCTAACC  | AGAGGGGACA | CACACCAGGT  | GTCTCAATG  |
| 81781 | TATTGCCCTG  | AAACAGTTGT | GCTGGGACTG  | TGATGGGGGG | TGGCCATGTA  | GCCACCCCCA |
| 81841 | CCACCCCCAA  | GCCACTCTCT | CCAAGGAAAT  | CCTCCTAAAG | ATCCCTTTAC  | ATCCTCCATG |
| 81901 | TGGTGGGGAG  | GTTCTAGAGT | TGGGTGCATG  | TGTCTTCAGC | TACTGACAAT  | GCAGACCTTA |
| 81961 | GTTGGCACCT  | CGTCTTGACC | TATCTGTGTT  | GCTGTTCTTG | GCCTCCAGT   | GAAACTCCCC |
| 82021 | ATGGGCCATC  | CAGTTGGGGT | GCAGTGTGGC  | CACCCCTTGG | CAGGTTCCTG  | CCTTGCTGGA |
| 82081 | GAGCACAGGG  | CCCTCCTGGC | TCTTGTAATA  | CACCTCCCAT | GGTACAGAGA  | GGCCAGCAGT |
| 82141 | GATGTGAGGC  | CCAACCTCCC | TCCATGGTGT  | TCCCAAGCAG | CTCCCTTTCT  | GGGTCAAGG  |
| 82201 | GGTGGCAAG   | ACAGTGCAGC | GTCCAATTTT  | TGACTCAAGC | CGGGCCTGGC  | TATCGCAGCT |
| 82261 | CTGCACTGTG  | TGTGACAGCA | AGGCAACTCA  | CCCAGTCCCG | TGGCAGTGAC  | CGTGTCCGAG |
| 82321 | GAAGCCTCCT  | CACACCCTCT | GTCTCAAGGA  | CTCTGGCATT | TAGCTGGACT  | TGCTGTAGCT |
| 82381 | CTGAGCCTTT  | CTGCCATTGC | CATCACCTTG  | TCAGAAACTC | AGGCCGAATC  | TGCACTCAGA |
| 82441 | GTTGTGCCCA  | GGCAGTTGAG | CCAACACTTG  | CTCAGCGATA | TTGTACATG   | ACAAGGCAT  |
| 82501 | GTCACCACTG  | GGCATCTGCG | GTAGCGCAGT  | GTGCGCTGGA | TGGACCCGGA  | GGGTGTCTGT |
| 82561 | GTCATGCTAG  | TGCTAGTGAT | GGGAGCCCCG  | TGAGCCCAT  | CCCCGCCCTC  | CCATGCCCTC |
| 82621 | AGCAGCTGCC  | TGGGGACAGC | CAATGGCCTG  | GGTGTCTTCT | AGGCTACCAT  | ATGGCTTCCA |
| 82681 | GGAAACTCGA  | GAACCTTTCT | CTCCCTTGCC  | TACACTCTTC | ACACAGGCCT  | GTGCTGGCCA |
| 82741 | GCGGTGGGGA  | TCCGGCATTG | CTATCTTAGG  | TGCAGAGAGT | GACTGACTCA  | TTGCAGGCCT |
| 82801 | GGGAGATAAG  | ACTGATGGCC | CAGCCAGCAA  | GATGTATGGA | TTTCTCAGAG  | GCAGTGGCCT |
| 82861 | CTGTCAATTG  | CCTCAGGAAA | TGCTGGTGAT  | TCTGGTGGCC | TGAGGTCAAT  | GCATGTCAAC |
| 82921 | GTGGCCAAC   | TGCCTTATAA | ACTTTTTTTC  | TGGACAATTG | CGTGCACCTG  | CCTGTAACAG |
| 82981 | TGTCCTGTTG  | TTTATGATGC | AGAAATAGGT  | GTTTTTAAAG | CCTATTGATT  | TTGGTACTAT |
| 83041 | TAATGTGGTC  | AGGAACCTTC | TCAGTCTTTC  | TTGTTTGGGG | TGAGCTGTGC  | CTTCTAAAC  |
| 83101 | AGGAACCCAA  | GACACCCCA  | AAAGCTGCTC  | ACCAGCACTG | CCAGCCTCCC  | TCTTACCAAG |
| 83161 | TAGCACCCGT  | TCAGGACATT | CTGCGAAAGG  | CATTTGCCCA | GAGTTGGGA   | GGAAGGAAAT |

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| SEQ ID NO | Sequence                                                           |
|-----------|--------------------------------------------------------------------|
| 83221     | GTAACATTTT GGGGCACCTA CCATATGCCA GGCACCAGGC TAAACGTGTT CACACAAATT  |
| 83281     | CTCTTACTAA CCCTCACCAT CCTTCTACAA GACAACTAG TATCTTCATC TTGGGGTTCA   |
| 83341     | AGATGAGGAA ATGGAGGCTC AGAGAGGTTG AATGAATGCC GGTGCCTGGA TATGAACCCC  |
| 83401     | ATCTGCCTGA CTCGCAACC CAGGCAAAGT CTTTCCTTGA ACTTCCAGC AGCCACTGCT    |
| 83461     | TAGACACAGC CTCCACAACC ATGGCTCAGC AGCAAATTGC TTCTCTGACC TCACTCAGCC  |
| 83521     | TGTGTGTCTT TGTGAGTGA GGCATTGAGG ACCCTGGTCC CAAAGTGGAG AAAGTCTTTC   |
| 83581     | CTACTAGGTC ATAGCTACAC CTGCATGTGG GTGCTGTGCC TTTTGTTTAG TGAACTTTAA  |
| 83641     | TCACCAGCAT CCTCAGCAAT GACATTGCA GAGAAGCCAG AGCTGAGGCA CCTTGGTATT   |
| 83701     | CTTGGGATGT GACTTTCCTG AATGTTTAAAG GGAAATGCC CGAAGGTACA GAGAGCTTGG  |
| 83761     | TTTCTAGTAA ACAATAACTG TCTTGCTTTT ACCCCCTTTC ATTTGCTGAC ACATACACCA  |
| 83821     | GCTGAAGAAG CAGGCATTGG AGACACCCCC AGCCTGGAAG ACGAAGCTGC TGGTCACGTG  |
| 83881     | ACCCAAGGTC AGTGAAGTGG AATTGCCTGC CATGACTTGG GGGTTGGGGG GAGGGACATG  |
| 83941     | GGGTGGGGTC TGCCCTGAAA AGATCATTTG GACCTGAGCT CTAATTCACA AGTCCAGGAG  |
| 84001     | ATTTTAGGGA GTTGGTTCTT ATCAAAGGTT GGCTACTCAG ATATAGAAAG AGCCCTAGTG  |
| 84061     | GTTTTTTTCT AATACCATTT CTGGGTAATT CCTAAGGCAT TTAGTGTTC T GAAAGATGCT |
| 84121     | AGCCTTGTC AGCCTGGGAG TTGAGAATGA ATGTCTAACA GAAACTCTAG GCCGGGCCTG   |
| 84181     | GTGGCTCACG CCTCTAATCC CAGCACTATG GGAGAGCCAG GTGGGCAGAT CACCTGAGGT  |
| 84241     | CAGGAGTTTG AGACCAGCCT GGCCAACATG TGAATCCTG TCTCACTACA AATAAAAAAA   |
| 84301     | TTAGCCGGGT GTGGTGGTAG GTGCCATAAA TCCCAGCTAC TCAGGAGGCT GAGGCAGGAC  |
| 84361     | AATCCCTCGA ACCCAGGAGG TGGACGTTGC AGTGAGCCGA GATCGCATCA TTGCACTCCA  |
| 84421     | GCCTGGGCAA CAAAAGCAAA ACTCCGTCTC AAAAAAAAAA AAGAACTCA AATATGTGTG   |
| 84481     | ACAGGCGATT CTCACCTGAG GCTGCCCTGT GGCTGATCCA GGAGCAAGGC CTTAACCATG  |
| 84541     | TCATCCCAAA GCGATTGCTT GTAACTTTC TTCTGTGCAG CCTCAACCC TTATTATGAT    |
| 84601     | TTTCTTCTCA GGAACCAAA TGTCTGTATT AAGAAAGGCA GCTTTGTGTA ATCATTATC    |
| 84661     | ATAAATATCT TAAGAAAAAT CCTAGAGATT CCTAATTTTA GGAATGGGA GACCTATGGT   |
| 84721     | ACTGATATAA TGTGGGCTGG GCTTGTTC TGTCTATTGC TAGATAAATG AACTTGAGAG    |
| 84781     | CCTACTGTAA AATGTGGAAG CTTCTAGATT GCAGAAGGGC TGGAAAGACA CTGTTCTTTT  |
| 84841     | CTCCCGAGTC ATGGGATCTG TCCAGTATTT AGAGCTGCCT CTGAGGCCAT CTGATTCTAG  |
| 84901     | GAGACTCTGC CTCGTTGAGG ATATTTTGAG GCCTAACTAC ACATTCTGCT CCCAGAGAG   |
| 84961     | GTCACAGCCT ATAGCAGGCT GATGTTTCTC ATGTCACATG GCACAGAAAG GCACATTTTC  |
| 85021     | GTTCTCAGGC TAACAAAGAG CTTCAAAAAC TATTAGAAGG GACAGTGGCT ATAAGAGAA   |
| 85081     | AACCTCAGTC AATGTGTGAA ATTAAGTAGG AACCTGGCTC CTGTTTCTTT TAGGTGATGT  |
| 85141     | TTTTCAGCTT AGGTAAAACT AGAGGCTTTG ATAAAGCATG ACCTCTAGAA ATCATTGCTT  |
| 85201     | TTCAATAATG GAAGTGGGTT TGAGTTTTTT CTACTGATTG TTAGTGACAG TGATGCTTAC  |
| 85261     | ATGCCCCCAG AACATATTCC ATGCAACAAA AAAAGCCAG GTACCCGCTC TTGCTGGGAA   |
| 85321     | CTTGACTTTT GTGCTCACTG AATTTTAAAG TTTCTGACAG CAGCCTGGAA TCATGGAGGG  |
| 85381     | ATAAAGTACC TATTAGTAAG ATGAAAAAAG GTGTTTCAGG TTGGAGCTGC AGTCTGTTGA  |
| 85441     | GAGTAAGCTA TGGGAAGGCC TGTATACGAG GGGTGGACTT TTCTTCTGTA AGTGTCCAGA  |
| 85501     | GACCAGGCCT CCTGAAGAGG GCATGGGGGC TTAAGTTACC TGGACTACTG TGTTTACAAT  |
| 85561     | ACTCATTTAT CTTGAAGTCC TCCTAACCCC TGAGAATTGC TACATTTAGT ATTTGCTGAG  |
| 85621     | TACTTCCTAG CATCCTAGGG AATCAATAGA ACATTCTCCC AACCAGGCTG GGTGCGGTGG  |
| 85681     | CTCATGTCTG TAATCCCAGC ACTTTGGGAG GCCAAGGTAG GCAGATCCCT TGAGGCCAGG  |
| 85741     | AGTGCAAGAC TAGCCTGGCT GACATGGTGA AACCCTGCTC TTACTAAAAA TACAAAAGTT  |
| 85801     | AGCCAGGCAT GGTGGTACAC ACCTGTAATC CCAGCTACAT GGGAGGAGTA GGAGGCAGGA  |
| 85861     | GAATTGCTTG AACCTGGGAG GTGGAGGTTG CTGTGAGCCG AGATCATGCC ACTGCACTCC  |
| 85921     | AGCCTGGGCG ACAGAGTGAG TGAGACTCTG TTTAAAAAAA AAAAAAAAAA GAACATTCTC  |
| 85981     | CTAACCTGCG TTCTTCTCCT AGGGGTGTAA TTAATCATGT CAGTTTCTCTC ATTGATACAC |
| 86041     | ACACACACAC ACTACAATCC TGTATCCATT ACTTTTCAAG GTACATTTAC TATTTACGTT  |
| 86101     | TGGGGTCTCT GTCTCTTTTT TAATAGTGTT TCTTAAAGTC TTGTATTATA TCAGAGTACA  |
| 86161     | GTAAACATCC AGTCAAGAGC ACTCTAGTAA GCTCTAGGAG GAAAGCGACT TCCGGAAGGC  |
| 86221     | AGTGGAGACC TGTCTGTGTT GGGCAGCATA GGGGCAGCCC CTGCCTCTGG TCAGTTCTGG  |
| 86281     | CGCTCAGGCT CAGGGTTGCC TCTGGGCTGT TCTTCCCAGA GACTGACAAA GGGCTCCCAT  |
| 86341     | AAGGCACCTG CAGAGCCTGT GAGAAGCTGA AGTCAATGTT TTCCTGACAC CAGTTGATCT  |
| 86401     | GTGCAGGATC CATTGATTTA ACCACCTGCT GTGTGGCATG CACTGTGGTC GATGCCAGGA  |
| 86461     | ACAGGAATTG GAGGGGCCCA TGAGCATGGC CAGTATCACA GGCTGGAGGT GCTGCTGCGC  |
| 86521     | CTTGACCGGG CCTCTTGGGG ATGAGCCCAT GTCACCACC TTGCCTCCGA TGGGGTCGGG   |
| 86581     | CCCACAGGTT ACCTTTGTGT GTCCATGACC ACACCTTCTC CCCCAGCTC ATCCAAATCT   |
| 86641     | CTTTCTTTTC CAAGCCCTG AATCCTTCAG GGCTGCAGGT TTTGTTTAAA GCAGAGCTGG   |
| 86701     | TGAGTTGCAT AGGTGTGTGC ATTGGGACTA GATGGGGTGT TCAAGAGAGT GGGAGTTAAA  |
| 86761     | AAACATAAAG GGTATTTATT AGGAGAACCA AGGAGTGTA TTTCTCTGTT CTTAATATGC   |
| 86821     | GGCCAGGTTA ATGAATGTCA CGTGAATGAA CCAGAAAAAA ATGAAGTGTG CCCTTGATCA  |
| 86881     | GCTGGGTTGG TGTGCAGCAA GCTGTGTGAC CAGGGGACAG CAGTGGTCTC GAGGGCCGTC  |
| 86941     | ACTGTCTGCC GTGCAGAGCC CTTCTCCCA CGGGGGCCTA CCTCACCTGT GCCAAGGGCT   |
| 87001     | TGTCTGTGGT CAGTGACCTG GATAGATCTG AATGGGGCTT CTTTTCGAG GAGTCTTATG   |
| 87061     | GCAGGTCTCT CAGTAAAGAC TCCATTCTTG ATGATCACAC ATTTTGGATT TTCCAAATCT  |
| 87121     | GTCAAGAAAT GGGCTTGAGG CGGGGTTTGT GGGCACTAGT TTCACTGGTT TCATTTACCA  |
| 87181     | AAAAGGGGAG CAGAAGTCAA GTATGGTGGC TCATCCCTGT AATCCCAGAG GCAAGAGAAT  |
| 87241     | TGCTTGAGCC CAGGAGTTCG AGACCAGCCT GAGCAACATA AGGAGACCCC GTCTCCACAA  |
| 87301     | AATGAAAAAA TAACATTTTA GTCAAGCTG GTGGCATGCA TCTGTGGTCC CAGCTGCTTG   |
| 87361     | GGAGGGTGAG ATGGGAGGGT TGTGAGGCC CTGGAGTTAA AGTTGCAATG AGCTGTGATT   |
| 87421     | GCACCACTGC ACTTAGCCT GGGTGACAGA ACGAGACCTC GTCTCAAAAA AAAAAAAAAA   |
| 87481     | GAAAGAAAGA AAGGAAAAAA AAAACTCATG CCTGTAATCC CAGCACTTTG GGGACCGGGG  |
| 87541     | TGGGCAGATC ACGAGGTCAG GAGATCAAGA CTATCCTGGC CAACATGGTG AAACCCCGTT  |
| 87601     | TCTACTAAAA ATACAAAAAT TAGCCAGGTG TGGTGGCAGC TGCCTGTAAT CCCAGTTACT  |

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SEQ ID NO Sequence

|       |             |            |             |             |             |             |
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| 87661 | CGGGAGGCTG  | AGGCAGGAGA | ATCGCTTGAA  | CCAGGGAGTC  | AGAGGTGCA   | GTGAGCTGAG  |
| 87721 | ATCGTGCCAC  | TGTACTCCAG | CCTGGGCGAC  | AGAGTGAGAC  | TCTGTCTCAA  | ACCAAAAAAA  |
| 87781 | AGGGGTGGGG  | GGCGGGGGCA | GGAGAACAGT  | GAGAGGTAGG  | GAGAGGAAAG  | GGGATTCTCG  |
| 87841 | CTACACCCAA  | ACCAGATACC | ATCTAGAGGC  | TAGAATCTTT  | GGGAGGCTCA  | AATTCCTTAG  |
| 87901 | AAAGCAGGAG  | AAGCTTCTGT | AGCCCTCCCG  | CTTTCCAGT   | AGATTAAGCC  | CAGGGCGGCT  |
| 87961 | CCAGATGTGT  | GACATGCTCT | GTGCCCAACC  | AGAGCCCATC  | ATAGGCAGAG  | GAATAACACC  |
| 88021 | CACACCAGAA  | GGGCCCTCGG | AGGTCACCAC  | GTCCAAGAAC  | CCTCTTTACA  | GATGAGGAAA  |
| 88081 | CTGAGGCCCA  | GAGAGGGGAG | AGCCACCTAG  | CGAGCTGGTG  | GCGGCTAGAC  | CAGGAGAGCT  |
| 88141 | GTCATTCCAA  | GCAAGCAAAG | GCAACGAGAC  | GAGCCCAGAG  | CTGTGCTCCC  | ATCTCTTTGT  |
| 88201 | TAGGGGGGCT  | GGGATGCCCT | CTCAGTGTCA  | TTTTGTCCAG  | GATGATGCTC  | CCTCTCTTAA  |
| 88261 | GCGATTAAAT  | CGCCCTTGCT | AACCTTTTGC  | TATCGCTGCC  | TCTTCAAACC  | AGAGGAGTTG  |
| 88321 | AGAGTTCCCG  | GCCGGCAGAG | GAAGGCGCCT  | GAAAGGCCCC  | TGGCCAATGA  | GATTAGCGCC  |
| 88381 | CACGTCCAGC  | CTGGACCTTG | CGGAGAGGCC  | TCTGGGGTCT  | CTGGGCCGTG  | CCTCGGGGAG  |
| 88441 | GAAGAGCCCA  | AAGCTCCCGT | CCCGCTGACC  | GCGAGCCTTC  | CTCAGCACCG  | TCCCGTTTGC  |
| 88501 | CCAGCGCCTC  | CTCCAACAGG | AGGCCCTCAG  | GAGCCCTCCC  | TGGAGTGGGG  | ACAAAAAGGC  |
| 88561 | GGGGACTGGG  | CCGAGAAGGG | TCCGGCCCTT  | CCGAAGCCCG  | CCACCCTGCG  | GTATCTCCAC  |
| 88621 | ACAGAGCCCT  | AAAGTGTGTA | GGTGGTCCAG  | GAAGGCTTCC  | TCCGAGAGCC  | AGGCCCCCCA  |
| 88681 | GGTCTGAGCC  | ACCAGCTCAT | GTCCGGCATG  | CCTGGGGCTC  | CCCTCCTGCC  | TGAGGGCCCC  |
| 88741 | AGAGAGGCCA  | CACGCCAACC | TTCCGGGACA  | GGACCTGAGG  | ACACAGAGGG  | CGGCCGCCAC  |
| 88801 | CGCCCTGAGC  | TGCTCAAGCA | CCAGCTTCTA  | GGAGACTCTG  | ACCAGGAGGG  | GCCGCCGCTG  |
| 88861 | AAGGGGCGAG  | GGGGCAAAGA | GAGGCCGGGG  | AGCAAGGAGG  | AGGTGGATGA  | AGACCGCGAC  |
| 88921 | GTCGATGAGT  | CCTCCCCCCA | AGACTCCCTT  | CCCTCCAAGG  | CCTCCCGCAG  | CCAAGATGGG  |
| 88981 | CAAGCCTCCG  | AGACAGCCGC | CAGAGAAGCC  | ACCAGCATCC  | CAGGCTTCCC  | AGCGGAGGGT  |
| 89041 | GCCATCCCCC  | TCCCTGTGGA | TTCTCTCTCC  | AAAGTTTCCA  | CAGAGATCCC  | AGCCTCAGAG  |
| 89101 | CCCGACGGGC  | CCAGTGTAGG | GCGGGCCAAA  | GGGCAGGATG  | CCCCCTGGA   | GTTCACGTTT  |
| 89161 | CACGTGGA    | TCACACCCAA | CGTGCAAG    | GAGCAGGCGC  | ACTCGGAGGA  | GCATTTGGGA  |
| 89221 | AGGGTGCAT   | TTCCAGGGGC | CCCTGGAGAG  | GGGCCAGAGG  | CCCGGGGCCC  | CTCTTTGGGA  |
| 89281 | GAGGACACAA  | AAGAGGCTGA | CCTTCCAGAG  | CCCTCTGAAA  | AGCAGCCTGC  | TGCTGCTCCG  |
| 89341 | CGGGGGAAGC  | CCGTGAGCCG | GGTCCCTCAA  | CTCAAAGGTC  | TGTGTCTTGA  | GCTTCTTCGC  |
| 89401 | TCCTTCCCTG  | GGGACCTCCC | AGGCCCTCCA  | GGCTGCGGGC  | ACTGCCACTG  | AGCTTCCAGG  |
| 89461 | CCTCCCGACT  | CCTGCTGCTT | CTGACGTTCC  | TAGGACGCCA  | CTAAATCGAC  | ACCTGGGTGC  |
| 89521 | AGCTGCTCCA  | CTCCCTCGGC | CTCCTCCCGT  | GCTCAGGCTG  | TGGCCGCACG  | CGCCCTCAC   |
| 89581 | GCTTGCCCGC  | CACTCTGCAT | GTCAACAGCA  | CCCCCGCTCC  | GTGCTACCCA  | CCTTGTTTGA  |
| 89641 | CTCTCTGCGC  | ACTTGATTGT | TCCACAACGG  | CCCATCAGCC  | CACAGGAGGT  | TTGGTGGGTG  |
| 89701 | CCTTCCACCG  | ACAGGATGAC | GGGTGCCCTC  | ATGGTGTCTA  | GAACCTCTCA  | ACCCTCCCAT  |
| 89761 | GTAGGCATAA  | GCAGCCCCAC | TTTGCAGATG  | AGGAAACGGA  | GGCTCAGAGA  | AGTACAGTAA  |
| 89821 | CTTGCCGAAG  | GCCAATGAGT | AGTAAGTGAC  | AGAGCCAGGT  | TTGGGATCCA  | GGTAGGTTGT  |
| 89881 | CTCTGAAGA   | CACGCCTGTC | CTGCATCCCA  | CAACGCCTCC  | CAGGAGGTGC  | TGGAGTGTGG  |
| 89941 | ACGCCTAACA  | CAGAGATGTG | CAGGGCACAC  | ACAGCAGGTG  | ACACACACAG  | CATCCAGAGG  |
| 90001 | TGGCCAGAG   | CTCATGCTGT | GCCTTTGGCC  | CAGTGCCCTG  | CCCCACCCA   | CTCTGCCTTG  |
| 90061 | TGGCAGGAAG  | ACAAGGAGCA | GACACAAGAT  | CTCCCTGGTC  | CACATGCCAC  | CACCTCCCTC  |
| 90121 | TGCAGAGGAC  | AAGGGGATCC | TCATGCTGGC  | ATTGGAGGGG  | GTTGAGCAGG  | GCCCACCTTG  |
| 90181 | AGCCCTCAGG  | AGCACGACCA | CAGCAGCCCT  | GCAGGGAGGG  | ATTGGTGGGA  | GGAGAGTCCC  |
| 90241 | AAGTATCAGG  | GAGAGGAGAG | TTGGTGTCCC  | ACAGGAGACC  | TCAGAGCCAC  | AAGGCCGAGCT |
| 90301 | TGTTTCATAA  | TTTGGGACCC | TTAGCATTTC  | ACAGTTATTT  | GCAGAGCCCA  | GAAATGGATG  |
| 90361 | TTACTGAAGC  | TCACAGTTGC | AAGCATCTGT  | TAAATTTTAA  | TTAGATTTTA  | CTTTTAGGGA  |
| 90421 | AAACTTTGAA  | ATGCTATAAA | GAAGCCTGTG  | TTTAAAGATT  | AAGACAGAGG  | CTGGGGGCGA  |
| 90481 | TGGCTCACGC  | CTGTAATCTC | AGCACTTTGG  | GAGGCCAAGG  | CAGGTGGATC  | ATTTGAGGTT  |
| 90541 | AGGAGTTCGA  | GACCAGCCTG | GCCAACATGG  | TGAGACCCTG  | TCTCTACTAA  | AATTACAAAA  |
| 90601 | AATTAGTCTG  | GCCTGGTGGC | GGGCACCTGT  | AGTCCCAGCT  | ATTGGGGAGG  | CTGAAGCAGG  |
| 90661 | ATAAGTGCTT  | GAACCCAGGA | GGCAGAGGTT  | ACAGTGAGCC  | AAGATCACAC  | CACCTGTACCC |
| 90721 | TAAGCCTGGG  | CGACAGAGTG | AGACTCTGTC  | TCAAAAAATA  | AAATAAAATA  | AAGTTAAGAG  |
| 90781 | AGAAAAAAT   | ATATCCTATA | TCCTTTGTTA  | AATTCCAAAA  | CAGTAGGGGA  | CAAATAACTG  |
| 90841 | ACTTGACAGG  | TTACTACAAT | ATTTCTTGAA  | ATGATGTTTT  | CTTGAATACT  | GGCCTACTAG  |
| 90901 | AGGTTTCATAG | GTGTGTTTGG | ATTAAAAAAG  | AGTTTCCATGG | CCCAGTGACT  | GGGGGAAAAA  |
| 90961 | AATAAAGGAC  | TAAAGTAAGT | TAAACAGGCT  | TTTCTGCTGC  | AGGACTTGTC  | AGAGCCTTTA  |
| 91021 | ATGTACTAAT  | GGCCATTGTG | ACCCTCTGAG  | AAGGTCACAG  | AGTGGGTTTC  | CCAACTTAC   |
| 91081 | TTGATTCTAC  | CTGCTAACAT | TTCTTGAGAG  | AAGTTTGGGA  | AATGCCGATT  | TAGCAGATTG  |
| 91141 | TTTTGTGTG   | CCGTGGATGG | TGCTGGTTGA  | TGTGGGCAAA  | ACAAAGAACCA | CGTGAGTCAG  |
| 91201 | ATCCGCTTGG  | GGCTCTTACT | AAAGTGCAGG  | TTCCAGGTG   | CCACTTTAGG  | CTTACAGACC  |
| 91261 | CAGTTGTGGG  | GTAAGCCTGG | GAGTCTTTTA  | GCAGGTGATT  | CTGCCACATA  | GTATAGTTGG  |
| 91321 | AAAACCTCTG  | GGCATACTCA | TTGCTGGTCC  | CTCTAGAAAT  | CAGGTTGACA  | ATAGCCAATG  |
| 91381 | AGAAGCTCCA  | AGAGACCCAG | TTGTCCATGG  | GGTAGAGGGA  | ATGTGATATT  | GAAACCAAAG  |
| 91441 | AAGAAAAATCT | ATGATCAGTT | TTCAGCAGTG  | ACTGTCAAGA  | GAAGGAGAAG  | GGTGAGTTAG  |
| 91501 | CGCTGATGCT  | GGCTGACAGG | TCAGCGGGTT  | GGTTTCACCA  | AGGAGTGTGA  | TGAAGGCTGA  |
| 91561 | TGTTGTCTGT  | GGGAATGTAT | GATGGTAACT  | GGTTGTAGC   | TAATTTGGGG  | AAGCAGTGAG  |
| 91621 | AATTCGTGCC  | CTTTGAAGAC | CAGTAAGTGG  | CAAGAAACCC  | ACCAGGCCCTG | GCTCAGGGCT  |
| 91681 | GGGCTGGGCT  | TGGCTCGTCT | CAGAGCAGCT  | GGGGCTGGTG  | GCCAAAGCCA  | CCATTAGTGA  |
| 91741 | GGGGCAGGCC  | CTGGGGGTAC | AACCAAGCAAC | TAGGGGACAA  | AGACAAACCT  | GCCAGCCTCT  |
| 91801 | CCTATTCTGG  | AGGCGTGTGA | CCAGAAATGG  | AGATGGGTTG  | GTCAGCATAA  | GATGGCCAGG  |
| 91861 | AAGGTGGA    | TCAGGACTGC | TGGCAATCTA  | GCCACATGGG  | CAGGGGAGCC  | GGGTGGTTCC  |
| 91921 | AGGCAGTTTC  | CAAGGCCAAG | AGGGTGAGCA  | GGCACCTCAC  | AGGGAATCAG  | GGCCAAGCCT  |
| 91981 | GGCTGCAGTG  | TGGAGACAAT | GCACCCACCC  | CCATCCTTGG  | ATCTTGACAG  | AGGCTGGGTC  |
| 92041 | CTCACTGAGC  | TACCAACATC | CATGGCCCTG  | AGGCTTTTAA  | AACACCCATC  | CATGGAGTGG  |

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| 92101 | GGCTGGTCCC  | AGTGGGGTGA | GGCTGACCCT | GGCAGAAACA  | GGGCAGGAGC  | CTGTGGGTGA  |
| 92161 | GGGAGACTGC  | ACCTTCCTTA | GATAGCCTCC | ATGCCATCAT  | GTCCCCGTGA  | CAGTTTCTGC  |
| 92221 | TGCGTCCCTT  | CTGCATGGTC | CCACCCTCGG | CCAGCCTGCT  | GCCCCCTCTT  | GCCAGGTTGC  |
| 92281 | GCTAATCAGT  | GACCCCACTG | TGCTGTGTTG | ATACTAACAA  | TGCGAGGCCT  | AGCAGATTCA  |
| 92341 | AGGGAAAAGA  | GAACCAACTG | GGTTTCCACC | AGACCCAAC   | AAACAAACAT  | GGACCTATCC  |
| 92401 | CAGAGAAATC  | CAGCTTCACC | ACAGCTGGCT | TTCTGTGAAC  | AGTGAAAATG  | GAGTGTGACA  |
| 92461 | AGCATTCTTA  | TTTTATATTT | TATCAGCTCG | CATGGTCAGT  | AAAAGCAAAG  | ACGGGACTGG  |
| 92521 | AAGCGATGAC  | AAAAAAGCCA | AGGTAAGCTG | ACGATGCCAC  | GGAGCTCTGC  | AGCTGGTCAA  |
| 92581 | GTTTACAGAG  | AAGCTGTGCT | TTATGTCTGA | TTCATTCTCA  | TATATAATGT  | GGGGAGTATT  |
| 92641 | TGTCACATAA  | GTACAGCTGT | CATTTAAAGT | GCTTTGTATT  | TTGGGGCAGG  | CTTTTAAAAA  |
| 92701 | GTCCAGCATT  | TATTAGTTTT | GATACTTACC | CCAGGGAAGA  | GCAGTTGGCA  | GGTTCATGAA  |
| 92761 | GTCATGCTCC  | TAATTCCAGC | TTTCTTAGTG | TACTTTCAGT  | GAGACCTGTA  | CAGTAAATGA  |
| 92821 | AGGTGTGTTT  | GAAAACCAAA | CCCAGGACAG | TAAATGAAGG  | TGTTGTTGAA  | AACCAGCCCT  |
| 92881 | ACGACAGTAA  | ATGAAGCCAT | CTTCTCACTG | CATAAACTGC  | ACCCAGATCT  | TCGCCCATCC  |
| 92941 | TTCTCAGTAT  | TTCACTTCAC | CCATTGTTTA | CTGTCTCAAT  | GACTGGGGAA  | ATGTCTGGGG  |
| 93001 | AAATGCTCCC  | GTAATTGCAC | AGTGGCGTTT | TTCCCTGGAA  | ATCCCCACAT  | GGCTCTAGAT  |
| 93061 | ACGACCTGAT  | TTTCTTAAAG | GTATCTAAAA | TTTCCAGCAT  | AAATTCTGTC  | TGAAACACCT  |
| 93121 | GAATTTTAAAT | CAGTACTGGA | GCCCGGAGGG | CATCTCCAGT  | TGCCACATAG  | CTCTGAGCAT  |
| 93181 | TCAGTGGTGT  | GTTGAGGGCT | GCTCCCGGAA | GTGCTGCAG   | AGTCAGGGCT  | CCCCAGCCTC  |
| 93241 | ATCTAGTAGG  | GCAGTGAAG  | GGCCTGTGGG | GATTTGGAGA  | GCTGGCCTGG  | GTCTCTGAAG  |
| 93301 | TGATAGTGAC  | AGCTGCTTGT | CAATCACGGT | GCACATTTAG  | TGCTGGGGGC  | AGGGGGCAGG  |
| 93361 | GAATACCAGC  | CTCATGTCAT | CATGCATTCA | TTTGTCTCTT  | CCTTCATTCA  | TTTATTTCAGT |
| 93421 | ACACATGGGT  | ACAACATCCC | TGCCCTGGAG | TGCCCCAGAG  | TCTAGGGAGG  | GGAAAGATCT  |
| 93481 | ATTACCCCTGG | GCCTCGGCCA | GCTGGGGAGT | GCTGCTGGTG  | GAGAGGGGCC  | GTGTGCAGCG  |
| 93541 | AGGGAAGGAG  | GAGTCGTCAA | TACCCCCACC | CCAGCTTTGC  | TTTCTTGTC   | TCAGCCCCAG  |
| 93601 | GGCCCCAGCC  | TGTGTCCCTC | CTCTCCCAT  | GCTACTTCAT  | CTCCTGGGTC  | CTCCTTACCA  |
| 93661 | AGCCTGACCA  | CACAGAGGGC | CTTGGCCGCT | TCCATGGGGA  | ATTGGAAAGC  | AATAAGATAG  |
| 93721 | CATCCCCCTAG | AAGCCCAGTG | AAGTCTGGGA | CAGGACCCCT  | CTCTGAGCTC  | TGACTTGCTC  |
| 93781 | TTGGAACAC   | TTGAGGCTT  | AGCCTCCCA  | CTTTGTTTCC  | CGAGAGTGTG  | ACCTGTTCCC  |
| 93841 | CTCCAAACAC  | CCCCTTCTCC | TCCAGGGCCA | TGCCACCCCG  | TCAAAATCCC  | CCACGGGCAG  |
| 93901 | GACGAAGTGT  | GGGTGTGAGT | CACCATCTAT | CCTGCATCCT  | GGTTCACGGG  | CCCCCCCCAG  |
| 93961 | CCCCCGCTCT  | ATAGGGACAG | GCGTGCAGAC | ACCCGTCCTT  | GGCTGCTTCC  | TCTTGTGGA   |
| 94021 | TGGGTTCAAA  | AGTAAGCAGT | GTTGTTTACA | CTGACAAACT  | GAAGAAAAAA  | GAAGAAAGAGA |
| 94081 | TAACATTGGA  | GGCTTGGCAC | AGTGGCTCAT | GCCTGTAATC  | CCAGCACTTT  | GGGAGGCTAA  |
| 94141 | GTGTGGGAGA  | TGTCCCCAGC | CCAAGAGTTC | TAGACCAGCC  | TGGGCAACAT  | AGCAAGACCC  |
| 94201 | CATCTCAAAA  | AAAAAATTTA | ATTGGCCAGG | CAGAGGTGGG  | AGGATCACTT  | GAACCCAAAG  |
| 94261 | GGTGGAGGCT  | GCAGTGAGCC | GTGATGGCAC | CAGTGCACCT  | CAGCCAGGGC  | AACAGAGGGA  |
| 94321 | GACCTGTGCT  | CTAAAAACAA | CAAAACAAAC | AACAAACAAA  | AGAGTTAACA  | TTGGCCAGAT  |
| 94381 | TAGGATTAC   | CAGATAGTGT | TAATATTAGT | TTGATTTGAG  | ACTTTAATCA  | GAAGGCACAT  |
| 94441 | GTGTGGTGGG  | GGTGGGTGTA | ACCTAAGTCA | GGTAGAATCT  | TTCCAATCTG  | GGGGGGGCAC  |
| 94501 | ACTCTCTGAT  | GTAGCCATAT | GAGTCTGTCA | GTGTGGTGGG  | AGAGGCCATG  | GGTTAATGGG  |
| 94561 | CAGGTAAAAA  | AGCACCTTGC | CTGGAATTGA | GTAGAAAGTA  | AGGCCCTTCA  | GACCCCGTGA  |
| 94621 | CACACTTGGG  | GACATTTTCT | TGAGTAACAT | CCTAAGATTG  | ATGTACCTTG  | ATGATCTCCA  |
| 94681 | TTCACTTACT  | CATGTGAAGC | ACCTTTAAAC | CAGTCGTCTC  | CAAATTACAG  | GGCACAGTAA  |
| 94741 | CATCCAACAG  | GCTGGAGAAA | GAACGTACTA | GAACCTTCCAT | TCCTTTTTC   | TGTCTCTTTC  |
| 94801 | TAAAGCTTTT  | GTCAAGGCCA | GGCGCGGTGG | CTCAGCCTTG  | TAATCCAGC   | ACTTTGGGAG  |
| 94861 | GCCGAGACGG  | GTGGATCACG | AGGTGAGGAG | ATCGAGACCA  | TCTTGCTTAA  | CACAGTGAAA  |
| 94921 | CCCCATCTCT  | ACTAAAAATA | CAAAAAACG  | AGCCGGGCGT  | GGTGGTGGGC  | GCCTGTAGTC  |
| 94981 | CCAGCTACTC  | GGGAGGCTGA | GGCAGGAGAA | TGGCGTGAAC  | CCAGGAGGCA  | GAGCTTGCAG  |
| 95041 | TGAGCCGAGA  | TTGCACCACT | GCAGTCCAGC | CTGGGCGACA  | GAGCGAGACT  | CCGTCTCAAA  |
| 95101 | AAAGAAAAAG  | AAAAAGAAAA | AGAACTGTGA | TTGGGGAGGA  | CGGTCACTTT  | CCTGTCTTTA  |
| 95161 | CTGATCAGAA  | GGGATATTAA | GGGTACCTGA | TTCAAACAGC  | CTGGAGATCA  | CTGCTTTCAA  |
| 95221 | CCATTACCTG  | CCTTATTTAT | TTTTAGTTAC | TGTCCTTTTT  | TCAGTTTGTT  | TCCTCTCTCC  |
| 95281 | ATGTGCTGAC  | TTTTATTTTG | ATTTTATTTA | TGTTTATGTT  | TAAGACATCC  | ACACGTTTCT  |
| 95341 | CTGCTAAAC   | CTTGAAAAAT | AGGCCTTGCC | TTAGCCCCAA  | ACACCCCACT  | CCTGGTAGCT  |
| 95401 | CAGACCTCTT  | GATCCAACCC | TCCAGCCCTG | CTGTGTGCC   | AGAGCCACCT  | TCCTCTCCTA  |
| 95461 | AATACGTCTC  | TTCTGTCACT | TCCCGAACTG | GCAGTTCTGG  | AGCAAAGGAG  | ATGAACTCA   |
| 95521 | AGGTAAGGAA  | ACCACCTTTG | AAAAGAACCA | GGCTGCTCTG  | CTGTGGTTTG  | CAATGTGGG   |
| 95581 | GTTTGTGTTT  | TTGTTTTTTA | GCCTCAAGAA | CCTTTCTTCA  | AATGAGTTCT  | GGCATAGAAG  |
| 95641 | CACCGTGTA   | AATAGTTAGA | ATTCTGGGCA | AAGGGGAAAA  | GAGAGCTGGG  | GGCCATCCCT  |
| 95701 | CTCAGACCCC  | CACAGGCTCT | CATAGCAGCA | GCTCTTAAGA  | CACCTGGTGG  | GACCTTGGTT  |
| 95761 | TCGAAATCGC  | TACTCTAAGG | CTGGGCACGG | TGGCTCACAC  | CTGTAATCCC  | AGCTCTTTAG  |
| 95821 | GAGGCCGAGG  | AGGGTGGATC | ACCTGAGATC | AGGAGTTGGA  | GACCAAGCCTG | GCTAATCATG  |
| 95881 | CAAAACCCCTG | TCTCTACTAA | AAATACAAAA | ATTAGCCGGG  | CGTGGTGGTA  | TGCTGGTGG   |
| 95941 | TAATCGCAGC  | TACTCGGGAG | GCTGAGGCAC | AAGGATTGCT  | TGAACCCAG   | AGGCAGAGGT  |
| 96001 | TGATAGTTAG  | TCCAGCTTGG | GCGACAGAGC | AAGACCCCTG  | CGCAAAAAAT  | GTTTAAAAAA  |
| 96061 | CAAAACCCAAA | ATTGCTACTC | TCATTGGGTT | CCTTTGCCCA  | TTCTGTGATT  | TGGCAAGAGA  |
| 96121 | AATGCTTCCA  | GATTGCCCTG | ATCTGGGTAG | GACAGCATCA  | CGCCATAGCA  | ACACTGCCCT  |
| 96181 | GTGAGCTCAC  | TGCCCCCTCA | ACTAGCTTGT | GGTCCTTGTT  | TAATGTGCTG  | TTCTTTTTTG  |
| 96241 | AGTTTGTGTT  | ATGTCTAAGG | GTCATCTGCT | GGGTAACGGA  | ACCCAGGGAC  | TGCCCTAGTC  |
| 96301 | CCTAGACTGT  | GCCATGCCCG | ACTCTGCCAG | CTTTGTGAGT  | GATGCTGGTG  | CTCGCTCTCT  |
| 96361 | CGGGTGCTCG  | CCTGGTCTGA | GCACACCCAA | GGAGTTCCTG  | AGGCCTTAGG  | GTTGTTTGGC  |
| 96421 | AGAGAATGAA  | AGAACACGAC | CTAGCTCTCT | TTAGCATCTT  | TGGTCAGGTT  | CAACACTGCC  |
| 96481 | CCCAGGGGCC  | TCTGGTGGAG | CCAACCACCA | TCAGCCAAAT  | AAATCCATAA  | TTAGAGTCAG  |

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| 96541  | AAAATGGATG  | TCTGCATATG  | TGTAGTGCAC | TAATGTCCTG  | CCGATGATTG  | ACATGGAGTG  |
| 96601  | GAGAGTGACC  | TGATCATTGC  | TGTGAGCTCT | GCTGGCCTTG  | GCACAACCTCA | TGCTGATAAC  |
| 96661  | TAATGCACAC  | AGTTCCTCTG  | GGAGGAAATG | TCCTCAGGGA  | ACTTGAGGTT  | TGGGTGGGGA  |
| 96721  | TGTGGGTTTG  | TGTGCCCAGC  | AAGCCCTTGT | GGTTGTAGCA  | GACACTAGTG  | GCATCTAGGA  |
| 96781  | GGCAAAGGGT  | CACCCCAGTC  | TTAGCCACGT | TTTGAGTCAA  | GGTGGCGGAG  | TGGGGCTGGT  |
| 96841  | GTTGACTCTT  | GGTGGCAGTA  | ACTTTTCCCA | ATGGTGAAAA  | ACCCCTCTAT  | CATGTTTCAT  |
| 96901  | TTACAGGGGG  | CTGATGGTAA  | AACGAAGATC | GCCACACCGC  | GGGGAGCAGC  | CCCTCCAGGC  |
| 96961  | CAGAAGGGCC  | AGGCCAACGC  | CACCAGGATT | CCAGCAAAAA  | CCCCGCCCGC  | TCCAAAGACA  |
| 97021  | CCACCCAGCT  | CTGGTAAGAA  | GAACGTTCTC | TTGAATCTTA  | GAGGAAGCTG  | AAGCTCTCAG  |
| 97081  | AGGTACAGCC  | TTCATTTTAT  | GAGGCCCTAG | GCCACTGAGA  | ATGAATAACC  | CCTGGCAGCT  |
| 97141  | GGTCAGCAGC  | TTGCAGTTTA  | CTAAGCACTG | GAGTCTTCAT  | TGCCTTCTCA  | GTCTTTTGA   |
| 97201  | TTTCTGAGGC  | AAATGTTGAA  | TCCCTACCTT | TTTTTTTTTT  | TTTCTTTTGA  | GACAGAGTTT  |
| 97261  | CGCTTTTGT   | ATCCAGGCCG  | GAGTGCAGTG | GTGTGATCTC  | AGCTCACTGC  | ATCCTCCACC  |
| 97321  | TCCCAAGTTC  | AAGCGATTCT  | CCTACCTCAG | CCTCCCTAGT  | AGCTGGGATT  | ACAGGCACCT  |
| 97381  | GCCACTATGC  | CCGGCTAATT  | TTTTGTATTT | TTAGTAGAGA  | CAGGGTTTCA  | CCATGTTGGC  |
| 97441  | CAGGCTGGTC  | TCGAACGCCT  | GACCTCAGGT | GATCCACCTG  | CCTCGGCCTC  | CCAAAGTGCT  |
| 97501  | GGGATTACAG  | CGATGAGCCA  | CCACTCCAG  | CCTGAATCCT  | CACTTTTTAT  | CAATGAAGAA  |
| 97561  | ATTGAGGCTG  | ATTCTGCAGC  | ATGATAAAAA | AAAATACAGA  | AAAAGGAAAA  | AAAAGAAAGA  |
| 97621  | AATCGAGCCT  | CTGAGAGTTT  | GCTTGACTGA | GTCTAACACG  | CTCATTTTAA  | ACCCGAGGAA  |
| 97681  | AATGCAGTCA  | CATGACTACT  | AAGTGGCAGC | TCTCGGAGCC  | TCTCTGGCCC  | CAAGTCCAGG  |
| 97741  | GTTCCATAGA  | GGCAGCCCCA  | GCATGGCATG | TTTTCACTCC  | CCAAATGAGA  | CTCTGGAGAC  |
| 97801  | AAATGCTCTC  | GGAGACAGAG  | CAGCAGCCTG | GATAAGTCAC  | AATGGGTGAC  | GTCACTCAGG  |
| 97861  | TCCCAAGCCC  | TGGGCAGCTT  | AACCTGCTAG | GGACGTTAGG  | AGTCTGCTGC  | AAAACCTGAG  |
| 97921  | GGTCTTAGCT  | GAGCAGTCAC  | AGGCTGGGCC | CGTTGCCCTG  | GGCTCCTGTG  | AGTAAAACCC  |
| 97981  | AGTCAATTTT  | GAGTACCAG   | TAAGGCATCC | ATTGAGTTAT  | TTTGAGCCCA  | GGAGTGCTAT  |
| 98041  | TAAGAACAGT  | CGCGCTGGG   | CGTGGTGGCT | CATGCCTGTA  | ATCCCAGCAC  | TTTGGGAGGC  |
| 98101  | CAAGTGGGGC  | GGATCACCTG  | AGGTCAGGAG | TTCGAGACCA  | GCTTGGCCAA  | CATGGCAAAA  |
| 98161  | CCCCGTCTCT  | AATAAAAAATA | CAAAATAAAT | AGCTGGGCGT  | GGTGGCGGGC  | GCCTGTAATC  |
| 98221  | CCAGCTTCTC  | AGGAGGGTGA  | GGAAGGAGAA | TCACTTGAAC  | CCAGGAGGCA  | GAGGTTGCAG  |
| 98281  | TGAGCTGAGA  | TCGCACCATT  | GCACTCCAGC | CTGGATGACA  | AAAGTGAGAT  | TCCTTCTCAA  |
| 98341  | AAAAAATAAA  | AAAAAAACAG  | TCGTCCTCTT | TGGGGATTAG  | GGACAGCCTG  | CCTGCCTGCC  |
| 98401  | CGAGCAGTCT  | TCTCTTCCAT  | TGCCCCAGTG | AAGTATTCCA  | GGCCCCCTGG  | TTTAGACTCT  |
| 98461  | GCACCATGTA  | GGGGTGTCTG  | ACCTGCACTT | GCTCCTTGGT  | GGCACGGGCA  | GCCTATGGCA  |
| 98521  | CTTGCTGCGG  | GCTGTGACCA  | AAGCCTGGCC | TGGATCTTGG  | ATCTTGGTGA  | CTCTGCTTCT  |
| 98581  | CCCTGGCCTG  | AGGGAGCTGC  | CCAGAGCCTG | CCCACCACCT  | GCTGCGTGTC  | TTTGGGTTGG  |
| 98641  | CATTTCTCGC  | ACACATGCCG  | TGCAGTGGCA | CCCCCAAGGA  | TGGCCATTCA  | CTAAGGCCCA  |
| 98701  | TTGTTTTTGT  | CTTTTCGCTT  | CGTGTTTTCT | GGCCTGGTGT  | TTTTCTCATA  | TACATGTGAT  |
| 98761  | CGAGGGATAA  | TTCCAGAAAT  | TTTGACAGGA | TTTTAAGTAG  | CGTTTGGATC  | CTGCTGTTTT  |
| 98821  | TTTTTCACTT  | AACATCGGGC  | CAGTTGACTC | ACACTCTGTT  | TTTTGTGTTT  | GTTTTTTTGA  |
| 98881  | GACCGAGTCT  | CACGTGTGCA  | CCCAGGCTGA | AGTGCAGTGG  | CACAATCTTG  | GCATACTGCA  |
| 98941  | GGAGGTCTGA  | GGAGGCAGAG  | AGCAGTTTTT | CTGCCTCAGC  | CTCCTGAGTA  | GCTGGGACTA  |
| 99001  | CAGGCACAGG  | CCACCACGCC  | CTGCTAATTT | TTGTATTTTT  | AGTAAAGACA  | GGGTTTCACC  |
| 99061  | ATTTTGGCCA  | GCCTAGTCTC  | GAACCTCCTG | CCTCAAGTGA  | TCCGCCACCT  | TCGGCCTCCC  |
| 99121  | CAAGTGCTGG  | GATTACAGGG  | GACTCACACT | TTGTAACAAC  | CTGAAACAAC  | GTGATGCATT  |
| 99181  | TCCCTTTGGG  | TCTTACCTGC  | TCTTCGGTGG | CTGCCTGCAG  | TGGGAGAGAC  | CCTCCCCCTT  |
| 99241  | GGGCCCTTCG  | ACCTTGTTTC  | AGAATGGGGC | CCCTGCTGGG  | CCAGCTGTGG  | GTGCCTGCCA  |
| 99301  | CGTGAAGGAC  | TCATTAAGGC  | CCTGTTTAAG | CCTGATGATA  | ATAAGGCTTT  | CGTGGATTTT  |
| 99361  | TCTCTTTAAG  | CGACTAAGCA  | AGTCCAGAGA | AGACCACCCC  | CTGCAGGGCC  | CAGATCTGAG  |
| 99421  | AGAGGTACTC  | GGGAGCCTAC  | TTCCGCTGGG | GCAGCCTCCC  | TTTGCGTGTG  | TGGCCATTCA  |
| 99481  | CTGGCTTTGT  | TTTCTAGAGC  | CGGGAGGACC | CTTTTCTGCA  | ATGCAGGGGT  | CACACAGGGT  |
| 99541  | TCGCAGCCTG  | AAGATGGAGC  | AGTCCGAATT | CTCTTCCCTG  | TGCAGTTTGC  | GCAGCTGTGT  |
| 99601  | TTGCTGATG   | GGCTTTCTAA  | TCCTGTGTGC | TCTCCTTGAC  | TTGAGGGACA  | ATGGCATTAC  |
| 99661  | AGGCATGAGC  | CACCATGCCT  | GGCTGTCTCC | CTATGTTTCA  | GATGAAGACA  | TAGGCTTAAG  |
| 99721  | GAGGTCAGGT  | GACTTGCCCA  | CGACCCTCT  | GTAATAAGA   | GGCATGAAAA  | GTATTTGGAG  |
| 99781  | CCACCACCAC  | CAAGCCCACT  | GGTCACCTTG | GGTCTCTGAA  | GTCAGGGAGG  | CAGGAGGATG  |
| 99841  | GCGAGATGCA  | GGAGGCAGAG  | AGGCTGAGCC | TGGAGGCCCT  | GGAGGCCGAG  | GCCCCATCTG  |
| 99901  | TTGTTTCCTT  | ATGTGGAAAA  | TAAGAGGCTT | CGTTTGTCTT  | ATTGCCACAG  | AGCGTACTAC  |
| 99961  | TTCAAGGAACA | TCCAAGACAT  | GGAAATCCGC | AGGGCACGGT  | GGCTCACGTC  | TATAATCCCG  |
| 100021 | GCATTTTGGG  | AGGTTGAGGT  | GGGAGAATCG | CTTGAGGCCA  | GAAGTTCAAG  | ACCAGCCTGA  |
| 100081 | GCAACATAGT  | CAGACCCCGT  | CTCTATAAAA | AACATTATTT  | TTAAAAAAGA  | CATGGAAGTC  |
| 100141 | AAATTCTAAA  | AACTGGTGCT  | GGCTGGGTGC | GGTGGCTCAT  | GCCTATAATC  | CCAGCACTTT  |
| 100201 | GGGAGGCCGA  | GGCGGGTGGG  | TCACCTGAGG | TCAGGAGTTC  | AAGACCAGCG  | TGGCCAAACAT |
| 100261 | GGTAAACCTT  | CTACTAAAGA  | AATCTTTACT | GAAAATAACA  | AAATCCAGTC  | TCTACTAAAA  |
| 100321 | TAAGTCTCTA  | CTAAAAATAC  | AAAAATTAGC | CAGGCGTGGT  | GCTGCACACC  | TGTAATATCA  |
| 100381 | GCTACTCGGG  | AGGCTGAGGC  | AGGAGACTCG | CTTGATCCCA  | TGCAGCGGAG  | GTTGCAGTGA  |
| 100441 | GCCGAGATCA  | CGCCATTGCA  | CTCCAGCCTG | GGCATCAGAA  | TAAGACTCCG  | TCTCAAAAAA  |
| 100501 | AAAACCACAA  | AAAACAAAAA  | CAACAACAAA | AGAAAACTAG  | TGCTTATTCT  | TCACTGGCCA  |
| 100561 | AGCTGCCCAT  | TGGCTACATG  | GGTGCTTCAA | AGAGCTGCCC  | TTCTCCAGGT  | CTGGCCAGCA  |
| 100621 | GCTATGTGTT  | ACAGCAAAATG | CTGGGGCAG  | CGGCAGGGGC  | ATTGCTGCGG  | GAAGCTTCTG  |
| 100681 | GACTTGCAGG  | AAAGCTAAGT  | TCTCAGACTG | CAGGGGAGCT  | AAGCACACCT  | CGGCACAGGG  |
| 100741 | TGAGGCCTGC  | GGTCTCAGA   | CTTCAGTCTT | TGTGGAGCTT  | GAGAAAAATG  | AGGCTTTGCA  |
| 100801 | GGTCCCACCC  | CTAGAGATTCT | TGCTCTATCC | ACTCTTGAAG  | GGGATCGAGA  | AATTTGCATT  |
| 100861 | TTGCAACTCC  | CACCTTTCTC  | CTTGAAAGCT | CCGGAGATTCT | TGACGCAAGG  | TTCCGTGGGC  |
| 100921 | CACACTTTGG  | AAAATACAGA  | CCCATGAGAT | AGAATACCAG  | ACTGTTGAAG  | TGTAACGGGG  |

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| 100981 | GCCTGGGAAG | TGCAGTAACA  | GAAGCAAGTT  | TGAGGGTAAA  | GGACACCCAG | AGGAGGGAGG  |
| 101041 | GACAGCATCT | GCATGGAGAG  | GAGAAGAGAC  | CCCCCAGCAG  | CTTCCAGGGT | GTGGAAGGG   |
| 101101 | TGCGCTAGTA | ACTGCTATGC  | ATGGCAGGTG  | GGGAAGTGT   | CGTCAGGGCA | CAGCAGCATG  |
| 101161 | AAGCGGTATG | GCTCGTGTGG  | ACAGCTAGGG  | ACAGGCAGGC  | GTGGAGCAGG | CATCCTGTTC  |
| 101221 | TGAAGGCCAA | ATCCACACAGA | GGAGCCAGGG  | TGCTGGCAGG  | AGCCCTGAAC | TAGCCGAACA  |
| 101281 | GCTGAACAGC | TGAACATTCA  | CCCTGTGGGG  | AAAGGGTCAG  | AAGCGTCCAG | GCTTGAGGGC  |
| 101341 | ACAGCTGGGT | CTCGTCACTG  | CATCACCCCT  | ATTTAGGATA  | AAGGCCCTGA | AGAATTGTAT  |
| 101401 | TAGAGGTTGG | CAGAAGCATAT | CTACCACCTC  | CTGGAGCCAC  | GCTGGCCGCA | GGGATTATAA  |
| 101461 | TTATTTCCAT | TTTCAAATTA  | AGGCCCTCTGA | GCTCAGAGAG  | GGGAAGTTAC | TTGTCTGAGG  |
| 101521 | CCACACAGCT | TGTTGGAGCC  | CATCTCTTGA  | CCCCAAGACT  | GTGGAGCCGA | GTGGGCCACC  |
| 101581 | TCTCTGGGAG | CGGTATTGG   | ATGGTGGTTG  | ATGGTTTTCC  | ATTGCTTTCC | TGGGAAAGGG  |
| 101641 | GTGTCTCTGT | CCCTAAGCAA  | AAAGGCAGGG  | AGGAAGAGAT  | GCTTCCCCAG | GGCAGCCGTC  |
| 101701 | TGCTGTAGCT | GCGCTTCCAA  | CCTGGCTTCC  | ACCTGCCTAA  | CCCAGTGGTG | AGCCTGGGAA  |
| 101761 | TGGACCCACG | GGACAGGCAG  | CCCCCAGGGC  | CTTTTCTGAC  | CCCACCCACT | CGAGTCTCTG  |
| 101821 | CTTCACTCCC | TTCCCTTCTT  | CCCAGGTGAA  | CCTCCAAAAT  | CAGGGGATCG | CAGCGGCTAC  |
| 101881 | AGCAGCCCCG | GCTCCCCAGG  | CACCTCCCGC  | AGCCGCTCCC  | GCACCCCGTC | CCTTCCAAAC  |
| 101941 | CCACCCACCC | GGGAGCCCAA  | GAAGGTGGCA  | GTGGTCCGTA  | CTCCACCCAA | GTCCCGCTCT  |
| 102001 | TCCGCCAAGA | GCCGCCTGCA  | GACAGCCCCC  | GTGCCCATGC  | CAGACCTGAA | GAATGTCAAG  |
| 102061 | TCCAAGATCG | GCTCCACTGA  | GAACCTGAAG  | CACCAGCCGG  | GAGGCGGGAA | GGTGAGAGTG  |
| 102121 | TGCTGTGGTG | CGTGGAGGTG  | TGGGGGGCTG  | CGCCTGGAGG  | GGTAGGGCTG | TGCTGGAAG   |
| 102181 | GGTAGGGCTG | CGCCTGGAGG  | TGCGCGGTTG  | AGCGTGGAGT  | CGTGGGACTG | TGCATGGAGG  |
| 102241 | TGTGGGGCTC | CCCGCACCTG  | AGCACCCCCG  | CATAACACCC  | CAGTCCCTCT | TGGACCTCT   |
| 102301 | TCAAGGAAGT | TCAGTTCTTT  | ATTGGGCTCT  | CCACTACACT  | GTGAGTGCCG | TCCTCAGGCG  |
| 102361 | AGAGAACGTT | CTGGCTCTTC  | TCTTGCCCCCT | TCAGCCCCCTG | TTAATCGGAC | AGAGATGGCA  |
| 102421 | GGGCTGTGTC | TCCACGGCCG  | GAGGCTCTCA  | TAGTCAGGGC  | ACCCACAGCG | GTTCCTCCAC  |
| 102481 | TGCCCTTCTG | GCAGAAATACA | CTGCCACCCA  | TAGGTCAGCA  | TCTCCACTCG | TGGGCCATCT  |
| 102541 | GCTTAGGTTG | GGTTCTCTCT  | GATTCTGGGG  | AGATTGGGGG  | TTCTGTTTTG | ATCAGCTGAT  |
| 102601 | TCTTCTGGGA | GCAAGTGGGT  | GCTCGCGAGC  | TCTCCAGCTT  | CCTAAAGGTT | GAGAAGCACA  |
| 102661 | GACTTCTGGG | GCCTGGCCTG  | GATCCCTTTC  | CCCATTCCTG  | TCCCTGTGCC | CCTCGTCTGG  |
| 102721 | GTGCGTTAGG | GCTGACATAC  | AAAGCACCCAC | AGTGAAAGAA  | CAGCAGTATG | CCTCTCACT   |
| 102781 | AGCCAGGTGT | GGGCGGGTGG  | GTTCCTTCCA  | AGGCCCTCTCT | GTGGCCGTGG | GTAGCCACCT  |
| 102841 | CTGTCTCTCA | CCGCTGCAGT  | CTTCCCTCTG  | TGTGTGCTCC  | TGGTAGCTCT | GCGCATGCTC  |
| 102901 | ATCTTCTTAT | AAGAACACCA  | TGGCAGCTGG  | GCGTAGTGGC  | TCACGCCTAT | AATCCCAGCA  |
| 102961 | CTTTGGGAGG | CTGAGGCGAG  | CAGATCACGA  | GGTCAGGAGT  | TCGAGACCAA | CCTGACCAAC  |
| 103021 | AGGGTGAAGA | CTCGTCTCTA  | CTAAAATATC  | AAAAATACCT  | GGGCGTGGTG | GTGGTGCGCG  |
| 103081 | CCTATAATCC | CAGCTACTCA  | GGAGGCTGAG  | GCAGGAGAAT  | CGCTTGAAAC | CAGGAGGCAG  |
| 103141 | AGGTTGCAGT | GAGCCGAGAT  | AGTGCCACTG  | CACCTCCAGT  | TGAGCAACAG | AGCGAGACTC  |
| 103201 | TGTCTCAAAA | CAAAATAAAA  | CAAAACCAAAA | AAACCCACCA  | TGGCTTAGGG | CCCAGCCTGA  |
| 103261 | TGACCTCATT | TTTCACTTAG  | TCACCTCTCT  | AAAGGCCCTG  | TCTCCAAATA | GAGTCACATT  |
| 103321 | CTAAGGTACG | GGGGTGTGTG  | GGAGGGGGGT  | TAGGGCTTCA  | ACATGTGAAT | TTGCGGGGAC  |
| 103381 | CACAATTACG | CCAGGACCC   | CGCTCCCGCC  | ACCCAGCACT  | GGGAGAGTGG | GGAAGGGTGA  |
| 103441 | AGAGGAGGCT | GGGGGTGAGA  | AGGACCACAG  | CTCACTCTGA  | GGCTGCAGAT | GTGCTGGGCC  |
| 103501 | TTCTGGGCAC | TGGGCCTCGG  | GGAGCTAGGG  | GGCTTTCTGG  | AACCTTGGGC | CTGCGTGTCA  |
| 103561 | GCTTGCCTTC | CCACGCGAGG  | CGCTCTCCAC  | ACCATTGAAG  | TTCTTATCAC | TTGGGTCTGA  |
| 103621 | GCCTGGGGCA | TTTGGACGGA  | GGGTGGCCAC  | CAGTGCACAT  | GGGCACCTTG | CCTCAAACCC  |
| 103681 | TGCCACCTTC | CCCCACCCAG  | GATCCCCCTT  | GCCCCGAAAC  | AAGCTTGTGA | GTGCAGTGTG  |
| 103741 | ACTATCCATC | GGGATGGAAA  | TGGACGGTCG  | GGTTAAAGG   | GACGCATGTG | TAGACCTTGC  |
| 103801 | CTCTGTGCAT | CAGGCCCTCT  | TTGAGAGTCC  | CTGCGTGCCA  | GGCGGTGCAC | AGAGGTGGAG  |
| 103861 | AAGACTCGGC | TGTGCCCCAG  | AGCACCTCCT  | CTCATCGAGG  | AAAGGACAGA | CAGTGGCTCC  |
| 103921 | CTGTGGCTTG | TGGGGACAAG  | GGCAGAGCTC  | CCTGGAACAC  | AGGAGGGAGG | GAAGGAAGAG  |
| 103981 | AACATCTCAG | AATCTCCCTC  | CTGATGGCAA  | ACGATCCGGG  | TTAAATTAAG | GTCCGGCCTT  |
| 104041 | TTCTGTGTCA | GGCATGTGGA  | GCTTGTAGTG  | GAAGAGGCTC  | TCTGGACCTT | CATCCACCAC  |
| 104101 | AGTGGCCTGG | TTAGAGACCT  | TGGGGAAATA  | ACTCACAGGT  | GACCCAGGGC | CTCTGTCTCT  |
| 104161 | TACCGCAGCT | GAGGGAAACT  | GTCCCTGCGCT | TCCACTGGGG  | ACAATGCGCT | CCCTCGTCTC  |
| 104221 | CAGACTTTCC | AGTCTCTATT  | CGGTTCTCGA  | AAGTCGCTTC  | CAGAAGCCCC | ATCTTGGGAC  |
| 104281 | CACCTGTGCT | TTCAATCTCC  | AGGGTGCCCT  | GCCTTGCTGC  | TGCCCAAGAC | CCCAGAGGGG  |
| 104341 | CCCTCACTGG | CCTTTCTCTG  | CTTTTCTCCC  | ATTGCCACCC  | CATGCACCCC | CATCTGTCTC  |
| 104401 | CAGCACCACG | ACTGCCATCC  | AGGATCTCCT  | CAAGTCACAT  | AACAAGCAGC | ACCCACAAGG  |
| 104461 | TGCTCTCCTT | CCCTAGCCT   | GAATCTGCTG  | CTCCCCGTCT  | GGGGTTCCCT | GCCCATGCAC  |
| 104521 | CTCTGGGGGC | CCCTGGGTTT  | TGCCATACCC  | TGCCCTGTGT  | CCCATGGTGG | GGAATGTCCT  |
| 104581 | TCTCTCTCTA | TCTCTCTCCT  | TCCCTTAAAT  | CCAAGTTCAG  | TTGCCATCTC | CTCCAGGAAG  |
| 104641 | CTCTCTCTGA | TTCCCTCTCT  | TCTTCTTAAA  | GCCCCGTGTA  | ACTCTGACCA | CACCTGAGCAT |
| 104701 | GTGTCTGCTG | CTCCCTAGTC  | TGGGCCATGA  | GTGAGGGTGG  | AGGCCAAGTC | TCATGCATTT  |
| 104761 | TTGCAGCCCC | CACAAGACTG  | TGCAGGTGGC  | CGGCCCTCAT  | TGAATGCGGG | GTAAATTTAA  |
| 104821 | CTCAGCCTCT | GTGTGAGTGG  | ATGATTCAGG  | TTGCCAGAGA  | CAGAACCCTC | AGCTTAGCAT  |
| 104881 | GGGAAGTAGC | TTCCCTGTGT  | ACCCCTGAGT  | CATCTGAGGT  | TGGCTTGGAA | GGTGTGGGCA  |
| 104941 | CCATTGGGCC | CAGTTCTTAC  | AGCTCTGAAG  | AGAGCAGCAG  | GAATGGGGCT | GAGCAGGGAA  |
| 105001 | GACAACTTTT | CATTGAAGGC  | CCCTTTTCAGG | GCCAGAAGTC  | TCCCTCCACC | CCTGCAGCTG  |
| 105061 | CCCTGCCTCT | GCCCATGAGG  | GGTGAGAGTC  | AGGCCAGCTC  | ATGCCAAGTG | TAGAAAGGGG  |
| 105121 | CAGATGGGAG | CCCCAGGTTA  | TGACGTCACC  | ATGCTGGGTG  | GAGGCAGCAC | GTCCAAATCT  |
| 105181 | ACTAAAGGGT | TAAAGGAGAA  | AGGGTGACTT  | GACTTTTCTT  | GAGATATTTT | GGGGGACGAA  |
| 105241 | GTGTGGAAAA | GTGGCAGAGG  | ACACAGTCAC  | AGCCTCCCTT  | AAATGCCAGG | AAAGCCTAGA  |
| 105301 | AAAATTGTCT | GAAACTAAAC  | CTCAGCCATA  | ACAAAGACCA  | ACACATGAAT | CTCCAGGAAA  |
| 105361 | AAAGAAAAAG | AAAATGTCTA  | TACAGGGTCC  | ATGCACAAGA  | GCCTTTAAAA | TGACCGCTG   |



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| 105421 | AAGGGTGTCA  | GGCCTCCTCC  | TCCTGGACTG | GCCTGAAGGC  | TCCACGAGCT  | TTTGCTGAGA  |
| 105481 | CCTTTGGGTC  | CCTGTGGCCT  | CATGTAGTAC | CCAGTATGCA  | GTAAGTGCTC  | AATAAATGTT  |
| 105541 | TGGCTACAAA  | AGAGGCAAAG  | CTGGCGGAGT | CTGAAGAATC  | CCTCAACCGT  | GCCGGAACAG  |
| 105601 | ATGCTAACAC  | CAAAGGGAAA  | AGAGCAGGAG | CCAAGTCACG  | TTTGGGAACC  | TGCAGAGGCT  |
| 105661 | GAAAACCTGC  | GCAGATTGCT  | GCAAATCATT | GGGGGAAAAA  | CGGAAAACGT  | CTGTTTTCCC  |
| 105721 | CTTTGTGCTT  | TTCTCTGTTT  | TCTTCTTTGT | GCTTTTCTCT  | GTTTTCAGGA  | TTTGCTACAG  |
| 105781 | TGAACATAGA  | TTGCTTTGGG  | GCCCCAAATG | GAATTATTTT  | GAAAGGAAAA  | TGCAGATAAT  |
| 105841 | CAGGTGGCCG  | CACCTGGAGCA | CCAGCTGGGT | AGGGGTAGAG  | ATTGCAGGCA  | AGGAGGAGGA  |
| 105901 | GCTGGGTGGG  | GTGCCAGGCA  | GGAAGAGCCC | GTAGGCCCCG  | CCGATCTTGT  | GGGAGTCGTG  |
| 105961 | GGTGGCAGTG  | TTCCCTCCAG  | ACTGTAAAAA | GGAGCACCTG  | GCGGGAAGAG  | GGAAATCTTT  |
| 106021 | TAAACATCAT  | TCCAGTGCCC  | GAGCCTCCTG | GACCTGTTGT  | CATCTTGAGG  | TGGGCCTCCC  |
| 106081 | CTGGGTGACT  | CTAGTGTGCA  | GCCTGGCTGA | GACTCAGTGG  | CCCTGGGTTC  | TTACTGCTGA  |
| 106141 | CACCTACCTC  | CAACCTCAAC  | CACCTGCGGC | TCCTGTGCAC  | CCTGATCCAG  | TGGCTCATTT  |
| 106201 | TCCACTTTCA  | GTCCCAGCTC  | TATCCCTATT | TGCAGTTTCC  | AAGTGCCCTG  | TCCTCAGTCA  |
| 106261 | GCTCAGACCC  | AGCCAGGCCA  | GCCCCTGGTT | CCCACATCCC  | CTTTGCCAAG  | CTCATCCCCG  |
| 106321 | CCCTGTTTGG  | CCTGCGGGAG  | TGGGAGTGTG | TCCAGACACA  | GAGACAAAGG  | ACCAGCTTTT  |
| 106381 | AAAACATTTT  | GTTGGGGCCA  | GGTGTGGTGG | CTCACACCTA  | ATCCCAACAC  | CTGGGGAGGC  |
| 106441 | CAAGGCAGAA  | GGATCACTTG  | AGTCCAGGAG | TTCAAGACCA  | GCCTGGGCAA  | CATAGGGAGA  |
| 106501 | CCCTGTCTCT  | ACAATTTTTT  | TTTTAATTAG | CTGGGCCTGT  | TGGCACTCTC  | CTGTAGTTCC  |
| 106561 | AGCTACTCTA  | GAGGCTGAGG  | TGGGAGGACT | GCTTGAGCCT  | GGGAGGTCAG  | GGCTGCAATG  |
| 106621 | AGCCATGTTC  | ACACCACTGA  | ACGCCAGCCT | GGGCGAGACC  | CTGTATCAAA  | AAAGTAAAGT  |
| 106681 | AAAAATGAATC | CTGTACGTTA  | TATTAAGGTG | CCCCAAATG   | TACTTAGAAG  | GATTTTCATAG |
| 106741 | TTTTAAATAC  | TTTTGTTATT  | TAAAAAATTA | AATGACTGCA  | GCATATAAAT  | TAGGTTCTTA  |
| 106801 | ATGGAGGGGA  | AAAAGAGTAC  | AAGAAAAGAA | ATAAGAATCT  | AGAAACAAAG  | ATAAGAGCAG  |
| 106861 | AAATAAACCA  | GAAAACACAA  | CCTTGCACTC | CTAACTTAAA  | AAAAAAAATG  | AAGAAAACAC  |
| 106921 | AACCCAGTAA  | ACAAATATA   | ACAGCATTAA | GAGCTGGCTC  | CTGGCTGGGC  | GCGGTGGCGC  |
| 106981 | ATGCCTGTAA  | TCCCAACACT  | TTGGGAGGCC | GATGCTGGAG  | GATCACTTGA  | GACCAGGAGT  |
| 107041 | TCAAGGTTGC  | AGTGAGCTAT  | GATCATACCA | CTACACCCTA  | GCCTGGGCAA  | CACAGTGAGA  |
| 107101 | CTGAGACTCT  | ATTAAAAAAA  | AAATGCTGGT | TCCTTCCTTA  | TTTCATTCCCT | TTATTCATTG  |
| 107161 | ATTCAGACAA  | CATTTATGGG  | GCACCTCTGA | GCACCAGGCT  | CTGTGCTAAG  | AGCTTTTGCC  |
| 107221 | CCCAGGGTCC  | AGGCCAGGGG  | ACAGGGGCAG | GTGAGCAGAG  | AAACAGGGCC  | AGTCACAGCA  |
| 107281 | GCAGGAGGAA  | TGTAGGATGG  | AGAGCTTGGC | CAGGCAAGGA  | CATGCAGGGG  | GAGCAGCCTG  |
| 107341 | CACAAGTCAG  | CAAGCCAGAG  | AAGACAGGCA | GACCCCTGTT  | TGGGACCTGT  | TCAGTGGCCT  |
| 107401 | TTGAAAGGAC  | AGCCCCCACC  | CGGAGTGCTG | GGTGAGGAG   | CTGAAGGAGG  | ATAGTGAAC   |
| 107461 | CTGCAACGCT  | GGAGCTCTTC  | AGAGCAAAAG | CAAAATAAAC  | AACCTGAGGC  | AGCTGGGGCA  |
| 107521 | GCAGAGGGTG  | TGTGTTTCAGC | ACTAAGGGGT | GTGAAGCTTG  | AGCGCTAGGA  | GAGTTCACAC  |
| 107581 | TGGCAGAAGA  | GAGGTTGGGG  | CAGCTGCAAG | CCTCTGGACA  | TGCCCCGACA  | GGACAGAGGG  |
| 107641 | TGGTGGACGG  | TGGCCCTGAA  | GAGAGGCTCA | GTTCACTGGG  | CAGTGGCCGT  | GGGAGTGCTG  |
| 107701 | AAGCAGGCAG  | GCTGTCGGCA  | TCTGCTGGGG | ACGGTTAAGC  | AGGGGTGAGG  | GCCCAGCCTC  |
| 107761 | AGCAGCCCTT  | CTTGGGGGGT  | CGCTGGGAAA | CATAGAGGAG  | AACCTGAAGAA | GCAGGGAGTC  |
| 107821 | CCAGGGTCCA  | TGCAGGGCGA  | GAGAGAAGTT | GCTCATGTGG  | GGCCCCAGGT  | GCAGGATCAG  |
| 107881 | GAGAAGCTGG  | GACCCCTGTGA | CTGCCAGCGG | GGAGAAGGGG  | GTGTGCAGGA  | TCATGCCCAG  |
| 107941 | GGAGGGGCCC  | AGGGGCCCAA  | GCATGGGGGG | GCCTGGTTGG  | CTCTGAGAAG  | ATGGAGCTAA  |
| 108001 | AGTCAAGCTT  | TCCGAGGATG  | TCCAGGCCAA | TAGTTGGGAT  | GTGAAGACGT  | GAAGCAGCAC  |
| 108061 | AGAGCCTGGA  | AGCCCAGGAT  | GGACAGAAAC | CTACCTGAGC  | AGTGGGGCTT  | TGAAAGCCTT  |
| 108121 | GGGGCGGGGG  | GTGCAATATT  | CAAGATGGCC | ACAAGATGGC  | AATAGAATGC  | TGTAACCTTC  |
| 108181 | TTGGTTCCTG  | GCCGCAGCCT  | GGGTGGCTGC | TTCTTCCCTC  | GTGTGTATTG  | ATTGTGTTCT  |
| 108241 | CTTTTTTGAG  | ACAGAGTCTT  | GCTGGGTTGC | CCAGGCTGGA  | GTGCAGTGGT  | GCGATCATAG  |
| 108301 | CTCACTGCAG  | CCTTGAAGTC  | CTGAGCTCAA | GAGATCCTTC  | CACCTCAGCC  | TCCTTGAGTAG |
| 108361 | TTGGGACCAC  | AGGCTTGAC   | CACAGTGCCC | AACTAATTTC  | TTATATTTTT  | TGTAGAGATG  |
| 108421 | GGGTTTCACT  | GTGTGCGCCA  | GGATGGTCTT | GAACTCCTGG  | GCTCAAGTGA  | TCCTCCTGCC  |
| 108481 | TCAGCCTCGC  | AAATTGCTGG  | GATTACAGGT | GTGAGCCACC  | ATGCCCCGAC  | TTCTCTTTTT  |
| 108541 | AAGGGCGTGT  | GTGTGTGTGT  | GTGTGTGTGT | GCGCACTCTC  | GTCTTCACCT  | TCCCCAGGCC  |
| 108601 | TTGCTCTGTC  | TCTACCCAGT  | CACCTCTGCC | CATCTCTCCG  | ATCTGTTTCT  | CTCTCCTTTT  |
| 108661 | ACCCCTCTTT  | CCTCCCTCCT  | CATACACCAC | TGACCATTTAT | AGAGAAGTGA  | GTATTTCTAAA |
| 108721 | AATACATTTT  | ATTTATTTAT  | TTTGAGACAG | AGTCTCACTC  | TGTCACCCAG  | GCTGGAGTGC  |
| 108781 | AGTGGTGCAA  | TCTCGGCTCA  | CTGCAACCTC | CGCCTCCAG   | GTTGAAGCAA  | CTCTCCTGCC  |
| 108841 | TCAGCCTCCC  | TAGTAGCTGG  | GATTACAAGC | ACACACCACC  | ATGCCTAGCA  | AATTTTTTATA |
| 108901 | TTTTTAGTAG  | AGGAGGGGTG  | TCACCATGTT | TGCCAAGCTG  | GTCTCAAAC   | CCTGGCCTCA  |
| 108961 | GGTGATCTGC  | CTACCTTGGT  | CTCCCAAAGT | GCTGGGATTA  | CAGGTGTGAG  | CCACCACGCC  |
| 109021 | TGCCCTTAAA  | AATACATTAT  | ATTTAATAGC | AAAGCCCCAG  | TTGTCACTTT  | AAAAAGCATC  |
| 109081 | TATGTAGAAC  | ATTTATGTGG  | AATAAATACA | GTGAATTTGT  | ACGTGGAATC  | GTTTGCTCTC  |
| 109141 | CCTCAATCAG  | GGCCAGGGAT  | GCAGGTGAGC | TTGGGCTGAG  | ATGTCAGACC  | CCACAGTAAG  |
| 109201 | TGGGGGGCAG  | AGCCAGGCTG  | GGACCCCTCT | CTAGGACAGC  | TCTGTAACCT  | TGAGACCCCTC |
| 109261 | CAGGCATCTT  | TTCTGTATAC  | TCAGTGCTTC | TGAAAAATCT  | GTGTGAATCA  | AATCATTTTA  |
| 109321 | AAGGAGCTTG  | GGTTCAATCA  | TGTTTAAAGG | ACAGTGTAAG  | TAATTTCTGAA | GGTGACTCTA  |
| 109381 | CCCTGTTATT  | TGATCTCTTC  | TTTGGCCAGC | TGACTTAACA  | GGACATAGAC  | AGGTTTTTCT  |
| 109441 | GTGTCACTTC  | CTAAGCTGAT  | CACCTTGGAC | TTGAAGAGGA  | GGCTTGTGTG  | GGCATCCAGT  |
| 109501 | GCCCACCCCG  | GGTTAAACTC  | CCAGCAGAGT | ATTGCACTGG  | GCTTGCTGAG  | CCTGGTGAGG  |
| 109561 | CAAAGCACAG  | CACAGCGAGC  | ACCAGGCAGT | GCTGGAGACA  | GGCCAAGTCT  | GGGCCAGCCT  |
| 109621 | GGGAGCCAAC  | TGTGAGGCAC  | GGACGGGGCT | GTGGGGCTGT  | GGGGCTGCAG  | GCTTGGGGCC  |
| 109681 | AGGGAGGGAG  | GGCTGGGCTC  | TTTGGAACAG | CCTTGAGAGA  | ACTGAACCCA  | AACAAAACCA  |
| 109741 | GATCAAGGTC  | TAGTGAGAGC  | TTAGGGCTGC | TTTGGGTGCT  | CCAGGAAATT  | GATTAAACCA  |
| 109801 | AGTGAGACACA | CACCCCCAGC  | CCCACCTCAC | CACAGCCTCT  | CCTTCAGGGT  | CAAACTCTGA  |

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SEQ ID NO Sequence

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| 109861 | CCACAGACAT  | TTCTCCCTCG  | ACTAGGAGTT  | CCCTGGATCA  | AAATTGGGAG  | CTTGCAACAC  |
| 109921 | ATCGTTCTCT  | CCCTTGATGG  | TTTTTGTCTAG | TGTCTATCCA  | GAGCTGAAGT  | GTAATATATA  |
| 109981 | TGTTACTGTA  | GCTGAGAAAT  | TAAATTTTCAG | GATTCTGATT  | TCATAATGAC  | AACCATTCTCT |
| 110041 | CTTTTCTCTC  | CCTTCTGTAA  | ATCTAAGATT  | CTATAAACGG  | TGTTGACTTA  | ATGTGACAAAT |
| 110101 | TGGCAGTAGT  | TCAGGTCTGC  | TTTGTAATA   | CCCTTGTGTC  | TATTGTAAAA  | TCTCACAAAG  |
| 110161 | GCTTGTGTCC  | TTTTTTGTGG  | GGTTAGAACA  | AGAAAAAGCC  | ACATGGAAAA  | AAAATTTCTT  |
| 110221 | TTTTTTTTTT  | TTGTTTGCTT  | GTTTTTTTGA  | GACAGAGTTT  | CACCTCTGTCG | CCCAGGCTGG  |
| 110281 | AGTGCAGTGG  | TGCGATCTCC  | GCCCACTGCA  | AGCTCCACCT  | CCCGGGTTCA  | TGCTATTCTC  |
| 110341 | CTGTCTCAGC  | CTCCCAAGTA  | GCTGGGACTG  | CAGGTGCCCCG | CCACCACACC  | TGGCTAATTT  |
| 110401 | TTTTGTATTT  | TTAGTAGAGA  | CGGGGTTTCA  | CCGTGTTAGC  | CAGGATGGTC  | TCAATCTCCT  |
| 110461 | GACCTCGTCA  | TCTGCCTGCC  | TCGGCCTCCC  | AAAGTGCTGA  | GATTACAGGC  | GTGAGCCACC  |
| 110521 | GTGCCCGGCC  | AGAAAAAAC   | ATTTCTAAGT  | ATGTGGCAGA  | TACTGAATTA  | TTGCTTAATG  |
| 110581 | TCCTTTGATT  | CATTGTGTTA  | ATTTCTTTAA  | TGGATTAGTA  | CAGAAAACAA  | AGTTCTCTTC  |
| 110641 | CTTGAAAAAC  | TGGTAAGTTT  | TCTTTGTCTAG | ATAAGGAGAG  | TTAAATAACC  | CATGACATTT  |
| 110701 | CCCTTTTTGC  | CTCGGCTTCC  | AGGAAGCTCA  | AAGTTAAATG  | TAATGATCAC  | TCTTGTAATT  |
| 110761 | ATCAGTGTGT  | ATGCCCTTCC  | CTTCTTCTAA  | TGTTACTCTT  | TACATTTTCC  | TGCTTTATTA  |
| 110821 | TGTGTGTGTG  | TTTCTAATTC  | TAAAGCTGTT  | CCACTCTTTT  | CTGAAAGCAG  | GCAAACTTTC  |
| 110881 | TAAGCCTTAT  | CCACTGAAAA  | GTTATGAATA  | AAAAATGATC  | GTCAAGCCTTA | CAGGTGCTGA  |
| 110941 | GGCTACTCCA  | GAGGCTGAGG  | CCAGAGGACC  | ACTTGAGCCC  | AGGAATTTGA  | GACCTGGGCT  |
| 111001 | TGGCAGCATAT | GCAAGACTCT  | ATCTCCATTA  | AAACTATTTT  | TTTTTATTTA  | AAAAATAATC  |
| 111061 | CGCAAAGAAG  | GAGTTTATGT  | GGGATTCTCT  | AAAAATCGGAG | GGTGGCATGA  | ATTGATTCAA  |
| 111121 | AGACTTGTGC  | AGAGGGCGAC  | AGTGACTCCT  | TGAGAAGCAG  | TGTGAGAAAG  | CCTGTCCCAC  |
| 111181 | CTCCTTCCGC  | AGCTCCAGCC  | TGGGCTGAGG  | CACTGTCTCA  | GTGTCTCTTT  | GCTGGCAGGA  |
| 111241 | GAGAAATTTCA | ACATTCACCA  | AAAAGTAGTA  | TTGTTTTTAT  | TAGGTTTATG  | AGGCTGTAGC  |
| 111301 | CTTGAGGACA  | GCCCAGGACA  | ACTTTGTTGT  | CACATAGATA  | GCCTGTGGCT  | ACAAACTCTG  |
| 111361 | AGATCTAGAT  | TCTTCTGTGG  | CTGCTTCTGA  | CCTGAGAAAG  | TTGCGGAACC  | TCAGCGAGCC  |
| 111421 | TCACATGGCC  | TCCTTGTCTCT | TAACTGTGGG  | ACGGTGGGCA  | AGAAAGGTGA  | TGTGGCACTA  |
| 111481 | GAGATTTATC  | CATCTCTAAA  | GGAGGAGTGG  | ATTGTACATT  | GAAACACCAG  | AGAAGGAATT  |
| 111541 | AAAGAGGAAG  | AATTTAGATA  | TCTAAAAATG  | TAGGTCAGGC  | GCTCCTGTGT  | TGATTGCAGG  |
| 111601 | GCTATTTCACA | ATAGCCAAGA  | TTTGGAAGCA  | ACCCAAGTGT  | CCATCAACAG  | ACAAATGGAT  |
| 111661 | AAAGAAAAATG | TGGTGCATAT  | ACACAATGGA  | ATACTATTCA  | GCCATGAAAA  | AGAATGAGAA  |
| 111721 | TCTGTCTATT  | GAAACAACAT  | GGATGGAACT  | GGAGGACATT  | ATGTTAAGTG  | AAATAAGCCA  |
| 111781 | GACAGAAGGA  | CAGACTTCAC  | ATGTTCTCAC  | ACATTTGTGG  | GAGCTAAAAA  | TTAAACTCAT  |
| 111841 | GGAGATAGAG  | AGTAGAAGGA  | TGGTTACCAG  | AGGCTGAGGA  | GGGTGGAGGG  | GAGCAGGGAG  |
| 111901 | AGAGTAGGGA  | TGGTTAATGG  | GTACAAAAAC  | GTAGTTAGCA  | TGCATAGATC  | TAGTATTGGA  |
| 111961 | TAGCACAGCA  | GGGTGACGAC  | AGCCAACAGT  | AATTTATAGT  | ACATTTAAAA  | ACAACTAAAA  |
| 112021 | GAGTGTAAAT  | GGACTGGCTA  | ACATGGTGAA  | ACCCCGTCTC  | TACTAAAAAT  | ACAAAAATTA  |
| 112081 | TCTGGGCATG  | GTGGCTCACG  | CCTGTAATCC  | CAGCACTTTG  | GGAGGCCGAG  | GCGGGCCGAT  |
| 112141 | CACGAGGTCA  | GGAGATCGAG  | ACCATCCTAG  | CTAACATGGT  | GAAACCCCGT  | CTCTACTACA  |
| 112201 | AATACAAAAA  | AAAGAAAAAA  | TTAGCCGGGC  | ATGGTGGTGG  | GCGCCTGTAG  | TCCCAGCTAC  |
| 112261 | TCTGGAGGCT  | GAGGCAGGAG  | AATGGCGTGA  | ACCCGGGAGG  | CGGAGCTTGC  | AGTGAGCCGA  |
| 112321 | GATCGCGCCA  | CTGCACTCCA  | GCCTGGGCGA  | CAAGGCAAGA  | TTCTATCTCA  | AAAAAATAAA  |
| 112381 | AATAAAATAA  | AATAAAATAA  | TAAAAATAAA  | TAAAAATAAA  | TAAAAATAAT  | AAAAATAAAT  |
| 112441 | AAAATGTTATA | ATTGGAATGT  | TTATAACACA  | AGAAATGATA  | AATGCTTGAG  | GTGATAGATA  |
| 112501 | CCCCATTTCAC | CGTGATGTGA  | TTATTGCACA  | ATGTATGTCT  | GTATCTAAAT  | ATCTCATGTA  |
| 112561 | CCCCACAAGT  | ATATACACCT  | ACTATGTACC  | CATATAAATT  | TAAAAATTA   | AAATTATAAA  |
| 112621 | ACAAAAATAA  | ATAAGTAAAT  | TAAATGTAG   | GCTGGACACC  | GTGGTTTACG  | CCTGTAATCC  |
| 112681 | CAGTGCTTTG  | TGAGGCTGAG  | GTGAGAGAAT  | CACCTTGAGCC | CAGGAGTTTG  | AGACCGGCCT  |
| 112741 | GGGTGACATA  | GCGAGACCCC  | ATCATCACAA  | AGAAATTTTA  | AAAATTAGCT  | GGGCGTGGTA  |
| 112801 | GCACATACCG  | GTAGTTCCAG  | CTACTTGGGA  | GACCGAGGCA  | GGAGGATTGC  | TTGAGCCACG  |
| 112861 | GAGTTTAAAG  | CTGCAGTGAG  | CTACGATGGC  | GCCACTGCAT  | TCCAGCCTGG  | GTGACAGAGT  |
| 112921 | GAGAGCTTGT  | CTCTATTTTA  | AAAATAATAA  | AAAGAAATAA  | TAAAAATAAA  | TTAAATATGA  |
| 112981 | AATATGTGCA  | TGTTAGAAAA  | AATACACCCA  | TCAGCAAAAA  | GGGGGTAAAG  | GAGCGATTTT  |
| 113041 | AGTCATAAAT  | GGAGAGATGC  | AGAATAAGCC  | AGCAATGCAG  | TTTCTTTTAT  | TTTGGTCAAA  |
| 113101 | AAAAATAAGC  | AAAACAATGT  | TGTAAACACC  | CAGTGCTGGC  | AGCAATGTGG  | TGAGGCTGGC  |
| 113161 | TCTCTCACCA  | GGGCTCACAG  | GGAAAACTCA  | TGCAACCCTT  | TTAGAAAAGC  | ATGTGGAGAG  |
| 113221 | TTGTACCGAG  | AGGTTTGTAGA | ATATTTATATA | CTTTGACCCA  | GAAATTTCTAT | TCTAGGACTC  |
| 113281 | TGTGTATATGA | AAATAACCCA  | TCATATGGAA  | AAAGCTCCTT  | TCAGAAAAGAG | GTTCATGGGA  |
| 113341 | GGCTGTTTGT  | ATTTTTTTTT  | TCTTTGCATC  | AAATCCAGCT  | CCTGCAGGAC  | TGTTTGTATT  |
| 113401 | ATTGAAGTAC  | AAAGTGAAT   | CAATACAAAT  | GTTGGATAGC  | AGGGGAACAA  | TATTCACAAA  |
| 113461 | ATGGAATGGG  | ACATAGTATT  | AAACATAGTG  | CTTCTGATGA  | CCGTAGACCA  | TAGACAAATG  |
| 113521 | TTAGGATATG  | ATATCACTTC  | TTTTGTGTGT  | TTTTGTATT   | TGAGACGAAG  | TCTCATTTCTG |
| 113581 | TACCCAGGC   | TGGAGTTTCAG | TGGCGCCATC  | TCAGCTCACT  | GCAACCTCCA  | TCTCCGGGCT  |
| 113641 | TCAAGCTATT  | CTCCTTCTCT  | AACCTCCCCA  | GTAGCTGGGT  | TGCGCACCAC  | CATGCCTGGC  |
| 113701 | TAACTTTTGT  | ATTTTGTAGTA | CAGACGGGGT  | TTCAACACGT  | TGGCCAGGCT  | GCTCTTGAAC  |
| 113761 | TCTTGACGCT  | AGGTGATCCA  | CCAGCCTTGA  | CCTCCCCAAG  | TGCTAGGATT  | ACAGGAGCCA  |
| 113821 | CTGTACCCAG  | CCTAGGATAT  | GATATCACTT  | CTTAGAGCAA  | GATACAAAA   | TGCATGTGCA  |
| 113881 | CAATAATTTCT | ACCAAGTATA  | GGTATACAGG  | GGTAGTTATA  | TATAAATGAG  | ACTTCAAGGA  |
| 113941 | AATACAAACA  | AATGCAATCG  | TGATTGTGTT  | AGGGTGGTAA  | GAAAAACGGT  | TTTGTCTTGA  |
| 114001 | TGAGCTCTGT  | TTTTTAAAT   | CGTTATATTT  | TCTAATAAAA  | ATACATAGTC  | TTTTGAAGGA  |
| 114061 | ACATAAAAGA  | TTATGAAGAA  | ATGAGTTAGA  | TATTGATTCC  | TATTGAAGAT  | TCAGACAAAT  |
| 114121 | AAAATTAAGG  | GGAAAAAAGA  | CGGGATGAAC  | CAGAAGTCAG  | GCTGGAGTTC  | CAACCCAGGA  |
| 114181 | TCCGACAGCC  | CAGGCTGATG  | GGGCCTCCAG  | GGCAGTGGTT  | TCCACCCAGG  | ATTTCTCAAAA |
| 114241 | GAGCCACTGA  | GGTCTCAGTG  | CCATTTTCAA  | GATTTTCGAA  | GCGGCCTGGG  | CACGGCTGGT  |

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| SEQ ID NO | Sequence                                                           |
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| 114301    | CCTTCACTGG GATCACCACCT TGGCAATTAT TTACACCTGA GACGAATAAA AACCAGAGTG |
| 114361    | CTGAGATTAC AGGCATGGTG GCTTACGCTT GTAATCGGCT TTGGGAAGCC GAGGTGGGCT  |
| 114421    | GATTGCTTGA GCCCAGGAGT TTCAAACCTAT CCTGGACAAC ATAGCATGAC CTCGTCTCTA |
| 114481    | CAAAAAATAC AAAAAATTG CCAGGTGTGG TGGCATGTGC CTGTGGTCCC AGCTACTTGG   |
| 114541    | GAGGCTGAAG TAGGAGAATC CCCTGAGCCC TGGGAAGTCG AGGCTGCACT GAGCCGTGAT  |
| 114601    | GGTGTCACTG CACTCCAGCC TGGGTGACAA AGTGAGACCC TATCTCACAA AGAAAAAATA  |
| 114661    | CAAAACAAAA AACCCAAAGC ACACTGTTTC CACTGTTTCC AGAGTTCCCT AGAGGAAAGG  |
| 114721    | TCACCGGTG AGGAAGACGT TCTCACTGAT CTGGCAGAGA AAATGTCCAG TTTTCCAAAC   |
| 114781    | TCCCTAAACC ATGGTTTTCT ATTTCATAGT TCTTAGGCAA ATTGGTAAAA ATCATTCTCT  |
| 114841    | ATCAAAACGC TGATATTTTC ACACCTCCCT GGTGTCTGCA GAAAGAACCT TCCAGAAATG  |
| 114901    | CAGTCGTGGG AGACCCATCC AGGCCACCCC TGCTTATGGA AGAGCTGAGA AAAAGCCCCA  |
| 114961    | CGGGAGCATT TGCTCAGCTT CCGTTACGCA CCTAGTGGCA TTGTGGGTGG GAGAGGGCTG  |
| 115021    | GTGGGTGGAT GGAAGGAGAA GGCACAGCCC CCCCTTGCG AGACAGAGCC CTCGTACAGA   |
| 115081    | AGGGACACCC CACATTTGTC TTCCCCACAA AGCGGCCTGT GTCCTGCCTA CGGGGTACAG  |
| 115141    | GCTTCTCAAA CCTGGCTGTG TGTGAGAATC ACCAGGGGAA CTTTTCAAAA CTAGAGAGAC  |
| 115201    | TGAAGCCAGA CTCCTAGATT CTAATTCTAG GTCAGGGCTA GGGGCTGAGA TTGTAATAAT  |
| 115261    | AAGTCGAGGC TTCTGATGCC CGGCAGGCTT GAGAACAGCC GCAGGGAGTT CTCTGGGAAT  |
| 115321    | GTGCCGTGGG GTCTAGCCAG GTGTGAGTGG AGATGCCGGG GAACTTCCTA TTACTIONCTC |
| 115381    | GTCAGTGTGG CCGAACACAT TTTTCACTTG ACCTCAGGCT GGTGAACGCT CCCCTCTGGG  |
| 115441    | TTTTCAGGCT CACGATGCCA TCCTTTTGTG AAGTGAGGAC CTGCAATCCC AGCTTCGTAA  |
| 115501    | AGCCCGCTGG AAATCACTCA CACTTCTGGG ATGCCTTCAG AGCAGCCCTC TATCCCTTCA  |
| 115561    | GCTCCCTTGG GATGTGACTC AACCTCCCGT CACTCCCGAG ACTGCCTCTG CCAAGTCCGA  |
| 115621    | AAGTCGAGGC ATCCTTGCGA GCAAGTAGGC GGGTCCAGGG TGGCGCATGT CACTCATCGA  |
| 115681    | AAGTGGAGGC GTCCTTGCGA GCAAGCAGGC GGGTCCAGGG TGGCGTGTCA CTCATCCTTT  |
| 115741    | TTTCTGGCTA CCAAGGTGTC AGATAATTAA TAAGAAGCTG GATCTTAGCA ACGTCCAGTC  |
| 115801    | CAAGTGTGGC TCAAAGGATA ATATCAAAAC CGTCCCGGGA GGCGGCAGT TGAGTACCTT   |
| 115861    | CACACGTCCC ATGCGCCGTG CTGTGGCTTG AATTATTAGG AAGTGGTGTG AGTGCGTACA  |
| 115921    | CTTGCAGAGC ACTGCATAGA ATAAATCCTT CTGGGGCTCT CAGGATCTGG CTCGCGACCTC |
| 115981    | TGGGTGAATG TAGCCCCGCT CCCCACATTC CCCCACACGG TCCACTGTTT CCAGAAGCCC  |
| 116041    | CTTCTCTATA TTCTAGGAGG GGGTGTCCCA GCATTTCTGT GTCCCCCAGC CTGCGCAGGC  |
| 116101    | TGTGTGGACA GAATAGGGCA GATGACGGAC CCTCTCTCCG GACCCTGCCT GGGAAAGCTA  |
| 116161    | GAATACCCAT CAAAGTCTCC TTCCACTCAT GCCCAGCCCT GTCCCCAGGA GCCCCTAGC   |
| 116221    | CCATTGGAAG TTGGGCTGAA GGTGGTGGCA CCTGAGACTG GGCTGCCGCC TCCTCCCCCG  |
| 116281    | ACACCTGGGC AGGTGTGAGT TGAGTGGCTC CACTGTGGAC AGGTGACCCG TTTGTCTGA   |
| 116341    | TGAGCGGACA CCAAGGTCTT ACTGTCCTGC TCAGCTGCTG CTCCTACACG TTCAAGGCAG  |
| 116401    | GAGCCGATTC CTAAGCCTCC AGCTTATGCT TAGCCTGCGC CACCCTCTGG CAGAGACTCC  |
| 116461    | AGATGCAGAA AGCCAAACCA AAGTGCAGCA GGTCCCTCTG CCCAGCGTTG AGGTGTGGGA  |
| 116521    | TGAGAAATGT GCTTTTGGCC CTTTATGATT TGGCTGCCTC TTGCCAGGAG TGGTGGCTCG  |
| 116581    | TGCCCTGTAAT TCCAGCACTT TGGGAGACTA AGGCGGGAGG TTCGCTTGAG CCCAGGAGTT |
| 116641    | CAAGACCAGC CTGGGCAACA ATGAGACCCC TGTGTCTACA AAAAGAAATTA AAATTAGCCA |
| 116701    | GGTGTGGTGG CACGCACCTG TAGTCCCAGC TACTTGGGAG GCTGAGGTGG GAGGATTGCC  |
| 116761    | TGAGTCCGGG AGGCGGAAGT TGCAAGGAGC CATGATCGCG CCACTGCACT TCAACCTAGG  |
| 116821    | CAACAGAGTG AGACTTTGTC TCAAAAAACA ATCATATAAT AATTTTAAAA TAAATAGATT  |
| 116881    | TGGCTTCCCT TAAATGTCCC CGGGGACTCC GTGCATCTTC TGTGGAGTGT CTCCTGAGA   |
| 116941    | TTCCGGGACTC AGATCCTCAA GTGCAACTGA CCCACCCGAT AAGCTGAGGC TTCATCATCC |
| 117001    | CCTGGCCGGT CTATGTGAGC TGGGCACCCG AGGCTCCTCT CCCACCACTG CTCTTGGTCA  |
| 117061    | GCTGAAAGCA AACTGTTAAC ACCCTGGGGA GCTGGACGTA TGAGACCTTT GGGGTGGGAG  |
| 117121    | GCGTTGATTT TTGAGAGCAA TCACCTGGCC CTGGCTGGCA GTACCGGGAC ACTGCTGTGG  |
| 117181    | CTCCGGGGTG GGCTGTCTCC AGAAAAATGCC TGGCCTGAGG CAGCCACCCG CATCCAGCCC |
| 117241    | AGAGGGTTTA TTCTTGCAAT GTGCTGCTGC TTCCTGCCCT GAGCACCTGG ATCCCGGCTT  |
| 117301    | CTGCCCTGAG GCCCCTTGAG TCCACAGGT AGCAAGCGCT TGCCCTGCGG CTGCTGCATG   |
| 117361    | GGGCTAAGTA ACGCTTCTTC ACCAGTGTCT GCTAAGTGTG TCCTCTGTCT CCCACGCCCT  |
| 117421    | GCTCTCCTGT CCCCCAGTT TGTCTGCTGT GAGGGGACAG AAGAGGTGTG TGC CGCCCCC  |
| 117481    | ACCCCTGCCC GGGCCCTTGT TCCTGGGATT GCTGTTTTCA GCTGTTTGAG CTTTGATCCT  |
| 117541    | GGTTCTCTGG CTTCTCTCAA GTGAGCTCGG CCAGAGGAGG AAGGCCATGT GCTTTCTGTT  |
| 117601    | TGAAGTCAAG TCTGGTGCCC TGGTGGAGGC TGTGCTGCTG AGGCGGAGCT GGGGAGAGAG  |
| 117661    | TGCACACGGG CTGCGTGGCC AACCCCTCTG GGTAGCTGAT GCCCAAAGAC GCTGCACTGC  |
| 117721    | CCAGGACATC TGGGACCTCC CTGGGGCCCG CCCGTGTGTC CCGCGCTGTG TTCATCTGCG  |
| 117781    | GGCTAGCCGT TGACCCGCGC TGTGCTCGTC TGCGGGCTAG CCTGTGTCCC GCGCTCTGCT  |
| 117841    | TGTCTGCGGT CTAGCCTGTG ACCTGGCAGA GAGCCACCAG ATGTCCCGGG CTGAGCACTG  |
| 117901    | CCCTCTGAGC ACCTTCACAG GAAGCCCTTC TCCTGGTGAG AAGAGATGCC AGCCCTTGGC  |
| 117961    | ATCTGGGGGC ACTGGATCCC TGGCCTGAGC CCTAGCCTCT CCCAGCCTG GGGGCCCTTT   |
| 118021    | CCCAGCAGGC TGGCCCTGCT CTTCTCTTAC CTGGGACCCCT TCTGCCCTCT GGCTGGACCC |
| 118081    | TGGAAGCTCT GCAGGGCCTG CTGTCCCCCT CCCTGCCCTC CAGGTATCCT GACCACCGGC  |
| 118141    | CCTGGCTCCC ACTGCCATCC ACTCCTCTCC TTTCTGGCCG TTCCTGGTTC CCTGTCCCAG  |
| 118201    | CCCCCTCCC CCTCTCACGA GTTACCTCAC CCAGGCCAGA GGAAGAGAGG AAGGAGGCC    |
| 118261    | TGGTCATACC AGCACGTCTT CCCACCTCCC TCGGCCCTGG TCCACCCCTT CAGTGTGGC   |
| 118321    | CTCAGAGCAC AGCTCTCTCC AAGCCAGGCC GCGCGCCATC CATCTCTCTT GTCCCCAAC   |
| 118381    | GTCCTTGCCA CAGATCATGT CCGCCCTGAC ACACATGGGT CTCAGCCATC TCTGCCCCAG  |
| 118441    | TTAACTCCCC ATCCATAAAG AGCACATGCC AGCTGACACC AAAATAATTC GGGATGGTTC  |
| 118501    | CAGTTTAGAC CTAAGTGGAA GGAGAAACCA CCACCTGCCC TGCACCTTGT TTTTGGTGA   |
| 118561    | CCTTGATAAA CCATCTTCAG CCATGAAGCC AGCTGTCTCC CAGGAAGCTC CAGGGCGGTG  |
| 118621    | CTTCTCGGG AGCTGACTGA TAGGTGGGAG GTGGCTGCCC CCTTGCACCT TCAGGTGACC   |
| 118681    | CCACACAAGG CCACTGTCTG AGGCCCTGGG GACTCCAGGA ATGTCAATCA GTGACCTGCC  |

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| SEQ ID NO | Sequence                                                            |
|-----------|---------------------------------------------------------------------|
| 118741    | CCCCAGGCC                                                           |
| 118801    | GCCACTGTGG AGTCCCTACA GCGTGCCCC CACAGGGGAG CTGGTTCTTT GACTGAGATC    |
| 118861    | AGCTGGCAGC TCAGGGTCAT CATTCCCAGA GGGAGCGGTG CCCTGGAGGC CACAGGCCTC   |
| 118921    | CTCATGTGTG TCTGCGTCCG CTCGAGCTTA CTGAGACACT AAATCTGTTG GTTCTGCTG    |
| 118981    | TGCCACCTAC CCACCCTGTT GGTGTTGCTT TGTTCCTATT GCTAAAGACA GGAATGTCCA   |
| 119041    | GGACACTGAG TGTGCAGGTG CCTGCTGGTT CTCACGTCCG AGCTGCTGAA CTCGCTGGG    |
| 119101    | TCCTGCTTAC TGATGGTCTT TGCTCTAGTG CTTTCCAGGG TCCGTGGAAG CTTTTCTGG    |
| 119161    | AATAAAGCCC ACGCATCGAC CCTACAGCG CCTCCCCTCT TTGAGGCCCA GCAGATACCC    |
| 119221    | CACCTCTGCC TTTCCAGCAA GATTTTTCAG ATGCTGTGCA TACTCATCAT ATTGATCACT   |
| 119281    | TTTTTCTTCA TGCCTGATTG TGATCTGTCA ATTTTCATGTC AGGAAAGGGA GTGACATTTT  |
| 119341    | TACACTTAAG CGTTTGCTGA GCAAATGTCT GGGTCTTGCA CAATGACAAT GGGTCCCTGT   |
| 119401    | TTTTCCCAGA GGCTCTTTTG TTCTGCAGGG ATTGAAGACA CTCCAGTCCC ACAGTCCCCA   |
| 119461    | GCTCCCTCGG GGCAGGGTTG GCAGAAATTC GACAACACAT TTTTCCACCC TGACTAGGAT   |
| 119521    | CTGCTCACTA TGGCAGCTGG GAACCACTGT CCAATAAGGG CCTGGGCTTA CACAGCTGCT   |
| 119581    | TCTCATTGAG TTACACCTTT AATAAAATAA TCCCATTTTA TCCTTTTTGT CTCTCTGTCT   |
| 119641    | TCCTCTCTCT CTGCTTTTCC TCTTCTCTCT CCTCTCTCT CATCTCCAGG TGCAAAATAGT   |
| 119701    | CTACAAAGCC GTTGACCTGA GCAAGGTGAC CTCCAAGTGT GGCTCATTAG GCAACATCCA   |
| 119761    | TCATAAACCA GGTAGCCCTG TGAAGGTGA GGGTTGGGAC GGGAGGGTGC AGGGGGTGA     |
| 119821    | GGAGTCCTGG TGAGGCTGGA ACTGCTCCAG ACTTCAGAA GGGCTGGAAA GGATATTTTA    |
| 119881    | TGTAGACCTA CATCAAGGAA AGTGTGAGT GTGAAACTTG CGGGAGCCCA GGAGGCGTGG    |
| 119941    | TGGCTCCAGC TCGCTCTGTC CCAGGCCATG CTGCCCCAAGA CAAGGTGAGG CGGGAGTGAA  |
| 120001    | GTGAAATAAG GCAGGCACAG AAAGAAAGCA CATATTCTCG GCCGGGCGCT GTGGCTCACG   |
| 120061    | CTGTCTCTTA TGGCAGCTGG GGAGGCCAAG GTGGGTGGAT CATGAGGTCA GGAGATTGAG   |
| 120121    | ACCATCTCTG CTAACACAGT GAAACCCCGT CTCTACTAAA AATACAAAAA ATTAGCCGGG   |
| 120181    | CGTGTGTGTG GGCCTCTGTA GTCCCAAGTA CTCGGGAGGC TGAGGCAGGA AAATGGCGTG   |
| 120241    | AACCCGGAAG GCGGAGCTTG CAGTGAGCGG AGTGAGCAGA GATCGCGCCA CTGCACTCCA   |
| 120301    | GCCTGGGCGA CAGAGCGAGA CTCCTCTCTA AAAAAAAAAA GCACATGTTT TCGCTTCTTT   |
| 120361    | GTGGGATCCA GGAGATAGAG AATAGAAGGA TGGTTACCAG AGGCTGGGAA GGGTAGTGAG   |
| 120421    | GGGATGGTGG GGGATGGTTC AATGGGTACA AAAAAAATAG AATAAGACCT AGTATTTGAT   |
| 120481    | AGTGAACACG GGTGACTATA GTCAATAATA ATTTAATTGT ACATTTAAAA ATAATAAAAA   |
| 120541    | GATAGCCGGG TGCAGTGGCT TACGTCTGTA ATCCAGTAGT TTTGGGAGGC TGAGGTGGGC   |
| 120601    | GTTTGAGACC AGCCTGGCCA ACATGGTGAA ACCCCATCTC TACTAAAAAT ACAAAAAATTA  |
| 120661    | GCCAGGCATG GTGGCGGGCG CCTGTAATCC CAGCTACTCG GGAGGCTGAG GCAGGAGAAT   |
| 120721    | CACTTGAACC TGGGAGGCAG AGGTTGCACT GAGCCGAGAT CTTGCCACTG CACTCCAGCC   |
| 120781    | TGGGTGACAG TGAACCTCCG TCTCAAAAAA AAAAAATAAA ATACAGCTGG GCACGGTGGC   |
| 120841    | TCACGCTGTG AATCCAGCA CTTTGGGAGG CCGAGGCGAG CGGATCACAA GGTCAGGAGA    |
| 120901    | TATAGACCAT CCTGGCTAAC ACGGTGAAAC CCGGTCTCTA CTAAAAATAC AAAAAATTAG   |
| 120961    | CACGCGCTGG TGGCAGGTGC CTATAGTCCC AGCTACTCAC AAGGCTGAGG CAGGAGAATG   |
| 121021    | GCATGAACCT GGGAGGCGGA GCTTGCACTG AGCCGAGATT GTGCCACTGC ACTCCAGCCT   |
| 121081    | GGGCGAGAGA GTGAGACTCC GTCTCAAAAC AAAAAACAAA ACAAAAACAA AAACAAACAC   |
| 121141    | ACCAACAAAA CTTAAAGAA TATAAATGGA TTGTTGTGTA CACAAAGGAC AAATGTTTGA    |
| 121201    | GGGGATGGAT ACCCCATTTT CCATGATGTG ATTATTATAC ATTGTGTGTC TGTATCAAAA   |
| 121261    | CATCTCATGA GCCCATAAAA TATATACACC TAACTATGTA CCCACAAAAA TTAAAAAAAT   |
| 121321    | ATATTTTTTA AGGTGAAGAG GGAGGCGAGA TGCTGGCCTT AACCCCTAAC CCGTTGTTCT   |
| 121381    | CCCTGCAAGC TGTCACAGG GCCTCTCAGA CTCGAGGTTT AGCTATATGG ATGCATGAGC    |
| 121441    | TTGGTCCCCA GCCAACATGG GAGACACTTC ACCATCCGCA GCAGCTACAG CACAGGAACC   |
| 121501    | CTGGGTCACT GCCATGTCCC CTCTGTGACT TTGTTTAAAC AGAAAAATGAT GCTCTGGGCC  |
| 121561    | GGCTGTGGTG GCCCACACCT ATAATCCCAG CACCTTGGGA GGCGGGGGTG GGCAGATTGC   |
| 121621    | CTGAGGTGAG GAGTTGGAGA TCAAGCTGGC CGACATGGCG AAACCCCATG TCTACTAAAA   |
| 121681    | ATACAAAAAC TAGCCAGGCA TGGTGGCACA TGCCTGTAAT CCCAGTACT TGGGAGGCTG    |
| 121741    | AAGCAGGAGA ATCACTTGAA CCCAGGAGGC AGAGGCTGAG TGAGCCAAGA TCGTGCCAAT   |
| 121801    | GCACCTCCAG TTGGGTGAGG GAGTGAGACT CCGTCTCAAA AAAAAAAAAA AAGAAAAGAA   |
| 121861    | AAGAAAAGAA AGTGATCCTA CTGGAACCAT GCTTACTCCC CTCCCCACCT CACACTGTGT   |
| 121921    | AGAAATTAGT GCTGTCGGCC AGGCGCGGTG GCTCATGCCT GTAATCGCAG CACTTTGGGA   |
| 121981    | GGCCAAGGCA GCGGATCAC GAGGTGAGGA GATCAAGACC ATCTTGCTTA ACACAGTGAA    |
| 122041    | ACCCTGTCTC TACTAAAAAT ACAAAAAATT AGCCGGGCAT GGTGGCAGGC ACCTGTAGTC   |
| 122101    | CCAATACTTT GGGAGGCTGA GGCAGGAGAA TGGCATGAAC CTGGGAGGCG GAGCTTGCAG   |
| 122161    | TGAGCCAAGA TCGCGCCACT GCATACCAGC CTAGGTGACA GAGTGAGACT CAGCAAAAAA   |
| 122221    | AGAAAGAAAG AAAGAAAGAA ATCAGTGCTG TCTATACTTC TTTCTGCACT GATGGAAATA   |
| 122281    | TTCTGTATCT GTGCTGTCCA GTATAGTAGC CACTAGCTAC ATGTGGCACT TGAAACATGG   |
| 122341    | CTGGTACAGT TGAGGAAGAG TGGCTGCCAT ATCGGACGAC ACAGCTATAG ATTCTGTCAC   |
| 122401    | CCCACCCCGA GAGTCCAGAG CGGGGACTTC TGCCTTAGGC CCTATTGAGG GCTGATTTTT   |
| 122461    | ACTTGAACCC TTAAGTGTGGG AAGAGAAGGC CATGAGAAGT TCAAGTCTAGA ATGTGACTCC |
| 122521    | TTATTTTCTG GCTCCCTTGG ACACCTTGTG GGAATTTAGTC TCCCTGTGGA AAGTATTCCA  |
| 122581    | CAAGTGGTGC CACCACCCCA GCTGTGAGAG CAGCTGGGAG CTGCTTTTGT CATCTTTCCC   |
| 122641    | TGGAAGTCTC TGTGGGCTGT CTCTTCTCTA TGCCTTGTCC CATGCTTGGG CATGGTGTCA   |
| 122701    | AGCGTCAGGA GGGAGAAAGG GTCCCTATTT ATTTATTTAG AGAGGGACCC TTCTTCTGTT   |
| 122761    | CCAGGCTGCG AGTGCACTGG TCGATCTCTG GCTCACTGCA ACCTCCGCTT CCTGGGTCTA   |
| 122821    | ACAGGATCTC CTGCCTCAGC CTCCCTGAGTA GCTGAGATTA CAGGCACATG CCAACATGCC  |
| 122881    | TGGCTAATTT TTTTTTTTTT TTTTTTTTTT TTTTTTTTGG ATGGAGTTGT ACTCTCATTG   |
| 122941    | CCAGGCTGCG AATGTAATGG CACAATCTCG GCTCACTGCA ACCTCCACCT CTGGGATTCA   |
| 123001    | AGCAATTCTC CTGTCTCAGC TTCCCAAGTA GCTGGGATTA CAGGTGCCCG CCACCATGCT   |
| 123061    | CAACTAATTT TTGTATTTT AGACGAGGTT TCACCATGTT GGTGAGACTG               |
| 123121    | GTCTCAAACCT CCTGACCTCA GGTGATCCAC CTGCTCCGCG CTCCCAAGT GCTAGGATTA   |

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| 123181 | CAGGCATGAG  | CCACCACGCC  | CGGCCTGAAA  | GGGTTCCTTAT | TTAGTGTGCA | TTTGACATT  |
| 123241 | CAATTTAATT  | CCAAGGTCTT  | GTGGGGTCAT  | GGTTTACAGG  | ATGTTGATAT | AGAAAAGACT |
| 123301 | TCACTTAATG  | GGCCGGGCGC  | AGTGGGTCAT  | GCCTGTAATC  | CCAGCACTTT | GGGAGGCCGA |
| 123361 | GGCAGGCAGA  | TCAGGAGGTC  | AGGAGATTGA  | GACCATCCTG  | GCTAACACAG | TGAAACCCCA |
| 123421 | TCTCTACTGA  | AAATACAAAA  | AATTAGCTGG  | GCGTGGTGGC  | AGGCACCTGT | AGTCCAGCC  |
| 123481 | ACTCGGTTGG  | CTGAGGCAGG  | AGAATGGCAT  | GAACCCGGGA  | GGCGGAGCTT | GCAGTGAGCA |
| 123541 | GAGACCATTG  | CACTGCACTC  | CAGCCTGGGC  | GACAGAGCAA  | GACTCTGTCT | CAAGAAAAAA |
| 123601 | AAAAAATAAA  | CAGACTTTAC  | TTACTGGAAG  | CCAACCAATG  | TATATTTAGA | GTAATTTTTC |
| 123661 | CTGGGCTGAG  | CTGTCAATTA  | CTTTTGCAGT  | ATCTCAAGAA  | GAAGAGTTTA | CAGTGTAAT  |
| 123721 | ATTTGATGCA  | CACTTTGATT  | ATATAGATGA  | AGCAAACTAT  | TTTCAAGAGC | TTTGCAAGGA |
| 123781 | CTTACTTGTA  | TCCAAACACC  | ATTCTAAAGG  | AGTCTACCT   | ACTTCTAAAG | GCTGGTCTCT |
| 123841 | ACTTGGAACC  | ACTTGCTTGG  | CCCTGGTTCA  | AGTCTGTCTG  | CAAACCTGGA | AGTCTGTCTA |
| 123901 | TTGTCTTCTT  | CCCTCCAGAG  | CAGTGGCACC  | CAATCTAATT  | TTTGCTGTGC | CCCAGCAGCC |
| 123961 | CTGGCAGT    | TGCCCTGTAG  | ACTGCAGACC  | TCATGTAATG  | TATGTTAAGT | CCACAGAACC |
| 124021 | ACAGAAGATG  | ATGGCAAGAT  | GCTCTTGTGT  | GTGTTGTGTT  | CTAGGAGGTG | GCCAGGTGGA |
| 124081 | AGTAAATATG  | GAGAAGCTTG  | ACTTCAAGGA  | CAGAGTCCAG  | TCGAAGATTG | GGTCCCTGGA |
| 124141 | CAATATCAC   | CACGTCCTTG  | GCGGAGGAAA  | TAAAAAGGTA  | AAGGGGTAG  | GGTGGGTTGG |
| 124201 | ATGCTGCCCT  | TGGGTATATG  | GGCATTAAATC | AAGTTGAGTG  | GACAAAGGCT | GGTCCAGTTC |
| 124261 | CCAGAGGAGG  | AAAACAGAGG  | CTTCTGTGTT  | GACTGGCTGG  | ATGTGGGCCC | TCAGCAGCAT |
| 124321 | CCAGTGGGTC  | TCCACTGCCT  | GTCTCAATCA  | CCTGGAGCTT  | TAGCACGTTT | CACACCTGGG |
| 124381 | CCCCAACCTG  | GAGAGGCTGA  | CCAATGGGTC  | TCAGGGGCAG  | CTCGGTTGCT | GGAGTTTTTG |
| 124441 | TTTTTATTTA  | TTTTTATGTA  | TTTAAAGGCA  | GGTCTCTGTA  | TTAGTCCATT | CTCACACTGC |
| 124501 | TAATAAAGAC  | ATACCCAAGA  | CTGGGTAATT  | TATAAAGGAA  | AGAGGTTTAA | TGGACTCACA |
| 124561 | GTTCCACATG  | GCTGGGGAGG  | CCTCAAAATC  | ATGGCGGAAG  | GCAAAGGAGA | AGCAAAGGCA |
| 124621 | TTTCTTACAT  | GGCGACAGGC  | AAGAGAGCGT  | GTGCAGGGGA  | ACTCCCATTT | ATAAAACCAT |
| 124681 | CAGACTTCAT  | GAGATTTATT  | CACATATCAT  | AGAACAGCAT  | GGGAAAGACC | CGCCCCCATG |
| 124741 | ATTCAGTTAC  | CTCCCACTGG  | GTCCCTCCCA  | TGACACATGG  | AATTATGGGA | GCTACAATTC |
| 124801 | AAGATGAGAT  | TTGGGTGGGG  | ACACAGCCAA  | ACCATATCAG  | TCTCCCTCTG | TCATCCAGGC |
| 124861 | TGGAGTGCAC  | TGGCATGATC  | TCGGCTCACT  | GCAGCCTCTA  | CCTCCCTGGG | TCAGGTGATC |
| 124921 | TTCCCACTCT  | AGCCTCCAG   | GTAGCTGGAA  | CTACAGGTAC  | CTGCCACTAT | GCCTGGCTAA |
| 124981 | ATATTTTGTA  | TTTCTGTG    | AGACGAGGTT  | TTGCCACGTT  | GCCCAGGCTG | GTCTTGAACT |
| 125041 | CCTGAGGTCA  | AGCAATATGC  | CCACCTCGGC  | CTCCCAAGGT  | GCTGGGATTA | CAGGTGTGAG |
| 125101 | CCACAGTGCT  | CGGCCTAAGT  | CACCTGCAGT  | TTTAAAGCTC  | CCAGGTGATT | CTTCAGTGCA |
| 125161 | GTCAAAAGTG  | AGAACTGGCT  | GGGTGCGGTG  | GCTCATGCCT  | GTAATCCAG  | CACCTTGGGA |
| 125221 | GGCGAAGGTC  | GGCAGATGGC  | TTGAGGTCAG  | GAGTTCAAGA  | CACGCCTGGC | CAACATGGTA |
| 125281 | AAACCCCATC  | TCTACTAAAA  | ATACAAAAGT  | TAGCTGGGTG  | TGGTGGTGCG | TGCCTGTAAT |
| 125341 | CCCAGCTACT  | TGGGAGGCTG  | AGGCATGAGA  | ATTGCTTGAA  | CCCAGGGGAC | AGAGGTTGTA |
| 125401 | GTGAGCCGAG  | ATCGTGCCAC  | TGCACTCCAG  | CCTGGGCAAC  | AGAGTGAGAT | TCCATCTCAC |
| 125461 | AAAAAATAAA  | AAAAAAGCGA  | GAACCACTGT  | CCTAGGCCCT  | GATGTTTGCA | GGCAACTAAA |
| 125521 | AAAGGAAGTG  | GACATCCCCA  | GTCAGCTGTG  | GCGCACCAAG  | AACAAGTCAT | GGGAACATAA |
| 125581 | CTTAATTTTC  | TAAATGGGTT  | ACTAGGCACT  | TAGAGCAAAA  | CAATGATGGC | GAAATCCTGA |
| 125641 | TTTCAGCAAA  | GCCTCTGCCT  | GCCTGTCTTG  | GAAGTATCCA  | CATGAGGCTG | CTGGGGCCTT |
| 125701 | GGTGTCCCCA  | GCAGTTTCTA  | GTCTCTAGGT  | CTTGCTGTGG  | GTGCTGTGTC | AGTGAGGGTG |
| 125761 | TGTGTGGCGC  | TGGGTGAGCT  | CTGTCTAGGC  | CTGGCACAGG  | ATGCGGTCTG | GTAGCTGCTG |
| 125821 | CTTCTCTTCT  | GCAGAAGCGC  | AGCCAAGCAC  | CCTCTGGGGT  | TTCAGGCCCC | CACCCAGCCT |
| 125881 | GAAGTTCCTG  | GAGTGGGTCA  | CTTCCCAACC  | TTCAAGGCTC  | CCCAGCAGCT | GACTGGGGAG |
| 125941 | TGGTGGAGGG  | AAAAGGGATT  | GTATTAGTCC  | GTTTTCACGC  | CGTGTATGAA | GACATACCCG |
| 126001 | ATACTGGGCA  | GTCTAAAAGA  | TAGAGGTCTG  | ATGGACTCAC  | AGTTCCACGT | GACTGGGGAG |
| 126061 | GCCTGACATC  | CATGGTGGAA  | GGTGAAAGGC  | TTGTCTCACA  | CGGTGGCAGA | CAAGAGAAAA |
| 126121 | GAGCTTTGTC  | AGGGGAACCT  | CCCTTTATAA  | AACCATCAGA  | TCTCGGGAGA | CTTATTCACT |
| 126181 | ATCATGAGAA  | CAGCACGGGA  | AAGACCCTCC  | TCTATGATTC  | AATTACCTCC | CACCAGGTCC |
| 126241 | CTCCCAACAAC | ATGTAGGAAT  | TGTGGGAACT  | ACAATTCAAG  | ATGACATTTG | GGTGGGGACA |
| 126301 | CAGCCAAACC  | ATATCAGGGC  | GTCCAGAAAA  | GGGTATAGGG  | TCTGAGACCC | AAGTCAGCAT |
| 126361 | GAGAAAGTAT  | GCTTCTCATG  | GTGGCCCACT  | TGGGTGGAAG  | TGGCAGCCGG | GCCGTCTTTC |
| 126421 | CACCAAGGCA  | CTCAAGTAGC  | AGCTGAGAGA  | CCCCTGCCCT  | GGCCAGTCCC | CGCCCTCCCC |
| 126481 | TCTTGCCACT  | GCCTCTGGTT  | CTGAACAGAT  | GGGCACCCCT  | ATCTTGATTT | TGTGATTAAT |
| 126541 | GTCTAACCAAT | GTAAGTTTGT  | GAGAAGGGTT  | TGCTGATACA  | GCCTTGCTGC | AGATGCTGCG |
| 126601 | AACTGTGGCC  | TGGGGCAGAC  | CTTACCTCCA  | GACACGCCCT  | GAGGCAGGGG | AGGGCACTGG |
| 126661 | CCCGTAGCTG  | GCCGAGAGCT  | CTCGGGTTGC  | GCGACAGGGA  | TACTTTTCAG | CGCTGGGTC  |
| 126721 | GCTATCCAAA  | GTGAGAAAAC  | GAGGAGGGAC  | CAGGAGGCTG  | TCCGCCCTCA | GAGATGTGGG |
| 126781 | GGCCAGGTCC  | AGTTATCTGG  | GGAAGCAGTA  | AGCTTCTCTG  | CTGTTTCTAA | CCCCAGGCTC |
| 126841 | CCCCTGGTCT  | AAGGCAGGGC  | CTCCAGCCCT  | CGGGGCACCT  | TAAAGATATC | TGGGCTGGGC |
| 126901 | CCCATCCCCA  | CAGTCTGACT  | GAGTGGGTCT  | GGATAGGGCC  | TGAGCATTGG | TGATTTCCTG |
| 126961 | GGTGAAAGGA  | GGCCCTCCAC  | AGTCTCTGGA  | AGCTTCTCTG  | TGTTAGGAAA | AGCTCTGGGC |
| 127021 | TTGACTCTGC  | TTTGAAAGTC  | AAGATCCGCA  | AATCCTCTCA  | GCCTCAGTTT | CTCCTTCAGC |
| 127081 | AAGATGAAAT  | GGAAATGCTG  | TACCTACGTC  | CCGGGGTGGT  | TGTGAGACCC | AAAAAAGACA |
| 127141 | ATGTTCTGGA  | AGGTTCTCTG  | TGCGTTGCAG  | TCCTCTAAGA  | ACCTGAGTTA | GAGCCACGCT |
| 127201 | GAGTCTCAGC  | TTCTTGCTCT  | CTTCTGTTTC  | AAATCTCGTC  | ATGTGATAGC | TCAGGAAGGG |
| 127261 | TAGGACGGGC  | CCTGCCCCCT  | ACTCAGAAAA  | CACCATCCTG  | GTCTGGGGTA | TCCCCGACGC |
| 127321 | ATTAGTCCCC  | TGTTTTTCCCA | GTGATTGAG   | AAAAATTGCT  | AACAAGCAGT | GGGGCACACC |
| 127381 | ACCAGCCTCC  | TGGGTTTCTT  | TCAGTTTGGG  | GATTTTGGGA  | CATTCCAGG  | AATGTCTTAA |
| 127441 | AAAACACTTC  | AAAAACATT   | AACATAAATA  | TTTTTATCAA  | AGCCTGTATT | AAATGGTCTT |
| 127501 | TCAAGAAAAAT | ACAGTAACAG  | GTCAAGCATG  | GTGGCTCATG  | CCTGAACACC | CAGCACTTTG |
| 127561 | GGAGGCCAAG  | GCAGGCAGAT  | CACCTGAAAT  | CAGGAGTTCA  | AGACCAACCT | GGCCAACACA |

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SEQ ID NO Sequence

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| 127621 | GCCAAATCCC | ATCTCTACAA  | AAAATACAAA | AATTAGCTGG | GTGTGGTGGC | ACACACCTGT  |
| 127681 | AGTCCCAGCT | ACTTGGGAGG  | CCGAGGCAGG | AGAATTGCTT | GATCCCGGAG | GCGGAGGTTG  |
| 127741 | CAGTGAGCTG | AGATCGTGCC  | ACTGCACTCC | AGCGTGGGTG | ACAAGGTGAA | TCTTTGTCTC  |
| 127801 | AAAAAAAAAA | AAAAAAAAAA  | GATAAAATAC | AGTATACAGT | AATAGAGAAC | AATCCTTTTT  |
| 127861 | TCAAAGTAGT | GACCCCAAAT  | GAACAAAATA | TGCATCTAGC | TTAAATGCGA | ACCTGGTTTT  |
| 127921 | CTCTACGCCC | ATTCAAGCCC  | CTGCAATAGG | GGCCCTTAC  | CCCGCATCCA | TGGACTCCTA  |
| 127981 | AAATTATATG | GAAAATGGCT  | GTGTGTGAGT | GTGGATGGAC | ATGTGCACAC | ATATTTTGGG  |
| 128041 | CTTTACCAAG | TGCTCAAAGA  | GCCTAGGACC | CAAAAAGGGC | TGAGAATGAC | CGTGTGCGCC  |
| 128101 | ACTTCAGGGT | CATCAGGAAT  | TGCTGTGCAC | TGCTCACTTC | TCCAGTGAAC | ACTTTCTGCT  |
| 128161 | TCTGTGTTTC | CTGGTATCCT  | TTGGGACTCC | TGGCTAGGTC | ATGTGTTTCT | CTACTTTCAA  |
| 128221 | AAGGGCTTCA | GCCAGGCACG  | ATGGCATGAG | CCTGTAGTCC | CAGTTGCTCT | GGAGGTTAAG  |
| 128281 | GTGGGAAGAT | TGCTTGAGCC  | CAGGAATTTG | AGGCCAGCCT | GGGCAAGTAG | ATAGGTAGAT  |
| 128341 | GATTGATAGA | TAGATAGATA  | GATAAATAGA | TGGATAGATA | AGTCGCTAGA | CAGTCATCCA  |
| 128401 | TCCACCCATG | CACACATAAA  | AAGGCCTTTG | TCATGTCTAG | TTTTGTGGCC | CACCTGCCAG  |
| 128461 | TGTTGCCACC | AGTTGTGTGCC | CCTCCAAACT | CATCAGTCAC | TGGCAAACAG | GAGGAATGTG  |
| 128521 | TGGCTCATGT | CTGGGCATCA  | GTGGCTGTGG | GAGACATCCT | TGATCTTCTC | CAGCTTCTCC  |
| 128581 | TCTCCACATT | TCCTTTGCAA  | TCTGGCAATA | TCTATATAAA | TAAATGTGTC | ATGCCCTTTG  |
| 128641 | ACCTAAGAGC | TTCACCTTCTA | GGACCCACTT | ACACGTGTGT | GACATGATGT | TCATACGGGT  |
| 128701 | TTATTATATC | GAGGTTGTTC  | ATACACACCA | TTGCCTGTAA | TCACTAAAGG | CGGGAGCAGC  |
| 128761 | TCCACATCCC | ATCCACAGAG  | GAGTAGATGC | CTTTTGTGAC | ATCCGTGGCG | ACGGAATACT  |
| 128821 | AAGCAGCCTG | TGTATCTATA  | CACTCACACG | TGTTTGTTTA | TGTTGTGAAT | ATCTCTGGAG  |
| 128881 | GGTACACAAG | AAACTTAAAA  | TGATCACTGT | CTCTGGGGAG | GGTACCTGGG | TGCCTGGGAG  |
| 128941 | GCCAGGTGAG | GAAGGAGTGG  | GCACAGGTAT | TACCAATTGG | AAGACAATAA | AAACAACAGC  |
| 129001 | TCCTGGCCAG | GCGCAGTGGC  | TCACGCCTGT | AATGGCAGCA | CTCTGAGAGG | CTGAGGCGGG  |
| 129061 | CAGATTGCTT | GCGTCCAGGA  | GTTCAGAGCC | AGCCTGGGCA | ACATAGCAAA | ACCCCGTTTC  |
| 129121 | TATTAATAAT | ACAAAAAATT  | AGCCAGGTGT | GGTGGCATGC | ACCTGTAAAT | CCAGCTACTC  |
| 129181 | GGGAGGCTGA | GGTGGGAGAA  | TCACCTGAGC | CTGGGAGGTC | AAGGCTGCAG | TGAGGTGAGA  |
| 129241 | TTGTGCCACC | GCACCTTAGC  | CTGGGCAGTA | GAGCAAGACC | CTGTCTCAAA | AACAAACAAA  |
| 129301 | AAACAGTCCC | TGGCACTCTG  | GGCCAGGCCT | GGCAGGGCAG | TTGGCAGGGC | TGGTCTTTCT  |
| 129361 | CTGGCACTTC | ATCTCACCCCT | CCCTCCCTTC | CTCTCTTTCG | AGATTGAAAC | CCACAAGCTG  |
| 129421 | ACCTTCCGCG | AGAACGCCAA  | AGCCAAGACA | GACCACGGGG | CGGAGATCGT | GTACAAGTGC  |
| 129481 | CCAGTGGTGT | CTGGGGACAC  | GTCTCCACGG | CATCTCAGCA | ATGTCTCCTC | CACCGGCAGC  |
| 129541 | ATCGACATGG | TAGACTCGCC  | CCAGCTCGCC | ACGCTAGCTG | ACGAGGTGTC | TGCCTCCCTG  |
| 129601 | GCCAAGCAGG | GTTTGTGATC  | AGGCCCTTGG | GGCGGTCAAT | AATGTGGAG  | AGGAGAGAAT  |
| 129661 | GAGGAGTGTG | GGAAAAAATA  | AGAATAATGA | CCCGGCCCCC | GGCCTCTGCC | CCAGCTGCTC  |
| 129721 | CCTCGCAGTT | CGGTAAATTG  | GTTAATCACT | TAACCTGCCT | TTGTCACTCG | GCTTTGGCTC  |
| 129781 | GGGACTTCAA | AATCAGTGAT  | GGGAGTAAGA | GCAAAATTCA | TCTTTCCAAA | TTGATGGGTG  |
| 129841 | GCCCTAGTAA | AAAAATTTTA  | AAAAAAAACA | TTCAAAAACA | TGGCCACATC | CAACATTTCC  |
| 129901 | TCAGGCAATT | CCTTTTGATT  | CTTTTTCCTT | CCCCCTCCAT | GTAGAAGAGG | GAGAAGGAGA  |
| 129961 | GGCTCTGAAA | GCTGCTTCTG  | GGGGATTTCA | AGGGACTGGG | GGTGCCAAAC | ACCTCTGGCC  |
| 130021 | TGTTTGTGGG | GGTGTACAG   | AGGCAGTGGC | AGCAACAAGG | GATTGTAAAC | TTGGTGTGTT  |
| 130081 | CGTGGAGCCA | CAGGCAGACG  | ATGTCAACCT | TGTGTGAGTG | TGACGGGGGT | TGGGGTGGGG  |
| 130141 | CGGGAGGCCA | CGGGGGAGGC  | CGAGGCAGGG | GCTGGGCAGA | GGGGAGAGGA | AGCACAAGAA  |
| 130201 | GTGGGAGTGG | GAGAGGAAGC  | CACGTGCTGG | AGAGTAGACA | TCCCCCTCCT | TGCGCTGGGG  |
| 130261 | AGAGCCAAGG | CCTATGCCAC  | CTGCAGCGTC | TGAGCGGCCG | CCTGTCTCTG | GTGGCCGGGG  |
| 130321 | GTGGGGGCTC | GCTGTGGGTC  | AGTGTGCCAC | CCTCTGCAGG | GCAGCCTGTG | GGAGAAGGGA  |
| 130381 | CAGCGGGTAA | AAAGAGAAGG  | CAAGCTGGCA | GGAGGGTGGC | ACTTCGTGGA | TGACCTCCTT  |
| 130441 | AGAAAAGACT | GACCTTGATG  | TCTTGAGAGC | GCTGGCCTCT | TCCTCCCTCC | CTGCAGGGTA  |
| 130501 | GGGGGCTCTG | GTTGAGGGGC  | TTCCCTCTGC | TCCACAGAAA | CCCTGTTTAA | TTGAGTCTGT  |
| 130561 | AAGGTTGGAA | CTGTGTCAT   | GATTTTGGCC | ACTTTGCAGA | CCTGGGACTT | TAGGGCTAAC  |
| 130621 | CAGTTCTCTT | TGTAAGGACT  | TGTGCTCTCT | GGGAGACGTC | CACCCGTTTC | CAAGCCTGGG  |
| 130681 | CCACTGGCAT | CTCTGGAGTG  | TGTGGGGGTC | TGGGAGGCAG | GTCCCGAGCC | CCCTGTCTCT  |
| 130741 | CCCACGGCCA | CTGCAGTCAC  | CCCGTCTGCG | CCGCTGTGCT | GTTGTCTGCC | GTGAGAGCCC  |
| 130801 | AATCACTGCC | TATACCCCTC  | ATCACACGTC | ACAATGTCCC | GAATTCCCAG | CCTACCCACC  |
| 130861 | CCTTCTCAGT | AATGACCTTG  | GTGGTTTGCA | GGAGGTACCT | ACTCCATACT | GAGGGTGAAA  |
| 130921 | TTAAGGGAAG | GCAAAGTCCA  | GGCACAAGAG | TGGGACCCCA | GCTCTCACT  | CTCAGTTCCA  |
| 130981 | CTCATCCAAC | TGGGACCCCT  | ACCACGAATC | TCATGATCTG | ATTTCGGTTC | CTGTCTCCTC  |
| 131041 | CTCCCGTCAC | AGATGTGAGC  | CAGGGCACTG | CTCAGCTGTG | ACCCTAGGTG | TTTCTGCCTT  |
| 131101 | GTTGACATGG | AGAGAGCCCT  | TTCCCTTGAG | AAGGCCCTGC | CCCTTCTGTG | GCTGAGCCCA  |
| 131161 | CAGCAGCAGG | CTGGGTGTCT  | TGGTTGTCTG | TGGTGGCACC | AGGATGGAAG | GGCAAGGCAC  |
| 131221 | CCAGGGCAGG | CCCACAGTCC  | CGCTGTCCCC | CACCTGACCC | CTAGCTTGTA | GCTGCCAAC   |
| 131281 | TCCACAGCAG | CCCAGCCCGC  | TGCTCAGCTC | CACATGCATA | GTATCAGCCC | TCCACACCCG  |
| 131341 | ACAAAGGGGA | ACACACCCCC  | TTGGAAATGG | TTCTTTTCCC | CCAGTCCCAG | CTGGAAGCCA  |
| 131401 | TGCTGTCTGT | TCTGCTGGAG  | CAGCTGAAAC | TATACATAGA | TGTTGCCCTG | CCCTCCCCAT  |
| 131461 | CTGCACCTCT | TTGAGTTGTA  | GTTGGATTGG | TCTGTTTATG | CTTGGATTCA | CCAGAGTGAC  |
| 131521 | TATGATAGTG | AAAAGAAAAA  | AAAAAAAAAA | AAAGGACGCA | TGTATCTTGA | AATGCTTGTA  |
| 131581 | AAGAGGTTTC | TAACCCACCC  | TCACGAGGTG | TCTCTACCCC | CCACACTGGG | ACTCGTGTGG  |
| 131641 | CCTGTGTGGT | GCCACCTCTG  | TGGGGCCTCC | CAAGTTTGTG | AAGGCTTTCC | TCAGCACCTG  |
| 131701 | GGACCCAATC | GAGACCAAGT  | TCTAGCAGCT | AAGGAGGCCG | TTAGCTGTGG | ACGAAGGCCCT |
| 131761 | GAAGCACAGG | ATTAGGACTG  | AAGCGATGAT | GTCCCCCTTC | CTACTTCCCC | TTGGGGCTCC  |
| 131821 | CTGTGTGAGG | GCACAGACTA  | GGTCTTGTGG | CTGGTCTGGC | TTGCGGCGCG | AGGATGGTTC  |
| 131881 | TCTCTGGTCA | TAGCCCGAAG  | TCTCATGGCA | GTCCCAAAGG | AGGCTTACAA | CTCCTGCATC  |
| 131941 | ACAAAGAAAA | GGAAGCCACT  | GCCAGCTGGG | GGGATCTGCA | GCTCCAGAAA | GCTCCGTGAG  |
| 132001 | CCTCAGCCAC | CCCTCAGACT  | GGGTTCTCTT | CCAAGCTCGC | CCTCTGGAGG | GCGAGCGCAG  |

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 132421 CTGAAAGGAA GTCTCTGGGC CCAGAACTCT CCACCAAGAG CCTCCCTGCC GTTCGCTGAG  
 132481 TCCCAGCAAT TCTCCTAAGT TGAAGGGATC TGAGAAGGAG AAGGAAATGT GGGGTAGATT  
 132541 TGGTGGTGGT TAGAGATATG CCCCCCTCAT TACTGCCAAC AGTTTCGGCT GCATTTCTTC  
 132601 ACGCACCTCG GTTCCTCTTC CTGAAGTTCT TGTGCCCTGC TCTTCAGCAC CATGGGCCTT  
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 133381 TCGCTTCCAC CTCTCATCT TGTCTCTCCA AGTAAAGCCA CGAGGTCGGG GCGAGGGCAG  
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 133621 TCTTCGGTAA TTCTGAGGGT GGGGGGAGGG ACATGAAATC ATCTTAGCTT AGCTTTCTGT  
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24

1 GCAGTCACCG CCACCCACCA GCTCCGGCAC CAACAGCAGC GCCGCTGCCA CCGCCACCT  
 61 TCTGCCGCGG CCACCAAGC CACCTTCTCC TCCTCCGCTG TCCTCTCCCG TCCTCGCCTC  
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 541 GTGGTCCAGG AAGGCTTCTC CCGAGAGCCA GGGCCCCCAG GTCTGAGCCA CCAGCTCATG  
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 2521 AATGAGAGAG TGTGGAATAA AAAAGAATAA TGACCCCGCC CCGCCCTCTC GCCCCAGCT  
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2641 CTCGGGACTT CAAAATCAGT GATGGGAGTA AGAGCAAATT TCATCTTTCC AAATTGATGG  
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 2761 TCCTCAGGCA ATTCTTTTGT ATTCTTTTTT CTTCCCCCTC CATGTAGAAG AGGGAGAAGG  
 2821 AGAGGCTCTG AAAGCTGCTT CTGGGGGATT TCAAGGGACT GGGGGTGCCA ACCACCTCTG  
 2881 GCGCTGTTGT GGGGTGTCTA CAGAGGCAGT GGCAGCAACA AAGGATTTGA AACTTGGTGT  
 2941 GTTCGTGGAG CCACAGGCAG ACGATGTCAA CTTGTGTGA GTGTGACGGG GGTGGGGTGT  
 3001 GGGCGGAGG CCACGGGGGA GGCCGAGGCA GGGGCTGGG AGAGGGGAGA GGAAGCACAA  
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 3361 GTAGGGGGCC TGAGTTGAGG GGCTTCCCTC TGCTCCACAG AAACCTCTGT TTATTGAGTT  
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 3541 GGGCCCACTG CATCTCTGGA GTGTGTGGGG GTCTGGGAGG CAGGTCCCGA GCCCCTGTCT  
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 6481 ATTTCTTCGG TAATTCTGAG GGTGGGGGGA GGGACATGAA ATCATCTTAG CTTAGCTTTC  
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 6601 GCTGTAAAG TGAATTTGGA AATAAGTTA TTACTCTGAT TAAA

25

1 MAEPRQEFV MEDHAGTYGL GDRKDQGGYT MHQDQEGDTD AGLKESPLQT PTEDGSEEPG  
 61 SETSDAKSTP TAEDVTAPLV DEGAPGKQAA AQPHTIEPEG TTAEAGIGD TPSLEDEAAG  
 121 HVTQEPESGK VVQEGFLREP GPPGLSHQLM SGMPGAPLLP EGPREATRQP SGTGPEDEB  
 181 QRHAPPELLKH QLLGDLHQEG PPLKAGGKKE RPKSKEEVDE DRDVEDESSPQ DSPPSKASPA  
 241 QDGRPPQTAA REATSPGFP ABGAIPLPVD FLSKVSTIEP ASEPDGSPSVG RAKGQDAPLE  
 301 FTFHVEITPN VQKEQAHSEE HLGRAAFPQA PEGPPEARP SLGEDTKHAD LPEPSEKQPA



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SEQ ID NO Sequence

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| 361 | AAPRGKPVSR | VPQLKARMVS | KSKDGTGSDD | KKAKTSTRSS  | AKTLKNRPCL | SPKHPTPGSS |
| 421 | DPLIQPSSPA | VCPEPPSSPK | YVSSVTSRTG | SSGAKEMKLLK | GADGKTKIAT | PRGAAPPQK  |
| 481 | QANATRIPA  | KTPPAKTPP  | SSATKQVQRR | PPPAGPRSER  | GEPPKSGDRS | GYSSPGSPGT |
| 541 | PGSRRTPSL  | PTPPTREPKK | VAVVRTPPKS | PSSAKSRLQT  | APVPMPLKN  | VKSKIGSTEN |
| 601 | LKHQPGGKV  | QIINKLDLS  | NVQSKCGSKD | NIKHVPGGGS  | VQIVYKPDV  | SKVTSKCGSL |
| 661 | GNHHKPGGG  | QIEVKSEKLD | FKDRVQSKIG | SLDNITHVPG  | GGNKIETHK  | LTFRENAKAK |
| 721 | TDHGAEIYK  | SPVVSQDTS  | RHLSNVSSST | SIDMVDSPQL  | ATLADEVSA  | LAKQGL     |

  

|    |      |            |            |            |            |            |            |
|----|------|------------|------------|------------|------------|------------|------------|
| 26 | 1    | GCAGTACCCG | CCACCCACCA | GCTCCGGCAC | CAACAGCAGC | GCCGCTGCCA | CCGCCCACCT |
|    | 61   | TCTGCCGCG  | CCACCACAGC | CACCTTCTCC | TCCTCCGCTG | TCCTCTCCCG | TCCTCGCCTC |
|    | 121  | TGTCGACTAT | CAGGTGAAC  | TGAACACAG  | ATGGCTGAGC | CCCGCCAGGA | GTTGGAAGTG |
|    | 181  | ATGGAAGATC | ACGCTGGGAC | GTACGGGTTG | GGGGACAGGA | AAGATCAGGG | GGGCTACACC |
|    | 241  | ATGCACCAAG | ACCAAGAGGG | TGACACGGAC | GCTGGCCTGA | AAGAATCTCC | CCTCGAGACC |
|    | 301  | CCCACTGAG  | ACGATCTGA  | GGAACCGGG  | TCTGAAACCT | CTGATGCTAA | GAGCACTCCA |
|    | 361  | ACAGCGGAAG | CTGAAGAAG  | AGGCATTGGA | GACACCCCA  | GCCTGGAAGA | CGAAGCTGCT |
|    | 421  | GGTACAGTGA | CCCAAGCTCG | CATGGTCAGT | AAAAGCAAAG | ACGGGACTGG | AAGCGATGAC |
|    | 481  | AAAAAAGCCA | AGGGGCTGA  | TGGTAAACG  | AAGATCGCCA | CACCGCGGG  | AGCAGCCCT  |
|    | 541  | CCAGGCCAGA | AGGGCCAGGC | CAACGCCACC | AGGATTCCAG | CAAAAACCCC | GCCCGCTCCA |
|    | 601  | AAGACACCA  | CCAGCTCTGG | TGAACCTCCA | AAATCAGGGG | ATCGCAGCGG | CTACAGCAGC |
|    | 661  | CCCGGCTCCC | CAGGCACTCC | CGGCAGCCGC | TCCCGCACCC | CGTCCCTTCC | AACCCACCCC |
|    | 721  | ACCCGGGAGC | CCAAGAAGGT | GGCAGTGGTC | CGTACTCCAC | CCAAGTCGCC | GTCTTCCGCC |
|    | 781  | AAGAGCCGCG | TGCAGACAGC | CCCCGTGCCC | ATGCCAGACC | TGAAGAATGT | CAAGTCCAAG |
|    | 841  | ATCGGCTCCA | CTGAGAACCT | GAAGCACACG | CCGGGAGGCG | GGAAGGTGCA | GATAATTAAT |
|    | 901  | AAGAAGCTGG | ATCTTAGCAA | CGTCCAGTCC | AAGTGTGGCT | CAAAGGATAA | TATCAAACAC |
|    | 961  | GTCCCGGGAG | CGCGCAGTGT | GCAAAATAGT | TACAAACAG  | TTGACCTGAG | CAAGGTGACC |
|    | 1021 | TCCAAGTGTG | GCTCATTAGG | CAACATCCAT | CATAAACAG  | GAGGTGGCCA | GGTGGAGTA  |
|    | 1081 | AAATCTGAGA | AGCTTGACTT | CAAGGACAGA | GTCCAGTCGA | AGATTGGGTC | CCTGGACAAT |
|    | 1141 | ATCACCACAG | TCCCTGGCGG | AGGAAATAAA | AAGATTGAAA | CCCACAAGCT | GACCTTCCGC |
|    | 1201 | GAGAACGCCA | AAGCCAAAGC | AGACCAAGCG | CGGAGATCG  | TGTACAAGTC | GCCAGTGGTG |
|    | 1261 | TCTGGGGACA | CGTCTCCACG | GCATCTCAGC | AATGTCTCCT | CCACCGGCAG | CATCGACATG |
|    | 1321 | GTAGACTCGC | CCCAGCTCGC | CACGCTAGCT | GACGAGGTGT | CTGCCCTCCT | GGCCAGCAG  |
|    | 1381 | GGTTTGTGAT | CAGGCCCTCG | GGGCGGTCAA | TAATTGTGGA | GAGGAGAGAA | TGAGAGAGTG |
|    | 1441 | TGGAATAAAT | AAGAATAATG | ACCCGGCCCC | CGCCCTCTGC | CCCCAGCTGC | TCCTCGCAGT |
|    | 1501 | TCGGTTAATT | GGTTAATCAC | TTAACCCTGT | TTTGTCACTC | GGCTTTGGCT | CGGGACTTCA |
|    | 1561 | AAATCAGTGA | TGGGAGTAAG | AGCAAATTC  | ATCTTTCCAA | ATTGATGGGT | GGGCTAGTAA |
|    | 1621 | TAAATATATT | AAAAAAAAC  | ATTCAAAAAC | ATGGCCACAT | CCAACATTTC | CTCAGGCAAT |
|    | 1681 | TCCTTTTGAT | TCTTTTCTCT | TCCCCCTCCA | TGTAGAAGAG | GGAGAAGGAG | AGGCTCTGAA |
|    | 1741 | AGCTGCTTCT | GGGGGATTTT | AAGGGACTGG | GGGTGCCAAC | CACCTCTGGC | CCTGTTGTGG |
|    | 1801 | GGGTGTCAAC | GAGGCAGTGG | CAGCAACAAA | GGATTGAAA  | CTTGGTGTGT | TCGTGGAGCC |
|    | 1861 | ACAGGCAGAG | GATGTCAACC | TTGTGTGAGT | GTGACGGGGG | TTGGGTGGGG | GCGGGAGGCC |
|    | 1921 | ACGGGGGAGG | CCGAGGCAGG | GGCTGGGCAG | AGGGGAGAGG | AAGCAACAAG | AGTGGGAGTG |
|    | 1981 | GGAGAGGAAG | CCACGTGCTG | GAGAGTAGAC | ATCCCCCTCC | TTGCCGCTGG | GAGAGCCAAG |
|    | 2041 | GCCTATGCCA | CCTGAGCGGT | CTGAGCGGCC | GCCTGTCTCT | GGTGGCCGGG | GGTGGGGGCC |
|    | 2101 | TGCTGTGGGT | CAGTGTGCCA | CCCTCTGCAG | GGCAGCCTGT | GGGAGAAGGG | ACAGCGGGTA |
|    | 2161 | AAAAGAGAAG | GCAAGCTGGC | AGGAGGGTGG | CACCTCTGTG | ATGACCTCCT | TAGAAAAGAC |
|    | 2221 | TGACCTTGAT | GTCTTGAGAG | CGCTGGCCTC | TTCTCTCTCT | CCTGCAAGGT | AGGGGGCCTG |
|    | 2281 | AGTTGAGGGG | CTTCCCTCTG | CTCCACAGAA | ACCCTGTTTT | ATTGAGTTCT | GAAGGTTGGA |
|    | 2341 | ACTGCTGCCA | TGATTTTGGC | CACCTTGCAG | ACCTGGGACT | TTAGGGCTAA | CCAGTTCTCT |
|    | 2401 | TTGTAAAGAC | TTGTGCCTCT | TGGGAGACGT | CCACCCGTTT | CCAAGCCTGG | GCCACTGGCA |
|    | 2461 | TCTCTGGAGT | GTGTGGGGGT | CTGGGAGGCA | GGTCCCGAGC | CCCTGTCTCT | TCCACCGGCC |
|    | 2521 | ACTGCAGTCA | CCCCGTCTGC | GCCGCTGTGC | TGTTGTCTGC | CGTGAGAGCC | CAATCACTGC |
|    | 2581 | CTATACCCCT | CATCACACGT | CACAATGTCC | CGAATTCCTA | GCCTCACCAC | CCCTTCTCAG |
|    | 2641 | TAATGACCCT | GGTTGGTTGC | AGGAGGTACC | TACTCCATAC | TGAGGGTGAA | ATTAAGGGAA |
|    | 2701 | GGCAAGTCC  | AGGCACAAGA | GTGGGACCCC | AGCCTCTCAC | TCTCAGTTCC | ACTCATCCAA |
|    | 2761 | CTGGGACCC  | CACCACGAAT | CTCATGATCT | GATTCCGGTT | CCTGTCTCCT | CCTCCCGTCA |
|    | 2821 | CAGATGTGAG | CCAGGGCACT | GCTCAGCTGT | GACCCTAGGT | GTTTCTGCCT | TGTTGACATG |
|    | 2881 | GAGAGAGCCC | TTTCCCTTGA | GAAGGCCTGG | CCCCTTCTCT | TGCTGAGCCC | ACAGCAGCAG |
|    | 2941 | GCTGGGTGTC | TTGGTTGTCA | GTGGTGGCAC | CAGGATGGAA | GGGCAAGGCA | CCCAGGGCAG |
|    | 3001 | GCCCACAGTC | CCGCTGTCCC | CCACTTGAC  | CCTAGCTTGT | AGCTGCCAAC | CTCCACAGCA |
|    | 3061 | GCCCAGCCCG | CTGCTCAGCT | CCACATGCAT | AGTATCAGCC | CTCCACACCC | GACAAAGGGG |
|    | 3121 | AAACACACCC | CTTGGAAATG | GTTCTTTTCC | CCCAGTCCCA | GCTGGAAGCC | ATGCTGTCTG |
|    | 3181 | TTCTGTGTTG | GCAGCTGAAC | ATATACATAG | ATGTTGCCCT | GCCTCCCCA  | TCTGCACCTT |
|    | 3241 | GTGAGTTTGT | AGTTGGATT  | GTCTGTTTAT | GCTTGGATT  | ACCAGAGTGA | CTATGATAGT |
|    | 3301 | GAAAGAGAAA | AAAAAAGAAA | AAAAGGACGC | ATGTATCTTG | AAATGCTTGT | AAAGAGGTTT |
|    | 3361 | CTAACCCACC | CTCACGAGGT | GTCTCTCACC | CCCACACTGG | GACTCGTGTG | GCCTGTGTGG |
|    | 3421 | TGCCACCCCT | CTGGGGCCTC | CCAAGTTTGT | AAAGGCTTTC | CTCAGCACCT | GGGACCCCA  |
|    | 3481 | AGAGACCAGC | TTCTAGCAGC | TAAGGAGGCC | GTTCAGCTGT | GACGAAGGCC | TGAAGCACAG |
|    | 3541 | GATTAGGACT | GAAGCGATGA | TGTCCCTTTC | CCTACTTCCC | CTTGGGGCTC | CCTGTGTGAG |
|    | 3601 | GGCAGACAGT | AGGTCTTGTG | GCTGGTCTGG | CTTGGCGGCG | GAGGATGGTT | CTCTCTGGTC |
|    | 3661 | ATAGCCCGAA | GTCTCATGGC | AGTCCCAAAG | GAGGCTTACA | ACTCTGCAT  | CACAAGAAAA |
|    | 3721 | AGGAAGCCAC | TGCCAGCTGG | GGGGATCTGC | AGCTCCAGCA | AGCTCCGTGA | GCCTCAGCCA |
|    | 3781 | CCCCTCAGAC | TGGGTTCCCT | TCCAAGCTCG | CCCTCTGGAG | GGGCAGCGCA | GCCTCCACCC |
|    | 3841 | AAGGGCCCTG | CGACCACAGC | AGGGATTGGG | ATGAATTGCC | TGTCCTGGAT | CTGCTCTAGA |
|    | 3901 | GGCCCAAGCT | GCCTGCCTGA | GGAAGGATGA | CTTGACAGT  | CAGGAGACAC | TGTTCCCAAA |

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SEQ ID NO Sequence

|    |      |             |             |             |             |             |             |
|----|------|-------------|-------------|-------------|-------------|-------------|-------------|
|    | 3961 | GCCTTGACCA  | GAGCACCTCA  | GCCCCTGAC   | CTTGACACAA  | CTCCATCTGC  | TGCCATGAGA  |
|    | 4021 | AAAGGGAAGC  | CGCCTTTGCA  | AAACATTGCT  | GCCTAAAGAA  | ACTCAGCAGC  | CTCAGGCCAA  |
|    | 4081 | ATTCTGGCAC  | TTCTGGTTTG  | GGTACAGTTA  | AAGGCAACCC  | TGAGGGACTT  | GGCAGTAGAA  |
|    | 4141 | ATCCAGGGCC  | TCCCCTGGGG  | CTGGCAGCTT  | CGTGTGCAGC  | TAGAGCTTTA  | CCTGAAAGGA  |
|    | 4201 | AGTCTCTGGG  | CCCAGAACTC  | TCCACCAAGA  | GCCTCCCTGC  | CGTTCGCTGA  | GTCCCAAGAA  |
|    | 4261 | TTCTCCTAAG  | TTGAAGGGAT  | CTGAGAAGGA  | GAAGGAAATG  | TGGGGTAGAT  | TTGGTGGTGG  |
|    | 4321 | TTAGAGATAT  | GCCCCCTCA   | TTACTGCCAA  | CAGTTTCGGC  | TGCATTTCTT  | CACGCACCTC  |
|    | 4381 | GGTTCCTCTT  | CCTGAAGTTC  | TTGTGCCCTG  | CTCTTCAGCA  | CCATGGGCCT  | TCTTATACGG  |
|    | 4441 | AAGGCTCTGG  | GATCTCCCCC  | TTGTGGGGCA  | GGCTCTTGGG  | GCCAGCCTAA  | GATCATGGTT  |
|    | 4501 | TAGGGTGATC  | AGTGTCTGGCA | GATAAATTGA  | AAAGGCACGC  | TGGCTTGTGA  | TCTTAAATGA  |
|    | 4561 | GGACAATCCC  | CCCAGGGCTG  | GGCACTCCTC  | CCCTCCCCCT  | ACTTCTCCCA  | CCTGCAGAGC  |
|    | 4621 | CAGTGTCTCT  | GGGTGGGCTA  | GATAGGATAT  | ACTGTATGCC  | GGCTCCTTCA  | AGCTGCTGAC  |
|    | 4681 | TCATCTTTATC | AATAGTTCCA  | TTTAAATTGA  | CTTCAGTGGT  | GAGACTGTAT  | CCTGTTTGCT  |
|    | 4741 | ATTGCTTTGT  | GTGCTATGGG  | GGGAGGGGGG  | AGGAATGTGT  | AAGATAGTTA  | ACATGGGCAA  |
|    | 4801 | AGGGAGATCT  | TGGGGTGCAG  | CACCTAAACT  | GCCTCGTAAC  | CCTTTTCATG  | ATTTCACCCA  |
|    | 4861 | CATTTTGCTAG | AGGGAGGGAG  | CAGCCACGGA  | GTTAGAGGCC  | CTTGGGGTTT  | CTCTTTTCCA  |
|    | 4921 | CTGACAGGCT  | TTCCCAGGCA  | GCTGGCTAGT  | TCATTCCCTC  | CCCAGCCAGG  | TGCAGCGCTA  |
|    | 4981 | GGAATATGGA  | CATCTGGTTG  | CTTTGGCCTG  | CTGCCCTCTT  | TCAGGGGTCC  | TAAGCCCACA  |
|    | 5041 | ATCATGCGCTC | CCTAAGACCT  | TGGCATCCTT  | CCCTCTAAGC  | CGTTTGGACC  | TCTGTGCCAC  |
|    | 5101 | CTCTCACACT  | GGCTCCAGAC  | ACACAGCCTG  | TGCTTTTGGG  | GCTGAGATCA  | CTCGCTTCAC  |
|    | 5161 | CCTCCTCATC  | TTTGTTCTCC  | AAGTAAAGCC  | ACGAGGTCCG  | GGCGAGGGCA  | GAGGTGATCA  |
|    | 5221 | CCTGCGGTGC  | CCATCTACAG  | ACCTGCAGCT  | TCATAAAACT  | TCTGATTTCT  | CTTCAGCTTTT |
|    | 5281 | GAAAAGGGTT  | ACCCTGGGCA  | CTGGCCTAGA  | GCCTCACCTC  | CTAATAGACT  | TAGCCCCATG  |
|    | 5341 | AGTTTGCCAT  | GTTGAGCAGG  | ACTATTTCTG  | GCACCTGCAA  | GTCCCATGAT  | TTCTTCGGTA  |
|    | 5401 | ATTCTGAGGG  | TGGGGGGAGG  | GACATGAAAT  | CATCTTAGCT  | TAGCTTTCTG  | TCTGTGAATG  |
|    | 5461 | TCATATAGT   | GTATTGTGTG  | TTTAAACAAA  | TGATTTACAC  | TGACTGTTCG  | TGTAAGAATG  |
|    | 5521 | AATTTGAAAA  | TAAAGTTATT  | ACTCTGATTA  | AA          |             |             |
| 27 | 1    | MAEPRQEFEV  | MEDHAGTYGL  | GDRKDQGGYT  | MHQDQEGDTD  | AGLKESPLQT  | PTEDGSEEPG  |
|    | 61   | SETSDAKSTP  | TAEAEAGIG   | DTPSLEDEAA  | GHVTQARMVS  | KSKDGTGSDD  | KKAKGADGKT  |
|    | 121  | KIATPRGAAP  | PGQKGQANAT  | RIPAKTPPAP  | KTPPSSGEP   | KSGDRSGYSS  | PGSPGTPGSR  |
|    | 181  | SRTPSLPTPP  | TREPKKVAVV  | RTPPKSPSSA  | KSRLQTAPVP  | MPDLKNVSK   | IGSTENLKHQ  |
|    | 241  | PGGGKVQIIN  | KKLDLSNVQS  | KCGSKDNIKH  | VPGGGSVQIV  | YKPVDLSKVT  | SKCGSLGNIH  |
|    | 301  | HKPGGQVEV   | KSEKLDKDR   | VQSKIGSLDN  | ITHVPGGNGK  | KIETHKLTER  | ENAKAKTDHG  |
|    | 361  | AEIVYKSPVV  | SGDTSRPHLS  | NVSTSGSIDM  | VDSPQLATLA  | DEVASLAKQ   | GL          |
| 28 | 1    | GCAGTCACCG  | CCACCCACCA  | GCTCCGGCAC  | CAACAGCAGC  | GCCGCTGCCA  | CCGCCACCTC  |
|    | 61   | TCTGCCGCCG  | CCACACAGC   | CACCTTCTCC  | TCCTCCCGTG  | TCCTCTCCCG  | TCCTCGCCTC  |
|    | 121  | TGTCGACTAT  | CAGGTGAAGT  | TTGAACCCAG  | ATGGCTGAGC  | CCCGCCAGGA  | GTTTGAAGTG  |
|    | 181  | ATGGAAGATC  | ACGCTGGGAC  | GTACGGGTG   | GGGGACAGGA  | AAGATCAGGG  | GGGCTACACC  |
|    | 241  | ATGCACCAAG  | ACCAAGAGGG  | TGACACGGAC  | GCTGGCCTGA  | AAGCTGAAGA  | AGCAGGCATT  |
|    | 301  | GGAGACACCC  | CCAGCCTGGA  | AGACGAAGCT  | GCTGGTCACG  | TGACCCAAGC  | TCGCATGGTC  |
|    | 361  | AGTAAAGACA  | AAGACGGGAC  | TGGAAGCGAT  | GACAAAAAAG  | CCAAGGGGGC  | TGATGGTAAA  |
|    | 421  | ACGAAGATCG  | CCACCCGCG   | GGGAGCAGCC  | CCTCCAGGCC  | AGAAGGGCCA  | GGCCAACGCC  |
|    | 481  | ACCAGGATTC  | CAGCAAAAC   | CCCGCCCGCT  | CCAAAGACAC  | CACCCAGCTC  | TGGTGAACCT  |
|    | 541  | CCAAAATCAG  | GGGATCGCAG  | CGGCTACAGC  | AGCCCCGGCT  | CCCCAGGCAC  | TCCCGGCAGC  |
|    | 601  | CGCTCCCGCA  | CCCCGTCCCT  | TCCAACCCCA  | CCACCCGGG   | AGCCCAAGAA  | GGTGGCAGTG  |
|    | 661  | GTCCGTACTC  | CACCCAGTC   | GCCGCTTTC   | GCCAAGAGCC  | GCCTGCAGAC  | AGCCCCCGTG  |
|    | 721  | CCCATGCCAG  | ACCTGAAGAA  | TGTCAAGTCC  | AAGATCGGCT  | CCACTGAGAA  | CCTGAAGCAC  |
|    | 781  | CAGCCGGGAG  | GCGGAAGGT   | GCAAAATAGTC | TACAAACCAG  | TTGACCTGAG  | CAAGGTGACC  |
|    | 841  | TCCAAGTGTG  | GCTCATTAGG  | CAACATCCAT  | CATAAACCCAG | GAGGTGGCCA  | GGTGAAGTA   |
|    | 901  | AAATCTGAGA  | AGCTTGACTT  | CAAGGACAGA  | GTCCAGTCGA  | AGATTGGGTC  | CCTGGACAAT  |
|    | 961  | ATCACCCACG  | TCCCTGGCGG  | AGGAAATAAA  | AAGATTGAAA  | CCCAACAAGCT | GACCTTCCGC  |
|    | 1021 | GAGAACGCCA  | AAGCCAAGAC  | AGACCACGGG  | GCGGAGATCG  | TGTACAAGTC  | GCCAGTGGTG  |
|    | 1081 | TCTGGGGACA  | CGTCTCCACG  | GCATCTCAGC  | AATGTCTCCT  | CCACCGGCAG  | CATCGACATG  |
|    | 1141 | GTAGACTCGC  | CCAGCTTCGC  | CACGCTAGCT  | GACGAGGTGT  | CTGCCCTCCT  | GGCCAAGCAG  |
|    | 1201 | GGTTTGTGAT  | CAGGCCCTTG  | GGGCGGTCAA  | TAATTGTGGA  | GAGGAGAGAA  | TGAGAGAGTG  |
|    | 1261 | TGGAAAAAAA  | AAGAATAATG  | ACCCGGCCCC  | CGCCCTCTGC  | CCCCAGCTGC  | TCCTCGCAGT  |
|    | 1321 | TCCGTTAATT  | GGTTAATCAC  | TTAACCTGCT  | TTTGTCACTC  | GGCTTTGGCT  | CGGCACTTCA  |
|    | 1381 | AAATCAGTGA  | TGGGAGTAAG  | AGCAAATTTT  | ATCTTTCCAA  | ATTGATGGGT  | GGGCTAGTAA  |
|    | 1441 | TAAATATATT  | AAAAAAAAC   | ATTCAAAAAC  | ATGGCCACAT  | CCAACATTTT  | CTCAGGCAAT  |
|    | 1501 | TCCTTTTGAT  | TCTTTTCTCT  | TCCCTCTCCA  | TGTAGAAGAG  | GGAGAAGGAG  | AGGCTCTGAA  |
|    | 1561 | AGCTGCTTCT  | GGGGGATTTT  | AAGGAGCTGG  | GGGTGCCAAC  | CACCTCTGGC  | CCTGTTGTGG  |
|    | 1621 | GGGTGTGACA  | GAGGCAGTGG  | CAGCAACAAA  | GGATTTGAAA  | CTTGGTGTGT  | TCGTGGAGCC  |
|    | 1681 | ACAGGCAGAC  | GATGTCAACC  | TTGTGTGAGT  | GTGACGGGGG  | TTGGGTGGG   | GCGGGAGGCC  |
|    | 1741 | ACGGGGGAGG  | CCGAGGCAGG  | GGCTGGGCAG  | AGGGGAGAGG  | AAGCACAAAG  | AGTGGGAGTG  |
|    | 1801 | GGAGAGGAAG  | CCACGTGCTG  | GAGAGTAGAC  | ATCCCCCTCC  | TTGCCGCTGG  | GAGAGCCAAG  |
|    | 1861 | GCCTATGCCA  | CCTGCAGCGT  | CTGAGCGGCC  | GCCTGTCTTT  | GGTGGCCGGG  | GGTGGGGGCC  |
|    | 1921 | TGCTGTGGGT  | CAGTGTGCCA  | CCCTCTGCAG  | GGCAGCCTGT  | GGGAGAAGGG  | ACAGCGGGTA  |
|    | 1981 | AAAAGAGAAG  | GCAAGCTGGC  | AGGAGGGTGG  | CACCTTCGTG  | ATGACCTCCT  | TAGAAAAGAC  |
|    | 2041 | TGACCTTGAT  | GTCTTGAGAG  | CGCTGGCCTC  | TTCTCCCTC   | CCTGCAGGGT  | AGGGGGCCTG  |
|    | 2101 | AGTTGAGGGG  | CTTCCCTCTG  | CTCCACAGAA  | ACCCTGTTTT  | ATTGAGTTCT  | GAAGGTTTGA  |
|    | 2161 | ACTGCTGCCA  | TGATTTTGGC  | CACCTTTGAC  | ACCTGGGACT  | TTAGGGCTAA  | CCAGTTCTCT  |
|    | 2221 | TTGTAAGGAC  | TTGTGCCTCT  | TGGGAGACGT  | CCACCCGTTT  | CCAAGCCTGG  | GCCACTGGCA  |

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SEQ ID NO Sequence

|      |                                                                               |             |             |             |            |             |            |
|------|-------------------------------------------------------------------------------|-------------|-------------|-------------|------------|-------------|------------|
| 2281 | TCTCTGGAGT                                                                    | GTGTGGGGGT  | CTGGGAGGCA  | GGTCCCCGAGC | CCCCTGTCCT | TCCCACGGCC  |            |
| 2341 | ACTGCAGTCA                                                                    | CCCCGTCTGC  | GCCGCTGTGC  | TGTTGTCTGC  | CGTGAGAGCC | CAATCACTGC  |            |
| 2401 | CTATACCCCT                                                                    | CATCACACGT  | CACAATGTCC  | CGAATTCCCA  | GCCTCACCAC | CCCTTCTCAG  |            |
| 2461 | TAATGACCCT                                                                    | GGTTGGTTGC  | AGGAGGTACC  | TACTCCATAC  | TGAGGGTGAA | ATTAAGGGAA  |            |
| 2521 | GGCAAAAGTCC                                                                   | AGGCACAAGA  | GTGGGACCCC  | AGCCTCTCAC  | TCTCAGTTCC | ACTCATCCAA  |            |
| 2581 | CTGGGACCCCT                                                                   | CACCACGAAT  | CTCATGATCT  | GATTTCGGTTC | CCTGTCCTCT | CCTCCCGTCA  |            |
| 2641 | CAGATGTGAG                                                                    | CCAGGGGCACT | GCTCAGCTGT  | GACCCTAGGT  | GTTTCTGCCT | TGTTGACATG  |            |
| 2701 | GAGAGAGCCC                                                                    | TTTCCCTCGA  | GAAGGCCTGG  | CCCCTTCTTG  | TGCTGAGCCC | ACAGCAGCAG  |            |
| 2761 | GCTGGGTGTC                                                                    | TTGGTTGTCA  | GTGGTGGCAC  | CAGGATGGAA  | GGGCAAGGCA | CCCAGGGCAG  |            |
| 2821 | GCCACACAGTC                                                                   | CCGCTGTCCC  | CCACTTGCAC  | CCTAGCTTGT  | AGCTGCCAAC | CTCCAGACA   |            |
| 2881 | GCCAGGCCC                                                                     | CTGCTCAGCT  | CCACATGCAT  | AGTATCAGCC  | CTCCACACCC | GACAAAGGGG  |            |
| 2941 | AACACACCCC                                                                    | CTTGGAAATG  | GTTCTTTTCC  | CCCAGTCCCA  | GCTGGAAGCC | ATGCTGTCTG  |            |
| 3001 | TTCTGCTGGA                                                                    | GCAGCTGAAC  | ATATACATAG  | ATGTTGCCCT  | GCCCTCCCCA | TCTGCACCTT  |            |
| 3061 | GTTGAGTTGT                                                                    | AGTTGGATTT  | GTCTGTTTAT  | GCTTGGATTCT | ACCAGAGTGA | CTATGATAGT  |            |
| 3121 | GAAAAGAAAA                                                                    | AAAAAAAAAA  | AAAAGGACGC  | ATGTATCTTG  | AAATGCTTGT | AAAGAGGTTT  |            |
| 3181 | CTAACCCACC                                                                    | CTCACGAGGT  | GTCTCTCACC  | CCCACACTGG  | GACTCGTGTG | GCCTGTGTGG  |            |
| 3241 | TGCCACCCCTG                                                                   | CTGGGGCCCTC | CCAAGTTTGT  | AAAGGCTTTC  | CTCAGCACCT | GGGACCCAAC  |            |
| 3301 | AGAGACCAGC                                                                    | TTCTAGCAGC  | TAAGGAGGCC  | GTTTCAGCTGT | GACGAAGGCC | TGAAGCACAG  |            |
| 3361 | GATTAGGACT                                                                    | GAAGCGATGA  | TGTCCTCTTC  | CCTACTTCCC  | CTTGGGGCTC | CCTGTGTCTG  |            |
| 3421 | GGCACAGACT                                                                    | AGGTCTTGTG  | GCTGGTCTGG  | CTTGGCGGCG  | GAGGATGGTT | CTCTCTGGTC  |            |
| 3481 | ATAGCCCGAA                                                                    | GTCTCATGGC  | AGTCCCAAAG  | GAGGCTTACA  | ACTCCTGCAT | CACAAGAAAA  |            |
| 3541 | AGGAAGCCAC                                                                    | TGCCAGCTGG  | GGGGATCTGC  | AGCTCCCAGA  | AGCTCCGTGA | GCCTCAGCCA  |            |
| 3601 | CCCCTCAGAC                                                                    | TGGGTTCTCT  | TCCAAGCTCG  | CCCTCTGGAG  | GGGCAGCGCA | GCCTCCACC   |            |
| 3661 | AAGGGCCCTG                                                                    | CGACCACAGC  | AGGGATTGGG  | ATGAATTGCC  | TGTCCTGGAT | CTGCTCTAGA  |            |
| 3721 | GGCCCAAGCT                                                                    | GCCTGCCTGA  | GGAAGGATGA  | CTTGACAAGT  | CAGGAGACAC | TGTTCCCAAA  |            |
| 3781 | GCCTTGACCA                                                                    | GAGCACCTCA  | GCCCGCTGAC  | CTTGACAAAA  | CTCCATCTGC | TGCCATGAGA  |            |
| 3841 | AAAGGGAAGC                                                                    | CGCCTTTGCA  | AAACATTGCT  | GCCTAAAGAA  | ACTCAGCAGC | CTCAGGCCCA  |            |
| 3901 | ATTCTGCCAC                                                                    | TTCTGGTTTG  | GGTACAGTTA  | AAGGCAACCC  | TGAGGGACTT | GGCAGTAGAA  |            |
| 3961 | ATCCAGGGCC                                                                    | TCCCCTGGGG  | CTGGCAGCTT  | CGTGTGCAGC  | TAGAGCTTTA | CCTGAAAGGA  |            |
| 4021 | AGTCTCTGGG                                                                    | CCCAGAACTC  | TCCACCAAGA  | GCCTCCCTGC  | CGTTCGCTGA | GTCCCAGCAA  |            |
| 4081 | TTCTCCTAAG                                                                    | TTGAAGGGAT  | CTGAGAAGGA  | GAAGGAAATG  | TGGGGTAGAT | TGTTGGTGGG  |            |
| 4141 | TTAGAGATAT                                                                    | GCCCCCTCA   | TTACTTGCCAA | CAGTTTCGGC  | TGCATTTCTT | CACGCACCTC  |            |
| 4201 | GGTTCCTCTT                                                                    | CCTGAAGTTC  | TTGTGCCCTG  | CTCTTCAGCA  | CCATGGGCCT | TCTTATACGG  |            |
| 4261 | AAGGCTCTGG                                                                    | GATCTCCCCC  | TTGTGGGGCA  | GGCTCTTGGG  | GCCAGCCTAA | GATCATGTGT  |            |
| 4321 | TAGGGTGATC                                                                    | AGTGTGGCA   | GATAAATTGA  | AAAGGCACGC  | TGGCTTGTGA | TCTTAAATGA  |            |
| 4381 | GGACAATCCC                                                                    | CCCAGGGCTG  | GGCACTCCTC  | CCCTCCCCTC  | ACTTCTCCCA | CCTGCAGAGC  |            |
| 4441 | CAGTGTCTCT                                                                    | GGGTGGGCTA  | GATAGGATAT  | ACTGTATGCC  | GGCTCCTTCA | AGCTGCTGAC  |            |
| 4501 | TCACTTTTATC                                                                   | AATAGTTTCA  | TTTAAATTGA  | CTTCAGTGGT  | GAGACTGTAT | CCTGTTTGCT  |            |
| 4561 | ATTGCTTGTT                                                                    | GTGCTATGGG  | GGGAGGGGGG  | AGGAATGTGT  | AAGATAGTTA | ACATGGGCAA  |            |
| 4621 | AGGGAGATCT                                                                    | TGGGTGTCAG  | CACTTAAACT  | GCCTCGTAAC  | CCTTTTCATG | ATTTCAACCA  |            |
| 4681 | CATTTGCTAG                                                                    | AGGGAGGGAG  | CAGCCACGGA  | GTTAGAGGCC  | CTTGGGGTTT | CTCTTTTCCA  |            |
| 4741 | CTGACAGGCT                                                                    | TTCCCAGGCA  | GCTGGCTAGT  | TCATTCCCTC  | CCCAGCCAGG | TGCATGGCGTA |            |
| 4801 | GGAAATATGGA                                                                   | CATCTGGTTG  | CTTTGGCCTG  | CTGCCCTCTT  | TCAGGGGTCC | TAAGCCCACA  |            |
| 4861 | ATCATGCTCT                                                                    | CCTAAGACCT  | TGGCATCCTT  | CCCTCTAAGC  | CGTTGGCACC | TCTGTGCCAC  |            |
| 4921 | CTCTCACACT                                                                    | GGCTCCAGAC  | ACACAGCCTG  | TGCTTTTGGG  | GCTGAGATCA | CTCGCTTCAC  |            |
| 4981 | CCTCCTCATC                                                                    | TTTGTCTTCC  | AAGTAAAGCC  | ACGAGGTGCG  | GGCGAGGGCA | GAGGTGATCA  |            |
| 5041 | CCTGCGTGTG                                                                    | CCATCTACAG  | ACCTGCAGCT  | TCATAAAACT  | TCTGATTTC  | CTTCAGCTTT  |            |
| 5101 | GAAAAGGGTT                                                                    | ACCCTGGGCA  | CTGGCCTAGA  | GCCTCACCTC  | CTAATAGACT | TAGCCCCATG  |            |
| 5161 | AGTTTGCCAT                                                                    | GTTGAGCAGG  | ACTATTTCTG  | GCACCTGCAA  | GTCCCATGAT | TCTTTCGGTA  |            |
| 5221 | ATTCGTAGGG                                                                    | TGGGGGGAGG  | GACATGAAAT  | CATCTTAGCT  | TAGCTTTCTG | TCTGTGAATG  |            |
| 5281 | TCTATATAGT                                                                    | GTATTGTGTG  | TTTTAACAAA  | TGATTTACAC  | TGACTGTTCG | TGTAAGAGTG  |            |
| 5341 | AATTTGGAAA                                                                    | TAAAGTTATT  | ACTCTGATTA  | AA          |            |             |            |
| 29   | 1                                                                             | MAEPRQEFEV  | MEDHAGTYGL  | GDRKDQGGYT  | MHQDQEGDTD | AGLKAEAEAGI | GDTPSLEDEA |
|      | 61                                                                            | AGHVTQARMV  | SKSKDGTGSD  | DKKAKGADGK  | TKIATPRGAA | PPGQKQGANA  | TRIPAKTPPA |
|      | 121                                                                           | PKTPPSSGEP  | PKSGDRSGYS  | SPGSPGTPGS  | RSRTPSLPTP | PTREPKKVAV  | VRTPPKSPSS |
|      | 181                                                                           | AKSRLQTAPV  | PMPDLKNVKS  | KIGSTENLKH  | QPGGGKVQIV | YKPVDLSKVT  | SKCGSLGNTH |
|      | 241                                                                           | HKPGGGQVEV  | KSEKLDFKDR  | VQSKIGSLDN  | ITHVPGGGNK | KIETHKLTFR  | ENAKAKTDHG |
|      | 301                                                                           | AEIVYKSPVV  | SGDTSRPHLS  | NVSSTGSDIM  | VDSPQLATLA | DEVSASLAKQ  | GL         |
| 30   | GCCTTTGCGAGCGTGACACAG                                                         |             |             |             |            |             |            |
| 31   | TTTCTTGTTGGGCAACTGGG                                                          |             |             |             |            |             |            |
| 32   | CTGGTGTGGGTGTTATTCTCT                                                         |             |             |             |            |             |            |
| 33   | eGs[eCm]o[eCm]oeToeTsdTsdGs[dCm]sdGsdAsdGs[dCm]sdGsdTsdGs[eCm]oeAo[eCm]seAseG |             |             |             |            |             |            |
| 34   | eTseToeTo[eCm]oeTsdTsdGsdTsdTsdGsdGsdGs[dCm]sdAsdAs[eCm]oeToeGseGseG          |             |             |             |            |             |            |
| 35   | [eCm]seToeGoeGoeTsdGsdTsdGsdGsdGsdTsdGsdTsdTsdAseToeTo[eCm]s[eCm]seT          |             |             |             |            |             |            |
| 36   | GTGGATCTAGTTCTCATCCA                                                          |             |             |             |            |             |            |

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| SEQ ID NO | Sequence                                                                     |
|-----------|------------------------------------------------------------------------------|
| 37        | GGATAGAATCCAGTTAATTA                                                         |
| 38        | TGTCACATCTAGAAACCACC                                                         |
| 39        | GCCTGGACAGAGGGATGAGC                                                         |
| 40        | AAGCTCCAGGTGATTGAGAC                                                         |
| 41        | TTCTGATACATTGTCAAAGT                                                         |
| 42        | AGCAGCTTGAAGGAGCCGGC                                                         |
| 43        | TGGCTGATGATACCATTGG                                                          |
| 44        | CTTTCTTGGAAAGTAT                                                             |
| 45        | TTTCTTGGAAAGTATA                                                             |
| 46        | TTCTTGGAAAGTATAT                                                             |
| 47        | TCTTGGAAAGTATATT                                                             |
| 48        | CTTGGAAAGTATATTC                                                             |
| 49        | GTGGATCTAGTTCTCA                                                             |
| 50        | TGGATCTAGTTCTCAT                                                             |
| 51        | GGATCTAGTTCTCATC                                                             |
| 52        | GATCTAGTTCTCATCC                                                             |
| 53        | ATCTAGTTCTCATCCA                                                             |
| 54        | TTCTGATACATTGTCA                                                             |
| 55        | TCTGATACATTGTCAA                                                             |
| 56        | CTGATACATTGTCAAA                                                             |
| 57        | TGATACATTGTCAAAG                                                             |
| 58        | GATACATTGTCAAAGT                                                             |
| 59        | AGCAGCTTGAAGGAGC                                                             |
| 60        | GCAGCTTGAAGGAGCC                                                             |
| 61        | CAGCTTGAAGGAGCCG                                                             |
| 62        | AGCTTGAAGGAGCCGG                                                             |
| 63        | GCTTGAAGGAGCCGGC                                                             |
| 64        | eGseToeGseGseAsdTsdCmsdTsdAsdGsdTsdTsdCmsdTsdCmseAseTo[eCm]s[eCm]seA         |
| 65        | eGseGoeAseTseAsdGsdAsdAsdTsdCmsdCmsdAsdGsdTsdTseAseAoeTseTseA                |
| 66        | eTseGoeTs[eCm]seAsdCmsdAsdTsdCmsdTsdAsdGsdAsdAsdAs[eCm]s[eCm]oeAs[eCm]s[eCm] |
| 67        | eGs[eCm]o[eCm]seTseGsdGsdAsdCmsdAsdGsdAsdGsdGsdGsdAseTseGoeAseGs[eCm]        |
| 68        | eAseAoeGs[eCm]seTsdCmsdCmsdAsdGsdGsdTsdGsdAsdTsdTseGseAoeGseAs[eCm]          |
| 69        | eTseTo[eCm]seTseGsdAsdTsdAsdCmsdAsdTsdTsdGsdTsdCmseAseAoeAseGseT             |
| 70        | eAseGo[eCm]seAseGsdCmsdTsdTsdGsdAsdAsdGsdGsdAsdGs[eCm]s[eCm]oeGseGs[eCm]     |
| 71        | eTseGoeGs[eCm]seTsdGsdAsdTsdGsdAsdTsdAsdCmsdCmsdAseTseToeTseGseG             |
| 72        | [CmL]s[TL]s[TL]sdTsdCmsdTsdTsdGsdGsdAsdAsdAsdGs[TL]s[AL]s[TL]                |
| 73        | [TL]s[TL]s[TL]sdCmsdTsdTsdGsdGsdAsdAsdAsdGsdTs[AL]s[TL]s[AL]                 |

-continued

| SEQ ID NO | Sequence                                                             |
|-----------|----------------------------------------------------------------------|
| 74        | [TL]s[TL]s[CmL]sdTsdTsdGsdGsdAsdAsdAsdGsdTsdAs[TL]s[AL]s[TL]         |
| 75        | [TL]s[CmL]s[TL]sdTsdGsdGsdAsdAsdAsdGsdTsdAsdTs[AL]s[TL]s[TL]         |
| 76        | [CmL]s[TL]s[TL]sdGsdGsdAsdAsdAsdGsdTsdAsdTsdAs[TL]s[TL]s[CmL]        |
| 77        | [GL]s[TL]s[GL]sdGsdAsdTsdCmsdTsdAsdGsdTsdTsdCms[TL]s[CmL]s[AL]       |
| 78        | [TL]s[GL]s[GL]sdAsdTsdCmsdTsdAsdGsdTsdTsdCmsdT[CM]s[AL]s[TL]         |
| 79        | [GL]s[GL]s[AL]sdTsdCmsdTsdAsdGsdTsdTsdCmsdTsdCms[AL]s[TL]s[CmL]      |
| 80        | [GL]s[AL]s[TL]sdCmsdTsdAsdGsdTsdTsdCmsdTsdCmsdAs[TL]s[CmL]s[CmL]     |
| 81        | [AL]s[TL]s[CmL]sdTsdAsdGsdTsdTsdCmsdTsdCmsdAsdTs[CmL]s[CmL]s[AL]     |
| 82        | [TL]s[TL]s[CmL]sdTsdGsdAsdTsdAsdCmsdAsdTsdTsdGs[TL]s[CmL]s[AL]       |
| 83        | [TL]s[CmL]s[TL]sdGsdAsdTsdAsdCmsdAsdTsdTsdGsdTs[CmL]s[AL]s[AL]       |
| 84        | [CmL]s[TL]s[GL]sdAsdTsdAsdCmsdAsdTsdTsdGsdTsdCms[AL]s[AL]s[AL]       |
| 85        | [TL]s[GL]s[AL]sdTsdAsdCmsdAsdTsdTsdGsdTsdCmsdAs[AL]s[AL]s[GL]        |
| 86        | [GL]s[AL]s[TL]sdAsdCmsdAsdTsdTsdGsdTsdCmsdAsdAs[AL]s[GL]s[TL]        |
| 87        | [AL]s[GL]s[CmL]sdAsdGsdCmsdTsdTsdGsdAsdAsdGsdGs[AL]s[GL]s[CmL]       |
| 88        | [GL]s[CmL]s[AL]sdGsdCmsdTsdTsdGsdAsdAsdGsdGsdAs[GL]s[CmL]s[CmL]      |
| 89        | [CmL]s[AL]s[GL]sdCmsdTsdTsdGsdAsdAsdGsdGsdAsdGs[CmL]s[CmL]s[GL]      |
| 90        | [AL]s[GL]s[CmL]sdTsdTsdGsdAsdAsdGsdGsdAsdGsdCms[CmL]s[GL]s[GL]       |
| 91        | [GL]s[CmL]s[TL]sdTsdGsdAsdAsdGsdGsdAsdGsdCmsdCms[GL]s[GL]s[CmL]      |
| 92        | [eCm]seToeTseTs[eCm]sdTsdTsdGsdGsdAsdAsdAsdGsdTsdAseTseAoeTseTs[eCm] |
| 93        | eGs[eCm]oeTseGs[eCm]sdAsdGsdGsdGsdAsdAsdAsdGsdAsdGseTseTo[eCm]seTseT |
| 94        | eGseAoeTseGseGsdAsdGsdAs[dCm]sdAsdGsdTsdGsdTsdGseTseAoeTs[eCm]seT    |

## SEQUENCE LISTING

Sequence total quantity: 94

SEQ ID NO: 1 moltype = RNA length = 20

FEATURE Location/Qualifiers

source 1..20

mol\_type = other RNA

organism = synthetic construct

SEQUENCE: 1

ctttcttgga aagtatatcc

20

SEQ ID NO: 2

moltype = RNA length = 20

FEATURE Location/Qualifiers

source 1..20

mol\_type = other RNA

organism = synthetic construct

SEQUENCE: 2

gctcagagct acagcaagtc

20

SEQ ID NO: 3

moltype = RNA length = 20

FEATURE Location/Qualifiers

source 1..20

mol\_type = other RNA

organism = synthetic construct

SEQUENCE: 3

gctgcaggga aagagttcctt

20

SEQ ID NO: 4

moltype = RNA length = 20

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|                       |                                                  |    |
|-----------------------|--------------------------------------------------|----|
| FEATURE               | Location/Qualifiers                              |    |
| source                | 1..20                                            |    |
|                       | mol_type = other RNA                             |    |
|                       | organism = synthetic construct                   |    |
| SEQUENCE: 4           |                                                  |    |
| gatggagaca gtgtgtatct |                                                  | 20 |
| SEQ ID NO: 5          | moltype = RNA length = 20                        |    |
| FEATURE               | Location/Qualifiers                              |    |
| source                | 1..20                                            |    |
|                       | mol_type = other RNA                             |    |
|                       | organism = synthetic construct                   |    |
| SEQUENCE: 5           |                                                  |    |
| gcagatgggg cccaaatcct |                                                  | 20 |
| SEQ ID NO: 6          | moltype = RNA length = 20                        |    |
| FEATURE               | Location/Qualifiers                              |    |
| source                | 1..20                                            |    |
|                       | mol_type = other RNA                             |    |
|                       | organism = synthetic construct                   |    |
| SEQUENCE: 6           |                                                  |    |
| gccacagtga cagcaaattc |                                                  | 20 |
| SEQ ID NO: 7          | moltype = RNA length = 20                        |    |
| FEATURE               | Location/Qualifiers                              |    |
| source                | 1..20                                            |    |
|                       | mol_type = other RNA                             |    |
|                       | organism = synthetic construct                   |    |
| SEQUENCE: 7           |                                                  |    |
| gccacagact cgagtatcag |                                                  | 20 |
| SEQ ID NO: 8          | moltype = RNA length = 20                        |    |
| FEATURE               | Location/Qualifiers                              |    |
| source                | 1..20                                            |    |
|                       | mol_type = other RNA                             |    |
|                       | organism = synthetic construct                   |    |
| SEQUENCE: 8           |                                                  |    |
| ccaaacgtga cttggctcct |                                                  | 20 |
| SEQ ID NO: 9          | moltype = RNA length = 20                        |    |
| FEATURE               | Location/Qualifiers                              |    |
| source                | 1..20                                            |    |
|                       | mol_type = other RNA                             |    |
|                       | organism = synthetic construct                   |    |
| SEQUENCE: 9           |                                                  |    |
| tatggacaca catgggtgta |                                                  | 20 |
| SEQ ID NO: 10         | moltype = RNA length = 20                        |    |
| FEATURE               | Location/Qualifiers                              |    |
| source                | 1..20                                            |    |
|                       | mol_type = other RNA                             |    |
|                       | organism = synthetic construct                   |    |
| SEQUENCE: 10          |                                                  |    |
| actaaagacg tgtcttttgg |                                                  | 20 |
| SEQ ID NO: 11         | moltype = RNA length = 20                        |    |
| FEATURE               | Location/Qualifiers                              |    |
| source                | 1..20                                            |    |
|                       | mol_type = other RNA                             |    |
|                       | organism = synthetic construct                   |    |
| SEQUENCE: 11          |                                                  |    |
| gtcatctcta ggtgacttcc |                                                  | 20 |
| SEQ ID NO: 12         | moltype = DNA length = 20                        |    |
| FEATURE               | Location/Qualifiers                              |    |
| source                | 1..20                                            |    |
|                       | mol_type = other DNA                             |    |
|                       | organism = synthetic construct                   |    |
| modified_base         | 1                                                |    |
|                       | mod_base = OTHER                                 |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a |    |
|                       | 3'-phosphorothioate linkage                      |    |
| modified_base         | 2                                                |    |
|                       | mod_base = OTHER                                 |    |
|                       | note = 2'-O-methoxyethyl 5-methyluridine         |    |
| modified_base         | 3                                                |    |

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|                       |                                                             |
|-----------------------|-------------------------------------------------------------|
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-O-methoxyethyl 5-methyluridine                    |
|                       | 4                                                           |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-O-methoxyethyl 5-methyluridine                    |
|                       | 5                                                           |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-O-methoxyethyl 5-methylcytidine with a            |
|                       | 3'-phosphorothioate linkage                                 |
|                       | 6                                                           |
|                       | mod_base = OTHER                                            |
| modified_base         | note = thymidine with a 3'-phosphorothioate linkage         |
|                       | 7                                                           |
|                       | mod_base = OTHER                                            |
| modified_base         | note = thymidine with a 3'-phosphorothioate linkage         |
|                       | 8                                                           |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
|                       | 9                                                           |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
|                       | 10                                                          |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
|                       | 11                                                          |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
|                       | 12                                                          |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
|                       | 13                                                          |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
|                       | 14                                                          |
|                       | mod_base = OTHER                                            |
| modified_base         | note = thymidine with a 3'-phosphorothioate linkage         |
|                       | 15                                                          |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
|                       | 16                                                          |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-O-methoxyethyl 5-methyluridine                    |
|                       | 17                                                          |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-O-methoxyethyladenosine                           |
|                       | 18                                                          |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-O-methoxyethyl 5-methyluridine with a             |
|                       | 3'-phosphorothioate linkage                                 |
|                       | 19                                                          |
|                       | mod_base = OTHER                                            |
| modified_base         | note = 2'-O-methoxyethyl 5-methyluridine with a             |
|                       | 3'-phosphorothioate linkage                                 |
|                       | 20                                                          |
|                       | mod_base = OTHER                                            |
| misc_feature          | note = 2'-O-methoxyethyl 5-methylcytidine                   |
|                       | 1..5                                                        |
| misc_feature          | note = RNA                                                  |
|                       | 6..15                                                       |
|                       | note = DNA                                                  |
| misc_feature          | 16..20                                                      |
|                       | note = RNA                                                  |
| SEQUENCE: 12          |                                                             |
| ctttcttgga aagtatattc | 20                                                          |
| SEQ ID NO: 13         | moltype = DNA length = 20                                   |
| FEATURE               | Location/Qualifiers                                         |
| source                | 1..20                                                       |
|                       | mol_type = other DNA                                        |
|                       | organism = synthetic construct                              |
| modified_base         | 1                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethylguanosine with a                    |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 2                                                           |
|                       | mod_base = OTHER                                            |

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|                       |                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------|
| modified_base         | note = 2'-O-methoxyethyl 5-methylcytidine<br>3<br>mod_base = OTHER                                       |
| modified_base         | note = 2'-O-methoxyethyl 5-methyluridine<br>4<br>mod_base = OTHER                                        |
| modified_base         | note = 2'-O-methoxyethyl 5-methylcytidine<br>5<br>mod_base = OTHER                                       |
| modified_base         | note = 2'-O-methoxyethyladenosine with a<br>3'-phosphorothioate linkage<br>6<br>mod_base = OTHER         |
| modified_base         | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>7<br>mod_base = OTHER                     |
| modified_base         | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>8<br>mod_base = OTHER                     |
| modified_base         | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>9<br>mod_base = OTHER                     |
| modified_base         | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate<br>linkage<br>10<br>mod_base = OTHER         |
| modified_base         | note = thymidine with a 3'-phosphorothioate linkage<br>11<br>mod_base = OTHER                            |
| modified_base         | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>12<br>mod_base = OTHER                    |
| modified_base         | note = 2'-deoxy 5-methylcytidine with a<br>3'-phosphorothioate linkage<br>13<br>mod_base = OTHER         |
| modified_base         | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>14<br>mod_base = OTHER                    |
| modified_base         | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>15<br>mod_base = OTHER                    |
| modified_base         | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate<br>linkage<br>16<br>mod_base = OTHER         |
| modified_base         | note = 2'-O-methoxyethyladenosine<br>17<br>mod_base = OTHER                                              |
| modified_base         | note = 2'-O-methoxyethyladenosine<br>18<br>mod_base = OTHER                                              |
| modified_base         | note = 2'-O-methoxyethylguanosine with a<br>3'-phosphorothioate linkage<br>19<br>mod_base = OTHER        |
| modified_base         | note = 2'-O-methoxyethyl 5-methyluridine with a<br>3'-phosphorothioate linkage<br>20<br>mod_base = OTHER |
| misc_feature          | note = 2'-O-methoxyethyl 5-methylcytidine<br>1..5<br>note = RNA                                          |
| misc_feature          | 6..15<br>note = DNA                                                                                      |
| misc_feature          | 16..20<br>note = RNA                                                                                     |
| SEQUENCE: 13          |                                                                                                          |
| gctcagagct acagcaagtc |                                                                                                          |
| 20                    |                                                                                                          |
| SEQ ID NO: 14         | moltype = DNA length = 20                                                                                |
| FEATURE               | Location/Qualifiers                                                                                      |
| source                | 1..20<br>mol_type = other DNA<br>organism = synthetic construct                                          |
| modified_base         | 1                                                                                                        |



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|                       |                                                             |
|-----------------------|-------------------------------------------------------------|
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethylguanosine with a                    |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 2                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |
| modified_base         | 3                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |
| modified_base         | 4                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethylguanosine                           |
| modified_base         | 5                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a            |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 6                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 7                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 8                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 9                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 10                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 11                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 12                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 13                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 14                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 15                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 16                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |
| modified_base         | 17                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |
| modified_base         | 18                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a            |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 19                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine with a             |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 20                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |
| misc_feature          | 1..5                                                        |
|                       | note = RNA                                                  |
| misc_feature          | 6..15                                                       |
|                       | note = DNA                                                  |
| misc_feature          | 16..20                                                      |
|                       | note = RNA                                                  |
| SEQUENCE: 14          |                                                             |
| gctgcaggga aagagttctt | 20                                                          |
| SEQ ID NO: 15         | moltype = DNA length = 20                                   |
| FEATURE               | Location/Qualifiers                                         |
| source                | 1..20                                                       |
|                       | mol_type = other DNA                                        |

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|                       |                                                             |
|-----------------------|-------------------------------------------------------------|
|                       | organism = synthetic construct                              |
| modified_base         | 1                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethylguanosine with a                    |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 2                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyladenosine                           |
| modified_base         | 3                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |
| modified_base         | 4                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethylguanosine                           |
| modified_base         | 5                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethylguanosine with a                    |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 6                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 7                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 8                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 9                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |
|                       | linkage                                                     |
| modified_base         | 10                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 11                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 12                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = thymine with a 3'-phosphorothioate linkage           |
| modified_base         | 13                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 14                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = thymidine with a 3'-phosphorothioate linkage         |
| modified_base         | 15                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 16                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |
| modified_base         | 17                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyladenosine                           |
| modified_base         | 18                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine with a             |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 19                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a            |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 20                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |
| misc_feature          | 1..5                                                        |
|                       | note = RNA                                                  |
| misc_feature          | 6..15                                                       |
|                       | note = DNA                                                  |
| misc_feature          | 16..20                                                      |
|                       | note = RNA                                                  |
| SEQUENCE: 15          |                                                             |
| gatggagaca gtgtgtatct | 20                                                          |
| SEQ ID NO: 16         | moltype = DNA length = 20                                   |

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|               |                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------|
| FEATURE       | Location/Qualifiers                                                                                       |
| source        | 1..20<br>mol_type = other DNA<br>organism = synthetic construct                                           |
| modified_base | 1<br>mod_base = OTHER<br>note = 2'-O-methoxyethylguanosine with a<br>3'-phosphorothioate linkage          |
| modified_base | 2<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methylcytidine                                        |
| modified_base | 3<br>mod_base = OTHER<br>note = 2'-O-methoxyethyladenosine                                                |
| modified_base | 4<br>mod_base = OTHER<br>note = 2'-O-methoxyethylguanosine                                                |
| modified_base | 5<br>mod_base = OTHER<br>note = 2'-O-methoxyethyladenosine with a<br>3'-phosphorothioate linkage          |
| modified_base | 6<br>mod_base = OTHER<br>note = thymidine with a 3'-phosphorothioate linkage                              |
| modified_base | 7<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                      |
| modified_base | 8<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                      |
| modified_base | 9<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                      |
| modified_base | 10<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                     |
| modified_base | 11<br>mod_base = OTHER<br>note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate<br>linkage          |
| modified_base | 12<br>mod_base = OTHER<br>note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate<br>linkage          |
| modified_base | 13<br>mod_base = OTHER<br>note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate<br>linkage          |
| modified_base | 14<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                     |
| modified_base | 15<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                     |
| modified_base | 16<br>mod_base = OTHER<br>note = 2'-O-methoxyethyladenosine                                               |
| modified_base | 17<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methyluridine                                        |
| modified_base | 18<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methylcytidine with a<br>3'-phosphorothioate linkage |
| modified_base | 19<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methylcytidine with a<br>3'-phosphorothioate linkage |
| modified_base | 20<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methyluridine                                        |
| misc_feature  | 1..5<br>note = RNA                                                                                        |
| misc_feature  | 6..15<br>note = DNA                                                                                       |
| misc_feature  | 16..20                                                                                                    |

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note = RNA
SEQUENCE: 16
gcagatgggg cccaaatcct                                20

SEQ ID NO: 17
FEATURE
source
    mol_type = DNA length = 20
    Location/Qualifiers
    1..20
    mol_type = other DNA
    organism = synthetic construct
modified_base
    1
    mod_base = OTHER
    note = 2'-O-methoxyethylguanosine with a
    3'-phosphorothioate linkage
modified_base
    2
    mod_base = OTHER
    note = 2'-O-methoxyethyl 5-methylcytidine
modified_base
    3
    mod_base = OTHER
    note = 2'-O-methoxyethyl 5-methylcytidine
modified_base
    4
    mod_base = OTHER
    note = 2'-O-methoxyethyladenosine
modified_base
    5
    mod_base = OTHER
    note = 2'-O-methoxyethyl 5-methylcytidine with a
    3'-phosphorothioate linkage
modified_base
    6
    mod_base = OTHER
    note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage
modified_base
    7
    mod_base = OTHER
    note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage
modified_base
    8
    mod_base = OTHER
    note = thymidine with a 3'-phosphorothioate linkage
modified_base
    9
    mod_base = OTHER
    note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage
modified_base
    10
    mod_base = OTHER
    note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage
modified_base
    11
    mod_base = OTHER
    note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate
    linkage
modified_base
    12
    mod_base = OTHER
    note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage
modified_base
    13
    mod_base = OTHER
    note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage
modified_base
    14
    mod_base = OTHER
    note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate
    linkage
modified_base
    15
    mod_base = OTHER
    note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage
modified_base
    16
    mod_base = OTHER
    note = 2'-O-methoxyethyladenosine
modified_base
    17
    mod_base = OTHER
    note = 2'-O-methoxyethyladenosine
modified_base
    18
    mod_base = OTHER
    note = 2'-O-methoxyethyl 5-methyluridine with a
    3'-phosphorothioate linkage
modified_base
    19
    mod_base = OTHER
    note = 2'-O-methoxyethyl 5-methyluridine with a
    3'-phosphorothioate linkage
modified_base
    20
    mod_base = OTHER
    note = 2'-O-methoxyethyl 5-methylcytidine
misc_feature
    1..5

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misc_feature      note = RNA
                  6..15
misc_feature      note = DNA
                  16..20
                  note = RNA
SEQUENCE: 17
gccacagtga cagcaaattc                20

SEQ ID NO: 18      moltype = DNA length = 20
FEATURE           Location/Qualifiers
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct
modified_base     1
                  mod_base = OTHER
                  note = 2'-O-methoxyethylguanosine with a
                    3'-phosphorothioate linkage
modified_base     2
                  mod_base = OTHER
                  note = 2'-O-methoxyethyl 5-methylcytidine
modified_base     3
                  mod_base = OTHER
                  note = 2'-O-methoxyethyl 5-methylcytidine
modified_base     4
                  mod_base = OTHER
                  note = 2'-O-methoxyethyladenosine
modified_base     5
                  mod_base = OTHER
                  note = 2'-O-methoxyethyl 5-methylcytidine with a
                    3'-phosphorothioate linkage
modified_base     6
                  mod_base = OTHER
                  note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage
modified_base     7
                  mod_base = OTHER
                  note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage
modified_base     8
                  mod_base = OTHER
                  note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage
modified_base     9
                  mod_base = OTHER
                  note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate
                    linkage
modified_base     10
                  mod_base = OTHER
                  note = thymidine with a 3'-phosphorothioate linkage
modified_base     11
                  mod_base = OTHER
                  note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate
                    linkage
modified_base     12
                  mod_base = OTHER
                  note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage
modified_base     13
                  mod_base = OTHER
                  note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage
modified_base     14
                  mod_base = OTHER
                  note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage
modified_base     15
                  mod_base = OTHER
                  note = thymidine with a 3'-phosphorothioate linkage
modified_base     16
                  mod_base = OTHER
                  note = 2'-O-methoxyethyladenosine
modified_base     17
                  mod_base = OTHER
                  note = 2'-O-methoxyethyl 5-methyluridine
modified_base     18
                  mod_base = OTHER
                  note = 2'-O-methoxyethyl 5-methylcytidine with a
                    3'-phosphorothioate linkage
modified_base     19
                  mod_base = OTHER
                  note = 2'-O-methoxyethyladenosine with a
                    3'-phosphorothioate linkage

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|                       |                                                             |    |
|-----------------------|-------------------------------------------------------------|----|
| modified_base         | 20                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine                           |    |
| misc_feature          | 1..5                                                        |    |
|                       | note = RNA                                                  |    |
| misc_feature          | 6..15                                                       |    |
|                       | note = DNA                                                  |    |
| misc_feature          | 16..20                                                      |    |
|                       | note = RNA                                                  |    |
| SEQUENCE: 18          |                                                             |    |
| gccacagact cgagtatcag |                                                             | 20 |
| SEQ ID NO: 19         | moltype = DNA length = 20                                   |    |
| FEATURE               | Location/Qualifiers                                         |    |
| source                | 1..20                                                       |    |
|                       | mol_type = other DNA                                        |    |
|                       | organism = synthetic construct                              |    |
| modified_base         | 1                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a            |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 2                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |    |
| modified_base         | 3                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyladenosine                           |    |
| modified_base         | 4                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyladenosine                           |    |
| modified_base         | 5                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyladenosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 6                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |    |
|                       | linkage                                                     |    |
| modified_base         | 7                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 8                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = thymidine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 9                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 10                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 11                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |    |
|                       | linkage                                                     |    |
| modified_base         | 12                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = thymidine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 13                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = thymidine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 14                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 15                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 16                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |    |
| modified_base         | 17                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |    |
| modified_base         | 18                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a            |    |
|                       | 3'-phosphorothioate linkage                                 |    |

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|                       |        |                                                             |
|-----------------------|--------|-------------------------------------------------------------|
| modified_base         | 19     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyl 5-methylcytidine with a            |
|                       |        | 3'-phosphorothioate linkage                                 |
| modified_base         | 20     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyl 5-methyluridine                    |
| misc_feature          | 1..5   |                                                             |
|                       |        | note = RNA                                                  |
| misc_feature          | 6..15  |                                                             |
|                       |        | note = DNA                                                  |
| misc_feature          | 16..20 |                                                             |
|                       |        | note = RNA                                                  |
| SEQUENCE: 19          |        |                                                             |
| ccaaacgtga cttggctcct |        | 20                                                          |
| SEQ ID NO: 20         |        | mol_type = DNA length = 20                                  |
| FEATURE               |        | Location/Qualifiers                                         |
| source                | 1..20  |                                                             |
|                       |        | mol_type = other DNA                                        |
|                       |        | organism = synthetic construct                              |
| modified_base         | 1      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyl 5-methyluridine with a             |
|                       |        | 3'-phosphorothioate linkage                                 |
| modified_base         | 2      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyladenosine                           |
| modified_base         | 3      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyl 5-methyluridine                    |
| modified_base         | 4      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethylguanosine                           |
| modified_base         | 5      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethylguanosine with a                    |
|                       |        | 3'-phosphorothioate linkage                                 |
| modified_base         | 6      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 7      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |
|                       |        | linkage                                                     |
| modified_base         | 8      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 9      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |
|                       |        | linkage                                                     |
| modified_base         | 10     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 11     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |
|                       |        | linkage                                                     |
| modified_base         | 12     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 13     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = thymidine with a 3'-phosphorothioate linkage         |
| modified_base         | 14     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 15     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 16     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethylguanosine                           |
| modified_base         | 17     |                                                             |
|                       |        | mod_base = OTHER                                            |

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|                       |                                                                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| modified_base         | note = 2'-O-methoxyethyl 5-methyluridine<br>18<br>mod_base = OTHER<br>note = 2'-O-methoxyethylguanosine with a<br>3'-phosphorothioate linkage |
| modified_base         | 19<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methyluridine with a<br>3'-phosphorothioate linkage                                      |
| modified_base         | 20<br>mod_base = OTHER<br>note = 2'-O-methoxyethyladenosine                                                                                   |
| misc_feature          | 1..5<br>note = RNA                                                                                                                            |
| misc_feature          | 6..15<br>note = DNA                                                                                                                           |
| misc_feature          | 16..20<br>note = RNA                                                                                                                          |
| SEQUENCE: 20          |                                                                                                                                               |
| tatggacaca catgggtgta | 20                                                                                                                                            |
| SEQ ID NO: 21         | moltype = DNA length = 20                                                                                                                     |
| FEATURE               | Location/Qualifiers                                                                                                                           |
| source                | 1..20<br>mol_type = other DNA<br>organism = synthetic construct                                                                               |
| modified_base         | 1<br>mod_base = OTHER<br>note = 2'-O-methoxyethyladenosine with a<br>3'-phosphorothioate linkage                                              |
| modified_base         | 2<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methylcytidine                                                                            |
| modified_base         | 3<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methyluridine                                                                             |
| modified_base         | 4<br>mod_base = OTHER<br>note = 2'-O-methoxyethyladenosine                                                                                    |
| modified_base         | 5<br>mod_base = OTHER<br>note = 2'-O-methoxyethyladenosine with a<br>3'-phosphorothioate linkage                                              |
| modified_base         | 6<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                                                          |
| modified_base         | 7<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                                                          |
| modified_base         | 8<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                                                          |
| modified_base         | 9<br>mod_base = OTHER<br>note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate<br>linkage                                               |
| modified_base         | 10<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                                                         |
| modified_base         | 11<br>mod_base = OTHER<br>note = thymidine with a 3'-phosphorothioate linkage                                                                 |
| modified_base         | 12<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                                                         |
| modified_base         | 13<br>mod_base = OTHER<br>note = thymidine with a 3'-phosphorothioate linkage                                                                 |
| modified_base         | 14<br>mod_base = OTHER<br>note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate<br>linkage                                              |
| modified_base         | 15<br>mod_base = OTHER<br>note = thymidine with a 3'-phosphorothioate linkage                                                                 |
| modified_base         | 16                                                                                                                                            |



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|                       |                                                             |    |
|-----------------------|-------------------------------------------------------------|----|
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |    |
| modified_base         | 17                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |    |
| modified_base         | 18                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methyluridine with a             |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 19                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 20                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine                           |    |
| misc_feature          | 1..5                                                        |    |
|                       | note = RNA                                                  |    |
| misc_feature          | 6..15                                                       |    |
|                       | note = DNA                                                  |    |
| misc_feature          | 16..20                                                      |    |
|                       | note = RNA                                                  |    |
| SEQUENCE: 21          |                                                             |    |
| actaaagacg tgtcttttgg |                                                             | 20 |
| SEQ ID NO: 22         | moltype = DNA length = 20                                   |    |
| FEATURE               | Location/Qualifiers                                         |    |
| source                | 1..20                                                       |    |
|                       | mol_type = other DNA                                        |    |
|                       | organism = synthetic construct                              |    |
| modified_base         | 1                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 2                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |    |
| modified_base         | 3                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |    |
| modified_base         | 4                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyladenosine                           |    |
| modified_base         | 5                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methyluridine with a             |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 6                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |    |
|                       | linkage                                                     |    |
| modified_base         | 7                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = thymidine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 8                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |    |
|                       | linkage                                                     |    |
| modified_base         | 9                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = thymidine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 10                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 11                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 12                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 13                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = thymidine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 14                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |

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modified\_base 15  
mod\_base = OTHER  
note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage

modified\_base 16  
mod\_base = OTHER  
note = 2'-O-methoxyethyl 5-methylcytidine

modified\_base 17  
mod\_base = OTHER  
note = 2'-O-methoxyethyl 5-methyluridine

modified\_base 18  
mod\_base = OTHER  
note = 2'-O-methoxyethyl 5-methyluridine with a  
3'-phosphorothioate linkage

modified\_base 19  
mod\_base = OTHER  
note = 2'-O-methoxyethyl 5-methylcytidine with a  
3'-phosphorothioate linkage

modified\_base 20  
mod\_base = OTHER  
note = 2'-O-methoxyethyl 5-methylcytidine

misc\_feature 1..5  
note = RNA

misc\_feature 6..15  
note = DNA

misc\_feature 16..20  
note = RNA

SEQUENCE: 22  
gtcatctcta ggtgacttcc

20

SEQ ID NO: 23 moltype = RNA length = 133781  
FEATURE Location/Qualifiers  
source 1..133781  
mol\_type = genomic RNA  
organism = Homo sapiens

SEQUENCE: 23

|             |            |             |            |             |             |      |
|-------------|------------|-------------|------------|-------------|-------------|------|
| gcagtcacgc  | ccacccacca | gctccggcac  | caacagcagc | gccgctgcc   | ccgcccacct  | 60   |
| tctgcgcgcg  | ccaccacagc | caccttctcc  | tcctccgctg | tcctctcccg  | tcctcgctc   | 120  |
| tgctgactat  | caggtaagcg | ccgcggctcc  | gaaatctgcc | tcgcgcgcgc  | cctctgtgca  | 180  |
| cccctgcgcg  | gccgcccctc | gccctccctc  | tcgcagact  | ggggcttcgt  | gcgcccggca  | 240  |
| tcggtcgggg  | ccacgcagcg | gccctccctc  | gctccctcgt | ctcggggggt  | ggggccaggg  | 300  |
| cggcctggaa  | agggacatga | gcaagggatg  | cacgcacgcg | tgagtgcgcg  | cgtgtgtgtg  | 360  |
| tgctggaggg  | tcttcaccac | cagattcgcg  | cagaccccag | gtggaggctg  | tgccggcagg  | 420  |
| gtggggcgcg  | gcgcgcgtga | cttgggggag  | ggggctgccc | ttcactctcg  | actgcagcct  | 480  |
| tttgccgcaa  | tgggcgtgtg | tgtgtgtgtg  | tgtgtgtgtg | tgtgtgtgtg  | tgtgtgtgga  | 540  |
| ggggtcgcgat | aacgaccccc | gaaaccgaat  | ctgaaatccg | ctgtccctgc  | cgctgttcgc  | 600  |
| catcagctct  | aagaaagacg | tggatcgggt  | tctagaaaag | atgactccct  | gcacgcccct  | 660  |
| ccctgcacct  | cccagcagct | gattccgaca  | gggccttcac | tgcccctgat  | tttaggcggg  | 720  |
| ggccggccccc | atcccccctt | cctccctcag  | aaacccctag | gggacatttg  | ggggctggga  | 780  |
| gaaatcgagg  | cgatggggag | gggtccacgc  | gctgtcactt | tagttgccct  | tccccctgcg  | 840  |
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SEQ ID NO: 25      moltype = AA length = 776
FEATURE           Location/Qualifiers
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                  mol_type = protein
                  organism = Homo sapiens

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HVTQEPESGK VVQEGFLREP GPPGLSHQLM SGMPGAPLLP EGPREATRQP SGTGPEDTEG 180
GRHAPELLKH QLLGDLHQEG PPLKGAGGKE RPKSKEEVDE DRDVESSPQ DSPPSKASPA 240
QDGRPPQTAA REATSIPGFP AEGAIPLPVD FLSKVSTEIP ASEPDGSPVG RAKGQDAPLE 300
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DPLIQPSSPA VCPPEPPSPK YVSSVTSRTG SSGAKEMKLK GADGKTKIAT PRGAAPPQK 480
GOANATRIPA KTPPAKTPP SSATKQVQRR PPPAGPRSER GEPKSGDRS GYSSPGSPGT 540
PGSRSTRPSL PTPPTREPCK VAVVRTPPKS PSSAKSRLQT APVPMPLDKN VKSKIGSTEN 600
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GNIHHKPGGG QVEVKSSEKLD FKDRVQSKIG SLDNITHVPG GGNKKIETHK LTFRENAKAK 720
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                  organism = Homo sapiens

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SEQ ID NO: 27      multype = AA  length = 412
FEATURE            Location/Qualifiers
source             1..412
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organism = Homo sapiens
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SRTPSLPTPP TREPKKVVAV RTPPKSPSSA KSRLQTAPVP MPDLKNVKSK IGSTENLKHQ 240
PGGGKVQIIN KKLDSLNVQS KCGSKDNIKH VPGGGSVQIV YKPVDSLKVT SKCGSLGNIH 300
HKPGGGQVEV KSEKLDKFDR VQSKIGSLDN ITHVPGGGNK KIETHKLTFR ENAKAKTDHG 360
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SEQ ID NO: 28      moltype = RNA length = 5372
FEATURE           Location/Qualifiers
source            1..5372
                  mol_type = genomic RNA
                  organism = Homo sapiens
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source                   1..352  
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                          organism = Homo sapiens

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modified\_base           1  
                          mod\_base = OTHER  
                          note = 2'-O-methoxyethylguanosine with a

-continued

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|                       |        |                                                             |
|-----------------------|--------|-------------------------------------------------------------|
| modified_base         | 2      | 3'-phosphorothioate linkage                                 |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyl 5-methylcytidine                   |
| modified_base         | 3      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyl 5-methylcytidine                   |
| modified_base         | 4      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyl 5-methyluridine                    |
| modified_base         | 5      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyl 5-methyluridine with a             |
|                       |        | 3'-phosphorothioate linkage                                 |
| modified_base         | 6      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = thymidine with a 3'-phosphorothioate linkage         |
| modified_base         | 7      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 8      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |
|                       |        | linkage                                                     |
| modified_base         | 9      |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 10     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 11     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 12     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |
|                       |        | linkage                                                     |
| modified_base         | 13     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 14     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = thymidine with a 3'-phosphorothioate linkage         |
| modified_base         | 15     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 16     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyl 5-methylcytidine                   |
| modified_base         | 17     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyladenosine                           |
| modified_base         | 18     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyl 5-methylcytidine with a            |
|                       |        | 3'-phosphorothioate linkage                                 |
| modified_base         | 19     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethyladenosine with a                    |
|                       |        | 3'-phosphorothioate linkage                                 |
| modified_base         | 20     |                                                             |
|                       |        | mod_base = OTHER                                            |
|                       |        | note = 2'-O-methoxyethylguanosine                           |
| misc_feature          | 1..5   |                                                             |
|                       |        | note = RNA                                                  |
| misc_feature          | 6..15  |                                                             |
|                       |        | note = DNA                                                  |
| misc_feature          | 16..20 |                                                             |
|                       |        | note = RNA                                                  |
| SEQUENCE: 33          |        |                                                             |
| gcctttgcga gcggtcacag |        | 20                                                          |
| SEQ ID NO: 34         |        | moltype = DNA length = 20                                   |
| FEATURE               |        | Location/Qualifiers                                         |
| source                |        | 1..20                                                       |
|                       |        | mol_type = other DNA                                        |

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|                       |                                                             |
|-----------------------|-------------------------------------------------------------|
|                       | organism = synthetic construct                              |
| modified_base         | 1                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine with a             |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 2                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |
| modified_base         | 3                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |
| modified_base         | 4                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |
| modified_base         | 5                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine with a             |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 6                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = thymidine with a 3'-phosphorothioate linkage         |
| modified_base         | 7                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 8                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = thymidine with a 3'-phosphorothioate linkage         |
| modified_base         | 9                                                           |
|                       | mod_base = OTHER                                            |
|                       | note = thymidine with a 3'-phosphorothioate linkage         |
| modified_base         | 10                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 11                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 12                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base         | 13                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |
|                       | linkage                                                     |
| modified_base         | 14                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 15                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base         | 16                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |
| modified_base         | 17                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethyl 5-methyluridine                    |
| modified_base         | 18                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethylguanosine with a                    |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 19                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethylguanosine with a                    |
|                       | 3'-phosphorothioate linkage                                 |
| modified_base         | 20                                                          |
|                       | mod_base = OTHER                                            |
|                       | note = 2'-O-methoxyethylguanosine                           |
| misc_feature          | 1..5                                                        |
|                       | note = RNA                                                  |
| misc_feature          | 6..15                                                       |
|                       | note = DNA                                                  |
| misc_feature          | 16..20                                                      |
|                       | note = RNA                                                  |
| SEQUENCE: 34          |                                                             |
| tttcttggtg ggcaactggg | 20                                                          |
| SEQ ID NO: 35         | moltype = DNA length = 20                                   |

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| FEATURE               | Location/Qualifiers                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| source                | 1..20<br>mol_type = other DNA<br>organism = synthetic construct                                        |
| modified_base         | 1<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methylcytidine with a 3'-phosphorothioate linkage  |
| modified_base         | 2<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methyluridine                                      |
| modified_base         | 3<br>mod_base = OTHER<br>note = 2'-O-methoxyethylguanosine                                             |
| modified_base         | 4<br>mod_base = OTHER<br>note = 2'-O-methoxyethylguanosine                                             |
| modified_base         | 5<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methyluridine with a 3'-phosphorothioate linkage   |
| modified_base         | 6<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                   |
| modified_base         | 7<br>mod_base = OTHER<br>note = thymidine with a 3'-phosphorothioate linkage                           |
| modified_base         | 8<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                   |
| modified_base         | 9<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                   |
| modified_base         | 10<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 11<br>mod_base = OTHER<br>note = thymidine with a 3'-phosphorothioate linkage                          |
| modified_base         | 12<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 13<br>mod_base = OTHER<br>note = thymidine with a 3'-phosphorothioate linkage                          |
| modified_base         | 14<br>mod_base = OTHER<br>note = thymidine with a 3'-phosphorothioate linkage                          |
| modified_base         | 15<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 16<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methyluridine                                     |
| modified_base         | 17<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methyluridine                                     |
| modified_base         | 18<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base         | 19<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base         | 20<br>mod_base = OTHER<br>note = 2'-O-methoxyethyl 5-methyluridine                                     |
| misc_feature          | 1..5<br>note = RNA                                                                                     |
| misc_feature          | 6..15<br>note = DNA                                                                                    |
| misc_feature          | 16..20<br>note = RNA                                                                                   |
| SEQUENCE: 35          |                                                                                                        |
| ctggtgtggg tggtattcct |                                                                                                        |

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|                       |                                |    |
|-----------------------|--------------------------------|----|
| SEQ ID NO: 36         | moltype = DNA length = 20      |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = other DNA           |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 36          |                                |    |
| gtggatctag ttctcatcca |                                | 20 |
| SEQ ID NO: 37         | moltype = DNA length = 20      |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = other DNA           |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 37          |                                |    |
| ggatagaatc cagttaatta |                                | 20 |
| SEQ ID NO: 38         | moltype = DNA length = 20      |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = other DNA           |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 38          |                                |    |
| tgtcacatct agaaaccacc |                                | 20 |
| SEQ ID NO: 39         | moltype = DNA length = 20      |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = other DNA           |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 39          |                                |    |
| gcctggacag agggatgagc |                                | 20 |
| SEQ ID NO: 40         | moltype = DNA length = 20      |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = other DNA           |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 40          |                                |    |
| aagctccagg tgattgagac |                                | 20 |
| SEQ ID NO: 41         | moltype = DNA length = 20      |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = other DNA           |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 41          |                                |    |
| ttctgataca ttgtcaaagt |                                | 20 |
| SEQ ID NO: 42         | moltype = DNA length = 20      |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = other DNA           |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 42          |                                |    |
| agcagcttga aggagccggc |                                | 20 |
| SEQ ID NO: 43         | moltype = DNA length = 20      |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..20                          |    |
|                       | mol_type = other DNA           |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 43          |                                |    |
| tggctgatga taccatttgg |                                | 20 |
| SEQ ID NO: 44         | moltype = DNA length = 16      |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..16                          |    |
|                       | mol_type = other DNA           |    |
|                       | organism = synthetic construct |    |
| SEQUENCE: 44          |                                |    |
| ctttcttggga aagtat    |                                | 16 |
| SEQ ID NO: 45         | moltype = DNA length = 16      |    |
| FEATURE               | Location/Qualifiers            |    |
| source                | 1..16                          |    |



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|                   |                                |    |
|-------------------|--------------------------------|----|
| SEQUENCE: 45      | mol_type = other DNA           |    |
| tttcttgga agtata  | organism = synthetic construct | 16 |
| SEQ ID NO: 46     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 46      |                                |    |
| ttcttgga gatat    |                                | 16 |
| SEQ ID NO: 47     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 47      |                                |    |
| tcttggaag tatatt  |                                | 16 |
| SEQ ID NO: 48     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 48      |                                |    |
| cttggaagt atattc  |                                | 16 |
| SEQ ID NO: 49     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 49      |                                |    |
| gtggatctag ttctca |                                | 16 |
| SEQ ID NO: 50     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 50      |                                |    |
| tggatctagt tctcat |                                | 16 |
| SEQ ID NO: 51     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 51      |                                |    |
| ggatctagtt ctcac  |                                | 16 |
| SEQ ID NO: 52     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 52      |                                |    |
| gatctagttc tcatcc |                                | 16 |
| SEQ ID NO: 53     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 53      |                                |    |
| atctagttct catcca |                                | 16 |
| SEQ ID NO: 54     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 54      |                                |    |
| ttctgataca ttgtca |                                | 16 |

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|                   |                                |    |
|-------------------|--------------------------------|----|
| SEQ ID NO: 55     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 55      |                                |    |
| tctgatacat tgtcaa |                                | 16 |
| SEQ ID NO: 56     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 56      |                                |    |
| ctgatacatt gtcaaa |                                | 16 |
| SEQ ID NO: 57     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 57      |                                |    |
| tgatacattg tcaaag |                                | 16 |
| SEQ ID NO: 58     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 58      |                                |    |
| gatacattgt caaagt |                                | 16 |
| SEQ ID NO: 59     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 59      |                                |    |
| agcagcttga aggagc |                                | 16 |
| SEQ ID NO: 60     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 60      |                                |    |
| gcagcttgaa ggagcc |                                | 16 |
| SEQ ID NO: 61     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 61      |                                |    |
| cagcttgaag gagccg |                                | 16 |
| SEQ ID NO: 62     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 62      |                                |    |
| agcttgaagg agccgg |                                | 16 |
| SEQ ID NO: 63     | moltype = DNA length = 16      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..16                          |    |
|                   | mol_type = other DNA           |    |
|                   | organism = synthetic construct |    |
| SEQUENCE: 63      |                                |    |
| gcttgaagga gccggc |                                | 16 |
| SEQ ID NO: 64     | moltype = DNA length = 20      |    |
| FEATURE           | Location/Qualifiers            |    |
| source            | 1..20                          |    |

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|               |                                                             |
|---------------|-------------------------------------------------------------|
|               | mol_type = other DNA                                        |
|               | organism = synthetic construct                              |
| modified_base | 1                                                           |
|               | mod_base = OTHER                                            |
|               | note = 2'-O-methoxyethylguanosine with a                    |
|               | 3'-phosphorothioate linkage                                 |
| modified_base | 2                                                           |
|               | mod_base = OTHER                                            |
|               | note = 2'-O-methoxyethylthymine                             |
| modified_base | 3                                                           |
|               | mod_base = OTHER                                            |
|               | note = 2'-O-methoxyethylguanosine with a                    |
|               | 3'-phosphorothioate linkage                                 |
| modified_base | 4                                                           |
|               | mod_base = OTHER                                            |
|               | note = 2'-O-methoxyethylguanosine with a                    |
|               | 3'-phosphorothioate linkage                                 |
| modified_base | 5                                                           |
|               | mod_base = OTHER                                            |
|               | note = 2'-O-methoxyethyladenosine with a                    |
|               | 3'-phosphorothioate linkage                                 |
| modified_base | 6                                                           |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |
| modified_base | 7                                                           |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |
|               | linkage                                                     |
| modified_base | 8                                                           |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |
| modified_base | 9                                                           |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |
| modified_base | 10                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |
| modified_base | 11                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |
| modified_base | 12                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |
| modified_base | 13                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |
|               | linkage                                                     |
| modified_base | 14                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |
| modified_base | 15                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |
|               | linkage                                                     |
| modified_base | 16                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-O-methoxyethyladenosine with a                    |
|               | 3'-phosphorothioate linkage                                 |
| modified_base | 17                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-O-methoxyethylthymine                             |
| modified_base | 18                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-O-methoxyethyl 5-methylcytidine with a            |
|               | 3'-phosphorothioate linkage                                 |
| modified_base | 19                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-O-methoxyethyl 5-methylcytidine with a            |
|               | 3'-phosphorothioate linkage                                 |
| modified_base | 20                                                          |
|               | mod_base = OTHER                                            |
|               | note = 2'-O-methoxyethyladenosine                           |
| misc_feature  | 1..5                                                        |
|               | note = RNA                                                  |
| misc_feature  | 6..15                                                       |
|               | note = DNA                                                  |

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|                       |                                                                      |    |
|-----------------------|----------------------------------------------------------------------|----|
| misc_feature          | 16..20                                                               |    |
|                       | note = RNA                                                           |    |
| SEQUENCE: 64          |                                                                      |    |
| gtggatctag ttctcatcca |                                                                      | 20 |
| SEQ ID NO: 65         | moltype = DNA length = 20                                            |    |
| FEATURE               | Location/Qualifiers                                                  |    |
| source                | 1..20                                                                |    |
|                       | mol_type = other DNA                                                 |    |
|                       | organism = synthetic construct                                       |    |
| modified_base         | 1                                                                    |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-O-methoxyethylguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 2                                                                    |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-O-methoxyethylguanosine                                    |    |
| modified_base         | 3                                                                    |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-O-methoxyethyladenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 4                                                                    |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage   |    |
| modified_base         | 5                                                                    |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-O-methoxyethyladenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 6                                                                    |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage          |    |
| modified_base         | 7                                                                    |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage          |    |
| modified_base         | 8                                                                    |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage          |    |
| modified_base         | 9                                                                    |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage            |    |
| modified_base         | 10                                                                   |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage  |    |
| modified_base         | 11                                                                   |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage  |    |
| modified_base         | 12                                                                   |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage          |    |
| modified_base         | 13                                                                   |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage          |    |
| modified_base         | 14                                                                   |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage            |    |
| modified_base         | 15                                                                   |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage            |    |
| modified_base         | 16                                                                   |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-O-methoxyethyladenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 17                                                                   |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-O-methoxyethyladenosine                                    |    |
| modified_base         | 18                                                                   |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage   |    |
| modified_base         | 19                                                                   |    |
|                       | mod_base = OTHER                                                     |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage   |    |

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|                       |                                                                              |    |
|-----------------------|------------------------------------------------------------------------------|----|
| modified_base         | 20                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethyladenosine                                            |    |
| misc_feature          | 1..5                                                                         |    |
|                       | note = RNA                                                                   |    |
| misc_feature          | 6..15                                                                        |    |
|                       | note = DNA                                                                   |    |
| misc_feature          | 16..20                                                                       |    |
|                       | note = RNA                                                                   |    |
| SEQUENCE: 65          |                                                                              |    |
| ggatagaatc cagttaatta |                                                                              | 20 |
| SEQ ID NO: 66         | moltype = DNA length = 20                                                    |    |
| FEATURE               | Location/Qualifiers                                                          |    |
| source                | 1..20                                                                        |    |
|                       | mol_type = other DNA                                                         |    |
|                       | organism = synthetic construct                                               |    |
| modified_base         | 1                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage           |    |
| modified_base         | 2                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethylguanosine                                            |    |
| modified_base         | 3                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage           |    |
| modified_base         | 4                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base         | 5                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethyladenosine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 6                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage          |    |
| modified_base         | 7                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 8                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage                    |    |
| modified_base         | 9                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage          |    |
| modified_base         | 10                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage                    |    |
| modified_base         | 11                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 12                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 13                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 14                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 15                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 16                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base         | 17                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                                    |    |
| modified_base         | 18                                                                           |    |

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|                       |                                                             |    |
|-----------------------|-------------------------------------------------------------|----|
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyladenosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 19                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a            |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 20                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |    |
| misc_feature          | 1..5                                                        |    |
|                       | note = RNA                                                  |    |
| misc_feature          | 6..15                                                       |    |
|                       | note = DNA                                                  |    |
| misc_feature          | 16..20                                                      |    |
|                       | note = RNA                                                  |    |
| SEQUENCE: 66          |                                                             |    |
| tgtcacatct agaaaccacc |                                                             | 20 |
| SEQ ID NO: 67         | moltype = DNA length = 20                                   |    |
| FEATURE               | Location/Qualifiers                                         |    |
| source                | 1..20                                                       |    |
|                       | mol_type = other DNA                                        |    |
|                       | organism = synthetic construct                              |    |
| modified_base         | 1                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 2                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |    |
| modified_base         | 3                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a            |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 4                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate  |    |
|                       | linkage                                                     |    |
| modified_base         | 5                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 6                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 7                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 8                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |    |
|                       | linkage                                                     |    |
| modified_base         | 9                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 10                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 11                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 12                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 13                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 14                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 15                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 16                                                          |    |
|                       | mod_base = OTHER                                            |    |

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|                       |                                                                              |    |
|-----------------------|------------------------------------------------------------------------------|----|
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage           |    |
| modified_base         | 17                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
| modified_base         | note = 2'-O-methoxyethylguanosine                                            |    |
|                       | 18                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethyladenosine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 19                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethylguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 20                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
| misc_feature          | note = 2'-O-methoxyethyl 5-methylcytidine                                    |    |
|                       | 1..5                                                                         |    |
| misc_feature          | note = RNA                                                                   |    |
|                       | 6..15                                                                        |    |
| misc_feature          | note = DNA                                                                   |    |
|                       | 16..20                                                                       |    |
|                       | note = RNA                                                                   |    |
| SEQUENCE: 67          |                                                                              |    |
| gcctggacag agggatgagc |                                                                              | 20 |
| SEQ ID NO: 68         | moltype = DNA length = 20                                                    |    |
| FEATURE               | Location/Qualifiers                                                          |    |
| source                | 1..20                                                                        |    |
|                       | mol_type = other DNA                                                         |    |
|                       | organism = synthetic construct                                               |    |
| modified_base         | 1                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethyladenosine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 2                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethyladenosine                                            |    |
| modified_base         | 3                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethylguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base         | 4                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base         | 5                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage           |    |
| modified_base         | 6                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage          |    |
| modified_base         | 7                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage          |    |
| modified_base         | 8                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 9                                                                            |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 10                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 11                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage                    |    |
| modified_base         | 12                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 13                                                                           |    |
|                       | mod_base = OTHER                                                             |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |    |
| modified_base         | 14                                                                           |    |

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|                       |                                                             |    |
|-----------------------|-------------------------------------------------------------|----|
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |    |
| modified_base         | 15                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |    |
| modified_base         | 16                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 17                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyladenosine                           |    |
| modified_base         | 18                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 19                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyladenosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 20                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |    |
| misc_feature          | 1..5                                                        |    |
|                       | note = RNA                                                  |    |
| misc_feature          | 6..15                                                       |    |
|                       | note = DNA                                                  |    |
| misc_feature          | 16..20                                                      |    |
|                       | note = RNA                                                  |    |
| SEQUENCE: 68          |                                                             |    |
| aagctccagg tgattgagac |                                                             | 20 |
| SEQ ID NO: 69         | moltype = DNA length = 20                                   |    |
| FEATURE               | Location/Qualifiers                                         |    |
| source                | 1..20                                                       |    |
|                       | mol_type = other DNA                                        |    |
|                       | organism = synthetic construct                              |    |
| modified_base         | 1                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate  |    |
|                       | linkage                                                     |    |
| modified_base         | 2                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylthymine                             |    |
| modified_base         | 3                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a            |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 4                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate  |    |
|                       | linkage                                                     |    |
| modified_base         | 5                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 6                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 7                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |    |
| modified_base         | 8                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 9                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |    |
|                       | linkage                                                     |    |
| modified_base         | 10                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 11                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |    |
| modified_base         | 12                                                          |    |



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|                       |                                                                              |
|-----------------------|------------------------------------------------------------------------------|
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage                    |
| modified_base         | 13                                                                           |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 14                                                                           |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage                    |
| modified_base         | 15                                                                           |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage          |
| modified_base         | 16                                                                           |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-O-methoxyethyladenosine with a 3'-phosphorothioate linkage         |
| modified_base         | 17                                                                           |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-O-methoxyethyladenosine                                            |
| modified_base         | 18                                                                           |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-O-methoxyethyladenosine with a 3'-phosphorothioate linkage         |
| modified_base         | 19                                                                           |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-O-methoxyethylguanosine with a 3'-phosphorothioate linkage         |
| modified_base         | 20                                                                           |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-O-methoxyethylthymine                                              |
| misc_feature          | 1..5                                                                         |
|                       | note = RNA                                                                   |
| misc_feature          | 6..15                                                                        |
|                       | note = DNA                                                                   |
| misc_feature          | 16..20                                                                       |
|                       | note = RNA                                                                   |
| SEQUENCE: 69          |                                                                              |
| ttctgataca ttgtcaaagt | 20                                                                           |
| SEQ ID NO: 70         |                                                                              |
| FEATURE               |                                                                              |
| source                | 1..20                                                                        |
|                       | mol_type = other DNA                                                         |
|                       | organism = synthetic construct                                               |
| modified_base         | 1                                                                            |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-O-methoxyethyladenosine with a 3'-phosphorothioate linkage         |
| modified_base         | 2                                                                            |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-O-methoxyethylguanosine                                            |
| modified_base         | 3                                                                            |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base         | 4                                                                            |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-O-methoxyethyladenosine with a 3'-phosphorothioate linkage         |
| modified_base         | 5                                                                            |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-O-methoxyethylguanosine with a 3'-phosphorothioate linkage         |
| modified_base         | 6                                                                            |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage          |
| modified_base         | 7                                                                            |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage                    |
| modified_base         | 8                                                                            |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage                    |
| modified_base         | 9                                                                            |
|                       | mod_base = OTHER                                                             |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |

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|                       |        |                                                                              |
|-----------------------|--------|------------------------------------------------------------------------------|
| modified_base         | 10     | mod_base = OTHER                                                             |
|                       |        | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 11     | mod_base = OTHER                                                             |
|                       |        | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 12     | mod_base = OTHER                                                             |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 13     | mod_base = OTHER                                                             |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 14     | mod_base = OTHER                                                             |
|                       |        | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 15     | mod_base = OTHER                                                             |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 16     | mod_base = OTHER                                                             |
|                       |        | note = 2'-O-methoxyethyl 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base         | 17     | mod_base = OTHER                                                             |
|                       |        | note = 2'-O-methoxyethyl 5-methylcytidine                                    |
| modified_base         | 18     | mod_base = OTHER                                                             |
|                       |        | note = 2'-O-methoxyethylguanosine with a 3'-phosphorothioate linkage         |
| modified_base         | 19     | mod_base = OTHER                                                             |
|                       |        | note = 2'-O-methoxyethylguanosine with a 3'-phosphorothioate linkage         |
| modified_base         | 20     | mod_base = OTHER                                                             |
|                       |        | note = 2'-O-methoxyethyl 5-methylcytidine                                    |
| misc_feature          | 1..5   | note = RNA                                                                   |
| misc_feature          | 6..15  | note = DNA                                                                   |
| misc_feature          | 16..20 | note = RNA                                                                   |
| SEQUENCE: 70          |        |                                                                              |
| agcagcttga aggagccggc |        | 20                                                                           |
| SEQ ID NO: 71         |        |                                                                              |
| FEATURE               |        |                                                                              |
| source                |        |                                                                              |
|                       | 1..20  | mol_type = other DNA                                                         |
|                       |        | organism = synthetic construct                                               |
| modified_base         | 1      | mod_base = OTHER                                                             |
|                       |        | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage           |
| modified_base         | 2      | mod_base = OTHER                                                             |
|                       |        | note = 2'-O-methoxyethylguanosine                                            |
| modified_base         | 3      | mod_base = OTHER                                                             |
|                       |        | note = 2'-O-methoxyethylguanosine with a 3'-phosphorothioate linkage         |
| modified_base         | 4      | mod_base = OTHER                                                             |
|                       |        | note = 2'-O-methoxyethyl 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base         | 5      | mod_base = OTHER                                                             |
|                       |        | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage           |
| modified_base         | 6      | mod_base = OTHER                                                             |
|                       |        | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 7      | mod_base = OTHER                                                             |
|                       |        | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                  |
| modified_base         | 8      |                                                                              |

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|                     |                                                                      |
|---------------------|----------------------------------------------------------------------|
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage            |
|                     | 9                                                                    |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage          |
|                     | 10                                                                   |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage          |
|                     | 11                                                                   |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage            |
|                     | 12                                                                   |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage          |
|                     | 13                                                                   |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage  |
|                     | 14                                                                   |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage  |
|                     | 15                                                                   |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage          |
|                     | 16                                                                   |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage   |
|                     | 17                                                                   |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-O-methoxyethylthymine                                      |
|                     | 18                                                                   |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage   |
|                     | 19                                                                   |
|                     | mod_base = OTHER                                                     |
| modified_base       | note = 2'-O-methoxyethylguanosine with a 3'-phosphorothioate linkage |
|                     | 20                                                                   |
|                     | mod_base = OTHER                                                     |
| misc_feature        | note = 2'-O-methoxyethylguanosine                                    |
|                     | 1..5                                                                 |
| misc_feature        | note = RNA                                                           |
|                     | 6..15                                                                |
| misc_feature        | note = DNA                                                           |
|                     | 16..20                                                               |
| misc_feature        | note = RNA                                                           |
| SEQUENCE: 71        |                                                                      |
| tggtgatga taccatttg | 20                                                                   |
| SEQ ID NO: 72       | moltype = DNA length = 16                                            |
| FEATURE             | Location/Qualifiers                                                  |
| source              | 1..16                                                                |
|                     | mol_type = other DNA                                                 |
|                     | organism = synthetic construct                                       |
| modified_base       | 1                                                                    |
|                     | mod_base = OTHER                                                     |
|                     | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage    |
| modified_base       | 2                                                                    |
|                     | mod_base = OTHER                                                     |
|                     | note = Locked thymine with a 3'-phosphorothioate linkage             |
| modified_base       | 3                                                                    |
|                     | mod_base = OTHER                                                     |
|                     | note = Locked thymine with a 3'-phosphorothioate linkage             |
| modified_base       | 4                                                                    |
|                     | mod_base = OTHER                                                     |
|                     | note = 2'-deoxythymine with a 3'-phosphorothioate linkage            |
| modified_base       | 5                                                                    |
|                     | mod_base = OTHER                                                     |
|                     | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage  |
| modified_base       | 6                                                                    |
|                     | mod_base = OTHER                                                     |

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|               |                                                                                       |
|---------------|---------------------------------------------------------------------------------------|
| modified_base | note = 2'-deoxythymine with a 3'-phosphorothioate linkage<br>7<br>mod_base = OTHER    |
| modified_base | note = 2'-deoxythymine with a 3'-phosphorothioate linkage<br>8<br>mod_base = OTHER    |
| modified_base | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>9<br>mod_base = OTHER  |
| modified_base | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>10<br>mod_base = OTHER |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>11<br>mod_base = OTHER |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>12<br>mod_base = OTHER |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>13<br>mod_base = OTHER |
| modified_base | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>14<br>mod_base = OTHER |
| modified_base | note = Locked thymine with a 3'-phosphorothioate linkage<br>15<br>mod_base = OTHER    |
| modified_base | note = Locked adenosine with a 3'-phosphorothioate linkage<br>16<br>mod_base = OTHER  |
| modified_base | note = Locked thymine                                                                 |

SEQUENCE: 72  
ctttcttgga aagtat 16

SEQ ID NO: 73  
FEATURE  
source

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| modified_base | moltype = DNA length = 16<br>Location/Qualifiers<br>1..16<br>mol_type = other DNA<br>organism = synthetic construct |
| modified_base | 1<br>mod_base = OTHER                                                                                               |
| modified_base | note = Locked thymine with a 3'-phosphorothioate linkage<br>2<br>mod_base = OTHER                                   |
| modified_base | note = Locked thymine with a 3'-phosphorothioate linkage<br>3<br>mod_base = OTHER                                   |
| modified_base | note = Locked thymine with a 3'-phosphorothioate linkage<br>4<br>mod_base = OTHER                                   |
| modified_base | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage<br>5<br>mod_base = OTHER                        |
| modified_base | note = 2'-deoxythymine with a 3'-phosphorothioate linkage<br>6<br>mod_base = OTHER                                  |
| modified_base | note = 2'-deoxythymine with a 3'-phosphorothioate linkage<br>7<br>mod_base = OTHER                                  |
| modified_base | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>8<br>mod_base = OTHER                                |
| modified_base | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>9<br>mod_base = OTHER                                |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>10<br>mod_base = OTHER                               |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>11<br>mod_base = OTHER                               |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>12<br>mod_base = OTHER                               |
| modified_base | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                                                         |

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|                    |    |                                                                   |
|--------------------|----|-------------------------------------------------------------------|
| modified_base      | 13 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |
| modified_base      | 14 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = Locked adenosine with a 3'-phosphorothioate linkage        |
| modified_base      | 15 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = Locked thymine with a 3'-phosphorothioate linkage          |
| modified_base      | 16 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = Locked adenosine                                           |
| SEQUENCE: 73       |    |                                                                   |
| tttcttggaag agtata |    | 16                                                                |
| SEQ ID NO: 74      |    |                                                                   |
| FEATURE            |    |                                                                   |
| source             |    |                                                                   |
|                    |    | mol_type = DNA length = 16                                        |
|                    |    | Location/Qualifiers                                               |
|                    |    | 1..16                                                             |
|                    |    | mol_type = other DNA                                              |
|                    |    | organism = synthetic construct                                    |
| modified_base      | 1  |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = Locked thymine with a 3'-phosphorothioate linkage          |
| modified_base      | 2  |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = Locked thymine with a 3'-phosphorothioate linkage          |
| modified_base      | 3  |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base      | 4  |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |
| modified_base      | 5  |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |
| modified_base      | 6  |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |
| modified_base      | 7  |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |
| modified_base      | 8  |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |
| modified_base      | 9  |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |
| modified_base      | 10 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |
| modified_base      | 11 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |
| modified_base      | 12 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |
| modified_base      | 13 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |
| modified_base      | 14 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = Locked thymine with a 3'-phosphorothioate linkage          |
| modified_base      | 15 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = Locked adenosine with a 3'-phosphorothioate linkage        |
| modified_base      | 16 |                                                                   |
|                    |    | mod_base = OTHER                                                  |
|                    |    | note = Locked thymine                                             |
| SEQUENCE: 74       |    |                                                                   |
| ttcttggaag gtatat  |    | 16                                                                |
| SEQ ID NO: 75      |    |                                                                   |
| FEATURE            |    |                                                                   |
| source             |    |                                                                   |
|                    |    | mol_type = DNA length = 16                                        |
|                    |    | Location/Qualifiers                                               |
|                    |    | 1..16                                                             |
|                    |    | mol_type = other DNA                                              |

|                  |                                                                   |    |
|------------------|-------------------------------------------------------------------|----|
|                  | organism = synthetic construct                                    |    |
| modified_base    | 1                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = Locked thymine with a 3'-phosphorothioate linkage          |    |
| modified_base    | 2                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base    | 3                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = Locked thymine with a 3'-phosphorothioate linkage          |    |
| modified_base    | 4                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |    |
| modified_base    | 5                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |    |
| modified_base    | 6                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |    |
| modified_base    | 7                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |    |
| modified_base    | 8                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |    |
| modified_base    | 9                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |    |
| modified_base    | 10                                                                |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |    |
| modified_base    | 11                                                                |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |    |
| modified_base    | 12                                                                |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |    |
| modified_base    | 13                                                                |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |    |
| modified_base    | 14                                                                |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = Locked adenosine with a 3'-phosphorothioate linkage        |    |
| modified_base    | 15                                                                |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = Locked thymine with a 3'-phosphorothioate linkage          |    |
| modified_base    | 16                                                                |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = Locked thymine                                             |    |
| SEQUENCE: 75     |                                                                   |    |
| tcttggaag tatatt |                                                                   | 16 |
| SEQ ID NO: 76    | moltype = DNA length = 16                                         |    |
| FEATURE          | Location/Qualifiers                                               |    |
| source           | 1..16                                                             |    |
|                  | mol_type = other DNA                                              |    |
|                  | organism = synthetic construct                                    |    |
| modified_base    | 1                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base    | 2                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = Locked thymine with a 3'-phosphorothioate linkage          |    |
| modified_base    | 3                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = Locked thymine with a 3'-phosphorothioate linkage          |    |
| modified_base    | 4                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |    |
| modified_base    | 5                                                                 |    |
|                  | mod_base = OTHER                                                  |    |
|                  | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |    |
| modified_base    | 6                                                                 |    |
|                  | mod_base = OTHER                                                  |    |

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|               |                                                                                       |
|---------------|---------------------------------------------------------------------------------------|
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>7<br>mod_base = OTHER  |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>8<br>mod_base = OTHER  |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>9<br>mod_base = OTHER  |
| modified_base | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>10<br>mod_base = OTHER |
| modified_base | note = 2'-deoxythymine with a 3'-phosphorothioate linkage<br>11<br>mod_base = OTHER   |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>12<br>mod_base = OTHER |
| modified_base | note = 2'-deoxythymine with a 3'-phosphorothioate linkage<br>13<br>mod_base = OTHER   |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>14<br>mod_base = OTHER |
| modified_base | note = Locked thymine with a 3'-phosphorothioate linkage<br>15<br>mod_base = OTHER    |
| modified_base | note = Locked thymine with a 3'-phosphorothioate linkage<br>16<br>mod_base = OTHER    |
| modified_base | note = Locked 5-methylcytidine                                                        |

SEQUENCE: 76  
cttgaaagt atattc

16

SEQ ID NO: 77  
FEATURE  
source

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| modified_base | moltype = DNA length = 16<br>Location/Qualifiers<br>1..16<br>mol_type = other DNA<br>organism = synthetic construct |
| modified_base | 1<br>mod_base = OTHER                                                                                               |
| modified_base | note = Locked guanosine with a 3'-phosphorothioate linkage<br>2<br>mod_base = OTHER                                 |
| modified_base | note = Locked thymine with a 3'-phosphorothioate linkage<br>3<br>mod_base = OTHER                                   |
| modified_base | note = Locked guanosine with a 3'-phosphorothioate linkage<br>4<br>mod_base = OTHER                                 |
| modified_base | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>5<br>mod_base = OTHER                                |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>6<br>mod_base = OTHER                                |
| modified_base | note = 2'-deoxythymine with a 3'-phosphorothioate linkage<br>7<br>mod_base = OTHER                                  |
| modified_base | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage<br>8<br>mod_base = OTHER                        |
| modified_base | note = 2'-deoxythymine with a 3'-phosphorothioate linkage<br>9<br>mod_base = OTHER                                  |
| modified_base | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage<br>10<br>mod_base = OTHER                               |
| modified_base | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage<br>11<br>mod_base = OTHER                               |
| modified_base | note = 2'-deoxythymine with a 3'-phosphorothioate linkage<br>12<br>mod_base = OTHER                                 |
| modified_base | note = 2'-deoxythymine with a 3'-phosphorothioate linkage                                                           |

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|                   |                                                                     |             |
|-------------------|---------------------------------------------------------------------|-------------|
| modified_base     | 13                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |             |
| modified_base     | 14                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = Locked thymine with a 3'-phosphorothioate linkage            |             |
| modified_base     | 15                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |             |
| modified_base     | 16                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = Locked adenosine                                             |             |
| SEQUENCE: 77      |                                                                     |             |
| gtggatctag ttctca |                                                                     | 16          |
| SEQ ID NO: 78     | moltype = DNA                                                       | length = 16 |
| FEATURE           | Location/Qualifiers                                                 |             |
| source            | 1..16                                                               |             |
|                   | mol_type = other DNA                                                |             |
|                   | organism = synthetic construct                                      |             |
| modified_base     | 1                                                                   |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = Locked thymine with a 3'-phosphorothioate linkage            |             |
| modified_base     | 2                                                                   |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |             |
| modified_base     | 3                                                                   |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |             |
| modified_base     | 4                                                                   |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |             |
| modified_base     | 5                                                                   |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |             |
| modified_base     | 6                                                                   |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |             |
| modified_base     | 7                                                                   |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |             |
| modified_base     | 8                                                                   |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |             |
| modified_base     | 9                                                                   |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |             |
| modified_base     | 10                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |             |
| modified_base     | 11                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |             |
| modified_base     | 12                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |             |
| modified_base     | 13                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |             |
| modified_base     | 14                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |             |
| modified_base     | 15                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |             |
| modified_base     | 16                                                                  |             |
|                   | mod_base = OTHER                                                    |             |
|                   | note = Locked thymine                                               |             |
| SEQUENCE: 78      |                                                                     |             |
| tggatctagt tctcat |                                                                     | 16          |



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|                  |                                                                     |
|------------------|---------------------------------------------------------------------|
| SEQ ID NO: 79    | moltype = DNA length = 16                                           |
| FEATURE          | Location/Qualifiers                                                 |
| source           | 1..16                                                               |
|                  | mol_type = other DNA                                                |
|                  | organism = synthetic construct                                      |
| modified_base    | 1                                                                   |
|                  | mod_base = OTHER                                                    |
|                  | note = Locked guanosine with a 3'-phosphorothioate linkage          |
| modified_base    | 2                                                                   |
|                  | mod_base = OTHER                                                    |
|                  | note = Locked guanosine with a 3'-phosphorothioate linkage          |
| modified_base    | 3                                                                   |
|                  | mod_base = OTHER                                                    |
|                  | note = Locked adenosine with a 3'-phosphorothioate linkage          |
| modified_base    | 4                                                                   |
|                  | mod_base = OTHER                                                    |
|                  | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base    | 5                                                                   |
|                  | mod_base = OTHER                                                    |
|                  | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base    | 6                                                                   |
|                  | mod_base = OTHER                                                    |
|                  | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base    | 7                                                                   |
|                  | mod_base = OTHER                                                    |
|                  | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |
| modified_base    | 8                                                                   |
|                  | mod_base = OTHER                                                    |
|                  | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |
| modified_base    | 9                                                                   |
|                  | mod_base = OTHER                                                    |
|                  | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base    | 10                                                                  |
|                  | mod_base = OTHER                                                    |
|                  | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base    | 11                                                                  |
|                  | mod_base = OTHER                                                    |
|                  | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base    | 12                                                                  |
|                  | mod_base = OTHER                                                    |
|                  | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base    | 13                                                                  |
|                  | mod_base = OTHER                                                    |
|                  | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base    | 14                                                                  |
|                  | mod_base = OTHER                                                    |
|                  | note = Locked adenosine with a 3'-phosphorothioate linkage          |
| modified_base    | 15                                                                  |
|                  | mod_base = OTHER                                                    |
|                  | note = Locked thymine with a 3'-phosphorothioate linkage            |
| modified_base    | 16                                                                  |
|                  | mod_base = OTHER                                                    |
|                  | note = Locked 5-methylcytidine                                      |
| SEQUENCE: 79     |                                                                     |
| ggatctagtt ctcac | 16                                                                  |

  

|               |                                                             |
|---------------|-------------------------------------------------------------|
| SEQ ID NO: 80 | moltype = DNA length = 16                                   |
| FEATURE       | Location/Qualifiers                                         |
| source        | 1..16                                                       |
|               | mol_type = other DNA                                        |
|               | organism = synthetic construct                              |
| modified_base | 1                                                           |
|               | mod_base = OTHER                                            |
|               | note = Locked guanosine with a 3'-phosphorothioate linkage  |
| modified_base | 2                                                           |
|               | mod_base = OTHER                                            |
|               | note = Locked adenosine with a 3'-phosphorothioate linkage  |
| modified_base | 3                                                           |
|               | mod_base = OTHER                                            |
|               | note = Locked thymine with a 3'-phosphorothioate linkage    |
| modified_base | 4                                                           |
|               | mod_base = OTHER                                            |
|               | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate |

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|                   |                                                                     |
|-------------------|---------------------------------------------------------------------|
|                   | linkage                                                             |
| modified_base     | 5                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 6                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |
| modified_base     | 7                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |
| modified_base     | 8                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 9                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 10                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base     | 11                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 12                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base     | 13                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |
| modified_base     | 14                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked thymine with a 3'-phosphorothioate linkage            |
| modified_base     | 15                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |
| modified_base     | 16                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked 5-methylcytidine                                      |
| SEQUENCE: 80      |                                                                     |
| gatctagttc tcatcc | 16                                                                  |
| SEQ ID NO: 81     | moltype = DNA length = 16                                           |
| FEATURE           | Location/Qualifiers                                                 |
| source            | 1..16                                                               |
|                   | mol_type = other DNA                                                |
|                   | organism = synthetic construct                                      |
| modified_base     | 1                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |
| modified_base     | 2                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked thymine with a 3'-phosphorothioate linkage            |
| modified_base     | 3                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |
| modified_base     | 4                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 5                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |
| modified_base     | 6                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |
| modified_base     | 7                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 8                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 9                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate         |

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|                   |                                                                     |    |
|-------------------|---------------------------------------------------------------------|----|
|                   | linkage                                                             |    |
| modified_base     | 10                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 11                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base     | 12                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 13                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 14                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |    |
| modified_base     | 15                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |    |
| modified_base     | 16                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked adenosine                                             |    |
| SEQUENCE: 81      |                                                                     |    |
| atctagttct catcca |                                                                     | 16 |
| SEQ ID NO: 82     | moltype = DNA length = 16                                           |    |
| FEATURE           | Location/Qualifiers                                                 |    |
| source            | 1..16                                                               |    |
|                   | mol_type = other DNA                                                |    |
|                   | organism = synthetic construct                                      |    |
| modified_base     | 1                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked thymine with a 3'-phosphorothioate linkage            |    |
| modified_base     | 2                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked thymine with a 3'-phosphorothioate linkage            |    |
| modified_base     | 3                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |    |
| modified_base     | 4                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 5                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 6                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 7                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 8                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 9                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base     | 10                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 11                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 12                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 13                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 14                                                                  |    |
|                   | mod_base = OTHER                                                    |    |

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|                   |                                                                                                                                                         |    |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| modified_base     | note = Locked thymine with a 3'-phosphorothioate linkage<br>15<br>mod_base = OTHER<br>note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base     | 16<br>mod_base = OTHER<br>note = Locked adenosine                                                                                                       |    |
| SEQUENCE: 82      |                                                                                                                                                         |    |
| tcttgataca ttgtca |                                                                                                                                                         | 16 |
| SEQ ID NO: 83     | moltype = DNA length = 16                                                                                                                               |    |
| FEATURE           | Location/Qualifiers                                                                                                                                     |    |
| source            | 1..16<br>mol_type = other DNA<br>organism = synthetic construct                                                                                         |    |
| modified_base     | 1<br>mod_base = OTHER<br>note = Locked thymine with a 3'-phosphorothioate linkage                                                                       |    |
| modified_base     | 2<br>mod_base = OTHER<br>note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage                                                              |    |
| modified_base     | 3<br>mod_base = OTHER<br>note = Locked thymine with a 3'-phosphorothioate linkage                                                                       |    |
| modified_base     | 4<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                                                                    |    |
| modified_base     | 5<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                                                                    |    |
| modified_base     | 6<br>mod_base = OTHER<br>note = 2'-deoxythymine with a 3'-phosphorothioate linkage                                                                      |    |
| modified_base     | 7<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                                                                    |    |
| modified_base     | 8<br>mod_base = OTHER<br>note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage                                                            |    |
| modified_base     | 9<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                                                                    |    |
| modified_base     | 10<br>mod_base = OTHER<br>note = 2'-deoxythymine with a 3'-phosphorothioate linkage                                                                     |    |
| modified_base     | 11<br>mod_base = OTHER<br>note = 2'-deoxythymine with a 3'-phosphorothioate linkage                                                                     |    |
| modified_base     | 12<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                                                                   |    |
| modified_base     | 13<br>mod_base = OTHER<br>note = 2'-deoxythymine with a 3'-phosphorothioate linkage                                                                     |    |
| modified_base     | 14<br>mod_base = OTHER<br>note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage                                                             |    |
| modified_base     | 15<br>mod_base = OTHER<br>note = Locked adenosine with a 3'-phosphorothioate linkage                                                                    |    |
| modified_base     | 16<br>mod_base = OTHER<br>note = Locked adenosine                                                                                                       |    |
| SEQUENCE: 83      |                                                                                                                                                         |    |
| tctgatacat tgtcaa |                                                                                                                                                         | 16 |
| SEQ ID NO: 84     | moltype = DNA length = 16                                                                                                                               |    |
| FEATURE           | Location/Qualifiers                                                                                                                                     |    |
| source            | 1..16<br>mol_type = other DNA<br>organism = synthetic construct                                                                                         |    |
| modified_base     | 1                                                                                                                                                       |    |

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|                   |                                                                     |
|-------------------|---------------------------------------------------------------------|
|                   | mod_base = OTHER                                                    |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |
| modified_base     | 2                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked thymine with a 3'-phosphorothioate linkage            |
| modified_base     | 3                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |
| modified_base     | 4                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |
| modified_base     | 5                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 6                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |
| modified_base     | 7                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base     | 8                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |
| modified_base     | 9                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 10                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 11                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |
| modified_base     | 12                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 13                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base     | 14                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |
| modified_base     | 15                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |
| modified_base     | 16                                                                  |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked adenosine                                             |
| SEQUENCE: 84      |                                                                     |
| ctgatacatt gtcaaa | 16                                                                  |
| SEQ ID NO: 85     | moltype = DNA length = 16                                           |
| FEATURE           | Location/Qualifiers                                                 |
| source            | 1..16                                                               |
|                   | mol_type = other DNA                                                |
|                   | organism = synthetic construct                                      |
| modified_base     | 1                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked thymine with a 3'-phosphorothioate linkage            |
| modified_base     | 2                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |
| modified_base     | 3                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |
| modified_base     | 4                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 5                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |
| modified_base     | 6                                                                   |
|                   | mod_base = OTHER                                                    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate         |

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|                   |                                                                     |    |
|-------------------|---------------------------------------------------------------------|----|
|                   | linkage                                                             |    |
| modified_base     | 7                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 8                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 9                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 10                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 11                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 12                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base     | 13                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 14                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 15                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 16                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked guanosine                                             |    |
| SEQUENCE: 85      |                                                                     |    |
| tgatacattg tcaaag |                                                                     | 16 |
| SEQ ID NO: 86     | moltype = DNA length = 16                                           |    |
| FEATURE           | Location/Qualifiers                                                 |    |
| source            | 1..16                                                               |    |
|                   | mol_type = other DNA                                                |    |
|                   | organism = synthetic construct                                      |    |
| modified_base     | 1                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 2                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 3                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked thymine with a 3'-phosphorothioate linkage            |    |
| modified_base     | 4                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 5                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base     | 6                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 7                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 8                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 9                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 10                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 11                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base     | 12                                                                  |    |

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|                   |                                                                     |    |
|-------------------|---------------------------------------------------------------------|----|
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 13                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 14                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 15                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 16                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked thymine                                               |    |
| SEQUENCE: 86      |                                                                     |    |
| gatacattgt caaagt |                                                                     | 16 |
| SEQ ID NO: 87     | moltype = DNA length = 16                                           |    |
| FEATURE           | Location/Qualifiers                                                 |    |
| source            | 1..16                                                               |    |
|                   | mol_type = other DNA                                                |    |
|                   | organism = synthetic construct                                      |    |
| modified_base     | 1                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 2                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 3                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |    |
| modified_base     | 4                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 5                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 6                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base     | 7                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 8                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 9                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 10                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 11                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 12                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 13                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 14                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 15                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 16                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked 5-methylcytidine                                      |    |
| SEQUENCE: 87      |                                                                     |    |
| agcagcttga aggagc |                                                                     | 16 |
| SEQ ID NO: 88     | moltype = DNA length = 16                                           |    |

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|                   |                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------|
| FEATURE           | Location/Qualifiers                                                                          |
| source            | 1..16<br>mol_type = other DNA<br>organism = synthetic construct                              |
| modified_base     | 1<br>mod_base = OTHER<br>note = Locked guanosine with a 3'-phosphorothioate linkage          |
| modified_base     | 2<br>mod_base = OTHER<br>note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |
| modified_base     | 3<br>mod_base = OTHER<br>note = Locked adenosine with a 3'-phosphorothioate linkage          |
| modified_base     | 4<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |
| modified_base     | 5<br>mod_base = OTHER<br>note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base     | 6<br>mod_base = OTHER<br>note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 7<br>mod_base = OTHER<br>note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |
| modified_base     | 8<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |
| modified_base     | 9<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |
| modified_base     | 10<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage        |
| modified_base     | 11<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage        |
| modified_base     | 12<br>mod_base = OTHER<br>note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage        |
| modified_base     | 13<br>mod_base = OTHER<br>note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage        |
| modified_base     | 14<br>mod_base = OTHER<br>note = Locked guanosine with a 3'-phosphorothioate linkage         |
| modified_base     | 15<br>mod_base = OTHER<br>note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage  |
| modified_base     | 16<br>mod_base = OTHER<br>note = Locked 5-methylcytidine                                     |
| SEQUENCE: 88      |                                                                                              |
| gcagcttgaa ggagcc | 16                                                                                           |
| SEQ ID NO: 89     | moltype = DNA length = 16                                                                    |
| FEATURE           | Location/Qualifiers                                                                          |
| source            | 1..16<br>mol_type = other DNA<br>organism = synthetic construct                              |
| modified_base     | 1<br>mod_base = OTHER<br>note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |
| modified_base     | 2<br>mod_base = OTHER<br>note = Locked adenosine with a 3'-phosphorothioate linkage          |
| modified_base     | 3<br>mod_base = OTHER<br>note = Locked guanosine with a 3'-phosphorothioate linkage          |
| modified_base     | 4<br>mod_base = OTHER<br>note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate         |



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|                   |                                                                   |
|-------------------|-------------------------------------------------------------------|
|                   | linkage                                                           |
| modified_base     | 5                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |
| modified_base     | 6                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |
| modified_base     | 7                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |
| modified_base     | 8                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |
| modified_base     | 9                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |
| modified_base     | 10                                                                |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |
| modified_base     | 11                                                                |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |
| modified_base     | 12                                                                |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |
| modified_base     | 13                                                                |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |
| modified_base     | 14                                                                |
|                   | mod_base = OTHER                                                  |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base     | 15                                                                |
|                   | mod_base = OTHER                                                  |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base     | 16                                                                |
|                   | mod_base = OTHER                                                  |
|                   | note = Locked guanosine                                           |
| SEQUENCE: 89      |                                                                   |
| cagcttgaag gagccg | 16                                                                |
| SEQ ID NO: 90     | moltype = DNA length = 16                                         |
| FEATURE           | Location/Qualifiers                                               |
| source            | 1..16                                                             |
|                   | mol_type = other DNA                                              |
|                   | organism = synthetic construct                                    |
| modified_base     | 1                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = Locked adenosine with a 3'-phosphorothioate linkage        |
| modified_base     | 2                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage        |
| modified_base     | 3                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base     | 4                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |
| modified_base     | 5                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage         |
| modified_base     | 6                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |
| modified_base     | 7                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |
| modified_base     | 8                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage       |
| modified_base     | 9                                                                 |
|                   | mod_base = OTHER                                                  |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage       |
| modified_base     | 10                                                                |

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|                   |                                                                     |    |
|-------------------|---------------------------------------------------------------------|----|
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 11                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 12                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 13                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base     | 14                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |    |
| modified_base     | 15                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 16                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked guanosine                                             |    |
| SEQUENCE: 90      |                                                                     |    |
| agcttgaagg agccgg |                                                                     | 16 |
| SEQ ID NO: 91     | moltype = DNA length = 16                                           |    |
| FEATURE           | Location/Qualifiers                                                 |    |
| source            | 1..16                                                               |    |
|                   | mol_type = other DNA                                                |    |
|                   | organism = synthetic construct                                      |    |
| modified_base     | 1                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 2                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked 5-methylcytidine with a 3'-phosphorothioate linkage   |    |
| modified_base     | 3                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked thymine with a 3'-phosphorothioate linkage            |    |
| modified_base     | 4                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage           |    |
| modified_base     | 5                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 6                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 7                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 8                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 9                                                                   |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 10                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 11                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage         |    |
| modified_base     | 12                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base     | 13                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate linkage |    |
| modified_base     | 14                                                                  |    |
|                   | mod_base = OTHER                                                    |    |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage          |    |
| modified_base     | 15                                                                  |    |

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|                   |                                                             |    |
|-------------------|-------------------------------------------------------------|----|
|                   | mod_base = OTHER                                            |    |
|                   | note = Locked guanosine with a 3'-phosphorothioate linkage  |    |
| modified_base     | 16                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = Locked 5-methylcytidine                              |    |
| SEQUENCE: 91      |                                                             |    |
| gcttgaagga gccggc |                                                             | 16 |
| SEQ ID NO: 92     | moltype = DNA length = 20                                   |    |
| FEATURE           | Location/Qualifiers                                         |    |
| source            | 1..20                                                       |    |
|                   | mol_type = other DNA                                        |    |
|                   | organism = synthetic construct                              |    |
| modified_base     | 1                                                           |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-O-methoxyethyl 5-methylcytidine with a            |    |
|                   | 3'-phosphorothioate linkage                                 |    |
| modified_base     | 2                                                           |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-O-methoxyethylthymine                             |    |
| modified_base     | 3                                                           |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate  |    |
|                   | linkage                                                     |    |
| modified_base     | 4                                                           |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate  |    |
|                   | linkage                                                     |    |
| modified_base     | 5                                                           |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-O-methoxyethyl 5-methylcytidine with a            |    |
|                   | 3'-phosphorothioate linkage                                 |    |
| modified_base     | 6                                                           |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |    |
| modified_base     | 7                                                           |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |    |
| modified_base     | 8                                                           |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base     | 9                                                           |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base     | 10                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base     | 11                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base     | 12                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base     | 13                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base     | 14                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-deoxythymine with a 3'-phosphorothioate linkage   |    |
| modified_base     | 15                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base     | 16                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate  |    |
|                   | linkage                                                     |    |
| modified_base     | 17                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-O-methoxyethyladenosine                           |    |
| modified_base     | 18                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate  |    |
|                   | linkage                                                     |    |
| modified_base     | 19                                                          |    |
|                   | mod_base = OTHER                                            |    |
|                   | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate  |    |

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|                       |                                                             |    |
|-----------------------|-------------------------------------------------------------|----|
|                       | linkage                                                     |    |
| modified_base         | 20                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |    |
| misc_feature          | 1..5                                                        |    |
|                       | note = RNA                                                  |    |
| misc_feature          | 6..15                                                       |    |
|                       | note = DNA                                                  |    |
| misc_feature          | 16..20                                                      |    |
|                       | note = RNA                                                  |    |
| SEQUENCE: 92          |                                                             |    |
| ctttcttgga aagtatattc |                                                             | 20 |
| SEQ ID NO: 93         | moltype = DNA length = 20                                   |    |
| FEATURE               | Location/Qualifiers                                         |    |
| source                | 1..20                                                       |    |
|                       | mol_type = other DNA                                        |    |
|                       | organism = synthetic construct                              |    |
| modified_base         | 1                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 2                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine                   |    |
| modified_base         | 3                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate  |    |
|                       | linkage                                                     |    |
| modified_base         | 4                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylguanosine with a                    |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 5                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a            |    |
|                       | 3'-phosphorothioate linkage                                 |    |
| modified_base         | 6                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 7                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 8                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 9                                                           |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 10                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 11                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 12                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 13                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 14                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 15                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage |    |
| modified_base         | 16                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate  |    |
|                       | linkage                                                     |    |
| modified_base         | 17                                                          |    |
|                       | mod_base = OTHER                                            |    |
|                       | note = 2'-O-methoxyethylthymine                             |    |
| modified_base         | 18                                                          |    |
|                       | mod_base = OTHER                                            |    |

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|                       |                                                                                 |
|-----------------------|---------------------------------------------------------------------------------|
|                       | note = 2'-O-methoxyethyl 5-methylcytidine with a<br>3'-phosphorothioate linkage |
| modified_base         | 19                                                                              |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate<br>linkage           |
| modified_base         | 20                                                                              |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-O-methoxyethylthymine                                                 |
| misc_feature          | 1..5                                                                            |
|                       | note = RNA                                                                      |
| misc_feature          | 6..15                                                                           |
|                       | note = DNA                                                                      |
| misc_feature          | 16..20                                                                          |
|                       | note = RNA                                                                      |
| SEQUENCE: 93          |                                                                                 |
| gctgcagggg aagagttctt | 20                                                                              |
| SEQ ID NO: 94         | moltype = DNA length = 20                                                       |
| FEATURE               | Location/Qualifiers                                                             |
| source                | 1..20                                                                           |
|                       | mol_type = other DNA                                                            |
|                       | organism = synthetic construct                                                  |
| modified_base         | 1                                                                               |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-O-methoxyethylguanosine with a<br>3'-phosphorothioate linkage         |
| modified_base         | 2                                                                               |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-O-methoxyethyladenosine                                               |
| modified_base         | 3                                                                               |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate<br>linkage           |
| modified_base         | 4                                                                               |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-O-methoxyethylguanosine with a<br>3'-phosphorothioate linkage         |
| modified_base         | 5                                                                               |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-O-methoxyethylguanosine with a<br>3'-phosphorothioate linkage         |
| modified_base         | 6                                                                               |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                     |
| modified_base         | 7                                                                               |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                     |
| modified_base         | 8                                                                               |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                     |
| modified_base         | 9                                                                               |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-deoxy 5-methylcytidine with a 3'-phosphorothioate<br>linkage          |
| modified_base         | 10                                                                              |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-deoxyadenosine with a 3'-phosphorothioate linkage                     |
| modified_base         | 11                                                                              |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                     |
| modified_base         | 12                                                                              |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage                       |
| modified_base         | 13                                                                              |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                     |
| modified_base         | 14                                                                              |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-deoxythymine with a 3'-phosphorothioate linkage                       |
| modified_base         | 15                                                                              |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-deoxyguanosine with a 3'-phosphorothioate linkage                     |
| modified_base         | 16                                                                              |
|                       | mod_base = OTHER                                                                |
|                       | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate                      |

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|               |                                                                              |
|---------------|------------------------------------------------------------------------------|
|               | linkage                                                                      |
| modified_base | 17                                                                           |
|               | mod_base = OTHER                                                             |
|               | note = 2'-O-methoxyethyladenosine                                            |
| modified_base | 18                                                                           |
|               | mod_base = OTHER                                                             |
|               | note = 2'-O-methoxyethylthymine with a 3'-phosphorothioate linkage           |
| modified_base | 19                                                                           |
|               | mod_base = OTHER                                                             |
|               | note = 2'-O-methoxyethyl 5-methylcytidine with a 3'-phosphorothioate linkage |
| modified_base | 20                                                                           |
|               | mod_base = OTHER                                                             |
|               | note = 2'-O-methoxyethylthymine                                              |
| misc_feature  | 1..5                                                                         |
|               | note = RNA                                                                   |
| misc_feature  | 6..15                                                                        |
|               | note = DNA                                                                   |
| misc_feature  | 16..20                                                                       |
|               | note = RNA                                                                   |

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SEQUENCE: 94  
gatggagaca gtgtgtatct

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1. A MAPT antisense oligonucleotide comprises a nucleic acid sequence selected from any one of SEQ ID NOs: 1-11, 30-32, or 36-63, wherein optionally one or more nucleotides are independently modified nucleotides, and wherein optionally one or more internucleotide linkages are modified internucleotide linkages.

2. The MAPT antisense oligonucleotide of claim 1, wherein the antisense oligonucleotide is single stranded.

3. The MAPT antisense oligonucleotide of claim 1, wherein the antisense oligonucleotide comprises one or more modified nucleotides.

4. The MAPT antisense oligonucleotide of claim 1, wherein the one or more modified nucleotides comprise a modified nucleobase.

5. The MAPT antisense oligonucleotide of claim 4, wherein the modified nucleobase is a 5-methylcytosine.

6. The MAPT antisense oligonucleotide of claim 5, wherein each C in the antisense oligonucleotide is a 5-methylcytosine.

7. The MAPT antisense oligonucleotide of claim 1, wherein the one or more modified nucleotides comprise a modified sugar.

8. The MAPT antisense oligonucleotide of claim 7, wherein the modified sugar is a 2'-O-methoxyethyl (2'-O-MOE) modified sugar, 2'-O-methyl modified sugar, 2'-fluoro modified sugar, or a locked nucleic acid (LNA).

9. The MAPT antisense oligonucleotide of claim 8, wherein the modified sugar is a 2'-O-MOE modified sugar.

10. The MAPT antisense oligonucleotide of claim 8, wherein the modified sugar is LNA.

11. The MAPT antisense oligonucleotide of claim 1, wherein the antisense oligonucleotide is 16-20 nucleotides in length.

12. The MAPT antisense oligonucleotide of claim 1, wherein the first to fifth nucleotides each comprise a 2'-O-MOE modified sugar, wherein the sixth to fifteenth nucleotides each comprise a 2'-deoxynucleotide, and wherein the sixteenth to twentieth nucleotides each comprise a 2'-O-MOE modified sugar.

13. The MAPT antisense oligonucleotide of claim 1, wherein the first to third nucleotides each comprise LNA,

wherein the fourth to thirteenth nucleotides each comprise a 2'-deoxynucleoside, and wherein the fourteenth to sixteenth nucleotides each comprise LNA.

14. The MAPT antisense oligonucleotide of claim 1, wherein the antisense oligonucleotide comprises one or more modified internucleotide linkages.

15. The MAPT antisense oligonucleotide of claim 14, wherein the one or more modified internucleotide linkages are phosphorothioate linkages.

16. The MAPT antisense oligonucleotide of claim 14, wherein the internucleotide linkages of the antisense oligonucleotide are sooooooooooooo, or sooooooooooooo, from 5' end to 3' end, wherein each s is a phosphorothioate linkage and each o is a phosphodiester linkage.

17. A MAPT antisense oligonucleotide comprises a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94.

18. The MAPT antisense oligonucleotide of claim 17, wherein the antisense oligonucleotide consists of a sequence selected from any one of SEQ ID NOs: 12-22, 33-35, or 64-94.

19. A pharmaceutical composition comprising the MAPT antisense oligonucleotide of claim 1 and a pharmaceutically acceptable carrier.

20. A pharmaceutical composition comprising the MAPT antisense oligonucleotide of claim 17 and a pharmaceutically acceptable carrier.

21. A method of reducing MAPT expression in a patient in need thereof, the method comprising administering to the patient an effective amount of the MAPT antisense oligonucleotide of claim 1.

22. A method of reducing MAPT expression in a patient in need thereof, the method comprising administering to the patient an effective amount of the MAPT antisense oligonucleotide of claim 17.

23. A method of treating a tauopathy in a patient in need thereof, the method comprising administering to the patient an effective amount of the MAPT antisense oligonucleotide of claim 1.

**24.** The method of claim **23**, wherein the tauopathy is selected from Alzheimer's disease (AD), frontotemporal dementia (FTD), frontotemporal dementia with parkinsonism linked to chromosome 17 (FTDP-17), frontotemporal lobar degeneration (FTLD), behavioral variant frontotemporal dementia (bvFTD), nonfluent variant primary progressive aphasia (nfvPPA), Parkinson's disease, Pick's disease (PiD), primary progressive aphasia-semantic (PPA-S), primary progressive aphasia-logopenic (PPA-L), multiple system tauopathy with presenile dementia (MSTD), neurofibrillary tangle (NFT) dementia, FTD with motor neuron disease, progressive supranuclear palsy (PSP), amyotrophic lateral sclerosis/parkinsonism-dementia complex (ALS-PDC), argyrophilic grain dementia (AGD), British type amyloid angiopathy, cerebral amyloid angiopathy, chronic traumatic encephalopathy (CTE), corticobasal degeneration (CBD), Creutzfeldt-Jakob disease (CJD), dementia pugilistica, diffuse neurofibrillary tangles with calcification, Down's syndrome, epilepsy, Gerstmann-Straussler-Scheinker disease, Hallervorden-Spatz disease, Huntington's disease, inclusion body myositis, lead encephalopathy, Lytico-Bodig disease, meningioangiomas, multiple system atrophy, myotonic dystrophy, Niemann-Pick disease type C (NP-C), non-Guamanian motor neuron disease with neurofibrillary tangles, postencephalitic parkinsonism, prion protein cerebral amyloid angiopathy, progressive subcortical gliosis, tangle only dementia, tangle-predominant dementia, ganglioglioma, gangliocytoma, subacute sclerosingpan encephalitis, tuberous sclerosis, lipofuscinosis, primary age-related tauopathy (PART), or globular glial tauopathies (GGT).

**25.** The method of claim **23**, wherein the MAPT antisense oligonucleotide is administered to the patient intrathecally, intravenously, subcutaneously, or via intracisternal magna injection.

**26.** A method of treating a tauopathy in a patient in need thereof, the method comprising administering to the patient an effective amount of the MAPT antisense oligonucleotide of claim **17**.

**27.** The method of claim **26**, wherein the tauopathy is selected from Alzheimer's disease (AD), frontotemporal dementia (FTD), frontotemporal dementia with parkinsonism linked to chromosome 17 (FTDP-17), frontotemporal lobar degeneration (FTLD), behavioral variant frontotemporal dementia (bvFTD), nonfluent variant primary progres-

sive aphasia (nfvPPA), Parkinson's disease, Pick's disease (PiD), primary progressive aphasia-semantic (PPA-S), primary progressive aphasia-logopenic (PPA-L), multiple system tauopathy with presenile dementia (MSTD), neurofibrillary tangle (NFT) dementia, FTD with motor neuron disease, progressive supranuclear palsy (PSP), amyotrophic lateral sclerosis/parkinsonism-dementia complex (ALS-PDC), argyrophilic grain dementia (AGD), British type amyloid angiopathy, cerebral amyloid angiopathy, chronic traumatic encephalopathy (CTE), corticobasal degeneration (CBD), Creutzfeldt-Jakob disease (CJD), dementia pugilistica, diffuse neurofibrillary tangles with calcification, Down's syndrome, epilepsy, Gerstmann-Straussler-Scheinker disease, Hallervorden-Spatz disease, Huntington's disease, inclusion body myositis, lead encephalopathy, Lytico-Bodig disease, meningioangiomas, multiple system atrophy, myotonic dystrophy, Niemann-Pick disease type C (NP-C), non-Guamanian motor neuron disease with neurofibrillary tangles, postencephalitic parkinsonism, prion protein cerebral amyloid angiopathy, progressive subcortical gliosis, tangle only dementia, tangle-predominant dementia, ganglioglioma, gangliocytoma, subacute sclerosingpan encephalitis, tuberous sclerosis, lipofuscinosis, primary age-related tauopathy (PART), or globular glial tauopathies (GGT).

**28.** The method of claim **26**, wherein the MAPT antisense oligonucleotide is administered to the patient intrathecally, intravenously, subcutaneously, or via intracisternal magna injection.

**29.** A method of reducing MAPT expression in a cell, the method comprising:

contacting the cell with the MAPT antisense oligonucleotide of claim **1**; and

incubating the cell for a time sufficient for degradation of MAPT mRNA, thereby reducing MAPT expression in the cell.

**30.** A method of reducing MAPT expression in a cell, the method comprising:

contacting the cell with the MAPT antisense oligonucleotide of claim **17**; and

incubating the cell for a time sufficient for degradation of MAPT mRNA, thereby reducing MAPT expression in the cell.

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